THE AREA FUNCTION OF THE EXCURSION SKELETON OF THE MÖBIUS SUM AND HORIZON-LOCALIZED CRITERIA FOR THE RIEMANN HYPOTHESIS

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ABSTRACT. Let $M(x) = \sum_{n \le x} \mu(n)$ be the Möbius sum, also called the Mertens function. Its integer zeros cut the walk $n \mapsto M(n)$ into excursions of constant sign, and this paper studies the Riemann hypothesis through the geometry of these excursions. To every completed excursion of length ℓ and height H we attach a triangular skeleton of height $\sqrt{H}$, and we compute exactly its area, its energy and its quadratic mass; the quadratic mass of an excursion equals minus a lag-weighted sum of the Möbius correlations internal to that excursion.

Five results carry the main weight. (i) Purely from disjointness, the number of excursions starting below X and crossing a fixed linear horizon λx_0 is at most $(\log X)/\log\lambda + O(1)$, while a power horizon x_0^A, $A > 1$, is crossed by only $O_A(\log\log X)$ excursions and every horizon $x_0\psi(x_0)$ with $\psi(x) \to \infty$ is crossed by $o(\log X)$ excursions. Combined with the known frequency of large oscillations of M, these bounds show that RH is equivalent to Littlewood's bound $M(t) \ll_\varepsilon t^{1/2+\varepsilon}$ imposed only on the initial part $x_0 < t \le \min\{x_1, \lambda x_0\}$ of every excursion $(x_0, x_1]$, for any fixed $\lambda > \lambda_0 := \exp(\pi/T_0)$, where $T_0 = 3 \cdot 10^{12}$ is a rigorously verified zero height. The older constant $\lambda_* := e^{\pi/\gamma_1}$ remains the uniform threshold obtained from Corollary 4 of Kaczorowski–Pintz alone. (ii) With $\Theta = \sup \Re\rho$, the absolute area of the skeleton up to x is $x^{1+\Theta/2+o(1)}$, and the Dirichlet series of $1/\zeta$ grouped by excursions has absolute block-convergence abscissa Θ. (iii) Inside the RH-equivalent window the values of M are explicit linear functions of μ on the prefix up to χu for every $\chi > \chi_0 := \lambda_0/2$; we split them into scale channels and into a smooth part and a large-prime part which, in the range of the split considered, have size $(1 + o(1))t/\log^2 t$ and opposite signs. On a completed excursion we also obtain a past-only forcing function $P_u(t)$ with an exact sign-coherent channel representation, and a finite Dirichlet transform that interpolates between the raw channel vector and a smoothed Möbius sum on the excursion window. (iv) At every zero v of M, $\sum_{n<v} M(n)^2 = -\sum_{h<v} hC_h^{[v]}$, and the corresponding signed weighted contribution lies at lags $h > v^\alpha$ for every $\alpha < \Theta$. On each completed excursion a dyadic cyclic Fejér argument gives an exact multiscale inequality forcing a positive signed correlation excess whenever the excursion height exceeds the square-root scale; consequently, failure of RH would force such an excess on an unbounded family of linearly short witness excursions. The matching quasi-linear upper bound is isolated as an open arithmetic target, not used as an input. (v) For cutoffs ending at an excursion boundary, and after the affine normalization $w(1) = 1$, the divisor equations compress the 2^J reflection orientations to at most $2N$ candidates. In a different, ambient statement, no $o(N)$ set of divisor rows uniformly controls the fixed-shift quadratic form on all squarefree-supported real vectors with relative error below $1/2$. Several natural strategies (termwise contraction, relaxed divisor constraints, fixed multiplicative depth) cannot by themselves reach the RH scale.

Finally, a unified RVM section constructs the pure midpoint-clock system first: the critical product $P_{\mathrm{RVM}}^{\mathrm{crit}}$, ζ_{RVM}, the exact prime-power/prime inversion $(J_{\mathrm{RVM}}, \pi_{\mathrm{RVM}})$, integer RVM-prime weights, and the critical RVM Mertens baseline. A critical-line homotopy then moves only the synthetic ordinates γ_k^{RVM} to the genuine ordinates γ_k, keeping $\Re s = 1/2$ throughout. The path is proved to be well defined, its induced explicit-formula family is C^1 in the distribution topology, the RVM endpoint is shown to converge uniformly on compact x-ranges, and the smoothed difference of the two endpoints is expressed exactly by the zero-count discrepancy $N_\zeta - N_{\mathrm{RVM}}$. Its endpoint equals ζ if and only if RH; this is explicitly classified as a divisor-based reformulation, not an independent proof. The same section finally treats the distinct true-β ordinate transport $P_{\mathrm{RVM}}^{\mathrm{ord}}$ and $M_{\mathrm{RVM}}^{\mathrm{ord}}$, whose horizontal divisor and pole abscissa remain those of the genuine zeros.

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Contents

1. Introduction 3
1.1. The zeta function and Riemann's hypothesis 4
1.2. Primes 4
1.3. The Möbius function and the Möbius sum 4
1.4. Relations between the Möbius sum and the primes 5
1.5. Oscillation 6
1.6. Liouville's function 6
1.7. Short intervals and Chowla's conjecture 6
1.8. The walk viewpoint 6
1.9. The method 7
1.10. Results 7
1.11. Status classification of the new material 8
1.12. Organization 9
Conventions 9
2. Notation and variables 9
RVM status and dependency audit (v55) 12
3. Excursions and the regularized skeleton 14
4. Exact formulas 16
5. Internal correlations of an excursion 17
6. Where ζ enters 19
7. Bounds for the area curves 20
8. Global criteria equivalent to RH 21
8.1. Exact logarithmic exponents from Pintz's mean and late-maximal-order theorem 23
8.2. Excursion blocks of $1/\zeta$ 25
9. Lacunary excursion families and horizon-localized criteria 27
9.1. Geometric counts 27
9.2. Arithmetic input: Kaczorowski–Pintz 29
9.3. Mathematical intuition: extremal zeros, witness density, and the width of the horizon 32
9.4. Horizon localization 34
9.5. What a genuinely sublinear horizon would require 35
9.6. Fixed-mask Catalan compression inside fractional square horizons 38
9.6.1. Selection energies: completed excursions and open prefixes 41
9.7. A deductive audit below the counting threshold 43
9.8. Wide peaks from cancellation in all short intervals 52
10. Arithmetic selection on the Catalan reflection fibre 54
11. Prefix structure of the localized window 58
11.1. The linear window and the one-step recursion 58
11.2. Complete excursions, past forcing, and channel transforms 67
11.3. The square range as an exact bilinear form of the prefix 73
11.4. Trial-division horizon of an excursion 73
11.5. Large-prime decomposition at level U^2 74
11.6. Local large-prime decomposition of an excursion 77
12. Two-point correlations and the energy of excursions 78
12.1. Signed lag averages and the weighted RH criterion 79
12.2. Lag localization of the energy of an excursion 81
12.3. Uniform two-point cancellation in sublinear intervals 82
12.4. Fejér reduction and the weaker local-variance target 83
12.5. Dyadic multiscale Fejér forcing on completed excursions 85

13. Extreme values in stochastic models of excursions 88
14. A one-frequency model for the net area curve 90
15. Column, row, and diagonal phase sums indexed by zeta zeros 90
15.1. The diagonal is ordinal, not arbitrary 90
15.2. The smooth clock and the obstruction for the actual ordinates 91
15.3. A scale benchmark and what it does not test 94
16. Numerical data 94
17. The RVM arithmetic system and its relation to the classical one 94
17.1. The midpoint Riemann–von Mangoldt clock 95
17.2. The critical RVM zeta function 96
17.3. The RVM prime-power count and prime count 96
17.4. Integer RVM primes: weighted and 0/1 realizations 97
17.5. The critical RVM Mertens object 99
17.6. Exact symmetric Hadamard normalization of the genuine completed factor 100
17.7. Exact defects relative to M, ζ, the ordinary primes and π 101
17.8. A Hamburger endpoint criterion 102
17.9. A critical-line ordinate homotopy from the RVM clock to the genuine ordinates 103
Quadratic transport and the corrected logarithmic interpolation 105
17.10. The true-β RVM ordinate transport 112
17.11. A multiplicative true-β transport with no derivative of zeta 112
17.12. An exact reflection defect 114
17.13. Horizontal zero energy encoded by the rigid product 115
17.14. The vertical resonant accumulation 116
17.15. Poisson geometry and the interpretation of rigid-product plots 118
17.16. The exact umbilical identity with the Mertens function 119
17.17. What the poles prove, and what they do not 120
17.18. A Gaussian-regularized criterion with explicit inversion 120
17.19. The unsmoothed transfer theorem that would complete the route 122
17.20. The residue-preserving model and the scope of the excursion theory 123
18. Conclusions 123
18.1. Status ledger 123
18.2. What the paper does 124
18.3. Horizon localization, area and arithmetic selection 124
18.4. Structural identities 125
18.5. Equivalent criteria and transported estimates 125
18.6. The pure critical RVM arithmetic system and the critical-line path 125
18.7. The rigid-ordinate transported baseline 126
18.8. Limits of the method and open directions 126
References 128

1. Introduction

The first part of this introduction (Sections 1.1–1.8) recalls the objects that the paper uses: the zeta function and the Riemann hypothesis in Riemann's form, the distribution of primes, the Möbius function and its summatory function, and the relations between them. The second part (Sections 1.9–1.12) describes the method and the results.

1.1. **The zeta function and Riemann's hypothesis.** For $\Re s > 1$ the Riemann zeta function is

$$\zeta(s) = \sum_{n\geq 1} \frac{1}{n^s} = \prod_p \Big(1 - \frac{1}{p^s}\Big)^{-1}, \tag{1.1}$$

the product running over the primes. The Euler product (1.1) is the analytic form of unique factorization, and it shows that $\zeta(s) \neq 0$ for $\Re s > 1$. Riemann [50] continued ζ meromorphically to $\mathbb{C}$, with a single simple pole at $s = 1$ of residue 1, and proved the functional equation

$$\xi(s) = \xi(1-s), \qquad \xi(s) := \tfrac{1}{2} s(s-1)\pi^{-s/2}\Gamma(\tfrac{s}{2})\zeta(s), \tag{1.2}$$

where ξ is entire. The zeros of ζ at $s = -2, -4, -6, \dots$ come from the poles of $\Gamma(s/2)$ and are called trivial; all other zeros, the nontrivial zeros $\rho = \beta + i\gamma$, are the zeros of ξ and lie in the critical strip $0 < \beta < 1$. Writing $s = \frac{1}{2} + it$, the function $\xi(\frac{1}{2} + it)$ is real for real t, and Riemann stated that it is very likely that all its zeros are real. This is the Riemann hypothesis (RH): every nontrivial zero satisfies $\beta = \frac{1}{2}$. Infinitely many zeros lie on the critical line (Hardy [14]); all zeros with $0 < \gamma \leq 3 \cdot 10^{12}$ lie on it [47]; the lowest positive ordinate is $\gamma_1 = 14.134725\dots$. Because the zero set is symmetric under $\rho \mapsto \overline{\rho}$ and $\rho \mapsto 1 - \rho$, the quantity

$$\Theta := \sup\{\Re\rho : \ \zeta(\rho) = 0,\ 0 < \Re\rho < 1\} \tag{1.3}$$

satisfies $\frac{1}{2} \leq \Theta \leq 1$, and RH is the statement $\Theta = \frac{1}{2}$. General accounts of the problem are [58, 7]; a spectral interpretation of the zeros through a trace formula on the adèle class space, which reduces RH to the validity of that trace formula, is due to Connes [6].

1.2. **Primes.** Let $\pi(x)$ be the number of primes up to x, $\mathrm{li}(x) = \int_2^x dt/\log t$, and let $\psi(x) = \sum_{n\leq x} \Lambda(n)$, where $\Lambda(n) = \log p$ if $n = p^k$ and $\Lambda(n) = 0$ otherwise. Taking the logarithmic derivative of (1.1),

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n\geq 1} \frac{\Lambda(n)}{n^s} \qquad (\Re s > 1). \tag{1.4}$$

Riemann's paper contains the explicit formula for $J(x) = \sum_{k\geq 1} \pi(x^{1/k})/k$,

$$J(x) = \mathrm{li}(x) - \sum_\rho \mathrm{li}(x^\rho) - \log 2 + \int_x^\infty \frac{dt}{t(t^2-1)\log t} \qquad (x > 1), \tag{1.5}$$

with the sum over ρ taken symmetrically; the analogous formula

$$\psi(x) = x - \sum_\rho \frac{x^\rho}{\rho} - \log 2\pi - \tfrac{1}{2}\log(1 - x^{-2}) \qquad (x > 1,\ x \neq p^k) \tag{1.6}$$

shows directly that every zero contributes an oscillation of size x^β [58, 39]. The prime number theorem $\pi(x) \sim x/\log x$, proved by Hadamard [13] and de la Vallée Poussin [59] from $\zeta(1+it) \neq 0$, is equivalent to $\psi(x) \sim x$. By the explicit formula, for every $\varepsilon > 0$ one has $\psi(x) - x \ll_\varepsilon x^{\Theta+\varepsilon}$, and RH is equivalent to $\pi(x) = \mathrm{li}(x) + O(\sqrt{x}\log x)$ [39]. The best unconditional error term comes from the Vinogradov–Korobov zero-free region [60, 11].

1.3. **The Möbius function and the Möbius sum.** The Möbius function, introduced by Möbius [38] in the inversion formula that bears his name, is $\mu(1) = 1$, $\mu(n) = (-1)^k$ if n is a product of k distinct primes, and $\mu(n) = 0$ if $p^2 \mid n$ for some prime p. It is multiplicative and is characterized by

$$\sum_{d\mid n} \mu(d) = \mathbf{1}_{n=1} \qquad (\mu * 1 = \delta), \tag{1.7}$$

equivalently by the Dirichlet series

$$\frac{1}{\zeta(s)} = \prod_p \Big(1 - \frac{1}{p^s}\Big) = \sum_{n\geq 1} \frac{\mu(n)}{n^s} = s\int_1^\infty M(x) x^{-s-1}\, dx \qquad (\Re s > 1), \tag{1.8}$$

where

$$M(x) = \sum_{n \le x} \mu(n) \tag{1.9}$$

is the *Möbius sum*. It is usually called the Mertens function, after Mertens [36], who tabulated it and conjectured $|M(x)| \le \sqrt{x}$; Stieltjes had claimed $M(x) = O(\sqrt{x})$ in 1885 [19]. Every recursion used below comes from (1.7): for $n \ge 2$, $\mu(n) = -\sum_{d|n,\, d<n} \mu(d)$, and, summing,

$$\sum_{k \le x} M(x/k) = 1 \qquad (x \ge 1) \tag{1.10}$$

[39, §3.2]. Squarefree integers have density $6/\pi^2$: $\sum_{n \le x} \mu(n)^2 = \frac{6}{\pi^2} x + O(\sqrt{x})$. The value $M(n)$ is the determinant of the $n \times n$ Redheffer matrix [48] and, up to sign, the Euler characteristic of a cell complex constructed by Björner [3]. Combinatorial algorithms based on (1.10) evaluate M quickly [8, 18], and Hurst [20] computed $M(x)$ for all $x \le 10^{16}$.

1.4. **Relations between the Möbius sum and the primes.** The Dirichlet series (1.4) and (1.8) give the convolution identities $\Lambda = \mu * \log$ and $\mu \log = -\mu * \Lambda$. Summed, they read

$$\psi(x) = \sum_{d \le x} \mu(d)\, T(x/d), \qquad \sum_{n \le x} \mu(n) \log n = -\sum_{d \le x} \mu(d)\, \psi(x/d), \tag{1.11}$$

with $T(y) = \sum_{m \le y} \log m = y \log y - y + O(\log y)$, and by partial summation

$$M(x) \log x = \sum_{n \le x} \mu(n) \log n + \int_1^x \frac{M(t)}{t}\, dt. \tag{1.12}$$

Thus information on ψ passes to M and back, with losses. At the qualitative level, the prime number theorem is equivalent to $M(x) = o(x)$ [27] and to $\sum_n \mu(n)/n = 0$ [31], and the Vinogradov–Korobov region gives $M(x) \ll x \exp(-c(\log x)^{3/5}(\log\log x)^{-1/5})$ for some absolute $c > 0$ [60, 11]. We also use $\sum_n \mu(n) \log n/n = -1$, which is the value at $s = 1$ of $-\frac{d}{ds}\zeta(s)^{-1} = \sum_n \mu(n)(\log n) n^{-s}$, the convergence at $s = 1$ being a consequence of the prime number theorem with error term.

At the quantitative level, both $\psi(x) - x$ and $M(x)$ are governed by Θ. For M the analogue of (1.6) is, if RH holds and all zeros are simple, for non-integer $x > 0$ [58, §14.27],

$$M(x) = \lim_{k \to \infty} \sum_{|\gamma| < T_k} \frac{x^\rho}{\rho \zeta'(\rho)} - 2 + \sum_{n \ge 1} \frac{(-1)^{n-1} (2\pi/x)^{2n}}{(2n)!\, n\, \zeta(2n+1)}, \tag{1.13}$$

along a sequence $T_k \to \infty$ on whose horizontal segments $1/\zeta(\varsigma + iT_k) \ll T_k^{\varepsilon}$; such a sequence exists under RH [58, §14.16]. The last series collects the residues at the trivial zeros, and the ρ-sum is not known to converge absolutely. In place of $\log x$ weights, the coefficients $1/\zeta'(\rho)$ now appear, which is why M is more sensitive than ψ to the zeros individually. For every $\varepsilon > 0$,

$$M(x) \ll_\varepsilon x^{\Theta + \varepsilon} \qquad \text{and} \qquad |M(x)| \ge x^{\Theta - \varepsilon} \text{ for arbitrarily large } x; \tag{1.14}$$

the upper bound follows from Perron's formula and the bound $1/\zeta(s) \ll |t|^{\varepsilon}$ to the right of $\Re s = \Theta$ (Littlewood [29], [58, §14.25], [45, §4]); the lower bound holds because otherwise $\int_1^\infty M(x) x^{-s-1} dx$ would continue $1/(s\zeta(s))$ analytically beyond $\Re s = \Theta$. This gives Littlewood's criterion: RH holds if and only if $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$ for every $\varepsilon > 0$ [29]. If moreover $|M(x)| \le C\sqrt{x}$ for all x, then $|1/(s\zeta(s))| \le C/(\varsigma - \frac{1}{2})$ for $\varsigma > \frac{1}{2}$, so every zero on the critical line is simple [58, §14.29]. Under RH the best known bound is $M(x) \ll \sqrt{x} \exp((\log x)^{1/2} (\log\log x)^{14})$ (Soundararajan [52], improving Maier–Montgomery [30]); under RH and a hypothesis on $\sum_{\gamma \le T} |\zeta'(\rho)|^{-2}$, Ng [40] obtained a limiting distribution for $e^{-y/2} M(e^y)$.

1.5. **Oscillation.** The Mertens conjecture is false (Odlyzko–te Riele [42]; effective form by Pintz [44], see also [26]); currently $\limsup M(x)/\sqrt{x} > 1.826$ and $\liminf M(x)/\sqrt{x} < -1.837$ [20]. Hence M changes sign, and vanishes, at arbitrarily large integers. Consequently, for every real x there exists a least integer zero $v > x$; whenever a later argument writes "the next zero after x", this existence is the input, while quantitative control of v/x comes separately from Corollary 9.11. Ingham [22] showed that $\limsup M(x)/\sqrt{x} = +\infty$ follows from RH and the linear independence of the ordinates. Kaczorowski and Pintz [25] proved a general sign-change theorem whose frequency is governed by the lowest singularity on the extremal boundary. Applied to $M_0 = M + 2$, it gives the extremal density γ_Θ/π (with the convention $\gamma_\Theta = \infty$ if the boundary is not attained), while their Corollary 4 records the uniform lower bound γ_1/π; the oscillations have size $x^{\Theta-\varepsilon}$, and every range $[Y^{1-\varepsilon}, Y]$ contains large values of both signs (Theorem 9.6). Pintz proved that the mean $\frac{1}{Y}\int_1^Y |M|$ tends to infinity [43], and recently that this mean and the late maximal order of $|M|$ are both $Y^{\Theta+o(1)}$ [45] (Theorem 8.10). These two inputs, the logarithmic frequency of large oscillations and their exact exponent, are the only deep information on M used in the paper.

1.6. **Liouville's function.** In this subsection only, $\lambda(n)$ and $L(x)$ denote Liouville's function and its summatory function; everywhere else λ is a horizon factor and L a block length. Liouville's function $\lambda(n) = (-1)^{\Omega(n)}$ has $\sum \lambda(n)n^{-s} = \zeta(2s)/\zeta(s)$; hence $\lambda = \mathbf{1}_\square * \mu$ and $\mu(n) = \sum_{d^2|n} \mu(d)\lambda(n/d^2)$, so with $L(x) = \sum_{n\le x} \lambda(n)$,

$$L(x) = \sum_{d\le\sqrt{x}} M(x/d^2), \qquad M(x) = \sum_{d\le\sqrt{x}} \mu(d)L(x/d^2), \tag{1.15}$$

and RH is equivalent to $L(x) \ll_\varepsilon x^{1/2+\varepsilon}$. Pólya's conjecture $L(x) \le 0$ is false [16]. Most correlation results below are proved for λ and μ by the same arguments; we state them for μ.

1.7. **Short intervals and Chowla's conjecture.** Chowla's conjecture [5] asserts, for distinct shifts $h_1, \dots, h_k$, $\sum_{n\le x} \mu(n+h_1)\cdots\mu(n+h_k) = o(x)$ (Cesàro form); the logarithmic form replaces $\sum_{n\le x}$ by $\sum_{n\le x} \frac{1}{n}$ and $o(x)$ by $o(\log x)$, and is implied by the Cesàro form. Sarnak [51] observed that Chowla implies Möbius disjointness from zero-entropy dynamics, and Tao proved the equivalence of the logarithmic Chowla and logarithmic Sarnak conjectures [55]. The known cases are the logarithmic two-point case (Tao [54]), the logarithmic odd-order cases (Tao–Teräväinen [56, 57]), and Cesàro statements averaged over the shifts: Matomäki–Radziwiłł–Tao [33] prove $\sum_{h\le K} |\sum_{n\le x} \mu(n)\mu(n+h)| = o(Kx)$ for every $K = K(x) \to \infty$. The two-point Cesàro case for a fixed shift remains open. These results rest on the Matomäki–Radziwiłł theorem [32] on multiplicative functions in almost all short intervals. For *all* short intervals, Matomäki–Teräväinen [35] prove $\sum_{x<n\le x+x^\theta} \mu(n) \ll x^\theta(\log x)^{-1/3+\varepsilon}$ for $\theta > 0.55$ and Fourier uniformity for $H \ge x^{3/5+\varepsilon}$, extended by Matomäki–Shao–Tao–Teräväinen [34]; primes in all short intervals are due to Huxley [21] and, with the Guth–Maynard large-value estimate, apply for every fixed exponent $\theta > 17/30$, i.e. to intervals of length x^θ [12]. The distinction between all intervals and almost all intervals is essential here, because the endpoints of excursions are selected by M itself.

1.8. **The walk viewpoint.** Denjoy [10] observed that, if the signs $\mu(n)$ on the squarefree integers behaved like independent coin tosses, $M(x)$ would be of size $x^{1/2+o(1)}$, which by Littlewood's criterion is exactly RH; this is a heuristic, not an equivalence. Wintner [61] introduced random multiplicative functions. Harper [15] proves low-moment and typical-size results exhibiting below-square-root scales in the specific random multiplicative models considered there; this is not a pointwise uniform bound for every realization. Areas and quadratic functionals of lattice excursions are classical [53, 41, 49, 23, 9, 4]; here they are applied to the deterministic walk $n \mapsto M(n)$, whose steps are $\mu(n) \in \{-1, 0, 1\}$.

1.9. **The method.** The integer zeros of M cut $[0,\infty)$ into zero runs and excursions $I_\nu = (x_{0,\nu}, x_{1,\nu}]$ on which M has constant sign σ_ν and does not vanish; we write ℓ_ν for the length and H_ν for the height of I_ν. Every analytic statement about M is then read through three kinds of excursion data.

Geometry. Each completed excursion carries a triangular skeleton of height $\sqrt{H_\nu}$ whose signed area is $\sigma_\nu\sqrt{H_\nu}\ell_\nu/2$. Its quadratic mass $Q_\nu = \sum_{n\in I_\nu} M(n)^2$ satisfies the exact bridge identity $Q_\nu = -\sum_h hC_h^{(\nu)}$, where $C_h^{(\nu)}$ are the Möbius correlations inside I_ν, and H_ν, ℓ_ν bound Q_ν and the linear mass sharply (Sections 3–5).

Counting. The excursions are disjoint intervals. Any family of disjoint intervals whose members satisfy $x_1 > \lambda x_0$ has at most $\log X/\log\lambda + O(1)$ members starting below X; if $x_1 > x_0^A$ with fixed $A > 1$ there are only $O_A(\log\log X)$ such members, and if $x_1 > x_0\psi(x_0)$ with $\psi(x)\to\infty$ there are $o(\log X)$. The Kaczorowski–Pintz theorem produces extremal witness density γ_Θ/π when RH fails. Since all zeros up to $T_0 = 3\cdot 10^{12}$ are on the critical line, this gives at least $(T_0/\pi - o(1))\log X$ large-witness excursions under $\Theta > 1/2$. Comparing the two counts localizes the large values of M (Section 9).

Arithmetic inside an excursion. Starting from a zero u of M, the values of M on $(u, 2u)$ are exact linear functions of μ on $[1, u]$, and on $(u, u^2]$ exact bilinear functions. For a completed excursion the no-crossing condition gives more: a past-only forcing function $P_u(t)$ is a sign-coherent sum of scaled values $M(t/k)$ throughout the excursion, and a finite Dirichlet transform of the channel vector redistributes the same information between scales and the excursion window. These identities, the recursion (1.10) and the factorization $n = ap$ with a large prime p describe how an excursion is built (Section 11).

1.10. **Results.** *Horizon localization* (Theorem 9.15, Corollaries 9.17 and 9.18). If a horizon function $\mathfrak{h}$ is crossed by at most $(\kappa + o(1))\log X$ excursions starting below X, with $\kappa < T_0/\pi$, then RH is equivalent to Littlewood's bound, with a constant depending only on ε, on the part of each excursion lying below its horizon. In particular, for every fixed $\lambda > \lambda_0$,

$$\mathrm{RH} \iff |M(t)| \le C_\varepsilon t^{1/2+\varepsilon} \ \text{ for } \ x_0 < t \le \min\{x_1, \lambda x_0\} \text{ on every excursion}, \tag{1.16}$$

where $T_0 = 3\cdot 10^{12}$ and $\lambda_0 = e^{\pi/T_0}$. The older $\lambda_* = e^{\pi/\gamma_1}$ is the balance point obtained from the uniform Corollary 4 density; the extremal Theorem 1 density together with the verified zero height yields the much smaller explicit linear factor λ_0. For M itself the excursions return quickly: $x_1 \le x_0^{1+o(1)}$ (Corollary 9.11).

Exact exponents (Corollaries 8.11 and 8.13, Theorem 8.16).

$$\sum_{n\le X}|M(n)| = X^{1+\Theta+o(1)}, \qquad \sum_{n\le X} M(n)^2 = X^{1+2\Theta+o(1)}, \tag{1.17}$$

$$\mathcal{E}(X) = X^{1+\Theta+o(1)}, \qquad \int_0^X |M_{\mathrm{s}}| = X^{1+\Theta/2+o(1)}, \tag{1.18}$$

where $\mathcal{E}$ is the chord energy of the skeleton and M_{s} the skeleton; and the series $\sum_I D_I(s)$ of $1/\zeta$ grouped by completed excursions has abscissa of absolute block convergence Θ. In particular the skeleton area satisfies $\int_0^X |M_{\mathrm{s}}| \ll X^{5/4+\varepsilon}$ for all $\varepsilon > 0$ if and only if RH holds.

The localized window (Section 11). For a zero u of M and $u < t \le \lambda u < 2u$,

$$M(t) = -\sum_{d\le \lambda u/2} \mu(d)\big(\lfloor t/d\rfloor - \lfloor u/d\rfloor\big), \tag{1.19}$$

so the RH-equivalent window depends only on μ up to χu, for every $\chi > \chi_0 = \lambda_0/2$. We decompose $M(t)$ into scale channels $M(t/k) - M(u/k)$ and prove a channel-descent inequality with an explicit loss. For a completed excursion $(u, v]$ we further define the past-forcing function $P_u(t)$ and prove

$$P_u(t) = \sum_{k\le q_u(t)} M(t/k), \qquad \sigma P_u(t) = \sum_{k\le q_u(t)} |M(t/k)| \quad (u < t < v), \tag{S1.1}$$

so the full no-crossing excursion gives a positive channel constraint on a quantity depending only on the past. Its channel vector also has the exact Dirichlet transform of Proposition 11.35. Finally we split M into a smooth and a large-prime part which, in the range of the split considered, both have size $(1+o(1))t/\log^2 t$, with opposite signs.

Correlations (Section 12). At every zero v of M, $\sum_{n<v} M(n)^2 = -\sum_{h<v} hC_h^{[v]}$. Averaged Chowla makes the lags $h \le \sqrt{v}$ negligible at the scale v^2, and Pintz's exponent makes the lags $h \le v^\alpha$, $\alpha < \Theta$, carry a vanishing proportion of $\sum_{n<v} M(n)^2$.

Effect of a violation of RH across Fejér scales. For a completed excursion let q be the number of nonzero increments, let m_{dy} be the number of dyadic scales after zero-padding to the next power of two, and put $\mathfrak{K}_I = \sum_{R \text{ dyadic}} F_R/R$. The cyclic boundary of a one-sign excursion has a favorable sign, and the dyadic inequality proved in Proposition 12.25 gives

$$\mathfrak{K}_I \ge \frac{H_I^2}{2m_{\mathrm{dy}}} - \frac{m_{\mathrm{dy}} q}{2}. \tag{1.20}$$

Consequently, if RH fails, then for every $\frac{1}{2} < \alpha < \Theta$ and every fixed $\lambda > \lambda_0$ an unbounded family of completed witness excursions with $v \le \lambda u$ satisfies $\mathfrak{K}_I \gg u^{2\alpha}/\log u$. Thus a violation of RH has a concrete signature inside the excursion formalism: a superlinear positive excess of signed Fejér correlations across dyadic scales. The converse estimate $\mathfrak{K}_I \ll_\lambda u(\log u)^A$ is not known here; it is isolated in Remark 12.27 as the open arithmetic estimate that would close this route.

Orientation and selection (Section 10). Reflecting whole excursions leaves M^2 unchanged. For a cutoff N ending at an excursion boundary and after imposing $w(1) = 1$, the central divisor equations compress the resulting 2^{J_N} reflection fibre to at most $2N$ candidates. Separately, on the ambient space of real vectors supported on squarefree integers, no $o(N)$ set of divisor rows uniformly controls a fixed-shift quadratic form with relative error below $1/2$.

Stochastic models (Section 13). Under a uniform tail hypothesis on a normalized excursion statistic, each level of lacunarity of the counted family removes one logarithm from the extreme-value scale.

1.11. **Status classification of the new material.** To prevent algebraic identities, imported theorems, divisor-based reformulations and open analytic steps from being conflated, the new material in this paper is classified throughout in four categories.

1. Exact algebraic identities. This class includes the excursion skeleton and its area and correlation identities, the prefix and channel formulae, the reflection fibre and its Gram selector, the finite prime-power/prime inversion in the RVM system, the exact symmetric Hadamard normalization, the distributional endpoint formula for the critical-line ordinate path, and the true-β transport identities. These statements require no unproved analytic estimate once their objects are defined.

2. Theorems proved using published results. This class includes the horizon and return-localization consequences after the hypotheses of the Kaczorowski–Pintz theorem are verified for the Mellin transform of M_0, the exact logarithmic exponents obtained from Pintz's 2026 theorem, and the short-interval and squarefree-pair consequences obtained from the cited Matomäki–Teräväinen and Mirsky estimates. The external input is identified at the point where it is used.

3. Equivalent criteria that explicitly encode the genuine zero divisor. The endpoint identification $\zeta_1 = \zeta$ for the critical-line ordinate homotopy, the true-β RVM transport, its horizontal energies, the Gaussian-regularized criterion, and the Hamburger endpoint condition for ζ_{line} belong here. They are useful exact reformulations and geometric diagnostics, but they are not independent proofs of RH because the genuine zero divisor is part of the input.

4. Hypotheses and open targets needed to complete a route. These include the one-excursion first-passage closure, the missing multiscale Fejér upper bound, the quantitative Lambert-clock estimates needed for the RVM square-root and unit-cell claims, the uniform stationary-phase remainder in

the ordinate-homotopy resonance calculation, the cell-positivity remainder, the unsmoothed bidirectional Mellin multiplier theorem, and the endpoint Dirichlet-series problem for ζ_{line}. Numerical observations are kept outside the proof chain.

No statement in Category 3, and no statement conditional on Category 4, is used below as an unconditional proof of RH.

1.12. **Organization.** Sections 3–5 set up the excursion geometry. Sections 6–8 relate it to ζ and give the area bounds, the global criteria, the exact exponents and the block series of $1/\zeta$. Section 9 contains the counting lemmas, the Kaczorowski–Pintz input, the horizon localization, the Catalan reference laws and the wide peaks. Section 10 treats the reflection fibre, Section 11 the localized window and the large-prime decomposition, Section 12 the correlations, and Section 13 stochastic models. After the one-frequency model, Section 15 compares column, row and ordinal diagonal zero-phase sums, proves convergence for the smooth zero-counting clock, and isolates the unresolved phase remainder for the actual ordinates. After the numerical data, Section 17 separates the exact critical RVM product and finite inversion identities from the quantitative Lambert-clock estimates that remain analytic targets. It then proves a critical-line homotopy from the RVM ordinates to the genuine ordinates, including a distributional C^1 transport, compact-uniform convergence at the RVM endpoint, and an exact route-independent formula for the smoothed endpoint difference. The endpoint equality with ζ is identified explicitly as RH. The section then treats the distinct true-β ordinate transport, identifies the horizontal zero energy carried by $P^{\mathrm{ord}}_{\mathrm{RVM}}$, records the resulting RH/Lindelöf divisor criteria, develops a Gaussian-regularized criterion with its inversion hypotheses made explicit, and isolates the unsmoothed multiplier theorem needed for pointwise transfer. The conclusions follow.

Conventions. $\rho = \beta + i\gamma$ runs over nontrivial zeros; $\Theta \in [\frac{1}{2}, 1]$ is defined in (1.3); $\gamma_1 = 14.134725\ldots$; RH means $\Theta = 1/2$. $M(x) = M(\lfloor x \rfloor)$ for real x and $M(x) = 0$ for $x < 1$. Statements taken from the literature carry the reference in their heading; heuristics are marked as such.

2. Notation and variables

Global symbols keep the meanings below. Symbols marked "local" are declared again where used and never leave that statement or subsection. Constants c, C are positive real numbers and may change from line to line. Unless a statement says otherwise, sequence and lag indices belong to $\mathbb{N} = \{1, 2, \ldots\}$, integer endpoints belong to $\mathbb{N}_0$, and physical-time and height parameters are real. Frequencies and phase exponents are real; zero ordinates are real and zero real parts lie in $(0, 1)$. The synthetic coefficients c_n^{RVM} and $\mu_{\mathrm{RVM}}(n)$ are real, not restricted to $\{-1, 0, 1\}$; weights in $(0, 1)$ are asserted only under the stated positivity hypotheses. In finite lattice blocks, L, H, u, v are integers whenever integer sums or first-arrival times use them; otherwise the stated floor convention applies. Exponents and loss parameters are real and fixed before the asymptotic limit; all o and O bounds are uniform only over the ranges expressly stated. The analytic variable is $s = \varsigma + it \in \mathbb{C}$, with $\varsigma, t \in \mathbb{R}$; powers of positive real numbers use their real logarithm. Test functions may be complex valued. Pairings with distributions are complex-linear integrals in the declared variable, not Hermitian inner products. For real-part kernels such as Q_φ, identities written using $\Re\widehat{g}_\varphi$ are first read for real-valued test functions and then extended complex-linearly. Improper integrals and series are assigned the indicated ordering or cutoff; meromorphic identities at common zeros or poles mean continuation after cancellation. A logarithm of a positive real product is real; a complex logarithm requires a simply connected zero-free domain. An inverse Mellin symbol alone does not assert the existence of a function or distribution.

Symbol	Meaning	Scope
$\rho = \beta + i\gamma$	Nontrivial zero of ζ; β denotes its real part in zero-mode discussions	Global

Symbol	Meaning	Scope
Θ	$\sup\{\Re\rho\}$	Global
γ_1	Lowest positive ordinate, $14.134725\ldots$	Global
$\lambda_*, \lambda_0, \chi_0$	e^{π/γ_1}, e^{π/T_0}, $\lambda_0/2$	Global
T_0, γ_Θ	Verified height $3\cdot 10^{12}$; first extremal ordinate (or ∞)	Global
θ_{sub}	Sublinear additive-horizon exponent in $u+u^{\theta_{\mathrm{sub}}}$	Section 9
$\varepsilon_{\mathrm{L}},\ H_{\mathrm{L}}(u),\ r_{\mathrm{L}}(u)$	Littlewood-level margin; $u^{1/2+\varepsilon_{\mathrm{L}}}$; first-passage distance to that level	Section 9
λ	Horizon factor, $\lambda>1$ (Liouville's $\lambda(n)$ only in Section 1.6)	Global
ψ	Function with $\psi(x)\to\infty$ in the superlinear horizon $x\psi(x)$	Section 9
$M,\ M_0$	Mertens function; $M_0=M+2$ (normalization of [25])	Global
$I_\nu=(x_{0,\nu},x_{1,\nu}]$	Completed excursion; also $(u_\nu,v_\nu]$	Global
σ_ν	Orientation ±1 of I_ν	Global
$\ell_\nu,\ H_\nu,\ \tau_\nu$	Length, height, first apex	Global
$a_\nu,\ b_\nu$	$\tau_\nu-x_{0,\nu}$, $x_{1,\nu}-\tau_\nu$	Global
$\mathcal{Z}=[p,q]$	Zero run	Section 4
$u(x),\ L_x$	Last zero $\le x$; terminal length $\lfloor x\rfloor-u(x)$	Global
$M_{\mathrm{s}},\ M_{\mathrm{sk}}$	Regularized / unregularized skeleton	Global
$\mathcal{A}_+,\ \mathcal{A}_-,\ \mathcal{A},\ \mathcal{A}_{\mathrm{abs}}$	Positive, signed negative, net, absolute area	Global
$\widetilde{A}$	$\int_0^x M$	Section 6
ch_ν	Chord of $\mathcal{A}$ over I_ν	Theorem 8.9
$\mathfrak{q}_\nu=q_\nu$	Number of active steps $\#\{n\in I_\nu:\mu(n)\neq0\}$; q in a general block	Global
$Q_\nu,\ \mathfrak{b}_\nu$	$\sum_{n\in I_\nu}M(n)^2$, $\sum_{n\in I_\nu}\lvert M(n)\rvert$	Global
$Q_{\mathrm{term}},\ \mathfrak{b}_{\mathrm{term}}$	Same for the terminal open prefix	Global
$B_1,\ B_2,\ \mathcal{E}$	$\sum_{n\le X}\lvert M\rvert$, $\sum_{n\le X}M^2$, chord energy	Global
$C_h^{(\nu)},\ C_h^{[v]},\ C_h^{(u,X]}$	Lag-h correlation in I_ν, in $(0,v]$, in $(u,X]$	Global
$C_g^{\langle h\rangle}$	Lag-g correlation in a truncated prefix of length h	Local
S_t	Oriented partial sum inside a declared block	Local
$\mathcal{F},\ E_{\mathcal{F}}(X)$	Family of excursions and its counting function	Section 9
$\mathfrak{h},\ \mathcal{F}_{\mathfrak{h}}$	Horizon and crossing family	Section 9
A	Power exponent in $x_1>x_0^A$	Section 9
η	Loss in the Kaczorowski–Pintz exponent $\Theta-\eta$	Global
κ	Coefficient of a crossing count $(\kappa+o(1))\log X$	Theorem 9.15
$\mathcal{S},\ L^{\#},\ r_I$	Squarefree clock, compressed length, support time	Catalan subsection
L	Length of a declared block (also support length $L^{\#}$ in Catalan lemmas)	Local
$p_{\mathrm{Cat}},\ p_{\mathrm{Cat}}^{\max},\ p_{\mathrm{open}}$	Reference-law tail weights	Catalan subsection
$J_N,\ E_\nu$	Number of completed excursions up to a zero cutoff N; active support of I_ν	Arithmetic selection
$\mathbf{z},\ \boldsymbol{\sigma}$	0/1 reflection vector; ±1 orientation vector	Arithmetic selection
$\mathfrak{D}_N,\ G_N$	Divisor-defect functional and its orientation Gram matrix	Arithmetic selection
$m_\nu,\ \phi_\nu,\ \mathscr{C}^{\mathrm{div}}$	Support centre, positive oriented increments, central divisor matrix	Arithmetic selection
$\mathbf{D}_N,\ \mathsf{A}_h,\ \mathfrak{S}_h$	Divisor matrix; fixed-shift form; pair density	Arithmetic selection
$\theta_{\mathrm{w}},\ \theta'$	Short-interval exponents >0.55	Wide peaks
$\Lambda,\ h_u,\ \omega_u$	Horizon factor, block length, Lipschitz modulus	Wide peaks
G	$(H-h_u)_+$	Proposition 9.56

Symbol	Meaning	Scope
μ_u	$\mu \mathbf{1}_{[1,u]}$	Section 11
$\Delta_k,\ \mathcal{R}_K$	Channel increment, deep remainder	Section 11
$P_u(t),\ q_u(t)$	Past-forcing function; local channel cutoff $\lfloor t/(u+1)\rfloor$	Section 11
$\Phi_s(n)$	Generalized Jordan weight $(\mathrm{id}_s * \mu)(n)$	Section 11
ϑ	Normalizing exponent in $\lvert M(y)\rvert / y^{\vartheta}$	Section 11
$c_y(t,u),\ K_y$	Kernel coefficients (argument y of $M(y)$)	Section 11
$\Upsilon,\ \mathcal{C}$	Multiplicity-weighted and coherent coefficients	Section 11
$\mathbf{a}=(a_d),\ D_{\mathbf{a}}$	Relaxed prefix vector and its divisor sum	Section 11
P	Prefix cutoff in layer statements	Section 11
$P^+(n)$	Largest prime factor	Section 11
$U,\ W,\ M_U^{\mathrm{sm}},\ M_U^{\mathrm{lp}},\ \widetilde{m}$	Large-prime level, t/U^2, smooth and large-prime parts, $\sum_{a\le y}\mu(a)/a$	Section 11
$\Omega(n),\ \omega(n)$	Prime factors with / without multiplicity	Global
$D_I(s)$	Dirichlet block of a completed excursion	Section 8
$\mathscr{P}_{m,j},\ \mathscr{D}_{\alpha_{\mathrm{ph}}}(N)$	Zero-ordinate phase matrix; weighted diagonal partial sum	Section 15
$C^{\mathrm{pts}}_{\alpha_{\mathrm{ph}}}(N),\ C^{\mathrm{path}}_{\alpha_{\mathrm{ph}}}(N)$	Terminal point average and path average of the diagonal phase sum	Section 15
$\bar{\gamma}_k^{\mathrm{RVM}},\ \widetilde{\gamma}_k^{\mathrm{RVM,rnd}},\ \chi_k^{\mathrm{RVM,ph}},\ \mathscr{R}_{\mathrm{RVM,ph}}(N)$	Smooth RVM clock; rounded artificial clock; unit complex phase factor; path remainder	Section 15
$\gamma_k^{\mathrm{RVM}},\ \rho_k^{\mathrm{RVM}},\ \mathcal{N}_{\mathrm{RVM}}$	Midpoint RVM ordinate, critical synthetic zero, and midpoint clock	Section 17
$P^{\mathrm{crit}}_{\mathrm{RVM}},\ \zeta_{\mathrm{RVM}},\ J_{\mathrm{RVM}},\ \pi_{\mathrm{RVM}}$	Pure critical RVM product, zeta, prime-power count, and prime count	Section 17
$J^{*,\mathbb{Z}}_{\mathrm{RVM}}$	Right-continuous integer-step extension $J^*_{\mathrm{RVM}}(\lfloor x\rfloor)$ for real $x\ge 1$	RVM discretization
$\Delta^{\mathrm{cont}}_{\mathrm{RVM}},\ \Delta^{\mathrm{step}}_{\mathrm{RVM}}$	Real count defects $J_{\mathbb{Z},\mathrm{RVM}} - J^*_{\mathrm{RVM}}$ and $J_{\mathbb{Z},\mathrm{RVM}} - J^{*,\mathbb{Z}}_{\mathrm{RVM}}$	RVM discretization
$c_n^{\mathrm{RVM}},\ \mathcal{P}_{\mathrm{RVM}},\ E_{\mathrm{RVM}}$	Real weight on integer support; conditional 0/1 RVM prime set; integer true-prime count defect	Section 17
$\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}},\ E^{\mathrm{ord}}_{\mathrm{RVM}}(T),\ Q^{\mathrm{ord}}_{\mathrm{RVM,a}}(T)$	True-β transport multiplier, horizontal energy and level count	Section 17
$M^{\mathrm{crit}}_{\mathrm{RVM}},\ P^{\mathrm{ord}}_{\mathrm{RVM}},\ M^{\mathrm{ord}}_{\mathrm{RVM}}$	Critical RVM Mertens baseline; true-β rigid product and transported Mertens object	Section 17
$\tau_k^{\mathrm{RVM}},\ \tau_{k,\kappa}^{\mathrm{RVM}}$	Real midpoint ordinate and real path ordinate; $k\in\mathbb{N}$, $0\le\kappa\le 1$	Quadratic transport
$E^{\mathrm{tr}}_{\mathrm{RVM}},\ A^{\mathrm{tr}}_{\mathrm{RVM}}$	Integer count discrepancy and nonnegative real quadratic-displacement sum	Quadratic transport
D_{RVM}	Real uniform bound $\sup_k \lvert\gamma_k - \gamma_k^{\mathrm{RVM}}\rvert$	Quadratic transport
$C_{\mathrm{RVM}}(\kappa)$	Real normalization defect; negative in the path interior for nonzero displacement	Logarithmic interpolation
c_{RVM}	Real vertical limit of $\log\lvert\zeta_{\mathrm{RVM}}\rvert$, when that limit is assumed	Conditional rigidity statement
$L_{\mathrm{RVM},\kappa}(s)$	Complex logarithm on the declared zero-free quadrant	Logarithmic interpolation
$K_{\mathrm{RVM,b}}(t)$	Nonnegative real interpolation-error majorant; real $b>0$	Logarithmic interpolation
$s=\varsigma+i\tau$	Complex variable	Global
$e(x),\ \xi$	$e^{2\pi i x}$; frequency	Section 12
K	Lag cutoff in averaged-Chowla statements	Section 12

Symbol	Meaning	Scope
$R,\ F_R,\ V_R$	Fejér scale and sums	Section 12
$\mathfrak{K}_I$	Signed dyadic multiscale Fejér sum $\sum_R F_R/R$ on a completed excursion	Section 12.5
ι	Exponent gap in $L \geq u^{3/5+\iota}$	Section 12
$(\mathfrak{X}, \mathcal{G}, \Pr),\ Z_k$	Probability space and excursion statistic	Section 13
$\mathcal{N}(X)$	Deterministic count bound	Section 13
$\kappa_0,\ \varpi$	Tail constant and exponent	Section 13

RVM status and dependency audit (v55). The table below uses the new four-row transport table, with its statements corrected to match the proofs in this manuscript. The original row identifiers 10, 10a, 10b and 9 are retained. The path parameter $\kappa \in [0,1]$ is real and local to the RVM homotopy. In the density-one statements, $s = \sigma + it \in \mathbb{C}$, $t \in \mathbb{R}$, and the real $\sigma \in (1/2, 1)$ is fixed before $t \to +\infty$.

No.	Statement	Previous status	Corrected current status
10	Loss of vertical rigidity at every $\kappa > 0$	First-order argument; unproved	Theorem under **P** and a proved RVM vertical limit. The corresponding disjunction also requires that limit. See Audit proposition A5.
10a	Monotone quadratic transport, including multiplicities; $A^{\mathrm{tr}}_{\mathrm{RVM}}(T) \sim T \log\log T/(\pi \log T)$	New result	Unconditional theorem, with cutoff and boundary estimates. See Audit proposition A3.
10b	Density-one affine logarithmic interpolation, uniformly in $\kappa \in [0,1]$, after subtracting $C_{\mathrm{RVM}}(\kappa)$	New result	Unconditional theorem. The normalization correction is generally nonzero; the original normalized path is not an exact logarithmic geodesic. See Audit proposition A4.
9	Propagation of a uniform vertical bound from $\kappa = 0$ to a positive path parameter	Claimed failure of uniform transport	Under the premises of row 10, no positive κ retains that bound. This conclusion concerns this bound along this path; it does not exclude every possible transfer argument or RH proof.

The sharp vertical-limit statement T_{RVM} is claimed in the supplied note, but its complete proof is not included here. It therefore remains an explicit premise of the rigidity statement. The unconditional quadratic transport and corrected interpolation do not depend on that premise.

What must be specified in the proposed three-hypothesis chain. Hypothesis 3c is stronger than a signed estimate of size $O(\log\log x)$, but its compatibility with 3a depends on which real-variable extension is used. For the continuous extension $\Delta^{\mathrm{cont}}_{\mathrm{RVM}} = J_{\mathbb{Z},\mathrm{RVM}} - J^{*}_{\mathrm{RVM}}$, 3a contradicts 3c by Audit proposition A1. If the RVM count is sampled only at the integers and extended instead by $J^{*,\mathbb{Z}}_{\mathrm{RVM}}(x) = J^{*}_{\mathrm{RVM}}(\lfloor x \rfloor)$, then $\Delta^{\mathrm{step}}_{\mathrm{RVM}} = J_{\mathbb{Z},\mathrm{RVM}} - J^{*,\mathbb{Z}}_{\mathrm{RVM}}$ has two atomic contributions at each integer and cancellation is possible. Thus step-version 3c is a legitimate open hypothesis, not a consequence of 3a or of an $O(\log\log x)$ signed bound. To use it in the factorization with ζ_{RVM}, hypothesis 3b must also include the continuous-to-step Mellin–Stieltjes correction, or else identify the integer-step object directly. Independently, the sharp vertical-growth statement T_{RVM} still needs its full proof in this document.

Audit proposition A1 (Continuous-extension weighted-variation obstruction). Suppose J^{*}_{RVM} is continuous at every integer $n > 2$, and $c^{\mathrm{RVM}}_n \sim 1/\log n$ with eventual positivity. Define the finite

counting sum

$$J_{\mathbb{Z},\mathrm{RVM}}(x) = \sum_{m\geq 1} m^{-1}\pi_{\mathrm{RVM}}(\lfloor x^{1/m}\rfloor). \tag{A1a}$$

and put

$$\Delta^{\mathrm{cont}}_{\mathrm{RVM}}(x) := J_{\mathbb{Z},\mathrm{RVM}}(x) - J^{*}_{\mathrm{RVM}}(x). \tag{A1b}$$

Then, for every real $1/2 < \sigma \leq 1$,

$$\int_1^\infty x^{-\sigma}|d\Delta^{\mathrm{cont}}_{\mathrm{RVM}}(x)| = \infty. \tag{S2.1}$$

Here the integral is interpreted as weighted total variation if $\Delta^{\mathrm{cont}}_{\mathrm{RVM}}$ is locally of bounded variation; lack of that regularity also prevents 3c from holding as a finite measure statement.

Proof. Let $\mathcal{E} = \{a^m : a, m \in \mathbb{N},\ a, m \geq 2\}$ be the perfect powers. At any integer $n > 2$ outside $\mathcal{E}$, only the $m = 1$ summand jumps, so the jump of $\Delta^{\mathrm{cont}}_{\mathrm{RVM}}$ is c^{RVM}_n. Total variation therefore dominates $\sum_{n\notin\mathcal{E},\ n>2} n^{-\sigma}|c^{\mathrm{RVM}}_n|$. For $\sigma > 1/2$,

$$\sum_{n\in\mathcal{E}} n^{-\sigma} \leq \sum_{a\geq 2}\sum_{m\geq 2} a^{-m\sigma} = \sum_{a\geq 2}\frac{a^{-2\sigma}}{1-a^{-\sigma}} < \infty. \tag{S2.2}$$

But $\sum_{n\geq 3} n^{-\sigma}/\log n = \infty$ when $\sigma \leq 1$. Removing the convergent perfect-power contribution leaves a divergent sum, and eventual positivity and asymptotic equivalence complete the proof. The required continuity is supplied here by compact-uniform convergence of the RVM explicit-formula series on $x > 1$ (away from the fixed cutoff). □

For comparison, the integer-step extension has, at a non-perfect-power integer $n > 2$, the jump

$$\Delta\Delta^{\mathrm{step}}_{\mathrm{RVM}}(n) = c^{\mathrm{RVM}}_n - \big(J^{*}_{\mathrm{RVM}}(n) - J^{*}_{\mathrm{RVM}}(n-1)\big). \tag{S2.2a}$$

No lower bound for the modulus of this difference follows from 3a. Hence Audit proposition A1 does not refute the step formulation of 3c; that formulation requires a separate weighted variation estimate, including the perfect-power jumps and the initial cells.

Absolute coefficients and finite exceptional weights. Write, for $\Re s > 1$,

$$f_{\mathrm{RVM}}(s) = \sum_{n\geq 2} c^{\mathrm{RVM}}_n \sum_{m\geq 1}\frac{n^{-ms}}{m}, \qquad e^{-f_{\mathrm{RVM}}(s)} = \sum_{n\geq 1}\mu_{\mathrm{RVM}}(n)n^{-s}. \tag{S2.3}$$

For nonnegative weights the coefficients of f_{RVM} are nonnegative, so expansion of the exponential gives the coefficientwise bound by $e^{f_{\mathrm{RVM}}}$. Finitely many real negative weights are handled by replacing them with their absolute values. Their finite Euler factors are holomorphic, nonvanishing and bounded on every $\Re s \geq a > 0$; they are not generally Dirichlet polynomials. If the identification hypothesis 3b is valid near 1 and h_{RVM} is holomorphic there, the residue-one pole of ζ_{RVM} yields

$$\sum_{n\leq x}|\mu_{\mathrm{RVM}}(n)| \leq x^c\sum_{n\geq 1}|\mu_{\mathrm{RVM}}(n)|n^{-c} \ll x^c(c-1)^{-1} \ll x\log x, \qquad c = 1+1/\log x, \quad x \geq 3. \tag{S2.4}$$

The weaker assumption $c^{\mathrm{RVM}}_n \sim 1/\log n$ alone gives only $\log\zeta_{\mathbb{Z},\mathrm{RVM}}(c) = (1+o(1))\log(1/(c-1))$, which does not justify the exact simple-pole bound.

Audit proposition A2 (A consistent conditional Perron route). Let $F_{\mathrm{RVM}}(s) = \sum_{n\geq 1} a^{\mathrm{RVM}}_n n^{-s}$ converge absolutely for $\Re s > 1$. Assume that it extends holomorphically to $\Re s > 1/2$, is bounded on each closed half-plane $\Re s \geq a > 1/2$, and satisfies $\sum_{n\geq 1}|a^{\mathrm{RVM}}_n|n^{-c} \ll (c-1)^{-1}$ as $c \downarrow 1$. Then $\sum_{n\leq x} a^{\mathrm{RVM}}_n = O_\varepsilon(x^{1/2+\varepsilon})$. This applies to $F_{\mathrm{RVM}} = 1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ only when these consistent analytic assumptions have independently been established.

Proof. Fix $a \in (1/2, 1)$. Absolute convergence on a line $\Re s = c > 1$ gives

$$a_n^{\mathrm{RVM}} = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} F_{\mathrm{RVM}}(c+it) n^{c+it}\, dt. \tag{S2.5}$$

Shift this averaging rectangle to $\Re s = a$. For each fixed n the horizontal terms divided by $2T$ tend to zero, by boundedness on the strip. Consequently $|a_n^{\mathrm{RVM}}| \ll_a n^a$. This is the local coefficient estimate that a global absolute-mass bound alone does not supply. Use truncated Perron at a half-integer $x \geq 3$, with $c = 1 + 1/\log x$ and $T = x$. Its absolute error is bounded by

$$\sum_{n\geq 1} |a_n^{\mathrm{RVM}}| (x/n)^c \min\{1, (T|\log(x/n)|)^{-1}\} \ll_a x^a \log x + \log x. \tag{S2.6}$$

Indeed, for $x/2 < n < 2x$ the coefficient bound and $|\log(x/n)| \gg |x-n|/x$ reduce this to a harmonic sum; the half-integer choice bounds $|x-n|$ below by $1/2$. Outside that range use absolute convergence and the assumed bound on the line c. There are no poles in the contour shift. The vertical integral on $\Re s = a$ is $O_a(x^a \log x)$ and the two horizontal integrals are $O_a(x^c/(T\log x)) = O_a(1/\log x)$. Thus the partial sum is $O_a(x^a \log x)$. Every real cutoff has the same partial sum as a half-integer cutoff within distance $1/2$. Taking $a = 1/2 + \min(\varepsilon/2, 1/4)$ absorbs the logarithm. □

3. Excursions and the regularized skeleton

Let μ be the Möbius function and

$$M(x) = \sum_{n\leq x} \mu(n), \qquad x \geq 0, \tag{3.1}$$

piecewise constant and right-continuous, with jumps $\mu(n) \in \{-1, 0, 1\}$. Throughout the paper, for real $x \geq 0$ we use the convention

$$M(x) = M(\lfloor x \rfloor), \tag{3.2}$$

and we set $M(x) = 0$ for $x < 1$. Thus expressions such as $M(t/k)$ always mean $M(\lfloor t/k \rfloor)$. Unless stated otherwise, every summation index is an integer; a real upper limit in a sum is interpreted through its floor. The segmentation of M by zero runs and the record of its extrema are standard computational objects; Hurst [20] computed $M(x)$ for all $x \leq 10^{16}$. The area and quadratic functionals of lattice excursions are classical [53, 41, 49, 23, 9, 4]; here they are applied to the deterministic Möbius walk.

Definition 3.1 (Excursions and zero runs)**.** An *excursion* is an interval

$$I_\nu = (x_{0,\nu}, x_{1,\nu}] \tag{3.3}$$

with integer endpoints, $M(x_{0,\nu}) = M(x_{1,\nu}) = 0$, $M(x_{0,\nu}+1) = \sigma_\nu \in \{\pm 1\}$, and $\sigma_\nu M(n) \geq 1$ for every integer $x_{0,\nu} < n < x_{1,\nu}$. Thus $x_{1,\nu}$ is the first return to zero after the departure from $x_{0,\nu}$. Consecutive excursions may be separated by a *zero run* $[x_{1,\nu}, x_{0,\nu+1}]$, a maximal set of consecutive integers on which $M = 0$; it may consist of a single point. The first excursion is $(0, 2]$. Since M assumes positive and negative values at arbitrarily large arguments ($M(x) = \Omega_\pm(\sqrt{x})$, Remark 7.6) and its integer jumps belong to $\{-1, 0, 1\}$, between two integers where M has opposite signs it must vanish at an integer. Hence M has arbitrarily large zeros and every excursion is finite. We also write $u_\nu = x_{0,\nu}$ and $v_\nu = x_{1,\nu}$.

Put

$$\ell_\nu = x_{1,\nu} - x_{0,\nu}, \qquad H_\nu = \max_{x_{0,\nu}<n<x_{1,\nu}} |M(n)|, \tag{3.4}$$

$$\tau_\nu = \min\{n \in I_\nu : |M(n)| = H_\nu\}, \qquad a_\nu = \tau_\nu - x_{0,\nu}, \qquad b_\nu = x_{1,\nu} - \tau_\nu. \tag{3.5}$$

Every excursion has $H_\nu \geq 1$ and $\ell_\nu \geq 2$.

Definition 3.2 (Completed excursions and open prefixes)**.** An excursion in the sense of Definition 3.1 is always *completed*: both $M(x_{0,\nu}) = 0$ and $M(x_{1,\nu}) = 0$ are part of the definition. A segment $(u, x]$ with $M(u) = 0$ and $M(n) \neq 0$ for $u < n \leq x$, with no return condition at x, is an *open prefix*; the terminal block of Definition 3.6 is one. Any estimate whose proof uses the backward bound $|M(u+r)| \leq x_{1,\nu} - u - r$ holds for completed excursions only and is never applied to an open prefix.

Lemma 3.3 (Height and step count)**.** *For every excursion,*

$$H_\nu \leq \min(a_\nu, b_\nu) \leq \frac{\ell_\nu}{2}. \tag{3.6}$$

Proof. The steps of M are 0 or ± 1, so at least H_ν steps are needed to reach height H_ν from zero and at least H_ν more to return to zero. □

Definition 3.4 (Crossing and touching zero runs)**.** The zero run $[x_{1,\nu}, x_{0,\nu+1}]$ between consecutive excursions is a *crossing run* if $\sigma_{\nu+1} = -\sigma_\nu$ and a *touching run* if $\sigma_{\nu+1} = \sigma_\nu$.

Definition 3.5 (Regularized skeleton)**.** The *skeleton* M_{s} is the continuous, piecewise linear function defined on every excursion by

$$M_{\mathrm{s}}(t) = \begin{cases} \sigma_\nu \sqrt{H_\nu} \dfrac{t - x_{0,\nu}}{a_\nu}, & x_{0,\nu} \leq t \leq \tau_\nu, \\ \sigma_\nu \sqrt{H_\nu} \dfrac{x_{1,\nu} - t}{b_\nu}, & \tau_\nu \leq t \leq x_{1,\nu}, \end{cases} \tag{3.7}$$

and by $M_{\mathrm{s}} \equiv 0$ on zero runs. Here "regularized" refers only to the normalization of the apex from H_ν to $H_\nu^{1/2}$; no analytic smoothing operation is intended. The *unregularized skeleton* M_{sk} is defined in the same way with peak $\sigma_\nu H_\nu$.

Definition 3.6 (Terminal block of a truncated computation)**.** Let $N \geq 2$ and $u = \max\{n \leq N : M(n) = 0\}$. If $u = N$ there is no terminal block. Otherwise $M(n) \neq 0$ for $u < n \leq N$, and since M has steps $0, \pm 1$ all these values have the sign of $M(N)$. Put

$$\sigma_{\mathrm{term}} = \operatorname{sgn} M(N), \qquad H_{\mathrm{term}} = \max_{u<n\leq N} |M(n)|, \qquad \tau_{\mathrm{term}} = \min\{n \in (u, N] : |M(n)| = H_{\mathrm{term}}\}. \tag{3.8}$$

On $[u, N]$ the regularized skeleton is the segment from $(u, 0)$ to $(\tau_{\mathrm{term}}, \sigma_{\mathrm{term}}\sqrt{H_{\mathrm{term}}})$ followed, if $\tau_{\mathrm{term}} < N$, by the segment ending at $(N, \sigma_{\mathrm{term}}\sqrt{|M(N)|})$. Its signed area is

$$\Delta\mathcal{A}_{\mathrm{term}}(u, N] = \frac{\sigma_{\mathrm{term}}}{2}\Big[\sqrt{H_{\mathrm{term}}}(N - u) + \sqrt{|M(N)|}(N - \tau_{\mathrm{term}})\Big]. \tag{3.9}$$

The terminal skeleton is provisional: after the next return to zero the completed excursion acquires a different descending side. Independently of it, for real x with last zero $u(x) \leq \lfloor x \rfloor$ define

$$\mathfrak{b}_{\mathrm{term}}(x) := \sum_{u(x)<n\leq\lfloor x\rfloor} |M(n)|, \qquad Q_{\mathrm{term}}(x) := \sum_{u(x)<n\leq\lfloor x\rfloor} M(n)^2, \qquad L_x := \lfloor x \rfloor - u(x). \tag{3.10}$$

These depend only on M.

Definition 3.7 (The three curves)**.** With $M_{\mathrm{s}}^{\pm} = \max(\pm M_{\mathrm{s}}, 0)$,

$$\mathcal{A}_+(x) = \int_0^x M_{\mathrm{s}}^+(t)\,dt, \qquad \mathcal{A}_-(x) = -\int_0^x M_{\mathrm{s}}^-(t)\,dt, \qquad \mathcal{A}(x) = \int_0^x M_{\mathrm{s}}(t)\,dt. \tag{3.11}$$

$\mathcal{A}_+$ is nondecreasing, $\mathcal{A}_-$ is nonincreasing, and both are constant on zero runs and on excursions of the opposite sign. The curve $\mathcal{A}_-$ records negative area with its sign; it is not the absolute value of the negative area.

Remark 3.8 (The family of g-skeletons). For any $g : \mathbb{N}_0 \to [0,\infty)$ with $g(0) = 0$ and $g(h) > 0$ for $h \geq 1$, one may replace the apex height $\sqrt{H}$ by $g(H)$. The signed area increment becomes $\frac{1}{2}\sigma_\nu g(H_\nu)\ell_\nu$. The correlation identities of Section 5 concern the Möbius increments and do not depend on g. The choice $g(H) = \sqrt{H}$ is a normalization.

4. Exact formulas

Lemma 4.1 (Increment per excursion)**.** *For every completed excursion,*

$$\mathcal{A}(x_{1,\nu}) - \mathcal{A}(x_{0,\nu}) = \sigma_\nu \frac{\sqrt{H_\nu}\,\ell_\nu}{2}, \qquad \sum_{n\in I_\nu} M_{\mathrm{s}}(n) = \sigma_\nu \sqrt{H_\nu}\frac{\ell_\nu}{2}. \tag{4.1}$$

Proof. The two pieces are triangles of height $\sqrt{H_\nu}$ and bases a_ν, b_ν. For the discrete sum, the ascending values at $j = 1,\dots,a_\nu$ sum to $\sqrt{H_\nu}\sum_{j\leq a_\nu} j/a_\nu = \sqrt{H_\nu}(a_\nu+1)/2$; the descending values at $\tau_\nu + r$, $r = 1,\dots,b_\nu$, sum to $\sqrt{H_\nu}\sum_{r\leq b_\nu}(b_\nu - r)/b_\nu = \sqrt{H_\nu}(b_\nu - 1)/2$. □

Proposition 4.2 (Explicit form of the three curves)**.** *Let $x \in I_\nu$. Then*

$$\mathcal{A}(x) = \begin{cases} \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\dfrac{(x - x_{0,\nu})^2}{2a_\nu}, & x \leq \tau_\nu, \\[2ex] \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\left(\dfrac{a_\nu}{2} + (x-\tau_\nu) - \dfrac{(x-\tau_\nu)^2}{2b_\nu}\right), & x \geq \tau_\nu. \end{cases} \tag{4.2}$$

At excursion endpoints,

$$\mathcal{A}(x_{1,k}) = \sum_{\nu\leq k}\sigma_\nu\frac{\sqrt{H_\nu}\ell_\nu}{2}, \qquad \mathcal{A}_+(x_{1,k}) = \sum_{\substack{\nu\leq k\\ \sigma_\nu=+1}}\frac{\sqrt{H_\nu}\ell_\nu}{2}, \qquad \mathcal{A}_-(x_{1,k}) = -\sum_{\substack{\nu\leq k\\ \sigma_\nu=-1}}\frac{\sqrt{H_\nu}\ell_\nu}{2}. \tag{4.3}$$

The curves are piecewise quadratic and C^1.

Proof. Integrate (3.7); the endpoint formulas follow from Lemma 4.1. □

A point y is a *strict inflection point* of a function if there exists $\delta > 0$ such that the function is strictly convex on one side of y and strictly concave on the other, within distance δ. The second derivative need not exist at y.

Lemma 4.3 (Critical geometry of $\mathcal{A}$)**.** *Let $\mathcal{Z} = [p,q]$ be the zero run between an excursion of sign σ_- ending at p and the next excursion of sign σ_+ starting at q.*

(i) *Every apex τ_ν is a strict inflection point.*
(ii) *If $\sigma_+ = -\sigma_-$, then $\mathcal{Z}$ is an extremal plateau; it is a strict local extremum exactly when $p = q$.*
(iii) *If $\sigma_+ = \sigma_-$, then $\mathcal{A}$ has no strict local extremum on $\mathcal{Z}$ and is monotone through the run; when $p < q$ it has a constant plateau on $[p,q]$. Its concavity has sign $-\sigma_-$ immediately to the left and $+\sigma_+$ immediately to the right. If $p = q$ the point is a strict inflection; if $p < q$, no point of $\mathcal{Z}$ is a strict inflection, and the change of concavity is carried by the whole plateau.*
(iv) *Strict local extrema occur exactly at isolated crossing zeros, and plateaus of extrema exactly at crossing runs of positive length; every point of such a plateau is a non-strict local extremum.*

Proof. On the descending side of the left excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_-$ and $\operatorname{sgn}\mathcal{A}'' = -\sigma_-$. On the ascending side of the right excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_+$ and $\operatorname{sgn}\mathcal{A}'' = \sigma_+$. Inside an excursion $\mathcal{A}' = M_{\mathrm{s}} \neq 0$, so no extremum occurs there. On $\mathcal{Z}$, $\mathcal{A}' = 0$. The assertions follow, and at every apex the sign of $\mathcal{A}''$ reverses. □

5. Internal correlations of an excursion

For an integer $L \geq 1$ and a real block $x_1, \ldots, x_L$ (local notation: here x_j are the entries of the block, not excursion endpoints) put

$$S_t = \sum_{j=1}^{t} x_j, \qquad C_h = \sum_{j=1}^{L-h} x_j x_{j+h}. \tag{5.1}$$

Lemma 5.1 (Triangular identity)**.** *For every real finite sequence,*

$$\sum_{t=1}^{L-1} S_t^2 + \sum_{h=1}^{L-1} hC_h = S_L \sum_{t=1}^{L-1} S_t. \tag{5.2}$$

Proof. Expanding the partial sums,

$$\sum_{t=1}^{L} S_t^2 = \sum_{i,j=1}^{L} x_i x_j (L + 1 - \max(i,j)). \tag{5.3}$$

Write $\max(i,j) = \frac{i+j+|i-j|}{2}$. Since $\frac{1}{2}\sum_{i,j}(i+j)x_i x_j = S_L \sum_i i x_i$ and each unordered pair at lag $h = |i-j| \geq 1$ occurs twice,

$$\sum_{t=1}^{L} S_t^2 = (L+1)S_L^2 - S_L \sum_{i=1}^{L} i x_i - \frac{1}{2}\sum_{i,j=1}^{L} x_i x_j |i-j| = (L+1)S_L^2 - S_L \sum_{i=1}^{L} i x_i - \sum_{h=1}^{L-1} hC_h. \tag{5.4}$$

Moreover $\sum_i i x_i = LS_L - \sum_{t=1}^{L-1} S_t$. Hence $\sum_{t=1}^{L} S_t^2 = S_L^2 + S_L \sum_{t<L} S_t - \sum_h hC_h$; subtract S_L^2. □

For a bridge, $S_L = 0$, so the preceding identity gives

$$\sum_{t=1}^{L-1} S_t^2 = -\sum_{h=1}^{L-1} hC_h. \tag{5.5}$$

Remark 5.2 (Hermitian form). For complex x_j and $C_h = \sum_{j=1}^{L-h} x_j \overline{x_{j+h}}$, the same computation applies. The ordered pairs (i,j) and (j,i) at lag h contribute $x_i\overline{x_j} + x_j\overline{x_i}$, so each lag contributes $2\Re C_h$ to $\sum_{i,j} x_i\overline{x_j}|i-j|$, and the cross term becomes $\Re(\overline{S_L}\sum_i i x_i)$. Hence

$$\sum_{t=1}^{L-1} |S_t|^2 + \sum_{h=1}^{L-1} h\Re C_h = \Re\left(\overline{S_L}\sum_{t=1}^{L-1} S_t\right). \tag{5.6}$$

For a completed excursion orient the increments by

$$x_j = \sigma_\nu \mu(x_{0,\nu} + j), \qquad S_t = |M(x_{0,\nu} + t)|. \tag{5.7}$$

Then $S_t > 0$ for $1 \leq t < \ell_\nu$ and $S_{\ell_\nu} = 0$. Put

$$Q_\nu = \sum_{t=1}^{\ell_\nu - 1} S_t^2, \qquad C_h^{(\nu)} = \sum_{j=1}^{\ell_\nu - h} x_j x_{j+h}, \qquad \text{so that} \qquad Q_\nu = -\sum_{h=1}^{\ell_\nu - 1} hC_h^{(\nu)}. \tag{5.8}$$

Lemma 5.3 (Zeroth moment and extreme lag)**.** *With $q_\nu = \sum_j x_j^2$,*

$$\sum_{h=1}^{\ell_\nu - 1} C_h^{(\nu)} = -\frac{q_\nu}{2}, \qquad C_{\ell_\nu - 1}^{(\nu)} = -1, \tag{5.9}$$

and q_ν is even and at least 2*.*

Proof. $0 = (\sum_j x_j)^2 = q_\nu + 2\sum_h C_h^{(\nu)}$. The first oriented step is $+1$; immediately before the first return the path is at height 1, so the last step is -1. The bridge condition forces equally many positive and negative active steps. □

Proposition 5.4 (Quadratic decomposition by excursions)**.** *For real $x \geq 1$ let $u(x)$ and L_x be as in Definition 3.6, and for $1 \leq j \leq L_x$ put $y_j = \mu(u(x)+j)$ and $C_h^{\mathrm{term}} = \sum_{j=1}^{L_x-h} y_j y_{j+h}$. Then*

$$\sum_{n\leq x} M(n)^2 = \sum_{\nu: x_{1,\nu}\leq u(x)} Q_\nu + Q_{\mathrm{term}}(x), \tag{5.10}$$

$$Q_{\mathrm{term}}(x) = M(\lfloor x\rfloor)^2 + M(\lfloor x\rfloor)\sum_{t=1}^{L_x-1} M(u(x)+t) - \sum_{h=1}^{L_x-1} hC_h^{\mathrm{term}}, \tag{5.11}$$

with $Q_{\mathrm{term}}(x) = 0$ if $L_x = 0$. At a zero x of M,

$$\sum_{n\leq x} M(n)^2 = \sum_{x_{1,\nu}\leq x} Q_\nu = -\sum_{x_{1,\nu}\leq x}\sum_h hC_h^{(\nu)}. \tag{5.12}$$

Proof. Every integer up to $u(x)$ lies either in a zero run or in a unique excursion interior. Apply Lemma 5.1 to the terminal block and add its final term $S_{L_x}^2$. □

Lemma 5.5 (Elementary bound for an open terminal block)**.** *Let $u \in \mathbb{N}_0$ satisfy $M(u) = 0$ and let $L \geq 0$. Then $|M(u+r)| \leq r$ for $1 \leq r \leq L$, and*

$$\sum_{r=1}^{L} M(u+r)^2 \leq \frac{L(L+1)(2L+1)}{6}. \tag{5.13}$$

In particular $Q_{\mathrm{term}}(x) \ll L_x^3$. The bridge bound (5.21)*, whose leading coefficient is $1/12$ rather than $1/3$, applies only to completed excursions.*

Proof. $M(u+r) = \sum_{j\leq r}\mu(u+j)$ with $|\mu| \leq 1$; square and sum. □

Thus, in this decomposition, the quadratic sum is expressed through lag correlations internal to excursions only: no correlation between increments of different completed excursions appears on the right-hand side. (The global autocorrelations of μ do contain such pairs; their weighted total is recovered in Proposition 12.2.)

Let

$$\mathfrak{b}_\nu = \sum_{t=1}^{\ell_\nu-1} S_t = \sum_{n\in I_\nu} |M(n)|. \tag{5.14}$$

Cauchy–Schwarz gives

$$\mathfrak{b}_\nu^2 \leq (\ell_\nu - 1)Q_\nu, \tag{5.15}$$

with equality exactly for a flat excursion of height 1. To separate support from sign geometry, let $P_h^{(\nu)} = \#\{j : x_j x_{j+h} \neq 0\}$ and let $D_h^{(\nu)}$ count the opposite-sign pairs among them. Then

$$C_h^{(\nu)} = P_h^{(\nu)} - 2D_h^{(\nu)}, \qquad Q_\nu = \sum_{h=1}^{\ell_\nu-1} h\big(2D_h^{(\nu)} - P_h^{(\nu)}\big). \tag{5.16}$$

Proposition 5.6 (Sharp bounds at fixed length and height)**.** *Let $L \geq 2$ be an integer and let $(S_t)_{0\leq t\leq L}$ be integers with $S_0 = S_L = 0$, $S_t \geq 1$ for $0 < t < L$, $S_t - S_{t-1} \in \{-1, 0, 1\}$, and $H = \max S_t$. Then $L \geq 2H$ and*

$$H^2 + L - 2H \leq \mathfrak{b} \leq H(L-H), \tag{5.17}$$

$$\frac{2H^3+H}{3} + L - 2H \leq Q \leq \frac{2H^3+H}{3} + (L-2H)H^2. \tag{5.18}$$

All four bounds are attained. In particular

$$Q \geq \frac{2H^3+H}{3}, \qquad Q \geq L-1. \tag{5.19}$$

Proof. Before and after the first visit to H the path visits each level $1, \dots, H-1$. Together with one visit to H, these $2H-1$ designated interior times contribute $2(1+\cdots+(H-1))+H = H^2$ to $\mathfrak{b}$ and $2\sum_{j<H} j^2 + H^2 = (2H^3+H)/3$ to Q. The remaining $L-2H$ interior times lie at heights between 1 and H, which gives the four inequalities. The lower bounds are attained by inserting the $L-2H$ remaining steps as horizontal steps at height 1 into the tent path of height H; the upper bounds by inserting all of them as horizontal steps at height H. Both constructions keep $S_t \geq 1$ on $0 < t < L$ and have increments in $\{-1, 0, 1\}$. □

Corollary 5.7 (Sharp fixed-length bounds)**.** *For $L \geq 2$,*

$$\mathfrak{b} \leq \left\lfloor \frac{L^2}{4} \right\rfloor, \qquad Q \leq \begin{cases} L(L^2+2)/12, & L \text{ even}, \\ L(L^2-1)/12, & L \text{ odd}, \end{cases} \tag{5.20}$$

and in particular

$$Q \leq \frac{L^3+2L}{12}. \tag{5.21}$$

Both bounds are attained by the tent path, with one horizontal step at maximal height when L is odd.

Proof. Use $S_t \leq \min(t, L-t)$ and sum this bound and its square. □

6. Where ζ enters

The net area curve $\mathcal{A}$ depends nonlinearly on M. The dependence enters through $\sqrt{H_\nu}$ and through the selection of $x_{0,\nu}, \tau_\nu, x_{1,\nu}$; no closed expression of its Mellin transform in terms of ζ is known. The nodes $x_{0,\nu}, x_{1,\nu}$ are zeros of M. If RH holds and all zeros are simple, there is a sequence $T_k \to \infty$ such that, for non-integer x,

$$M(x) = \lim_{k\to\infty} \sum_{|\gamma|<T_k} \frac{x^\rho}{\rho\,\zeta'(\rho)} - 2 + \sum_{n\geq 1} \frac{(-1)^{n-1}(2\pi/x)^{2n}}{(2n)!\,n\,\zeta(2n+1)} \tag{6.1}$$

[58, §14.27]. For $\widetilde{A}(x) := \int_0^x M(t)\,dt$ one has $\widetilde{A}(1) = 0$ (the value of M at the single point 1 does not affect the integral). Integration by parts therefore gives, for $\Re s > 1$,

$$\int_1^\infty \widetilde{A}(x) x^{-s-2}\,dx = \frac{1}{s(s+1)\zeta(s)}. \tag{6.2}$$

The residues of $x^{s+1}/(s(s+1)\zeta(s))$ are

$$\sum_\rho \frac{x^{\rho+1}}{\rho(\rho+1)\zeta'(\rho)} - 2x + 12 + (\text{residues at the trivial zeros}), \tag{6.3}$$

where $x/\zeta(0) = -2x$ is the residue at $s=0$ and $1/((-1)\zeta(-1)) = 12$ the residue at $s=-1$. Under RH and simplicity of the zeros, the contour argument of [58, §14.27], with the extra decay $(s+1)^{-1}$, turns (6.3) into an identity for $\widetilde{A}(x)$ with the ρ-sum taken as $\lim_k \sum_{|\gamma|<T_k}$; absolute convergence of the ρ-sum needs moment bounds for $1/\zeta'(\rho)$ of the type assumed in [40]. Formula (6.3) is not used in any proof. The area of M_{sk} differs from $\widetilde{A}$ by explicit local terms.

Proposition 6.1 (Dipole identity)**.** *For each excursion let its* legs *be the integer sets $J = (u_\nu, \tau_\nu] \cap \mathbb{Z}$ and $(\tau_\nu, v_\nu] \cap \mathbb{Z}$, with midpoints $c_J = (u_\nu + \tau_\nu)/2$ and $(\tau_\nu + v_\nu)/2$. Then for every zero v_k,*

$$\int_0^{v_k} M_{\mathrm{sk}}(t)\,dt = \widetilde{A}(v_k) + \sum_{J\subset(0,v_k]} \sum_{m\in J} \mu(m)\,(m - c_J). \tag{6.4}$$

Proof. Since M is constant on each $[n, n+1)$, $\int_a^b M(t)\,dt = \sum_{n=a}^{b-1} M(n)$ for integers $a < b$. On an ascent leg, $\int_u^\tau M_{\mathrm{sk}} = \sigma_\nu H a/2$ and $\int_u^\tau M = \sum_{m=u+1}^{\tau} \mu(m)(\tau - m)$, while $\sigma_\nu H = \sum_{m=u+1}^{\tau} \mu(m)$; the difference is $\sum_m \mu(m)(a/2 - \tau + m) = \sum_m \mu(m)(m - c_J)$. On a descent leg, $\int_\tau^v M_{\mathrm{sk}} = \sigma_\nu H b/2$ and $\int_\tau^v M = b\,\sigma_\nu H + \sum_{m=\tau+1}^{v} \mu(m)(v-m)$ with $\sum_{m=\tau+1}^{v} \mu(m) = -\sigma_\nu H$; the difference is $\sum_m \mu(m)(m - (v - b/2))$, and $v - b/2 = (\tau + v)/2 = c_J$. On zero runs both integrands vanish. Sum over legs. □

7. Bounds for the area curves

Lemma 7.1 (Transfer)**.** *Let $x \geq 1$ and let $x^* \geq x$ be a zero of M (in a computation truncated at $N \geq x$, take $x^* = N$ and the terminal skeleton of Definition 3.6). If $|M(n)| \leq C_0\, n^{1/2}$ for all $1 \leq n \leq x^*$ (so necessarily $C_0 \geq 1$), then*

$$|M_{\mathrm{s}}(t)| \leq C_0^{1/2} t^{1/4} \quad (0 \leq t \leq x), \qquad \mathcal{A}_+(x) + |\mathcal{A}_-(x)| \leq \tfrac{4}{5}\, C_0^{1/2} x^{5/4}. \tag{7.1}$$

Proof. For $t \geq \tau_\nu$, $|M_{\mathrm{s}}(t)| \leq \sqrt{H_\nu} \leq C_0^{1/2} \tau_\nu^{1/4} \leq C_0^{1/2} t^{1/4}$. For $u_\nu \leq t \leq \tau_\nu$ the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$ lies below the chord of $C_0^{1/2} t^{1/4}$ over $[u_\nu, \tau_\nu]$, whose endpoints $C_0^{1/2} u_\nu^{1/4} \geq 0$ and $C_0^{1/2} \tau_\nu^{1/4} \geq \sqrt{H_\nu}$ (as $\tau_\nu \leq x^*$) are at least as high; that chord lies below $C_0^{1/2} t^{1/4}$ by concavity. Integrate. □

Lemma 7.2 (A concave majorant)**.** *For $y > e$ let $\phi(y) = (\log y)^{3/5} (\log\log y)^{-1/5}$ and, for $\kappa > 0$, $F(y) = y^{1/2} e^{-\kappa\phi(y)}$. There is $y_0 = y_0(\kappa)$ such that F is increasing and concave on $[y_0, \infty)$.*

Proof. With $y = e^s$, $s > 1$, and $g(s) = s^{3/5} (\log s)^{-1/5}$, one has $F = \exp(s/2 - \kappa g(s))$. Put $h(s) = \tfrac{1}{2} - \kappa g'(s)$. Then

$$y F'(y) = F\, h(s), \qquad y^2 F''(y) = F\big(h(s)^2 - h(s) - \kappa g''(s)\big). \tag{7.2}$$

Since $g'(s) \to 0$ and $g''(s) \to 0$, $h(s) \to \tfrac{1}{2}$ and $h^2 - h - \kappa g'' \to -\tfrac{1}{4}$. Hence $F' > 0$ and $F'' < 0$ for large y. □

Corollary 7.3. *From the Vinogradov–Korobov bound* [60]*: there is an absolute constant $c > 0$ with*

$$M(x) \ll x \exp\big(-c\,\phi(x)\big) \qquad (x > e), \tag{7.3}$$

one obtains

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll x^{3/2} \exp\big(-\tfrac{c}{2}\,\phi(x)\big). \tag{7.4}$$

Proof. Let K be such that $|M(y)| \leq K y\, e^{-c\phi(y)}$ for $y \geq y_1$, and let F be as in Lemma 7.2 with $\kappa = c/2$, so that $\sqrt{|M(y)|} \leq K^{1/2} F(y)$ for $y \geq y_1$. Only finitely many excursions start below $Y_0 := \max(y_0, y_1)$; they are finite, so they contribute $O(1)$ to the area. Consider an excursion with $u_\nu \geq Y_0$. Its apex satisfies $\sqrt{H_\nu} = \sqrt{|M(\tau_\nu)|} \leq K^{1/2} F(\tau_\nu)$; the bound is taken at the apex τ_ν, which lies in the excursion, and not at its future endpoint v_ν. For $t \geq \tau_\nu$ monotonicity of F gives $|M_{\mathrm{s}}(t)| \leq \sqrt{H_\nu} \leq K^{1/2} F(t)$. For $u_\nu \leq t \leq \tau_\nu$, $|M_{\mathrm{s}}(t)|$ is the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$. This chord lies below the chord from $(u_\nu, K^{1/2} F(u_\nu))$ to $(\tau_\nu, K^{1/2} F(\tau_\nu))$, which has both endpoints at least as high, and the latter lies below $K^{1/2} F$ on $[u_\nu, \tau_\nu] \subset [y_0, \infty)$ by concavity. Hence $|M_{\mathrm{s}}(t)| \leq K^{1/2} F(t)$ for all $t \geq Y_0$ outside the finitely many early excursions, and since F is increasing,

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \leq O(1) + K^{1/2} \int_{Y_0}^{x} F(t)\,dt \leq O(1) + K^{1/2} x F(x) = O(1) + K^{1/2} x^{3/2} e^{-\frac{c}{2}\phi(x)}. \tag{7.5}$$

□

Corollary 7.4. *If RH holds, then*

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll_\varepsilon x^{5/4+\varepsilon}. \tag{7.6}$$

Consequently, if $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \gg x^{5/4+\delta}$ along an unbounded sequence for some fixed $\delta > 0$, then RH is false. By Corollary 8.13 the converse also holds: (7.6) *is equivalent to RH.*

Proof. It suffices to treat $0 < \varepsilon < \frac{3}{2}$, so that $t \mapsto t^{1/4+\varepsilon/2}$ is concave. Under RH, $|M(y)| \le C_\varepsilon y^{1/2+\varepsilon}$, so $\sqrt{H_\nu} \le C_\varepsilon^{1/2} \tau_\nu^{1/4+\varepsilon/2}$. The chord argument of Lemma 7.1, with the concave majorant $C_\varepsilon^{1/2} t^{1/4+\varepsilon/2}$ in place of $C_0^{1/2} t^{1/4}$, gives $|M_{\mathrm{s}}(t)| \le C_\varepsilon^{1/2} t^{1/4+\varepsilon/2}$ and, after integration, $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll_\varepsilon x^{5/4+\varepsilon/2}$. □

Remark 7.5 (Endpoint-indexed bounds). From $\ell_\nu \ge 2H_\nu$ and $\sqrt{H_\nu}\ell_\nu \ll v_\nu^{5/4+\varepsilon}$ one deduces $H_\nu \ll v_\nu^{5/6+O(\varepsilon)}$, indexed by the future endpoint v_ν. For a general family of disjoint intervals this gives no bound at interior points. For M itself Corollary 9.11 gives $v_\nu \le u_\nu^{1+\varepsilon}$ for all late excursions, so the bound becomes $|M(t)| \ll t^{5/6+O(\varepsilon)}$; the sharp statement is the exact area exponent of Corollary 8.13.

Remark 7.6 (Oscillation of the heights). The Mertens conjecture is false [42, 26]; the current bounds are due to Hurst [20], who proves $\limsup M(x)/\sqrt{x} > 1.826054$ and $\liminf M(x)/\sqrt{x} < -1.837625$. The excursion containing a point n with $M(n) > 1.826054\sqrt{n}$ has $u_\nu < n$, so $H_\nu > 1.826054\, u_\nu^{1/2}$ for infinitely many positive excursions, and similarly with 1.837625 for negative ones. A lower bound indexed by v_ν does not follow, since v_ν may be much larger than n.

8. Global criteria equivalent to RH

The Mellin identity and the global L^1/L^2 equivalences below are consequences of Littlewood's criterion and partial summation; the excursion decompositions and pruning lemmas rewrite them in terms of excursions. For $\Re s > 1$,

$$\int_1^\infty M(x) x^{-s-1}\, dx = \frac{1}{s\,\zeta(s)}. \tag{8.1}$$

If $\int_1^\infty |M(x)| x^{-\varsigma-1} dx < \infty$ for every $\varsigma > 1/2$, the left side continues analytically to $\Re s > 1/2$, so ζ has no zeros there.

Proposition 8.1 (L^1 criterion). *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{v_\nu \le x} \mathfrak{b}_\nu + \mathfrak{b}_{\mathrm{term}}(x) = \sum_{n \le x} |M(n)| \ll x^{3/2+\varepsilon}. \tag{8.2}$$

Proof. RH gives $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$, hence the bound. Conversely, fix $\varsigma_0 > 1/2$ and $0 < \varepsilon < \varsigma_0 - 1/2$. Partial summation with $\sum_{n \le x} |M(n)| \ll x^{3/2+\varepsilon}$ gives $\sum_n |M(n)| n^{-\varsigma_0-1} < \infty$; since M is constant on $[n, n+1)$, $\int_1^\infty |M(x)| x^{-\Re s-1} dx \le \int_1^\infty |M(x)| x^{-\varsigma_0-1} dx < \infty$ uniformly for $\Re s \ge \varsigma_0$. Hence $F(s) := \int_1^\infty M(x) x^{-s-1} dx$ converges absolutely and uniformly there and is holomorphic on $\Re s > \varsigma_0$. By (8.1), $F = 1/(s\zeta(s))$ on $\Re s > 1$; by uniqueness of analytic continuation the meromorphic function $1/(s\zeta(s))$ coincides with F on $\Re s > \varsigma_0$, so it has no poles there, i.e. ζ has no zeros there. As $\varsigma_0 > 1/2$ is arbitrary, RH follows. □

Lemma 8.2 (Pruning). $\mathfrak{b}_\nu \le H_\nu(\ell_\nu - 1) \le \ell_\nu^2/2$, *hence*

$$\sum_{v_\nu \le x,\ \ell_\nu \le L_0} \mathfrak{b}_\nu \le \frac{x L_0}{2}. \tag{8.3}$$

Corollary 8.3 (Reduction to long excursions). *For fixed $\delta \ge 0$ and $L_0 = x^{1/2-\delta}$, RH holds if and only if*

$$\sum_{\substack{v_\nu \le x \\ \ell_\nu > L_0}} \mathfrak{b}_\nu + \mathbf{1}_{\{L_x > L_0\}} \mathfrak{b}_{\mathrm{term}}(x) \ll x^{3/2+\varepsilon} \quad \text{for every } \varepsilon > 0. \tag{8.4}$$

At most x/L_0 long excursions are involved.

Proof. If RH holds, the left side is at most $\sum_{n\le x}|M(n)| \ll x^{3/2+\varepsilon}$. Conversely, the completed excursions with $\ell_\nu \le L_0$ contribute at most $xL_0/2 \le x^{3/2}$ by Lemma 8.2, and if $L_x \le L_0$ the terminal block contributes $\mathfrak{b}_{\mathrm{term}}(x) \le \sum_{r\le L_x} r = L_x(L_x+1)/2 \le L_0^2 \le x$ when $L_0 \ge 1$, and nothing when $L_0 < 1$, since then $L_x = 0$. Hence $\sum_{n\le x}|M(n)| \ll x^{3/2+\varepsilon}$, and Proposition 8.1 applies. □

Remark 8.4 (The $\sqrt{x}$ frontier). The cut may depend on ε: with $L_0 = x^{1/2+\varepsilon/2}$ the discarded completed excursions contribute at most $xL_0/2 = O(x^{3/2+\varepsilon/2})$, while a short terminal block contributes $O(L_0^2) = O(x^{1+\varepsilon})$. Both are admissible in the $x^{3/2+\varepsilon}$ budget. The frontier of pruning is therefore $x^{1/2+o(1)}$; what requires new information is a cut x^θ with a fixed $\theta > 1/2$. For $\ell_\nu \asymp x$ the trivial bound gives $\mathfrak{b}_\nu \ll x^2$ against a budget $x^{3/2+\varepsilon}$, and there are only $O(1)$ such excursions.

Heuristic 8.5 (Renewal tail)**.** If excursion lengths behave like those of a recurrent random walk,

$$\#\{v_\nu \le x :\ \ell_\nu > L\} \asymp \sqrt{x}\, L^{-1/2}, \tag{8.5}$$

and with $\mathfrak{b}_\nu \asymp \ell_\nu^{3/2}$ the dyadic range $[L, 2L]$ contributes $\asymp \sqrt{x}\, L$ to (8.2), so the terminal scales $L \asymp x$ dominate the L^1 sum.

Theorem 8.6 (Correlation criterion)**.** *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{n\le x} M(n)^2 = \sum_{v_\nu \le x} Q_\nu + Q_{\mathrm{term}}(x) = -\sum_{v_\nu \le x}\sum_h h C_h^{(\nu)} + Q_{\mathrm{term}}(x) \ll x^{2+\varepsilon}. \tag{8.6}$$

Proof. Under RH, $|M(n)| \ll_\delta n^{1/2+\delta}$ for every $\delta > 0$; with $\delta = \varepsilon/2$, $M(n)^2 \ll_\varepsilon n^{1+\varepsilon}$ and the sum is $\ll x^{2+\varepsilon}$. Conversely, by Cauchy–Schwarz,

$$\sum_{n\le x}|M(n)| \le x^{1/2}\Big(\sum_{n\le x} M(n)^2\Big)^{1/2} \ll x^{3/2+\varepsilon/2}; \tag{8.7}$$

apply Proposition 8.1. □

Corollary 8.7 (Local sufficient conditions)**.** *Suppose that for every $\varepsilon > 0$ there is C_ε such that*

$$Q_\nu \le C_\varepsilon\, \ell_\nu^{2+\varepsilon} \quad \textit{for every completed excursion with } \ell_\nu > v_\nu^{1/2}. \tag{8.8}$$

Then RH holds.

Proof. Let $\varepsilon_0 \in (0,1)$, put $\varepsilon = \varepsilon_0/4$, and let C_ε be as in the hypothesis. Split the completed excursions at $\ell_\nu = x^{1/2}$. If $\ell_\nu \le x^{1/2}$, then (5.21) gives $Q_\nu \le \ell_\nu x/12 + \ell_\nu/6$, and disjointness bounds the total by $x^2/12 + x/6$. If $\ell_\nu > x^{1/2}$ then $\ell_\nu > v_\nu^{1/2}$ since $v_\nu \le x$, and the hypothesis bounds the total by $C_\varepsilon \sum \ell_\nu^{2+\varepsilon} \le C_\varepsilon(\sum \ell_\nu)^{2+\varepsilon} \le C_\varepsilon x^{2+\varepsilon}$. The terminal block is not a completed excursion, and (5.21) does not apply to it. By Lemma 5.5, $Q_{\mathrm{term}}(x) \le L_x(L_x+1)(2L_x+1)/6$, which is $O(x^{3/2})$ if $L_x \le x^{1/2}$. If $L_x > x^{1/2}$, then $M(\lfloor x\rfloor) \ne 0$. Let $I_\nu = (u(x), v_\nu]$ be the unique completed excursion whose open prefix contains $\lfloor x\rfloor$; since all summands are nonnegative, $Q_{\mathrm{term}}(x) \le Q_\nu$. Fix $A = 1+\varepsilon$. By Corollary 9.11, $v_\nu \le u(x)^A \le x^A$ once x is large, so $\ell_\nu \le x^A$. If $\ell_\nu > v_\nu^{1/2}$ the hypothesis gives $Q_\nu \le C_\varepsilon x^{A(2+\varepsilon)}$; otherwise (5.21) gives $Q_\nu \ll \ell_\nu^3 \le v_\nu^{3/2} \le x^{3A/2}$. In all cases $Q_{\mathrm{term}}(x) \ll_\varepsilon x^{(1+\varepsilon)(2+\varepsilon)} \le x^{2+4\varepsilon} = x^{2+\varepsilon_0}$. This step uses the arithmetic input of Corollary 9.11: for a general family of disjoint intervals, the future return of the terminal block is not controlled by x, and a bound for completed excursions would not control $Q_{\mathrm{term}}(x)$. Altogether $\sum_{n\le x} M(n)^2 \ll_{\varepsilon_0} x^{2+\varepsilon_0}$ for every $\varepsilon_0 \in (0,1)$, and Theorem 8.6 gives RH. □

Heuristic 8.8 (Ballistic excursions)**.** In a model where, after a zero of M, the next values of μ behave locally at random with squarefree density $6/\pi^2$, the pattern "k non-zero steps of one sign followed by k of the other" follows a given zero with probability $\ge c_0^{-k}$. If excursions with $H_\nu = k$ and $\ell_\nu \asymp k$ occur for unbounded k, then by (5.19) $Q_\nu/\ell_\nu^{2+\varepsilon} \gg k^{1-\varepsilon} \to \infty$, so the unpruned version

of (8.8) fails for every $\varepsilon \in (0,1)$. If, as the probability c_0^{-k} suggests, such excursions first occur at positions v_ν growing at least exponentially in k, then $\ell_\nu \ll v_\nu^{1/2}$ and they do not affect the pruned condition (8.8); this is why the pruning in Corollary 8.7 is needed.

Theorem 8.9 (Energy criterion)**.** *With u the last zero $\le x$, $H_{\mathrm{run}}(x) = \max_{u<n\le x}|M(n)|$ and $H_{\mathrm{run}}(x) = 0$ when $u = \lfloor x \rfloor$, define*

$$\mathcal{E}(x) := \frac{1}{4}\Big(\sum_{v_\nu \le x} H_\nu \ell_\nu + (x-u)H_{\mathrm{run}}(x)\Big). \tag{8.9}$$

For a completed excursion let ch_ν be the chord joining $(u_\nu, \mathcal{A}(u_\nu))$ to $(v_\nu, \mathcal{A}(v_\nu))$. By (4.1), $\mathrm{ch}'_\nu = \sigma_\nu\sqrt{H_\nu}/2$, so $\frac{1}{4}H_\nu\ell_\nu = \int_{I_\nu}(\mathrm{ch}'_\nu)^2$ is the Dirichlet energy of the affine interpolant ch_ν joining the endpoint values of $\mathcal{A}$ on I_ν; it is not the Dirichlet energy $\int_{I_\nu} M_{\mathrm{s}}^2$ of $\mathcal{A}$ itself. Then

$$\sum_{n\le x}|M(n)| \le 4\mathcal{E}(x) \le x\max_{n\le x}|M(n)|, \tag{8.10}$$

and RH holds if and only if $\mathcal{E}(x) \ll x^{3/2+\varepsilon}$ for every $\varepsilon > 0$.

Proof. $\mathfrak{b}_\nu \le H_\nu\ell_\nu$, $\mathfrak{b}_{\mathrm{term}}(x) \le (x-u)H_{\mathrm{run}}(x)$ and $\sum \ell_\nu + (x-u) \le x$; apply Proposition 8.1. □

8.1. **Exact logarithmic exponents from Pintz's mean and late-maximal-order theorem.** The criteria above are upper-bound equivalences. Theorems 2.1 and 2.2 of Pintz [45, Theorems 2.1–2.2, equations (2.14)–(2.15)] determine, in logarithmic scale, the mean modulus and late maximal order of $|M|$ in terms of the rightmost zeros of ζ. In [45], $D_M(x) = x^{-1}\int_0^x |M|$ (equivalently $\int_1^x$), $S_{M,\delta}(x) = \max_{x^{1-\delta}\le u\le x}|M(u)|$ for each fixed $\delta > 0$, and $Z(x) = \max_{\gamma>0} x^\beta/\gamma$; Theorem 2.2 there, equation (2.15), states $\log D_M(x) \sim \log S_{M,\delta}(x) \sim \log Z(x)$ as $x \to \infty$ for every fixed $\delta > 0$, and (6.1) there gives $\log Z(x) \sim \Theta \log x$ (both cases $\Theta < 1$ and $\Theta = 1$). Since $\Theta > 0$, this is the statement below. For $\Theta < 1$ the logarithmic-scale lower bound for D_M already follows from Pintz's inequality $D_M(Y) \ge Y^{\beta_0}/(6|\gamma_0|^3)$ for $Y > e^{|\gamma_0|+4}$, valid for every zero $\beta_0 + i\gamma_0$ [43], and the upper bound from the classical estimate $M(x) \ll_\varepsilon x^{\Theta+\varepsilon}$; the case $\Theta = 1$ is the new content of [45]. We record the consequence for the excursion quantities used in this paper.

Theorem 8.10 (Pintz, logarithmic mean and late maximal order [45, Theorems 2.1 and 2.2])**.** *Let $\Theta = \sup\{\Re\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\}$. Then*

$$D_M(X) := \frac{1}{X}\int_1^X |M(t)|\,dt = X^{\Theta+o(1)}. \tag{8.11}$$

Moreover, for every fixed $\delta > 0$,

$$S_{M,\delta}(X) := \max_{X^{1-\delta}\le t\le X}|M(t)| = X^{\Theta+o_\delta(1)}. \tag{8.12}$$

Here the equalities are in logarithmic scale, equivalently

$$\frac{\log D_M(X)}{\log X} \to \Theta, \qquad \frac{\log S_{M,\delta}(X)}{\log X} \to \Theta \quad \text{for each fixed } \delta > 0. \tag{8.13}$$

Corollary 8.11 (Exact exponents of the absolute mass, the quadratic mass and the chord energy of the Möbius sum)**.** *Put*

$$B_1(X) := \sum_{n\le X}|M(n)|, \qquad B_2(X) := \sum_{n\le X} M(n)^2. \tag{8.14}$$

Then

$$B_1(X) = X^{1+\Theta+o(1)}, \qquad B_2(X) = X^{1+2\Theta+o(1)}, \qquad \mathcal{E}(X) = X^{1+\Theta+o(1)}. \tag{8.15}$$

The $o(1)$ statements in (8.15) are global as $X \to \infty$: for every fixed $\varepsilon > 0$ the corresponding two-sided power bounds hold for all sufficiently large X. Hence the same exponent $B_2(v) = v^{1+2\Theta+o(1)}$

remains valid after restriction to any *sequence of integer zeros $v \to \infty$; no subsequence or limsup assertion is being substituted here.*

Consequently

$$\Theta = \lim_{X\to\infty}\left(\frac{\log B_1(X)}{\log X} - 1\right) = \frac{1}{2}\lim_{X\to\infty}\left(\frac{\log B_2(X)}{\log X} - 1\right) = \lim_{X\to\infty}\left(\frac{\log \mathcal{E}(X)}{\log X} - 1\right). \tag{8.16}$$

In particular RH is equivalent to any one of the three logarithmic exponents being respectively $3/2$*,* 2*, and* $3/2$*.*

Proof. Let $N = \lfloor X \rfloor$. Since $M(t) = M(n)$ on $[n, n+1)$,

$$\int_1^X |M(t)|\,dt = \sum_{n=1}^{N-1} |M(n)| + (X-N)|M(N)|, \tag{8.17}$$

so

$$B_1(X) = \int_1^X |M(t)|\,dt + O(|M(N)| + 1). \tag{8.18}$$

The boundary term is $O(X)$, whereas Theorem 8.10 gives $\int_1^X |M(t)|\,dt = X^{1+\Theta+o(1)}$ and $\Theta \geq 1/2$; hence $B_1(X) = X^{1+\Theta+o(1)}$. We first note that

$$\max_{n\leq X} |M(n)| = X^{\Theta+o(1)}. \tag{8.19}$$

The lower bound follows from $S_{M,\delta}$ with any fixed δ. For the upper bound, if $\Theta < 1$, fix once and for all $\delta \in (1-\Theta, 1)$, independent of X, and split the range $n \leq X$ at $X^{1-\delta}$. On $n < X^{1-\delta}$ the trivial bound gives $|M(n)| \leq X^{1-\delta} = o(X^{\Theta})$; on $X^{1-\delta} \leq n \leq X$, Theorem 8.10 with this fixed δ gives $|M(n)| \leq S_{M,\delta}(X) = X^{\Theta+o(1)}$. When $\Theta = 1$ the trivial bound $|M(n)| \leq n$ suffices.

Cauchy–Schwarz and (8.19) now give

$$\frac{B_1(X)^2}{X} \leq B_2(X) \leq X \max_{n\leq X} |M(n)|^2, \tag{8.20}$$

so both sides have exponent $1 + 2\Theta$. Finally (8.10) squeezes $4\mathcal{E}(X)$ between $B_1(X)$ and

$$X \max_{n\leq X} |M(n)|, \tag{8.21}$$

both of exponent $1 + \Theta$. □

Corollary 8.12 (Length localization of the area and quadratic mass)**.** *Fix $0 \leq \alpha < \Theta$. Among the completed excursions ending at most X, the total contribution of those with $\ell_\nu \leq X^\alpha$ is negligible:*

$$\sum_{\substack{v_\nu \leq X \\ \ell_\nu \leq X^\alpha}} \mathfrak{b}_\nu + \mathbf{1}_{\{L_X \leq X^\alpha\}}\mathfrak{b}_{\mathrm{term}}(X) = o(B_1(X)), \tag{8.22}$$

and

$$\sum_{\substack{v_\nu \leq X \\ \ell_\nu \leq X^\alpha}} Q_\nu + \mathbf{1}_{\{L_X \leq X^\alpha\}}Q_{\mathrm{term}}(X) = o(B_2(X)). \tag{8.23}$$

Thus, on the logarithmic scale forced by the zeta zeros, essentially all L^1 and L^2 mass is carried by excursion blocks longer than X^α for every $\alpha < \Theta$.

Proof. Lemma 8.2 bounds the first complete-excursion sum by $O(X^{1+\alpha})$, while the short terminal block contributes $O(X^{2\alpha})$. Both are $o(X^{1+\Theta+o(1)})$ when $\alpha < \Theta$. For the quadratic mass use (5.21) and disjointness:

$$\sum_{\ell_\nu \leq X^\alpha} Q_\nu \ll X^{1+2\alpha}, \qquad Q_{\mathrm{term}}(X) \ll X^{3\alpha} \quad (L_X \leq X^\alpha). \tag{8.24}$$

Because $\alpha < \Theta \leq 1$, both exponents are strictly below $1 + 2\Theta$. □

Corollary 8.13 (Exact exponent of the skeleton area)**.** *The proof uses the return localization (RL) of Corollary 9.11, which holds unconditionally. Put* $\mathcal{A}_{\mathrm{abs}}(x) := \mathcal{A}_+(x) + |\mathcal{A}_-(x)| = \int_0^x |M_{\mathrm{s}}(t)|\,dt$. *Then*

$$\mathcal{A}_{\mathrm{abs}}(x) = x^{1+\Theta/2+o(1)} \qquad (x \to \infty). \tag{8.25}$$

In particular RH $\iff$ $\mathcal{A}_{\mathrm{abs}}(x) \ll_\varepsilon x^{5/4+\varepsilon}$ *for every* $\varepsilon > 0$.

Proof. Upper bound. Fix $0 < \varepsilon < 2-\Theta$. By (8.19) there is C_ε with $H_\nu = |M(\tau_\nu)| \le C_\varepsilon \tau_\nu^{\Theta+\varepsilon}$. Since $0 < (\Theta+\varepsilon)/2 < 1$, the function $C_\varepsilon^{1/2} t^{(\Theta+\varepsilon)/2}$ is concave and increasing, and the chord argument of Lemma 7.1 gives $|M_{\mathrm{s}}(t)| \le C_\varepsilon^{1/2} t^{(\Theta+\varepsilon)/2}$, hence $\mathcal{A}_{\mathrm{abs}}(x) \ll_\varepsilon x^{1+(\Theta+\varepsilon)/2}$. As ε is arbitrary, $\limsup_x \log \mathcal{A}_{\mathrm{abs}}(x)/\log x \le 1 + \Theta/2$.

Lower bound. Let $\delta > 0$ and fix $\varepsilon \in (0,1)$ so small that

$$e(\varepsilon) := \frac{1 + \Theta - \varepsilon - \frac{1}{2}(1+\varepsilon)(\Theta+\varepsilon)}{1+\varepsilon} \ge 1 + \frac{\Theta}{2} - \frac{\delta}{2}; \tag{8.26}$$

this is possible because $e(\varepsilon) \to 1 + \Theta/2$ as $\varepsilon \downarrow 0$. For $X \ge 2$ let $v^*(X)$ be the least zero of M with $v^*(X) \ge \lfloor X \rfloor$. By (RL) (Corollary 9.11 with $A = 1+\varepsilon$) there is $X_1(\varepsilon)$ such that $v^*(X) \le X^{1+\varepsilon}$ for every $X \ge X_1(\varepsilon)$: if $M(\lfloor X \rfloor) = 0$ then $v^*(X) = \lfloor X \rfloor$; otherwise $v^*(X)$ ends the excursion containing $\lfloor X \rfloor$, whose start u satisfies $u < X$, and $v^*(X) \le u^{1+\varepsilon}$ once $u \ge U_{1+\varepsilon}$, which holds for all large X because the excursions starting below $U_{1+\varepsilon}$ are finitely many and finite. Every $n \le X$ with $M(n) \ne 0$ lies in a completed excursion ending at most $v^*(X)$. With $\mathfrak{b}_\nu \le H_\nu \ell_\nu$, $m^* := \max_{n \le v^*(X)} |M(n)|$ and Lemma 4.1,

$$B_1(X) \le \sum_{v_\nu \le v^*(X)} H_\nu \ell_\nu \le (m^*)^{1/2} \sum_{v_\nu \le v^*(X)} \sqrt{H_\nu}\,\ell_\nu = 2(m^*)^{1/2} \mathcal{A}_{\mathrm{abs}}(v^*(X)). \tag{8.27}$$

By Corollary 8.11 and (8.19) there is $X_2(\varepsilon)$ such that for all $X \ge X_2(\varepsilon)$, $B_1(X) \ge X^{1+\Theta-\varepsilon}$ and $m^* \le v^*(X)^{\Theta+\varepsilon} \le X^{(1+\varepsilon)(\Theta+\varepsilon)}$. Hence, for $X \ge \max(X_1, X_2)$, using monotonicity of $\mathcal{A}_{\mathrm{abs}}$,

$$\mathcal{A}_{\mathrm{abs}}(X^{1+\varepsilon}) \ge \mathcal{A}_{\mathrm{abs}}(v^*(X)) \ge \tfrac{1}{2} X^{(1+\varepsilon)e(\varepsilon)}. \tag{8.28}$$

Given $y \ge \max(X_1, X_2)^{1+\varepsilon}$, put $X = y^{1/(1+\varepsilon)}$; then $\mathcal{A}_{\mathrm{abs}}(y) \ge \frac{1}{2} y^{e(\varepsilon)} \ge \frac{1}{2} y^{1+\Theta/2-\delta/2} \ge y^{1+\Theta/2-\delta}$ for all large y. Since $\delta > 0$ is arbitrary, $\liminf_y \log \mathcal{A}_{\mathrm{abs}}(y)/\log y \ge 1 + \Theta/2$. □

Remark 8.14 (Inputs). The new part is the inequality $B_1(X) \le 2(m^*)^{1/2} \mathcal{A}_{\mathrm{abs}}(v^*(X))$ and its use. The exponent inputs are Corollary 8.11 (from [45], or from [43] and the classical upper bound) and the return localization (RL) of Corollary 9.11; for an arbitrary family of disjoint intervals the terminal block of $B_1(X)$ is not controlled by the area up to X. For the lim sup form of the lower bound the classical $B_1(X) = \Omega(X^{1+\Theta-\varepsilon})$ (Mellin transform and Proposition 8.1) suffices instead of [45].

8.2. **Excursion blocks of $1/\zeta$.**

Proposition 8.15 (Excursion blocks of $1/\zeta$)**.** *For a completed excursion* $I = (x_0, x_1]$ *put* $D_I(s) = \sum_{x_0 < n \le x_1} \mu(n) n^{-s}$. *Then*

$$D_I(s) = \sum_{n=x_0+1}^{x_1-1} M(n)\big(n^{-s} - (n+1)^{-s}\big). \tag{8.29}$$

For real $\varsigma > 0$, $D_I(\varsigma)$ *has the sign of the excursion and*

$$|D_I(\varsigma)| \ge \varsigma \mathfrak{b}_I x_1^{-\varsigma-1} \ge \varsigma H_I^2 x_1^{-\varsigma-1}. \tag{8.30}$$

For $\Re s > 1$,

$$\frac{1}{\zeta(s)} = \sum_I D_I(s), \tag{8.31}$$

zero runs contributing nothing.

Proof. For $a_n = \mu(n)$, $A(n) = M(n)$ and $f(n) = n^{-s}$, finite summation by parts gives

$$\sum_{n=x_0+1}^{x_1} a_n f(n) = A(x_1)f(x_1) - A(x_0)f(x_0+1) + \sum_{n=x_0+1}^{x_1-1} A(n)\big(f(n) - f(n+1)\big). \tag{8.32}$$

Since $M(x_0) = M(x_1) = 0$, this is (8.29). For real $\varsigma > 0$, all interior $M(n)$ have the excursion sign, and

$$n^{-\varsigma} - (n+1)^{-\varsigma} = \varsigma \int_n^{n+1} x^{-\varsigma-1}\, dx \geq \varsigma x_1^{-\varsigma-1}. \tag{8.33}$$

The bound $\mathfrak{b}_I \geq H_I^2$ is (5.17). If M vanishes at two consecutive integers the intervening $\mu(n)$ is zero, so every n with $\mu(n) \neq 0$ lies in a completed excursion. For $\Re s > 1$ the ordinary Dirichlet series is absolutely convergent, so summing the finite disjoint blocks gives (8.31). □

Theorem 8.16 (Abscissa of absolute excursion-block convergence)**.** *Let $D_I(s)$ be the completed-excursion blocks of Proposition 8.15. The series*

$$\sum_I D_I(s) \tag{8.34}$$

has abscissa of absolute convergence, with respect to the excursion blocks, exactly

$$\boxed{\varsigma_{a,\mathrm{exc}} = \Theta}. \tag{8.35}$$

Equivalently, the infimum of the real numbers ς_0 for which the series converges absolutely throughout the half-plane $\Re s > \varsigma_0$ is Θ. More precisely, it converges absolutely and locally uniformly for $\Re s > \Theta$ and there represents $1/\zeta(s)$, while for every real $\varsigma < \Theta$ with $\varsigma \neq 0$ one has

$$\sum_I |D_I(\varsigma)| = \infty. \tag{8.36}$$

No assertion is made on the boundary line $\Re s = \Theta$. Consequently,

$$\mathrm{RH} \iff \sum_I D_I(s) \text{ converges absolutely for every } \Re s > \tfrac{1}{2}. \tag{8.37}$$

This is block absolute convergence; the ordinary termwise Dirichlet series $\sum_n \mu(n)n^{-s}$ has abscissa of absolute convergence 1 *and, by $M(x) \ll_\varepsilon x^{\Theta+\varepsilon}$, abscissa of (conditional) convergence Θ. Grouping by completed excursions converts this conditional convergence into absolute convergence of the blocks.*

Remark 8.17. The isolated point $s = 0$ does not enter the definition of the abscissa. For completeness, every completed excursion satisfies $D_I(0) = \sum_{n\in I} \mu(n) = 0$.

Proof. Let $s = \varsigma + i\tau$. Since

$$n^{-s} - (n+1)^{-s} = s \int_n^{n+1} x^{-s-1}\, dx, \tag{8.38}$$

with the real positive branch of $\log x$, one has, for $\varsigma \geq -1$,

$$|n^{-s} - (n+1)^{-s}| \leq |s| n^{-\varsigma-1}. \tag{8.39}$$

Hence

$$\sum_I |D_I(s)| \leq |s| \sum_{n\geq 1} |M(n)| n^{-\varsigma-1}. \tag{8.40}$$

Let $K_0 \Subset \{\Re s > \Theta\}$ be compact and set $\varsigma_0 = \min_{s\in K_0} \Re s > \Theta$. Choose $0 < \varepsilon_0 < \varsigma_0 - \Theta$. From $B_1(X) = X^{1+\Theta+o(1)}$ we have $B_1(X) \ll_{\varepsilon_0} X^{1+\Theta+\varepsilon_0}$ for large X, and partial summation gives

$$\sum_{n\geq 1} |M(n)| n^{-\varsigma_0-1} < \infty. \tag{8.41}$$

Since $\sup_{s\in K_0}|s|<\infty$, the Weierstrass criterion gives normal convergence of the block series on K_0. Thus it is holomorphic in $\Re s>\Theta$. There are no zeros of ζ in that half-plane by the definition of Θ, and $1/\zeta$ extends holomorphically to $s=1$ with value 0; hence $1/\zeta$ is holomorphic throughout the connected half-plane. The identity already holds for $\Re s>1$, so the identity theorem extends it to all $\Re s>\Theta$.

For real $\varsigma>0$, order the completed excursions by their right endpoints. If V is a zero of M, the fixed sign of each excursion and finite Abel summation give

$$\sum_{I:\,x_1(I)\le V}|D_I(\varsigma)|=\sum_{n<V}|M(n)|\big(n^{-\varsigma}-(n+1)^{-\varsigma}\big). \tag{8.42}$$

Since M has arbitrarily large zeros (Definition 3.1, using $M(x)=\Omega_\pm(\sqrt{x})$), letting $V\to\infty$ through zeros and using monotone convergence gives

$$\sum_{I}|D_I(\varsigma)|=\sum_{n\ge1}|M(n)|\big(n^{-\varsigma}-(n+1)^{-\varsigma}\big), \tag{8.43}$$

even when both sides are infinite. If the right side were finite for some $0<\varsigma<\Theta$, then for $n\le X$

$$n^{-\varsigma}-(n+1)^{-\varsigma}=\varsigma\int_n^{n+1}t^{-\varsigma-1}\,dt\ge\varsigma(X+1)^{-\varsigma-1}, \tag{8.44}$$

so

$$B_1(X)\le\frac{(X+1)^{\varsigma+1}}{\varsigma}\sum_{n\ge1}|M(n)|\big(n^{-\varsigma}-(n+1)^{-\varsigma}\big)=O(X^{1+\varsigma}), \tag{8.45}$$

contradicting $B_1(X)=\Omega(X^{1+\Theta-\varepsilon})$ for every $\varepsilon>0$. For real $\varsigma<0$ the same Abel identity holds (the factor $n^{-\varsigma}-(n+1)^{-\varsigma}$ is negative, so each block again has constant sign), and $|n^{-\varsigma}-(n+1)^{-\varsigma}|=|\varsigma|\int_n^{n+1}x^{-\varsigma-1}dx\ge|\varsigma|\min\{1,(V+1)^{-\varsigma-1}\}$ for $n<V$. Hence $\sum_{I:\,x_1(I)\le V}|D_I(\varsigma)|\ge|\varsigma|\min\{1,(V+1)^{-\varsigma-1}\}B_1(V-1)$, which tends to infinity along zeros V: for $\varsigma\le-1$ because $B_1\to\infty$, and for $-1<\varsigma<0$ because $V^{-\varsigma-1}B_1(V-1)\ge V^{\Theta-\varsigma-\varepsilon}$ for large V. Therefore the block abscissa is exactly Θ. □

Remark 8.18 (Status of the result). The analytic input is classical: $\sum_n|M(n)|n^{-\varsigma-1}$ converges for $\varsigma>\Theta$ and diverges for $\varsigma<\Theta$; the case $\varsigma=\Theta$ is left open (upper bound $M(x)\ll x^{\Theta+\varepsilon}$; lower bound by the Mellin argument of Proposition 8.1). The new point is the excursion grouping: the fixed sign of each completed excursion turns Abel summation into the positive identity (8.43). We did not find this block formulation in the literature (absence of a match is weak evidence).

9. Lacunary excursion families and horizon-localized criteria

9.1. **Geometric counts.** The lemmas of this subsection use only that excursions are disjoint integer intervals ordered by their starting points, the next one starting at or after the end of the previous one. They therefore hold verbatim for every sequence of disjoint integer intervals of length at least 2, with starts in $\{0,2,3,\dots\}$, deterministic or random, and in particular pathwise for any stochastic model of excursions (Section 13). For a family $\mathcal{F}$ of excursions and $X\ge2$ put

$$E_{\mathcal{F}}(X):=\#\{\nu\in\mathcal{F}:x_{0,\nu}\le X\}. \tag{9.1}$$

For a function $\mathfrak{h}:[0,\infty)\to[0,\infty]$, the family of excursions *crossing the horizon* $\mathfrak{h}$ is

$$\mathcal{F}_{\mathfrak{h}}:=\{\nu:x_{1,\nu}>\mathfrak{h}(x_{0,\nu})\}. \tag{9.2}$$

For $A>1$ the excursion I_ν is *A-exceptional* if $x_{1,\nu}>x_{0,\nu}^A$, i.e. $\nu\in\mathcal{F}_{x^A}$, and *A-ordinary* otherwise; for $A=2$ we say *square-exceptional* and *square-ordinary*. For $\lambda>1$ it is *λ-long* if $x_{1,\nu}>\lambda x_{0,\nu}$. We write $E_A=E_{\mathcal{F}_{x^A}}$, $E_\square=E_2$ and $E_\lambda=E_{\mathcal{F}_{\lambda x}}$; the context separates E_A (A an exponent) from E_λ (λ a factor).

Lemma 9.1 (Power lacunarity)**.** *For every $A > 1$ and $X \geq 2$,*

$$E_A(X) \leq 2 + \frac{1}{\log A} \log \frac{\log X}{\log 2}. \tag{9.3}$$

In particular

$$E_{\square}(X) \leq 2 + \log_2 \frac{\log X}{\log 2}, \qquad E_{\square}(X) \leq 4 \log\log X \quad (X \geq e^e). \tag{9.4}$$

Proof. No excursion starts at 1, since $M(1) = 1$. List the A-exceptional starts ≥ 2 as $2 \leq a_1 < a_2 < \cdots$. The next excursion starts at or after the end of the preceding one, so $a_{j+1} \geq x_{1,j} > a_j^A$ and $\log a_j \geq A^{j-1} \log 2$. If $a_j \leq X$ then $j \leq 1 + \log(\log X/\log 2)/\log A$; add 1 for the excursion $(0, 2]$. For the second bound in (9.4), $\log_2(\log X/\log 2) = (\log\log X - \log\log 2)/\log 2 \leq 1.443 \log\log X + 0.529$. □

The next lemma is the partition of the square horizon $[x_0, x_0^2] = [x_0, cx_0^2] \cup [cx_0^2, x_0^2]$: the upper part is accessible only to a doubly lacunary family, for every fixed $c > 0$.

Lemma 9.2 (Fractional square horizon)**.** *For $0 < c \leq 1$ and $X \geq 2$,*

$$E_{\mathcal{F}_{cx^2}}(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > cx_{0,\nu}^2\} \leq \frac{2}{c} + 2 + \log_2 \frac{\log X}{\log 2}. \tag{9.5}$$

Consequently, the range $cx_0^2 < t \leq x_0^2$ of the excursions starting at most X is visited by at most $O_c(\log\log X)$ excursions.

Proof. At most $\lceil 2/c \rceil \leq 2/c + 1$ members start below $2/c$. List the others as $a_1 < a_2 < \cdots$ and put $b_j = ca_j \geq 2$. Then $a_{j+1} \geq x_{1,j} > ca_j^2$, so

$$b_{j+1} = ca_{j+1} > c^2 a_j^2 = b_j^2, \tag{9.6}$$

and $\log b_j \geq 2^{j-1} \log 2$. Since $b_j \leq cX \leq X$, the number of such j is at most $1 + \log_2(\log X/\log 2)$. An excursion containing a point $t > cx_0^2$ has $x_1 \geq t > cx_0^2$. □

Lemma 9.3 (Superlinear horizons)**.** *Let $\psi : [0, \infty) \to (0, \infty)$ satisfy $\psi(x) \to \infty$. Then*

$$E_{\mathcal{F}_{x\psi(x)}}(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > x_{0,\nu}\psi(x_{0,\nu})\} = o(\log X). \tag{9.7}$$

Proof. Given $K > 0$ choose x_K with $\psi(x) \geq e^K$ for $x \geq x_K$. At most $x_K + 1$ members start below x_K; we may take $x_K \geq 1$. For consecutive members $a_j < a_{j+1}$ starting at or above x_K, $a_{j+1} \geq x_{1,j} > a_j\psi(a_j) \geq e^K a_j$, so $\log a_j$ increases by more than K at each step and there are at most $1 + \log X/K$ of them. Hence $\limsup E(X)/\log X \leq 1/K$ for every K. □

Lemma 9.4 (Linear lacunarity)**.** *For every $\lambda > 1$ and $X \geq 2$,*

$$E_\lambda(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > \lambda x_{0,\nu}\} \leq 2 + \frac{\log(X/2)}{\log \lambda}. \tag{9.8}$$

Proof. For starts $a_j \geq 2$, $a_{j+1} \geq x_{1,j} > \lambda a_j$, hence $a_j > 2\lambda^{j-1}$ for $j \geq 2$; add 1 for $(0, 2]$. □

All four bounds are sharp up to constants for general families of disjoint intervals, as geometric and doubly geometric chains of intervals show.

9.2. **Arithmetic input: Kaczorowski–Pintz.** We separate the published statement, the discrete separation mechanism, its translation from $M_0 = M+2$ to M, and the consequences for excursions.

Lemma 9.5 (Verification of the Kaczorowski–Pintz Mellin input)**.** *Let*

$$M_0(x) := M(x) - \frac{1}{\zeta(0)} = M(x) + 2 \qquad (x \geq 1), \tag{S9.1}$$

and extend M_0 by 0 on $0 < x < 1$. Its Mellin transform in the convention of [25] *is, initially for $\Re s > 1$,*

$$F_0(s) := \int_0^\infty M_0(x) x^{-s-1}\, dx = \frac{1}{s\zeta(s)} + \frac{2}{s} = \frac{\zeta(s)^{-1} + 2}{s}. \tag{9.9}$$

The apparent singularity at $s = 0$ is removable because $\zeta(0) = -1/2$. At every nontrivial zero ρ of ζ, the term $2/s$ is holomorphic and cannot cancel the pole of $1/(s\zeta(s))$; division by s is nonzero there. Thus the singularities of F_0 on the rightmost boundary are exactly the nontrivial zeta zeros, with their multiplicities. Consequently the boundary abscissa in Theorem 1 of [25] *is Θ, and its lowest positive boundary frequency is*

$$\gamma_\Theta := \min\{\gamma > 0 : \zeta(\Theta + i\gamma) = 0\} \tag{S9.2}$$

when the boundary is attained, while the theorem's convention gives $\gamma_\Theta = \infty$ when it is not attained.

Proof. Absolute convergence of (9.9) for $\Re s > 1$ follows from $M_0(x) = O(x)$. The identity follows from $1/\zeta(s) = s\int_1^\infty M(x)x^{-s-1}\,dx$ and $\int_1^\infty 2x^{-s-1}\,dx = 2/s$. The meromorphic continuation and the singularity statements follow from that displayed formula. Since poles of $1/\zeta$ occur with real parts arbitrarily close to Θ, F_0 is not regular in any half-plane $\Re s > \Theta - \eta$; it is regular in $\Re s > \Theta$. These are precisely the boundary hypotheses needed in the application of Theorem 1 of [25]. In the case $\gamma_\Theta = \infty$, the proof of that theorem permits an arbitrary finite height parameter G: one chooses a boundary strip with no singularity of height at most G, and obtains the density G/π before letting G be arbitrary. □

Theorem 9.6 (Kaczorowski–Pintz [25, Theorem 1, Corollaries 1 and 4])**.** *Let $M_0(x) = M(x) - 1/\zeta(0) = M(x) + 2$. Put*

$$\gamma_\Theta := \min\{\gamma > 0 : \zeta(\Theta + i\gamma) = 0\}, \tag{S9.3}$$

when the boundary $\Re s = \Theta$ contains a zero, and set $\gamma_\Theta = \infty$ otherwise. Then:

(i) *By Lemma 9.5, Theorem 1 (equivalently Corollary 1) of* [25] *applies to M_0 with boundary parameter γ_Θ. It gives, in the extended-real sense,*

$$\liminf_{Y\to\infty} \frac{V(M_0, Y)}{\log Y} \geq \frac{\gamma_\Theta}{\pi}, \tag{9.10}$$

with combined oscillation of size $x^{\Theta-\eta}$ for every fixed $\eta > 0$. If $\gamma_\Theta = \infty$, the last sentence of the proof of Lemma 9.5 gives the precise interpretation: for every fixed finite $G > 0$ the same argument gives the lower density G/π.

(ii) *Corollary 4 of* [25, p. 178, (2.13)–(2.15)] *gives the uniform weaker bound*

$$\liminf_{Y\to\infty} \frac{V(M_0, Y)}{\log Y} \geq \frac{\gamma_1}{\pi}, \tag{9.11}$$

again with combined oscillation of size $x^{\Theta-\eta}$. This bound does not require the extremal boundary to be attained.

(iii) *The local statement (2.2)–(2.3) of Theorem 1 implies that, for every $\delta > 0$ and $\eta > 0$, all sufficiently large Y have two points $t_+, t_- \in [Y^{1-\delta}, Y]$ of opposite signs with $M_0(t_+) \geq t_+^{\Theta-\eta}$ and $M_0(t_-) \leq -t_-^{\Theta-\eta}$.*

Finally, the rigorous verification of Platt–Trudgian [47] *shows that every zero with* $0 < \gamma \le T_0 := 3 \cdot 10^{12}$ *lies on the critical line. Hence*

$$\Theta > \frac{1}{2} \quad \Longrightarrow \quad \gamma_\Theta > T_0 \ \ \textit{or} \ \ \gamma_\Theta = \infty. \tag{9.12}$$

The following elementary lemma removes any ambiguity about what a sign alternation must do to an integer-valued walk with unit jumps.

Lemma 9.7 (Separation of alternating witnesses)**.** *Let* $W : \mathbb{N}_0 \to \mathbb{Z}$ *satisfy*

$$|W(n+1) - W(n)| \le 1 \qquad (n \ge 0), \tag{S9.4}$$

and extend W *to* $[0,\infty)$ *by* $W(t) := W(\lfloor t \rfloor)$*. If*

$$0 \le t_1 < t_2 < \cdots < t_N, \qquad W(t_j) \ne 0, \qquad W(t_j)W(t_{j+1}) < 0, \tag{S9.5}$$

then for every $j < N$ *there is an integer* z_j *with*

$$\lfloor t_j \rfloor < z_j < \lfloor t_{j+1} \rfloor, \qquad W(z_j) = 0. \tag{9.13}$$

In particular $z_1 < \cdots < z_{N-1}$*, and the integers* $\lfloor t_1 \rfloor, \dots, \lfloor t_N \rfloor$ *lie in pairwise distinct connected components of* $\{n \in \mathbb{N} : W(n) \ne 0\}$*.*

Proof. Put $a_j = \lfloor t_j \rfloor$. Since W is constant on $[n, n+1)$, $W(a_j) = W(t_j)$, so $a_j < a_{j+1}$ and the signs alternate. Fix $j < N$ and, after replacing W by $-W$ if necessary, suppose $W(a_j) > 0 > W(a_{j+1})$. Define

$$z_j := \min\{n > a_j : W(n) \le 0\}. \tag{S9.6}$$

Then $a_j < z_j \le a_{j+1}$ and, by minimality, $W(z_j - 1) > 0 \ge W(z_j)$. Both values are integers and their difference has absolute value at most 1, so $W(z_j) = 0$. Since $W(a_{j+1}) < 0$, actually $z_j < a_{j+1}$, proving (9.13). Moreover $z_j < a_{j+1} < z_{j+1}$, and a zero strictly between two nonzero integers prevents them from lying in the same connected component. □

Lemma 9.8 (From M_0 to M)**.** *If* $|M_0(t)| > 2$*, then*

$$\operatorname{sgn} M(t) = \operatorname{sgn} M_0(t), \qquad |M(t)| \ge |M_0(t)| - 2. \tag{S9.7}$$

In particular, if $\alpha > 0$*,* $|M_0(t)| \ge t^\alpha$*, and* $t^\alpha \ge 4$*, then*

$$\operatorname{sgn} M(t) = \operatorname{sgn} M_0(t), \qquad |M(t)| \ge \frac{1}{2} t^\alpha. \tag{9.14}$$

Proof. Since $M = M_0 - 2$, $M_0 > 2$ gives $M > 0$, whereas $M_0 < -2$ gives $M < -4$. Also $|M| \ge |M_0| - 2$. If $|M_0(t)| \ge t^\alpha \ge 4$, then $|M(t)| \ge t^\alpha - 2 \ge \frac{1}{2} t^\alpha$. □

Corollary 9.9 (Separation of large alternating M_0-witnesses)**.** *Let* $0 \le t_1 < \cdots < t_N$ *satisfy*

$$M_0(t_j)M_0(t_{j+1}) < 0 \quad (j < N), \qquad |M_0(t_j)| > 2 \quad (j \le N). \tag{S9.8}$$

Then the signs of $M(t_j)$ *alternate and the points* $t_1, \dots, t_N$ *belong to* N *pairwise distinct excursions of* M*. More precisely, for every* $j < N$ *there is an integer* z_j *such that*

$$\lfloor t_j \rfloor < z_j < \lfloor t_{j+1} \rfloor, \qquad M(z_j) = 0. \tag{S9.9}$$

Proof. Lemma 9.8 gives $\operatorname{sgn} M(t_j) = \operatorname{sgn} M_0(t_j)$. Since $M(n+1) - M(n) = \mu(n+1) \in \{-1, 0, 1\}$, Lemma 9.7, applied with $W = M$, gives the separating zeros and the distinct excursions. □

Corollary 9.10 (Distinct large excursions)**.** *Assume* $\Theta > 1/2$ *and fix* $0 < \eta < \Theta - 1/2$*. For every fixed finite* $G < \gamma_\Theta$*,*

$$N_\eta(Y) \ge \left(\frac{G}{\pi} - o(1) \right) \log Y, \tag{9.15}$$

where $N_\eta(Y)$ *counts distinct excursions starting at most* Y *that contain a point* $t \le Y$ *with* $|M(t)| \ge \frac{1}{2} t^{\Theta - \eta}$*. In particular, by* (9.12)*, one may take* $G = T_0$*.*

Proof. Fix $G < \gamma_\Theta$. By Theorem 9.6(i), for every sufficiently large Y there are

$$N \geq \left(\frac{G}{\pi} - o(1)\right) \log Y \tag{S9.10}$$

ordered points $0 < t_1 < \cdots < t_N \leq Y$ at which the signs of M_0 alternate and $|M_0(t_j)| \geq t_j^{\Theta-\eta}$. Since $\Theta - \eta > 0$, discard the boundedly many initial witnesses for which $t_j^{\Theta-\eta} < 4$. Lemma 9.8 then gives

$$|M(t_j)| \geq \frac{1}{2} t_j^{\Theta-\eta}, \tag{S9.11}$$

and Corollary 9.9 places the remaining witnesses in pairwise distinct excursions. The excursion containing t_j begins before $t_j \leq Y$, and removing $O(1)$ points does not change the logarithmic lower bound. □

Corollary 9.11 (Multiplicative localization of the returns of the Möbius sum)**.** (Return localization, (RL).) *For every fixed $A > 1$ there exists U_A such that every excursion starting at $u \geq U_A$ ends at*

$$v \leq u^A. \tag{9.16}$$

Equivalently, M has only finitely many A-exceptional excursions; in particular only finitely many square-exceptional ones.

Proof. Put $\delta = 1 - 1/A$ and $Y = u^A$, so that $[Y^{1-\delta}, Y] = [u, u^A]$. Fix $\eta = \Theta/2$. Enlarge U_A so that $t^{\Theta/2} > 4$ throughout $[u, u^A]$ whenever $u \geq U_A$. Theorem 9.6(iii) supplies $t_+, t_- \in [u, u^A]$ with opposite signs for M_0 and magnitudes at least $t_\pm^{\Theta/2} > 4$. By Corollary 9.9, after ordering these two points there is an integer zero z of M strictly between their integer parts. Neither witness lies in $[u, u+1)$, where $M = 0$ and $M_0 = 2$, so $u < z \leq u^A$. The first return v of the excursion beginning at u therefore satisfies $v \leq z \leq u^A$, proving (9.16). □

Remark 9.12. A bare sign change of $M_0 = M + 2$ would not force a zero of M. The decisive input is a sequence of alternating witnesses whose magnitudes tend to infinity. Once they exceed 2, their signs transfer to M; the discrete unit-jump lemma then forces an integer zero between every consecutive pair. For the quantitative lower bound $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$ we use the harmless stronger threshold $t^{\Theta-\eta} \geq 4$.

Remark 9.13 (Where (RL) is used). Property (RL) rests on the local statement (2.2)–(2.3) of Theorem 1 of [25], applied directly to the Mellin transform of M_0: every sufficiently late interval $[Y^{1-\varepsilon}, Y]$ contains a sign change combined with oscillation of size $x^{\Theta-\varepsilon}$. A density of sign changes alone would not bound individual return times. It is used exactly in: the terminal-block step of Corollary 8.7; the lower bound of Corollary 8.13; the converse direction of Theorem 12.3; Corollary 9.36; Proposition 9.39; Remark 7.5; the implication from RH to the past-forcing bound in Proposition 11.33, where the return is localized by $v \leq u^{1+\delta}$; and the statements that power and fractional-square horizons are inactive for M (Remark 9.14). All other results are independent of (RL); in particular the horizon theorem uses only Theorem 9.6(i).

Remark 9.14 (Geometric counts and arithmetic localization). For the excursions of M itself Corollary 9.11 is stronger than Lemma 9.1: each A-exceptional family is finite. Consequently, for M itself the power horizons x_0^A and the fractional-square horizons cx_0^2 are inactive for all late excursions, and the corresponding RH-equivalences (Corollary 9.16, Corollary 9.17(i)) follow already from Corollary 9.11; the horizons with genuine content for M are $x_0\psi(x_0)$ and λx_0. The geometric lemmas remain separate results because they are elementary, independent of ζ, and valid for every ordered family of disjoint intervals. This is exactly what the localization theorem below and the extreme-value theorems of Section 13 use: the former needs only a count below $(T_0/\pi)\log X$ in the explicit verified-height form, the latter needs a count valid pathwise in a model where no analogue of Kaczorowski–Pintz is available.

9.3. **Mathematical intuition: extremal zeros, witness density, and the width of the horizon.** The horizon argument compares two quantities on logarithmic scale. The first is the number of large alternating values of the Möbius sum supplied by the oscillation theorem. The second is the number of disjoint excursions that can cross a prescribed multiplicative horizon. The admissible horizon is the point at which these two logarithmic counts balance.

There are two useful versions of the oscillation density. The uniform one is Corollary 4 of [25], recorded in (9.11):

$$\liminf_{Y\to\infty} \frac{V(M_0, Y)}{\log Y} \geq \frac{\gamma_1}{\pi}, \qquad M_0 = M + 2, \tag{9.17}$$

with combined oscillation of size $x^{\Theta-\eta}$. Theorem 1 of [25] contains the finer extremal parameter γ_Θ of Theorem 9.6: the least positive ordinate of a singularity on the boundary $\Re s = \Theta$, with the convention $\gamma_\Theta = \infty$ if the boundary is not attained. Applied to the Mellin transform of M_0, it gives the extremal density

$$\liminf_{Y\to\infty} \frac{V(M_0, Y)}{\log Y} \geq \frac{\gamma_\Theta}{\pi} \tag{9.18}$$

in the extended sense explained in Theorem 9.6. Thus Θ determines the size exponent of the witnesses, whereas γ_Θ determines the logarithmic frequency available from the extremal singularities. The bound γ_1/π is the unconditional uniform substitute obtained without locating those extremal singularities.

It is important not to read the extremal parameter as an instruction to add the frequencies of several zeros. Theorem 1 of [25] allows several singularities on the boundary, whereas its restricted Theorem 2 is designed to isolate a single singularity in an effective argument. One mechanism in the proof is repeated logarithmic integration. If

$$\delta f(x) := \int_0^x f(t)\,\frac{dt}{t}, \tag{S9.12}$$

then, for $\Re\rho > 0$,

$$\delta^{m_{\mathrm{sm}}}(x^\rho) = \frac{x^\rho}{\rho^{m_{\mathrm{sm}}}}. \tag{S9.13}$$

The smoothing order m_{sm} used there grows proportionally to $\log Y$. Hence two boundary modes with the same real part but $|\rho_2| > |\rho_1|$ acquire the relative damping factor

$$\left(\frac{|\rho_1|}{|\rho_2|}\right)^{m_{\mathrm{sm}}}. \tag{S9.14}$$

This explains why the general theorem can guarantee the frequency associated with the lowest relevant ordinate without summing all available frequencies. More frequencies do not, by themselves, force more sign changes: for example

$$\sin y + \epsilon \sin 3y = \sin y\,[1 + 3\epsilon - 4\epsilon \sin^2 y] \tag{S9.15}$$

has exactly the signs of $\sin y$ when $0 < \epsilon < 1$.

There is nevertheless a precedent for improving a minimum-frequency sign-change bound by a controlled combination of several zeros. Morrill–Platt–Trudgian [46] proved for $\psi(x) - x$ that the lower sign-change density exceeds γ_1/π by $1.867 \cdot 10^{-30}$. That result is specific to the prime-number-theorem error term. Transferring the same strategy to M would require, among other things, control of the zero coefficients associated with $1/(\rho\zeta'(\rho))$ for simple zeros, together with the truncation remainder and the transfer from the spectral approximation to actual signs. No such transfer is used in the present excursion argument.

This distinction becomes effective in the contradiction argument for RH. If $\Theta > 1/2$, every zero on the extremal boundary is off the critical line. Platt–Trudgian [47] rigorously verified that every

nontrivial zero with $0 < \gamma \le T_0$, where

$$T_0 = 3 \cdot 10^{12}, \tag{9.19}$$

lies on $\Re s = 1/2$. Hence (9.12) gives $\gamma_\Theta > T_0$ or $\gamma_\Theta = \infty$. Equivalently, for every fixed finite $G \le T_0$, failure of RH supplies at least

$$\left(\frac{G}{\pi} - o(1)\right) \log X \tag{9.20}$$

distinct large-witness excursions beginning below X; taking $G = T_0$ gives the explicit verified-height count used below. No attainment assumption is hidden here: if the boundary is not attained, Theorem 1 permits every fixed finite G.

On the geometric side, an excursion $(u, v]$ crossing a fixed linear horizon λu satisfies

$$\log \frac{v}{u} > \log \lambda. \tag{9.21}$$

The logarithmic budget for ordered disjoint excursions therefore bounds the number of such crossings by

$$\left(\frac{1}{\log \lambda} + o(1)\right) \log X. \tag{9.22}$$

Thus the verified-height witnesses cannot all lie beyond the horizon once

$$\frac{1}{\log \lambda} < \frac{T_0}{\pi}, \tag{9.23}$$

that is, once

$$\lambda > \lambda_0 := \exp\left(\frac{\pi}{T_0}\right). \tag{9.24}$$

The older constant $\lambda_* = \exp(\pi/\gamma_1)$ is the balance point obtained from the uniform Corollary 4 density. The quantity e^{π/γ_Θ} is the extremal balance point when $\gamma_\Theta < \infty$; it is not an explicitly known numerical constant. None of these is asserted to be an intrinsic barrier for M: they are balance points of the same counting mechanism with different available witness densities.

The improvement from λ_* to λ_0 does not change the scale of the window. For every fixed $\lambda_0 > 1$, the width $(\lambda_0 - 1)u$ is still linear in u. We therefore reserve $\theta_{\mathrm{sub}} \in (0, 1)$ for a genuinely sublinear additive horizon

$$u < t \le u + u^{\theta_{\mathrm{sub}}}. \tag{9.25}$$

A logarithmic count alone cannot reach this scale, because

$$\log \frac{u + u^{\theta_{\mathrm{sub}}}}{u} = \log(1 + u^{\theta_{\mathrm{sub}}-1}) = u^{\theta_{\mathrm{sub}}-1} + o(u^{\theta_{\mathrm{sub}}-1}) \longrightarrow 0. \tag{9.26}$$

Such a window consumes no fixed positive amount of the logarithmic budget. The next qualitative step must therefore use local information about the position of a witness inside one excursion.

The proof of Theorem 1 of [25] also indicates why this step is substantially harder. Its iterated smoothing order grows on the scale of $\log Y$, strongly damping nonleading singularities. Passing back to bounded smoothing order, and ultimately to M itself, removes that damping. In a formal simple-zero explicit formula the exposed coefficients are of the shape $1/(\rho\zeta'(\rho))$; multiple zeros would require the corresponding higher-order residue terms. No such uniform bounded-smoothing estimate is claimed here. The method pursued below instead keeps the problem inside a completed excursion, where divisor recursions, no-crossing constraints and channel identities are exact.

9.4. **Horizon localization.**

Theorem 9.15 (Horizon-localized RH criterion)**.** *Let $\mathfrak{h} : [0,\infty) \to [0,\infty]$ be a function, let $\mathcal{F}$ be a family of excursions, and suppose that for some $0 \le \kappa < T_0/\pi$*

$$E_{\mathcal{F}\cup\mathcal{F}_{\mathfrak{h}}}(X) \le (\kappa + o(1)) \log X \qquad (X \to \infty). \tag{9.27}$$

Then RH is equivalent to the following statement: for every $\varepsilon > 0$ there is C_ε such that

$$|M(t)| \le C_\varepsilon t^{1/2+\varepsilon} \qquad \text{for all } t \in I_\nu,\ \nu \notin \mathcal{F},\ t \le \mathfrak{h}(x_{0,\nu}). \tag{9.28}$$

No condition is imposed on the excursions in $\mathcal{F}$, nor on the part of any excursion lying beyond its horizon $\mathfrak{h}(x_0)$.

Proof. RH implies (9.28) by Littlewood's criterion. Conversely, suppose $\Theta > 1/2$ and fix $\eta, \varepsilon > 0$ with $1/2 + \varepsilon < \Theta - \eta$. By (9.15) with $G = T_0$ there are at least $(T_0/\pi - o(1)) \log X$ distinct excursions starting at most X, each containing a witness t with $|M(t)| \ge \frac{1}{2} t^{\Theta-\eta}$. Consider such an excursion ν with witness t. If $\nu \notin \mathcal{F}$ and $t \le \mathfrak{h}(x_{0,\nu})$, the witness lies in the range of (9.28). Otherwise $\nu \in \mathcal{F}$, or $t > \mathfrak{h}(x_{0,\nu})$, in which case $x_{1,\nu} > t > \mathfrak{h}(x_{0,\nu})$ (as $M(t) \ne 0$) and $\nu \in \mathcal{F}_{\mathfrak{h}}$. By (9.27) at least

$$\Big(\frac{T_0}{\pi} - \kappa - o(1)\Big) \log X \longrightarrow \infty \tag{9.29}$$

witnesses lie in the range of (9.28). They belong to distinct excursions, so they are unbounded, and at them $t^{\Theta-\eta}/t^{1/2+\varepsilon} \to \infty$, contradicting (9.28). □

The case $\mathfrak{h}(x) = \infty$ excludes a sparse family $\mathcal{F}$ of whole excursions; the case $\mathcal{F} = \emptyset$ truncates every excursion at its horizon.

Corollary 9.16 (Square horizon)**.** *For every fixed $A > 1$, RH is equivalent to* (9.28) *restricted to A-ordinary excursions. In particular*

$$\boxed{\text{RH} \iff |M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \text{ for all } t \text{ in excursions with } x_1 \le x_0^2.} \tag{9.30}$$

A separate application of Theorem 9.15, with no excursions discarded and horizon $\mathfrak{h}(x) = x^2$, gives the second RH-equivalent statement

$$|M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \quad (x_{0,\nu} < t \le \min\{x_{1,\nu}, x_{0,\nu}^2\}) \tag{9.31}$$

over all excursions. The two restricted statements are therefore two distinct consequences of the horizon theorem, not an elementary equivalence between the restrictions themselves.

Proof. Take $\mathcal{F} = \mathcal{F}_{x^A}$, $\mathfrak{h} = \infty$, respectively $\mathcal{F} = \emptyset$, $\mathfrak{h}(x) = x^2$, in Theorem 9.15; Lemma 9.1 gives $\kappa = 0$. □

The count $E_\square(X) = O(\log\log X)$ is thus what makes the square horizon admissible. The square horizon can be lowered.

Corollary 9.17 (Descending horizons)**.** *RH is equivalent to Littlewood's bound $|M(t)| \ll_\varepsilon t^{1/2+\varepsilon}$ on the ranges*

$$x_{0,\nu} < t \le \min\{x_{1,\nu}, \mathfrak{h}(x_{0,\nu})\} \tag{9.32}$$

over all excursions, for each of the following horizons:

(i) $\mathfrak{h}(x) = cx^2$, *for any fixed $c > 0$;*
(ii) $\mathfrak{h}(x) = x\psi(x)$, *for any $\psi : [0,\infty) \to (0,\infty)$ with $\psi(x) \to \infty$;*
(iii) $\mathfrak{h}(x) = \lambda x$, *for any fixed $\lambda > \lambda_0 := e^{\pi/T_0}$*

Here c, ψ and λ do not depend on the excursion. The same holds with "truncate at $\mathfrak{h}(x_0)$" replaced by "discard the excursions with $x_1 > \mathfrak{h}(x_0)$". In particular it is enough to control the excursions with $x_1 \le \frac{5}{4} x_0$.

Proof. Apply Theorem 9.15 with $\mathcal{F} = \emptyset$ (respectively $\mathcal{F} = \mathcal{F}_{\mathfrak{h}}$, $\mathfrak{h} = \infty$). For (i), $c \geq 1$ reduces to Lemma 9.1 and $0 < c < 1$ to Lemma 9.2; $\kappa = 0$. For (ii), Lemma 9.3 gives $\kappa = 0$. For (iii), Lemma 9.4 gives $\kappa = 1/\log\lambda$, and $1/\log\lambda < T_0/\pi$ exactly when $\lambda > \lambda_0$. □

At the level of the restricted statements, (i) and (ii) imply (iii) eventually, because their horizons are at least λx_0 for every fixed admissible $\lambda > \lambda_0$ once x_0 is large. The converse does not follow from inclusion of ranges; rather, each of (i)–(iii) is separately RH-equivalent by Theorem 9.15. The finitely many small excursions affect only the constant C_ε. The horizon has been lowered from x_0^2 to λx_0: the possible failure of Littlewood's bound, if any, is visible within a bounded multiplicative distance of the start of an excursion. Quantitatively:

Corollary 9.18 (Two-sided localization of large values)**.** *Suppose $\Theta > 1/2$, and fix $0 < \eta < \Theta - \frac{1}{2}$ and $\lambda > \lambda_0$ before letting $X \to \infty$. Among the excursions starting at $x_0 \leq X$, at least*

$$\Big(\frac{T_0}{\pi} - \frac{1}{\log\lambda} - o(1)\Big)\log X \tag{9.33}$$

contain a point t_* with $|M(t_*)| \geq \frac{1}{2}t_*^{\Theta-\eta}$ and

$$x_0 + \tfrac{1}{2}x_0^{\Theta-\eta} \leq t_* \leq \lambda x_0. \tag{9.34}$$

Proof. Of the excursions of Corollary 9.10, those whose witness exceeds λx_0 are λ-long, hence at most $\log X/\log\lambda + 2$ by Lemma 9.4. For the others, $M(x_0) = 0$ and unit steps give $|M(t_*)| \leq t_* - x_0$, so $t_* - x_0 \geq \frac{1}{2}t_*^{\Theta-\eta} \geq \frac{1}{2}x_0^{\Theta-\eta}$. □

Remark 9.19 (Dyadic erosion and the uniform transition). Fix an integer $j \geq 0$ before letting $X \to \infty$. Lemma 9.2 with $c = 2^{-j}$ gives

$$\#\left\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > \frac{x_{0,\nu}^2}{2^j}\right\} = O_{2^j}(\log\log X), \tag{9.35}$$

so Theorem 9.15 successively permits the horizons

$$x_0^2, \quad \frac{x_0^2}{2}, \quad \frac{x_0^2}{4}, \quad \frac{x_0^2}{8}, \ldots. \tag{9.36}$$

The constant in the fixed-c estimate depends on c; hence one may not set $j = j(x_0) \sim \log_2 x_0$ by that lemma alone. If instead $j(x) = \lfloor \log_2(x/\psi(x)) \rfloor$ with $\psi(x) \to \infty$, then the corresponding horizon $x^2/2^{j(x)}$ lies between $x\psi(x)$ and $2x\psi(x)$ and Lemma 9.3 supplies the uniform $o(\log X)$ crossing count. Thus the rigorous erosion chain is

$$x_0^2 \to \frac{x_0^2}{2} \to \frac{x_0^2}{4} \to \cdots \to x_0\psi(x_0) \to \lambda x_0, \qquad \lambda > \lambda_0. \tag{9.37}$$

9.5. **What a genuinely sublinear horizon would require.** The verified-height argument makes the fixed multiplicative factor extremely close to 1, but it does not change the scale of the window:

$$t \leq \lambda_0 u \qquad \Longleftrightarrow \qquad t - u \leq (\lambda_0 - 1)u. \tag{S9.16}$$

For fixed $\lambda_0 > 1$ this width is still linear in u. A genuinely new localization would have to place a forbidden Littlewood-size value in

$$u < t \leq u + u^{\theta_{\mathrm{sub}}}, \qquad \theta_{\mathrm{sub}} < 1. \tag{S9.17}$$

The logarithmic interval count cannot by itself do this, because

$$\log\frac{u + u^{\theta_{\mathrm{sub}}}}{u} = u^{\theta_{\mathrm{sub}}-1} + o\Big(u^{\theta_{\mathrm{sub}}-1}\Big) \longrightarrow 0. \tag{S9.18}$$

There is an elementary height barrier that distinguishes two possible targets. If a first arrival has height $H \asymp u^{\Theta-\eta}$, then unit jumps imply $t_H - u \geq H$, so any bound $t_H - u \leq u^{\theta_{\mathrm{sub}}}$ must

have $\theta_{\mathrm{sub}} \geq \Theta - \eta$. Trying to localize the full Kaczorowski–Pintz height therefore approaches the unknown exponent Θ itself.

The more flexible target is the first crossing of a Littlewood-forbidden level. For a completed excursion $I = (u, v]$, fix a local margin $\varepsilon_{\mathrm{L}} > 0$ and put

$$H_{\mathrm{L}}(u) := u^{1/2+\varepsilon_{\mathrm{L}}}, \qquad r_{\mathrm{L}}(u) := \min\{1 \leq r < v - u : |M(u+r)| \geq H_{\mathrm{L}}(u)\}, \tag{9.38}$$

with $r_{\mathrm{L}}(u) = \infty$ when this set is empty. Necessarily $r_{\mathrm{L}}(u) \geq H_{\mathrm{L}}(u)$. Hence the only possible sublinear range is

$$\frac{1}{2} + \varepsilon_{\mathrm{L}} \leq \theta_{\mathrm{sub}} < 1. \tag{S9.19}$$

The open first-passage objective is to prove, under $\Theta > 1/2$, that for some such pair $(\varepsilon_{\mathrm{L}}, \theta_{\mathrm{sub}})$ there is an unbounded sequence of excursion starts u for which

$$r_{\mathrm{L}}(u) \leq u^{\theta_{\mathrm{sub}}}. \tag{9.39}$$

This would be enough for the inverse direction of a sublinear Littlewood criterion: one would apply the assumed bound with any exponent strictly smaller than ε_{L}.

The next lemma records the exact energy scale required for such a crossing.

Lemma 9.20 (Prefix-energy extraction)**.** *Let $M(u) = 0$, let $L \geq 1$, and put*

$$E(u, L) := \sum_{r=1}^{L} M(u+r)^2. \tag{S9.20}$$

For every $H > 0$,

$$E(u, L) \geq LH^2 \quad \Longrightarrow \quad \max_{1 \leq r \leq L} |M(u+r)| \geq H. \tag{9.40}$$

If $L > H$ and

$$N_H(u, L) := \#\{1 \leq r \leq L : |M(u+r)| \geq H\}, \tag{S9.21}$$

then

$$N_H(u, L) \geq \frac{E(u, L) - LH^2}{L^2 - H^2} \tag{9.41}$$

whenever the numerator is positive.

Proof. If every value had modulus $< H$, then $E(u, L) < LH^2$, proving (9.40). Since $M(u) = 0$ and the jumps have modulus at most 1, $|M(u+r)| \leq r \leq L$. Hence

$$E(u, L) \leq N_H(u, L)L^2 + (L - N_H(u, L))H^2, \tag{S9.22}$$

which rearranges to (9.41). □

For $L = \lfloor u^{\theta_{\mathrm{sub}}} \rfloor$ and $H = H_{\mathrm{L}}(u)$, a sufficient energy threshold for a crossing is therefore

$$E\Big(u, \lfloor u^{\theta_{\mathrm{sub}}} \rfloor\Big) \geq u^{\theta_{\mathrm{sub}}+1+2\varepsilon_{\mathrm{L}}+o(1)}. \tag{9.42}$$

For this energy extraction argument, the exponent is essential: an estimate of size $u^{1+2\varepsilon_{\mathrm{L}}+\delta}$ would not suffice unless $\delta \geq \theta_{\mathrm{sub}}$. More quantitatively, if $E(u, L) \geq (1+\delta)LH^2$ with fixed $\delta > 0$, then

$$N_H(u, L) \gg_\delta u^{1+2\varepsilon_{\mathrm{L}}-\theta_{\mathrm{sub}}} \tag{S9.23}$$

in the above power regime. Because $\theta_{\mathrm{sub}} > 1/2 + \varepsilon_{\mathrm{L}}$, this exponent is smaller than θ_{sub}, as required by the trivial bound $N_H \leq L$.

The missing implication is not geometric:

$$\text{a large later KP witness} \not\Longrightarrow E\Big(u, u^{\theta_{\mathrm{sub}}}\Big) \text{ large} \tag{S9.24}$$

from the present identities alone. A unit-step path may remain low on the whole prefix and rise only later. Persistence estimates near a late peak also remain near that peak; transporting them back to the initial prefix would already contain most of the desired first-passage conclusion.

The complete-excursion viewpoint clarifies exactly what is gained. For an open prefix of length L inside an excursion of orientation σ, set

$$x_r := \sigma\mu(u+r), \qquad S_r := \sum_{j\le r} x_j = \sigma M(u+r) > 0, \tag{S9.25}$$

and

$$C_h^{(u,L]} := \sum_{j=1}^{L-h} x_j x_{j+h}. \tag{S9.26}$$

The general triangular identity gives the exact boundary formula

$$E(u,L) = S_L^2 + S_L \sum_{r=1}^{L-1} S_r - \sum_{h=1}^{L-1} hC_h^{(u,L]}. \tag{9.43}$$

The middle term is nonnegative because the prefix has one sign. When the block is the *complete* excursion, $S_L = 0$ and the boundary term disappears exactly, recovering the bridge identity $E = -\sum_h hC_h$. This is the concrete structural advantage of complete no-crossing excursions: they remove a boundary contribution rather than creating cancellation automatically.

A sufficient open correlational target for the prefix would therefore be

$$S_L^2 - \sum_{h<L} hC_h^{(u,L]} \ge c\,L\,u^{1+2\varepsilon_{\mathrm{L}}} \qquad (c>1) \tag{S9.27}$$

for some $L \le u^{\theta_{\mathrm{sub}}}$. No estimate currently proved in the paper supplies this lower bound on the adaptively selected prefix.

Remark 9.21 (Why the available short-interval estimate points the other way). The all-interval estimate of Matomäki–Teräväinen [35, Theorem 1.1] gives cancellation of μ on intervals of length x^θ for every fixed $\theta > 0.55$. It is an upper bound for increments of M, not a lower bound forcing a large value. In the mesoscopic notation of Lemma 9.55,

$$|M(b) - M(a)| \le \omega_u(b-a) + h_u. \tag{S9.28}$$

Thus, if $M(u) = 0$ and $|M(t)| = H$, it implies

$$t - u \ge \frac{H - h_u}{\omega_u}, \tag{S9.29}$$

when the right side is positive. It controls the width of a peak that already exists; it does not locate its first arrival near u.

Remark 9.22 (Scale required from the centered-record functional). Equation (9.122) also makes the sublinear scale transparent. At the Littlewood level $H_{\mathrm{L}}(u)$, if $\theta_{\mathrm{sub}} > 1/2 + \varepsilon_{\mathrm{L}}$, then $H_{\mathrm{L}}(u)/u = o(u^{\theta_{\mathrm{sub}}-1})$. Hence a bound

$$\mathfrak{C}(u, H_{\mathrm{L}}) \ll u^{\theta_{\mathrm{sub}}-1} \tag{S9.30}$$

would imply $t_H - u \ll u^{\theta_{\mathrm{sub}}}$. A constant upper bound for $\mathfrak{C}$ can yield only a fixed multiplicative horizon. This is why a sublinear theorem requires a quantitatively decaying centered cost, not merely an improvement of its constant.

The current proof target can therefore be stated without referring to a later extremal witness: prove directly, under failure of RH, that for some $1/2 + \varepsilon_{\mathrm{L}} < \theta_{\mathrm{sub}} < 1$ there is an unbounded sequence of complete excursions $(u, v]$ with $v - u > \lfloor u^{\theta_{\mathrm{sub}}} \rfloor$ and

$$E\Big(u, \lfloor u^{\theta_{\mathrm{sub}}} \rfloor\Big) > u^{\theta_{\mathrm{sub}}+1+2\varepsilon_{\mathrm{L}}}. \tag{9.44}$$

By Lemma 9.20, this is exactly strong enough to force the missing first crossing. Deriving (9.44) from the no-crossing constraints, the past-forcing identities of Section 11, or centered channel correlations is the open arithmetic step.

9.6. **Fixed-mask Catalan compression inside fractional square horizons.** The fractional-square localization has a useful interaction with the exact combinatorics of a completed excursion. The relevant Catalan time is not physical time: zero Möbius increments must first be removed. This support-time distinction is essential below.

Definition 9.23 (Support-time path and fixed-mask Catalan law)**.** Let $I = (x_0, x_1]$ be a completed excursion, and list the nonzero increments in I as

$$x_0 < a_1 < a_2 < \cdots < a_L \le x_1, \qquad \mu(a_j) \neq 0. \tag{9.45}$$

Put

$$r_I(t) := \#\{j : a_j \le t\}, \qquad x_0 < t \le x_1. \tag{9.46}$$

For the excursion I_ν, let $\sigma_\nu \in \{\pm 1\}$ be its orientation and define

$$S_0 = 0, \qquad S_r := \sigma_\nu \sum_{j=1}^{r} \mu(a_j) \qquad (1 \le r \le L). \tag{9.47}$$

Then $L = 2n$ is even and

$$S_0 = S_L = 0, \qquad S_r > 0 \quad (1 \le r < L), \qquad S_r - S_{r-1} \in \{-1, +1\}. \tag{9.48}$$

Thus $(S_r)_{0 \le r \le L}$ is a primitive Dyck path. The *fixed-mask primitive Catalan law* keeps the physical support $\{a_1, \dots, a_L\}$ fixed and assigns to those support positions a uniformly chosen primitive Dyck sign sequence of length L. We denote its probability by $\mathbb{P}_{\mathrm{pCat},I}$.

For the observed excursion and real $x_0 < t \le x_1$ one has, exactly,

$$|M(t)| = S_{r_I(t)}, \qquad r_I(t) \le t - x_0 \le t, \tag{9.49}$$

with $r_I(t) < t$ whenever $x_0 \ge 1$, i.e. for every excursion other than $(0, 2]$. The first equality uses the orientation of the excursion and the fact that M is constant across the zero increments.

We first record the global Catalan estimate, since it is useful in its own right and clarifies why a different estimate is needed at the square horizon.

Lemma 9.24 (Catalan maximum tail, classical)**.** *Let $L = 2n$ and let $H_{\max}$ be the maximum of a uniformly chosen Dyck path of length $2n$. For $1 \le h \le n$,*

$$\mathbb{P}_{\mathrm{Cat},L}\{H_{\max} \ge h\} \le (n+1)\frac{\binom{2n}{n+h}}{\binom{2n}{n}} \le (n+1)\exp\left(-\frac{h^2}{n+h}\right) \le \left(\frac{L}{2}+1\right)e^{-h^2/L}. \tag{9.50}$$

For a uniformly chosen primitive Dyck path of length $L = 2n$,

$$\mathbb{P}_{\mathrm{pCat},L}\{H_{\max} \ge h\} \le \frac{L}{2}\exp\left(-\frac{(h-1)^2}{L-2}\right), \qquad 2 \le h \le n. \tag{9.51}$$

Proof. There are $\mathrm{Cat}_n = (n+1)^{-1}\binom{2n}{n}$ Dyck paths of length $2n$ and $\binom{2n}{n}$ unrestricted bridges. Reflecting a bridge after its first visit to level h gives a bijection between bridges that reach h and unrestricted walks ending at $2h$, hence their number is $\binom{2n}{n+h}$. Therefore

$$\mathbb{P}_{\mathrm{Cat},2n}\{H_{\max} \ge h\} \le (n+1)\frac{\binom{2n}{n+h}}{\binom{2n}{n}}. \tag{9.52}$$

Moreover

$$\frac{\binom{2n}{n+h}}{\binom{2n}{n}} = \prod_{j=1}^{h}\frac{n-j+1}{n+j}, \tag{9.53}$$

and $\log(1-x) \le -x$ gives

$$\log\frac{\binom{2n}{n+h}}{\binom{2n}{n}} \le -\sum_{j=1}^{h}\frac{2j-1}{n+j} \le -\frac{h^2}{n+h}. \tag{9.54}$$

Since $n+h \le 2n = L$, (9.50) follows. A primitive path of length $2n$ has the form $+1$, followed by a Dyck path of length $2n-2$, followed by -1; its maximum is one plus the maximum of the interior Dyck path. Applying the preceding bound with $n-1$ and $h-1$ proves (9.51). □

Remark 9.25 (Why the maximum tail is not the square-horizon estimate). The exponent in (9.51) is of order H^2/L. For a general family of disjoint intervals retained by the square horizon one may have $L \asymp x_0^2$ (for M itself such excursions are finitely many by Corollary 9.11), while the relevant witness lies much earlier, at physical time $t \asymp x_0$. Then a violation of size $t^{1/2+\varepsilon}$ need not make H^2/L large. Thus the global maximum tail, although correct, is not the right Catalan quantity for combining with the fractional-square horizon. The correct quantity is the pointwise height at the support time $r_I(t)$.

Lemma 9.26 (Pointwise primitive-Catalan tail). *Let $L = 2n \ge 4$ and let (S_r) be a uniformly chosen primitive Dyck path of length L. For $1 \le r \le L-1$ and real $h \ge 1$,*

$$\mathbb{P}_{\mathrm{pCat},L}\{S_r \ge h\} \le 8n^{3/2} \exp\left(-\frac{h^2}{2r}\right) \le 3L^{3/2} \exp\left(-\frac{h^2}{2r}\right). \tag{9.55}$$

By time reversal the same estimate holds with r replaced by $L-r$; hence

$$\mathbb{P}_{\mathrm{pCat},L}\{S_r \ge h\} \le 3L^{3/2} \exp\left[-\frac{h^2}{2\min\{r, L-r\}}\right]. \tag{9.56}$$

If $h > \min\{r, L-r\}$ the probability is 0, so the bound is only relevant for $h \le \min\{r, L-r\}$.

Proof. Ignore both positivity and the endpoint constraint. Among all 2^L sign words, the proportion whose first r signs have sum at least h is at most $\exp(-h^2/(2r))$ by the elementary Chernoff bound for a Rademacher sum. Therefore the number of primitive Dyck paths satisfying $S_r \ge h$ is at most

$$2^L \exp\left(-\frac{h^2}{2r}\right). \tag{9.57}$$

The total number of primitive Dyck paths of length $2n$ is Cat_{n-1}. The standard central-binomial lower bound $\binom{2m}{m} \ge 4^m/(2\sqrt{m})$ gives

$$\mathrm{Cat}_{n-1} = \frac{1}{n}\binom{2n-2}{n-1} \ge \frac{4^{n-1}}{2n\sqrt{n-1}}. \tag{9.58}$$

Since $2^L = 4^n$, division gives the first inequality in (9.55); the second follows from $8n^{3/2} < 3(2n)^{3/2}$. Reversing the path and changing the signs of the reversed increments preserves the primitive Dyck class and carries height S_r to height S_{L-r}, which proves (9.56). □

For an observed excursion I define its pointwise Catalan reference-tail weight by

$$p_{\mathrm{Cat}}(I,t) := \mathbb{P}_{\mathrm{pCat},I}\left\{S_{r_I(t)} \ge |M(t)|\right\}. \tag{9.59}$$

This is a reference-tail probability under the fixed-mask Catalan law; it is not a p-value for a stochastic model of the arithmetic Möbius path.

Corollary 9.27 (Fractional-square Catalan rarity). *Fix $c > 0$ and $\varepsilon > 0$. Let $I = (x_0, x_1]$ be a completed excursion with $x_1 \le cx_0^2$, and keep its zero mask fixed. For all sufficiently large x_0,*

$$\mathbb{P}_{\mathrm{pCat},I}\left\{\exists\, t \in I \cap \mathbb{Z} :\ S_{r_I(t)} \ge \lceil t^{1/2+\varepsilon} \rceil\right\} \le 3c^{5/2} x_0^5 \exp\left(-\frac{1}{2}x_0^{2\varepsilon}\right). \tag{9.60}$$

In particular the right-hand side is smaller than x_0^{-A} for every fixed $A > 0$ once x_0 is large enough.

Proof. If $L = L^{\#}(I) = 2$ then $S_r \le 1$ and the event is empty for $x_0 \ge 1$; assume $L \ge 4$. At a support position $t = a_r$ one has $r \le t - x_0 < t$. Hence Lemma 9.26 with $h = \lceil t^{1/2+\varepsilon} \rceil$ gives

$$\mathbb{P}_{\mathrm{pCat},I}\{S_r \ge t^{1/2+\varepsilon}\} \le 3L^{3/2} \exp\left(-\frac{1}{2}t^{2\varepsilon}\right) \le 3L^{3/2} \exp\left(-\frac{1}{2}x_0^{2\varepsilon}\right). \tag{9.61}$$

There are at most L support positions. Since $L \le x_1 - x_0 \le x_1 \le cx_0^2$, one has $3L \cdot L^{3/2} = 3L^{5/2} \le 3(cx_0^2)^{5/2} = 3c^{5/2}x_0^5$, and the union bound gives (9.60). Between support positions the path is constant, so no additional physical times have to be counted. □

The preceding statement is purely combinatorial: it measures how far a large value lies in the tail of a reference law, not how the Möbius constraints select a path. Combined with the localization theorem it gives the following quantitative form.

Corollary 9.28 (Catalan cost of a Kaczorowski–Pintz witness under $\Theta > 1/2$)**.** *Suppose $\Theta > 1/2$, fix $0 < \eta < \Theta - \frac{1}{2}$, put $\delta = \Theta - \eta - \frac{1}{2}$, and fix $\lambda > \lambda_0$. Then there are infinitely many excursions $I = (x_0, x_1]$ with $x_1 \le \lambda x_0$ containing a point t with*

$$-\log p_{\mathrm{Cat}}(I,t) \;\ge\; \frac{1}{8}x_0^{2\delta} - O(\log x_0). \tag{9.62}$$

The same holds with $x_1 \le \lambda x_0$ replaced by $x_1 \le cx_0^2$ for any fixed $c > 0$.

Proof. By Corollary 9.10 and Lemma 9.4, take $G = T_0$. Since all zeros with ordinate at most T_0 lie on the critical line, failure of RH gives $G < \gamma_\Theta$; moreover $\lambda > \lambda_0 = e^{\pi/T_0}$ gives $G/\pi > 1/\log\lambda$. Thus at least $(G/\pi - 1/\log\lambda - o(1))\log X$ excursions starting at most X have $x_1 \le \lambda x_0$ and contain t with $h = |M(t)| \ge \frac{1}{2}t^{1/2+\delta}$; for cx_0^2 use Lemma 9.2 instead. With $r = r_I(t) < t$ and $L \le x_1 - x_0$, Lemma 9.26 gives $p_{\mathrm{Cat}}(I,t) \le 3L^{3/2}\exp(-h^2/(2r)) \le 3L^{3/2}\exp(-t^{2\delta}/8)$, and $t > x_0$, $L \le \lambda x_0$ (respectively $L \le cx_0^2$). □

Remark 9.29 (Selector bounds are one-sided). Lemma 9.26 gives only the upper-tail estimate

$$p_{\mathrm{Cat}}(I,t) \le 3L^{3/2}\exp\left(-\frac{|M(t)|^2}{2r_I(t)}\right), \tag{9.63}$$

and hence the one-sided implication

$$-\log p_{\mathrm{Cat}}(I,t) \ge \frac{|M(t)|^2}{2r_I(t)} - \log(3L^{3/2}). \tag{9.64}$$

No reverse inequality, and therefore no equivalence "up to polynomial factors", is proved here. Consequently a selector bound $-\log p_{\mathrm{Cat}}(I,t) \le \Phi(x_0)$ with $\Phi \ge 0$, $\Phi(x) = x^{o(1)}$ on excursions with $x_1 \le \lambda x_0$ (so that $\Phi(x) + O(\log x) \le x^{2\varepsilon}$ eventually, for every $\varepsilon > 0$) implies $|M(t)| \le t^{1/2}x_0^{o(1)}$ there, so RH follows from Corollary 9.17; this is a sufficient condition, not an independent equivalent formulation. A polynomial lower bound $p_{\mathrm{Cat}} \ge x_0^{-A}$ would similarly give $|M(t)| \ll \sqrt{t\log t}$ on the retained ranges. To turn the Catalan weight into a two-sided measure of deviation one would need a matching lower-tail estimate of the form $-\log p_{\mathrm{Cat}}(I,t) \ll |M(t)|^2/r_I(t) + \log L$, which is not proved here.

Corollary 9.30 (Catalan cost along a Littlewood-violating sequence)**.** *Let $t_j \to \infty$, $A_j \to \infty$ and $\varepsilon_0 > 0$ with $|M(t_j)| \ge A_j t_j^{1/2+\varepsilon_0}$, and let $I_j = (x_{0,j}, x_{1,j}]$ be the excursion containing t_j. If $x_{1,j} \le x_{0,j}^B$ for a fixed B, then*

$$-\log p_{\mathrm{Cat}}(I_j,t_j) \;\ge\; \frac{A_j^2}{2}\,t_j^{2\varepsilon_0} - O_B(\log x_{0,j}) \;\ge\; \frac{A_j^2}{2}\,x_{0,j}^{2\varepsilon_0} - O_B(\log x_{0,j}). \tag{9.65}$$

Proof. With $r = r_{I_j}(t_j) < t_j$ and $h = |M(t_j)|$, $h^2/(2r) \ge A_j^2 t_j^{1+2\varepsilon_0}/(2t_j)$; apply Lemma 9.26 with $L \le x_{1,j} \le x_{0,j}^B$. □

9.6.1. *Selection energies: completed excursions and open prefixes.* The support-time clock of Definition 9.23 is the squarefree-compressed clock

$$\mathcal{S}(x) = \#\{n \le x : \mu(n) \neq 0\}, \qquad L^{\#}(I) = \mathcal{S}(x_1) - \mathcal{S}(x_0) \le x_1 - x_0, \tag{9.66}$$

and $r_I(t) = \mathcal{S}(t) - \mathcal{S}(x_0)$. Two tail quantities are natural: one for the height of a completed excursion, and one for an open prefix, which does not require knowledge of the future return.

Definition 9.31 (Selection energies)**.** For a completed excursion I with height $H(I)$ and compressed length $L^{\#} = L^{\#}(I)$, put

$$p_{\mathrm{Cat}}^{\max}(I) := \mathbb{P}_{\mathrm{pCat},L^{\#}}\{H_{\max} \ge H(I)\}, \qquad \mathcal{I}_{\mathrm{Cat}}(I) := -\log p_{\mathrm{Cat}}^{\max}(I). \tag{9.67}$$

For $t \ge 1$ with $M(t) \neq 0$ let $a(t)$ be the last zero of M before t and $r = \mathcal{S}(t) - \mathcal{S}(a(t))$. After orientation, the compressed segment from $a(t)$ to t is a positive meander of length r: a ± 1 path $(S_k)_{k \le r}$ with $S_0 = 0$ and $S_k > 0$ for $1 \le k \le r$. Let $\mathcal{M}_r$ be the set of such paths. Since $S_r \equiv r \pmod 2$, the exact endpoint event $S_r = h$ has probability zero when the integer h has incompatible parity. Define

$$p_{\mathrm{open}}(t) := \mathbb{P}_{\mathcal{M}_r}\{S_r \ge |M(t)|\}, \tag{9.68}$$

the uniform measure on $\mathcal{M}_r$.

Lemma 9.32 (Height–entropy inequalities)**.** (i) *If $L^{\#} = 2$, the compressed path is $0, 1, 0$ and $H(I) = 1$. For every completed excursion with $L^{\#} \ge 4$,*

$$\mathcal{I}_{\mathrm{Cat}}(I) \;\ge\; \frac{(H(I)-1)^2}{L^{\#}-2} - \log\frac{L^{\#}}{2}. \tag{9.69}$$

(ii) *$\#\mathcal{M}_r = \binom{r-1}{\lfloor (r-1)/2 \rfloor} \ge 2^r/(4\sqrt{r})$, and for $h \ge 1$*

$$\mathbb{P}_{\mathcal{M}_r}\{S_r \ge h\} \le 4\sqrt{r}\, e^{-h^2/(2r)}, \qquad -\log p_{\mathrm{open}}(t) \;\ge\; \frac{M(t)^2}{2r} - \log(4\sqrt{r}). \tag{9.70}$$

Proof. (i) is (9.51). (ii) The count of positive meanders is classical (ballot theorem; the number ending at s is $\frac{s}{r}\binom{r}{(r+s)/2}$). From $\binom{2m}{m} \ge 4^m/(2\sqrt{m})$ and $\binom{2m+1}{m} = \frac{1}{2}\binom{2m+2}{m+1}$ one gets $\#\mathcal{M}_r \ge 2^r/(4\sqrt{r})$. Among all 2^r words the proportion with $S_r \ge h$ is at most $e^{-h^2/(2r)}$ by the Chernoff bound; divide. □

Proposition 9.33 (Selector criteria)**.** (i) Completed excursions. *Fix $\lambda > \lambda_0$. If there is A such that $p_{\mathrm{Cat}}^{\max}(I) \ge x_0^{-A}$ for every sufficiently large completed excursion with $x_1 \le \lambda x_0$, then $H(I) \ll_{A,\lambda} \sqrt{x_0 \log x_0}$ on that family, and RH holds.*

(ii) Open prefixes. *If there is A such that $p_{\mathrm{open}}(t) \ge t^{-A}$ for all large t with $M(t) \neq 0$, then $|M(t)| \ll_A \sqrt{t \log t}$, and RH holds.*

Proof. (i) If $L^{\#} = 2$ then $H(I) = 1$. Otherwise $L^{\#} \ge 4$ and, by Lemma 9.32(i), $(H-1)^2 \le (L^{\#}-2)(A \log x_0 + \log L^{\#})$, and $L^{\#} \le (\lambda - 1)x_0$. Littlewood's bound therefore holds on every excursion with $x_1 \le \lambda x_0$, and Corollary 9.17(iii), in the form that discards the λ-long excursions, gives RH. (ii) By Lemma 9.32(ii), $M(t)^2 \le 2r(A \log t + \log(4\sqrt{r}))$ with $r \le t$. □

Remark 9.34 (Uniform and extremal linear thresholds). Corollary 4 of [25] gives the uniform density γ_1/π and hence the historical balance point $\lambda_* := e^{\pi/\gamma_1}$. Theorem 1 is finer: if RH is false, its frequency parameter is the first ordinate γ_Θ on the extremal boundary, with $\gamma_\Theta = \infty$ when that boundary is not attained. For a finite extremal ordinate the corresponding balance point of the same counting mechanism is e^{π/γ_Θ}; this is a statement about the counting mechanism, not a universal arithmetic optimum for M.

The verified height $T_0 = 3 \cdot 10^{12}$ gives the explicit fixed substitute $\lambda_0 = e^{\pi/T_0}$ in the contradiction argument, because $\Theta > 1/2$ implies $\gamma_\Theta > T_0$ or ∞. Thus λ_* remains useful as the uniform

Corollary 4 benchmark and in the structural surplus audit below, whereas λ_0 is the strongest fixed linear factor used by the present horizon theorem.

Proposition 9.35 (Logarithmic-length budget)**.** *Let $\mathcal{F}$ be a family of excursions and let $u_1 < \cdots < u_N$ be the starts of its members with $2 \le u_j \le X$, with ends v_j, and assume $N \ge 1$. For an empty family the first sum is 0 and the last-member assertion is not used. The initial excursion $(0, 2]$ is explicitly excluded, since $\log(v/u)$ is not defined at $u = 0$. Then*

$$\sum_{j=1}^{N-1} \log \frac{v_j}{u_j} \le \log \frac{X}{2}. \tag{9.71}$$

If moreover $v_N \le u_N^A$ for some $A \ge 1$, then $\sum_{j=1}^{N} \log(v_j/u_j) \le A \log X$.

Proof. Disjointness gives $u_{j+1} \ge v_j$, so $\log(v_j/u_j) \le \log u_{j+1} - \log u_j$; the sum telescopes to at most $\log u_N - \log u_1 \le \log(X/2)$. The last term is at most $(A-1)\log u_N \le (A-1)\log X$. □

Lemmas 9.1–9.4 are the special cases in which every term is bounded below. The budget form shows what the horizon theorem actually measures.

Corollary 9.36 (Mean logarithmic length of Kaczorowski–Pintz excursions)**.** *Suppose $\Theta > 1/2$, fix $0 < \eta < \Theta - \frac{1}{2}$, and let $\mathcal{K}(X)$ be any set of distinct excursions starting at most X, each containing a point t with $|M(t)| \ge \frac{1}{2} t^{\Theta-\eta}$, such that $\#\mathcal{K}(X) \ge (\gamma_1/\pi - o(1)) \log X$, with $u_\nu \ge 2$ for every selected member (such sets exist, after deletion of the initial excursion, by Corollary 9.10). Then*

$$\frac{1}{\#\mathcal{K}(X)} \sum_{\nu \in \mathcal{K}(X)} \log \frac{v_\nu}{u_\nu} \;\le\; \frac{\pi}{\gamma_1} + o(1) = \log \lambda_* + o(1). \tag{9.72}$$

Proof. Discard the initial excursion $(0, 2]$, whose start is 0 and whose contribution to the count is $O(1)$. Fix $A > 1$ before taking $X \to \infty$. By Corollary 9.11, all but finitely many excursions satisfy $v \le u^A$. These exceptions form a fixed global finite set once A is fixed, so the sum of their individual finite values $\log(v/u)$ is $O_A(1)$. Proposition 9.35 gives a total of at most $A \log X + O_A(1)$. Since $\#\mathcal{K}(X) \ge (\gamma_1/\pi - o(1)) \log X$, it follows for every fixed $A > 1$ that

$$\limsup_{X \to \infty} \frac{1}{\#\mathcal{K}(X)} \sum_{\nu \in \mathcal{K}(X)} \log \frac{v_\nu}{u_\nu} \le \frac{A\pi}{\gamma_1}. \tag{9.73}$$

Only after this limit is taken do we let $A \downarrow 1$, obtaining π/γ_1. □

Corollary 9.37 (Verified-height mean logarithmic length)**.** *Assume $\Theta > 1/2$ and fix $0 < \eta < \Theta - 1/2$. There exist Kaczorowski–Pintz witness families $\mathcal{K}_0(X)$, all with starts $u_\nu \ge 2$, with*

$$\#\mathcal{K}_0(X) \ge (T_0/\pi - o(1)) \log X \tag{S9.31}$$

and

$$\frac{1}{\#\mathcal{K}_0(X)} \sum_{\nu \in \mathcal{K}_0(X)} \log \frac{v_\nu}{u_\nu} \le \frac{\pi}{T_0} + o(1). \tag{S9.32}$$

Proof. Use Corollary 9.10 with $G = T_0$ and repeat the proof of Corollary 9.36, using the same logarithmic budget and return localization. □

Remark 9.38 (Uniform-threshold surplus benchmark). If one deliberately uses only the uniform Corollary 4 witness density, the same logarithmic budget has balance point λ_*. In that uniform benchmark, excursions carrying the selected witnesses cannot all have $\log(v/u) > \log \lambda$ with $\lambda > \lambda_*$; Corollary 9.36 is its averaged form. Consequently, any proof that such excursions have mean logarithmic length at least $\pi/\gamma_1 + \delta$ for some fixed $\delta > 0$ would contradict Corollary 9.36 and prove RH, and a surplus $\delta > \log(\lambda_*/\lambda)$ among the excursions whose witness lies beyond λx_0 would lower that uniform benchmark to the chosen $\lambda < \lambda_*$. The active verified-height horizon of Corollary 9.17(iii)

is already λ_0, so this paragraph is retained as a structural audit of the surplus mechanism rather than as the present linear bottleneck. Kaczorowski–Pintz, as stated, gives neither: it supplies at least one return in each band $[Y^{1-\delta}, Y]$ but no upper separation between consecutive witnesses. Nor do short-interval estimates for μ supply it: $|M(x+h) - M(x)| = o(h)$ is compatible with M staying positive throughout $(x, x+h]$, whereas a return requires a decrement of size at least $M(x)$ in a prescribed direction. Energy bounds do not help either, since a flat excursion of height 1 has $\mathfrak{b}_I, Q_I \asymp \ell_I$.

9.7. **A deductive audit below the counting threshold.** The following audit is normalized to the uniform Corollary 4 benchmark λ_*. It identifies the exact information that would have been sufficient to push that uniform multiplicative horizon below λ_*. We make that requirement quantitative and record a matching no-go statement showing that it does not follow from the geometric inputs alone.

Put

$$d_1 := \frac{\gamma_1}{\pi}, \qquad a_* := \frac{1}{d_1} = \frac{\pi}{\gamma_1}, \qquad \lambda_* = e^{a_*}. \tag{9.74}$$

Assume $\Theta > 1/2$, fix $0 < \eta < \Theta - \frac{1}{2}$, and let $\mathcal{K}(X)$ be a Kaczorowski–Pintz witness family as in Corollary 9.36. For each $\nu \in \mathcal{K}(X)$ choose one witness t_ν in the corresponding excursion $(u_\nu, v_\nu]$.

Fix $1 < \lambda < \lambda_*$ and define the bad subfamily

$$\mathcal{B}_\lambda(X) := \{\nu \in \mathcal{K}(X) : t_\nu > \lambda u_\nu\}. \tag{9.75}$$

Proposition 9.39 (Positional surplus criterion)**.** *Suppose there is a fixed $\delta > \log(\lambda_*/\lambda)$ such that*

$$\frac{1}{\#\mathcal{B}_\lambda(X)} \sum_{\nu \in \mathcal{B}_\lambda(X)} \left(\log \frac{v_\nu}{u_\nu} - \log \lambda \right) \geq \delta - o(1) \tag{9.76}$$

whenever $\mathcal{B}_\lambda(X)$ is nonempty. Then

$$\#\mathcal{B}_\lambda(X) \leq \left(\frac{1}{\log \lambda + \delta} + o(1) \right) \log X < (d_1 - o(1)) \log X. \tag{9.77}$$

Consequently $\gg \log X$ witnesses satisfy $t_\nu \leq \lambda u_\nu$, and the horizon argument gives the same contradiction under $\Theta > 1/2$ as in Theorem 9.15.

Proof. Fix first an exponent $A > 1$. Return localization and Proposition 9.35 give

$$\sum_{\nu \in \mathcal{K}(X)} \log \frac{v_\nu}{u_\nu} \leq A \log X + O_A(1). \tag{9.78}$$

After taking $X \to \infty$ we may let $A \downarrow 1$, exactly as in the proof of Corollary 9.36; hence the total logarithmic budget is $(1 + o(1)) \log X$. By (9.76), every bad member consumes on average at least $\log \lambda + \delta - o(1)$ of this budget. Therefore

$$(\log \lambda + \delta - o(1)) \#\mathcal{B}_\lambda(X) \leq (1 + o(1)) \log X. \tag{9.79}$$

Since $\log \lambda + \delta > \log \lambda_* = a_* = 1/d_1$, the resulting coefficient is strictly smaller than d_1. Subtracting the bad excursions from $\#\mathcal{K}(X) \geq (d_1 - o(1)) \log X$ leaves a positive proportion, on the logarithmic scale, of witnesses covered by the shorter horizon. □

The surplus has the exact decomposition

$$\log \frac{v_\nu}{u_\nu} - \log \lambda = \log \frac{t_\nu}{\lambda u_\nu} + \log \frac{v_\nu}{t_\nu}. \tag{9.80}$$

Thus the existence of a few long excursions is irrelevant: the required surplus must hold uniformly on average over the bad witness subfamily.

Unit step size alone does not provide such a fixed surplus. If a witness beyond the horizon has height $H \asymp t^{\alpha}$ with $\alpha < 1$, and even if one knows $M(\lambda u) = o(H)$ in the relevant orientation, then the step bound gives only

$$t - \lambda u \geq H - o(H), \qquad v - t \geq H, \tag{9.81}$$

and consequently

$$\frac{v}{u} \geq \lambda + (2 + o(1))\frac{H}{u} = \lambda + o(1). \tag{9.82}$$

The gain is additive and sublinear, whereas Proposition 9.39 requires a fixed logarithmic gain.

Proposition 9.40 (Geometric inputs do not force positional surplus)**.** *Disjointness of the excursions, increments in $\{-1, 0, 1\}$, a localized return, a witness count $(d_1 - o(1)) \log X$, and a mean logarithmic length bounded by $a_* + o(1)$ do not imply that a positive proportion of witnesses lies before λu for any fixed $\lambda < \lambda_*$. This remains true even if the witness height is $\asymp t^{\alpha}$ for a fixed $1/2 < \alpha < 1$ and every excursion is eventually shorter than u^A for each fixed $A > 1$.*

Proof. Choose

$$1 < \lambda < c < r < \lambda_*, \qquad \frac{1}{2} < \alpha < 1, \tag{9.83}$$

and, starting from a sufficiently large integer u_1, define recursively

$$u_{j+1} = \lceil r u_j \rceil, \qquad v_j = u_{j+1}, \qquad t_j = \lfloor c u_j \rfloor. \tag{9.84}$$

For all sufficiently large j, one has $u_j < t_j < v_j$. Since both gaps $t_j - u_j$ and $v_j - t_j$ are linear in u_j, whereas $H_j := \lfloor t_j^{\alpha}/2 \rfloor = o(u_j)$, one can construct on $(u_j, v_j]$ a primitive bridge with increments in $\{-1, 0, 1\}$ that rises to height H_j by time t_j, stays strictly away from zero while inserting horizontal steps, and returns to zero at v_j.

Moreover,

$$\#\{j : u_j \leq X\} = \frac{\log X}{\log r} + O(1) > d_1 \log X + O(1), \tag{9.85}$$

while

$$\frac{1}{J}\sum_{j \leq J} \log \frac{v_j}{u_j} \longrightarrow \log r < a_*, \qquad t_j > \lambda u_j. \tag{9.86}$$

Finally $v_j \asymp r u_j$, so $v_j \leq u_j^A$ for every fixed $A > 1$ and all sufficiently large j. This is not a construction of the Möbius sum; it is a countermodel to any deduction using only the listed geometric and counting inputs. □

First-arrival record blocks and an exact positional cost. The preceding no-go statement isolates the missing arithmetic information but still treats the position of a witness only through its total excursion length. There is a second exact decomposition, internal to a first arrival, that keeps track of the successive record levels.

Definition 9.41 (First-arrival record decomposition)**.** Let $I = (u, v]$ be a completed excursion of orientation $\sigma \in \{-1, 1\}$, and let $t_H = u + h < v$ be the first point at which

$$S_r := \sigma M(u + r), \qquad S_0 = 0, \tag{9.87}$$

reaches the integer height $H \geq 1$. Thus $S_h = H$ and $S_r < H$ for $0 \leq r < h$. Put

$$\tau_j := \min\{r : S_r = j\} \quad (0 \leq j \leq H), \qquad L_j := \tau_{j+1} - \tau_j, \tag{9.88}$$

and, for $0 \leq r \leq L_j$,

$$T_{j,r} := S_{\tau_j + r} - j, \qquad d_{j,r} := -T_{j,r}. \tag{9.89}$$

Then $T_{j,0} = 0$, $T_{j,L_j} = 1$, and $T_{j,r} \leq 0$ for $0 < r < L_j$. Define the record-return cost

$$\mathcal{V}(u, H) := \sum_{j=0}^{H-1} \sum_{r < L_j} d_{j,r}(d_{j,r} + 1). \tag{9.90}$$

Proposition 9.42 (Record-return decomposition and depth rigidity)**.** *With the notation above, if* $R_r := \max_{0\le s\le r} S_s$*, then*

$$\mathcal{V}(u,H)=\sum_{r=1}^{h-1}(R_r-S_r)(R_r-S_r+1). \tag{9.91}$$

Moreover, writing $m_j := L_j - 1$*, the path* $T_{j,0},\dots,T_{j,m_j}$ *is a completed nonpositive bridge at level* j*, and the last step* $m_j \to L_j$ *creates the next record. If*

$$A_j := \sum_{r=1}^{m_j-1} d_{j,r}, \qquad Q_j := \sum_{r=1}^{m_j-1} d_{j,r}^2, \tag{9.92}$$

then

$$\mathcal{V}=\sum_{j=0}^{H-1}(Q_j+A_j). \tag{9.93}$$

If $B_j := \max_{0\le r\le m_j} d_{j,r}$*, then*

$$\mathcal{V}_j := Q_j + A_j \ge \frac{B_j(B_j+1)(2B_j+1)}{3}, \qquad \sum_{j<H} B_j^3 \le \frac{3}{2}\mathcal{V}. \tag{9.94}$$

Equality in the first inequality is attained by the triangular descent to B_j *and return to* 0*, with arbitrary additional waits at depth* 0 *but no positive-depth waits or extra oscillations.*

Proof. Lemma 5.1, applied to the increment block from τ_j to τ_{j+1}, gives

$$\sum_{r<L_j} T_{j,r}^2 + \sum_{g<L_j} gC_g^{[j]} = \sum_{r<L_j} T_{j,r}. \tag{9.95}$$

Hence $-\sum_{g<L_j} gC_g^{[j]} = \sum_{r<L_j} d_{j,r}(d_{j,r}+1)$, which is (9.90). On $\tau_j \le r < \tau_{j+1}$ one has $R_r = j$, which gives (9.91). Since the first arrival at $j+1$ must end with a $+1$ step from j, one has $T_{j,L_j-1} = 0$; this proves the bridge decomposition and $\mathcal{V}_j = Q_j + A_j$. A bridge reaching depth B_j and returning to zero by unit steps must visit $1,2,\dots,B_j-1,B_j,B_j-1,\dots,1$. Therefore

$$\mathcal{V}_j \ge B_j(B_j+1) + 2\sum_{d=1}^{B_j-1} d(d+1) = \frac{B_j(B_j+1)(2B_j+1)}{3}. \tag{9.96}$$

The last assertion and the cubic sum follow immediately. □

For the positional identity below, the zero runs must be truncated at the first-arrival endpoint. For $u < n \le u+h$ with $\mu(n) \ne 0$, let

$$g_h(n) := \max\{g \ge 0 : n+g \le u+h,\ \mu(n+1) = \cdots = \mu(n+g) = 0\}, \tag{9.97}$$

and let g_u be the initial zero-run length after u, truncated in the same way. Put

$$\Gamma_h := \sum_{u<n\le u+h} \mu(n) g_h(n). \tag{9.98}$$

For $1 \le r \le h$ write $x_r := S_r - S_{r-1} = \sigma\mu(u+r)$ and $d_r := R_r - S_r$, and partition the indices into

$$\begin{aligned} \mathcal{A} &:= \{r : x_r = 1,\ d_r = 0\}, \quad \mathcal{B} := \{r : x_r = 1,\ d_r \ge 1\},\\ \mathcal{C} &:= \{r : x_r = -1\}, \qquad\qquad \mathcal{D} := \{r : x_r = 0,\ d_r = 0\},\\ \mathcal{E} &:= \{r : x_r = 0,\ d_r \ge 1\}. \end{aligned} \tag{9.99}$$

Define the nonnegative excess

$$E'''(u,H) := \sum_{r\in\mathcal{C}\cup\mathcal{E}} \big(d_r(d_r+1)-2\big) + \sum_{r\in\mathcal{B}} d_r(d_r+1) + 2\sum_{r\in\mathcal{B}} g_h(u+r). \tag{9.100}$$

Proposition 9.43 (Exact first-arrival position identity)**.** *With the preceding notation,*

$$\boxed{h = \mathcal{V}(u,H) + H + \sigma\Gamma_h + g_u - E'''(u,H).} \tag{9.101}$$

In particular $E'''(u,H) \geq 0$.

Proof. Let $D := \#\mathcal{C}$. The nonzero increments have final displacement H, so their number is $H+2D$. Every zero in the prefix belongs either to the initial run after u or to the run following a nonzero increment; because g_h is truncated at $u+h$, the last record step contributes no zero beyond the prefix. Hence

$$h = H + 2D + \sum_{r \in \mathcal{A}\cup\mathcal{B}\cup\mathcal{C}} g_h(u+r) + g_u. \tag{9.102}$$

Also

$$\sigma\Gamma_h = \sum_{r\in\mathcal{A}} g_h(u+r) + \sum_{r\in\mathcal{B}} g_h(u+r) - \sum_{r\in\mathcal{C}} g_h(u+r). \tag{9.103}$$

Finally, grouping the positive-depth zero steps according to the preceding nonzero increment gives exactly

$$\mathcal{V} = 2D + 2\sum_{r\in\mathcal{C}} g_h(u+r) + E'''. \tag{9.104}$$

Combining the last three identities proves (9.101). Nonnegativity follows term by term from $d_r \geq 1$ on $\mathcal{B}\cup\mathcal{C}\cup\mathcal{E}$. □

Lemma 9.44 (Uniform cancellation of the truncated zero-run correlation)**.** *Fix* $\iota > 0$. *Uniformly for*

$$u^{3/5+\iota} \leq h \leq u, \tag{9.105}$$

one has

$$\Gamma_h = o(h) \qquad (u \to \infty). \tag{9.106}$$

Proof. Fix $0 < A < 1/3$ and choose $0 < c < A$. Put $P = c\log\log u$, $Q := \prod_{p\leq P} p^2$ and $K := 2\pi(P)+1$. Let $N_P(m)$ be the indicator that $p^2 \mid m$ for at least one prime $p \leq P$, and let $g_P(n)$ be the length of the consecutive $N_P = 1$ run following n. The function g_P is periodic modulo Q. Moreover no K consecutive integers are all covered by these prime squares: the number covered in such a block is at most

$$K\sum_{p\leq P}\frac{1}{p^2} + \pi(P) \leq K\left(\frac{\pi^2}{8} - \frac{3}{4}\right) + \pi(P) < K. \tag{9.107}$$

Thus $0 \leq g_P \leq K$.

The difference between the genuine zero runs and the small-square runs is charged to integers divisible by p^2 with $p > P$. Splitting an excess run into pieces of length K gives, uniformly in the stated range,

$$\sum_{\substack{u<n\leq u+h\\ \mu(n)\neq 0}} |g_h(n) - g_P(n)| \ll K\left(\frac{h}{P} + \sqrt{u}\right) + K^2 = o(h). \tag{9.108}$$

Here the K^2 term accounts for truncation at $u+h$.

Write the finite Fourier expansion $g_P(n) = \sum_{a \bmod Q} \widehat{g}_P(a) e(an/Q)$ with normalized coefficients. Parseval and Cauchy–Schwarz give $\sum_a |\widehat{g}_P(a)| \leq K\sqrt{Q}$. Therefore (12.23) yields

$$\left|\sum_{u<n\leq u+h} \mu(n) g_P(n)\right| \ll_{A,\iota} K\sqrt{Q}\,\frac{h}{(\log u)^A}. \tag{9.109}$$

By the prime number theorem, $\sqrt{Q} = \exp(\vartheta(P)) = (\log u)^{c+o(1)}$, while $K = (\log\log u)^{1+o(1)}$. Since $c < A$, this is $o(h)$; together with the preceding comparison it proves (9.106). □

Corollary 9.45 (First-arrival cost at the verified-height threshold)**.** *Assume $\Theta > 1/2$, fix $0 < \eta < \Theta - 1/2$, put $\alpha := \Theta - \eta$, and fix*

$$\lambda_0 < \Lambda < 2. \tag{9.110}$$

For the Kaczorowski–Pintz excursions supplied by Corollary 9.18, set $H_u := \lfloor u^\alpha/2 \rfloor$ and let $t_H = u+h$ be the first arrival at height H_u in the orientation of the excursion. Then $H_u = o(u)$, $t_H \le \Lambda u$, and

$$\frac{t_H}{u} \le 1 + \frac{\mathcal{V}(u, H_u)}{u} + o(1). \tag{9.111}$$

Consequently, if for some fixed $\varepsilon_* > 0$ an unbounded sequence of these prefixes satisfies

$$\mathcal{V}(u, H_u) \le (\lambda_0 - 1 - \varepsilon_*)u, \tag{9.112}$$

then

$$t_H \le (\lambda_0 - \varepsilon_* + o(1))u. \tag{9.113}$$

Thus, combined with Littlewood's bound on a linear excursion window, such a cost estimate would lower the sufficient localized horizon below λ_0; the cost estimate alone is not a proof of RH.

Proof. Corollary 9.18 supplies an unbounded family of excursions with a point $t_* \le \Lambda u$ and $|M(t_*)| \ge \frac{1}{2}t_*^\alpha \ge \frac{1}{2}u^\alpha$, so the first arrival at H_u exists before t_*. The classical estimate $\sum_{n\le x}\mu(n)^2 = (6/\pi^2)x + O(\sqrt{x})$ implies $g_u = O(\sqrt{u})$, but the interval calculation is worth recording. By the definition of the initial zero run, $(u, u+g_u]$ contains no squarefree integer, hence

$$0 = \sum_{u<n\le u+g_u} \mu(n)^2 = \frac{6}{\pi^2}g_u + O\big(\sqrt{u+g_u} + \sqrt{u}\big)\,. \tag{9.114}$$

Therefore $g_u \ll \sqrt{u+g_u} + \sqrt{u}$. If $g_u \le u$ this is immediately $O(\sqrt{u})$; if $g_u > u$, then the same inequality gives $g_u \ll \sqrt{g_u}$ for all sufficiently large u, which is impossible. Thus $g_u = O(\sqrt{u})$ uniformly in the regime used here.

If $h < u^{3/5+\iota}$ for some fixed sufficiently small $\iota > 0$, then $h/u = o(1)$ directly. Otherwise Lemma 9.44 and (9.101), with $E''' \ge 0$, give $h \le \mathcal{V} + o(u)$. This proves (9.111) in both cases. The last implication is immediate. If Littlewood's bound held on a fixed horizon $\lambda' u$ with $\max\{1, \lambda_0 - \varepsilon_*\} < \lambda' < \lambda_0$, then along this sequence it would give $H_u \ll u^{1/2+\epsilon}$, contradicting $H_u \asymp u^\alpha$ for $1/2 + \epsilon < \alpha$. □

Remark 9.46 (Exact limitation of the record-return cost)*.* The improvement in Corollary 9.45 is the coefficient: at the active verified-height threshold the sufficient scale is $\lambda_0 - 1$, not $(6/\pi^2)(\lambda_0 - 1)$. It does not make the unresolved estimate weaker than early arrival. In the linear regime $h \asymp u$, the same identities give

$$\frac{\mathcal{V}(u, H)}{u} = \frac{h}{u} + \frac{E'''(u, H)}{u} + o(1). \tag{9.115}$$

Hence a bound $\mathcal{V} \le (\lambda_0 - 1 - \varepsilon_*)u$ simultaneously forces early arrival and leaves a reserve for the positive retrogression excess E'''. An independent estimate of E''', or a signed estimate controlling $h + E'''$ jointly, is therefore required to advance this route.

There is a second way to remove the deterministic first-arrival drift before asking for arithmetic cancellation. It leads to centered correlations on the record blocks.

Proposition 9.47 (Centered record-block position bound)**.** *For a record block j, put*

$$x_{j,s} := \sigma\mu(u + \tau_j + s) \quad (1 \le s \le L_j), \qquad z_{j,s} := x_{j,s} - \frac{1}{L_j}. \tag{9.116}$$

Then $\sum_{s\le L_j} z_{j,s} = 0$. Let

$$B_{j,r} := \sum_{s=1}^{r} z_{j,s} = T_{j,r} - \frac{r}{L_j}, \qquad C^\circ_{j,g} := \sum_{s=1}^{L_j - g} z_{j,s}z_{j,s+g}. \tag{9.117}$$

The centered bridge energy satisfies

$$\mathcal{Q}_j^\circ := \sum_{r=1}^{L_j-1} B_{j,r}^2 = -\sum_{g=1}^{L_j-1} gC_{j,g}^\circ \geq \frac{(L_j-1)(2L_j-1)}{6L_j}. \tag{9.118}$$

Consequently, with $\mathcal{Q}^\circ := \sum_{j<H} \mathcal{Q}_j^\circ$,

$$\mathcal{Q}^\circ \geq \frac{h}{3} - \frac{H}{2} + \frac{1}{6}\sum_{j<H}\frac{1}{L_j}, \qquad h \leq 3\mathcal{Q}^\circ + \frac{3}{2}H. \tag{9.119}$$

For $H = o(u)$*, the same inequality may be compared with any chosen linear factor. In particular, relative to the historical uniform benchmark* λ_**, an estimate*

$$\mathcal{Q}^\circ \leq \left(\frac{\lambda_* - 1}{3} - \varepsilon_*\right) u \tag{9.120}$$

places the first arrival at $t_H \leq (\lambda_* - 3\varepsilon_* + o(1))u$*. This is retained as a structural benchmark; the active verified-height factor is* λ_0*.*

Proof. The centered increments form a completed bridge, so Lemma 5.1 gives the equality in (9.118). Since $T_{j,r} \leq 0$ before the next record, $B_{j,r} \leq -r/L_j$, and hence

$$\mathcal{Q}_j^\circ \geq \frac{1}{L_j^2}\sum_{r=1}^{L_j-1} r^2 = \frac{(L_j-1)(2L_j-1)}{6L_j}. \tag{9.121}$$

Summing and using $\sum_j L_j = h$ proves (9.119). The final implication follows from $H = o(u)$. □

Proposition 9.48 (Logarithmically weighted centered record cost)**.** *Let* $t_H = u + h$ *be a first arrival with integer* $u \geq 1$*, put* $a_j = u + \tau_j$*, and define*

$$\mathfrak{C}(u, H) := \sum_{j=0}^{H-1} \frac{\mathcal{Q}_j^\circ}{a_j}. \tag{S9.33}$$

Then the two finite bounds

$$\log\frac{t_H}{u} \leq 3\mathfrak{C}(u, H) + \frac{3H}{2u} \tag{9.122}$$

and

$$\log\frac{t_H}{u+H} \leq 4\mathfrak{C}(u, H) \tag{9.123}$$

hold. In particular, if $H = o(t_H)$ *and* $\mathfrak{C}(u, H) \leq b$ *along a sequence, then* $t_H/u = O_b(1)$*, hence* $H/u = o(1)$*, and therefore*

$$\log\frac{t_H}{u} \leq 3b + o(1). \tag{9.124}$$

Assume in addition that, for some fixed $1 < \lambda \leq \Lambda < 2$*,* $\lambda u \leq t_H \leq \Lambda u$ *and* $H = o(u)$*. Then*

$$\mathfrak{C}(u, H) \geq \frac{1}{3}\log\frac{t_H}{u} + \frac{t_H - u}{2t_H} - o(1). \tag{9.125}$$

Consequently, with

$$g(x) := \frac{1}{3}\log(1+x) + \frac{x}{2(1+x)}, \tag{S9.34}$$

which is strictly increasing for $x \geq 0$*, any asymptotic upper bound* $\mathfrak{C}(u, H) \leq g(\lambda_* - 1) - \delta$ *on such a family forces* $t_H/u \leq \lambda_* - c_\delta + o(1)$ *for some* $c_\delta > 0$*.*

Proof. From (9.118),

$$\mathcal{Q}_j^\circ = \frac{L_j}{3} - \frac{1}{2} + \frac{1}{6L_j} + \mathcal{X}_j, \qquad \mathcal{X}_j = \sum_{r<L_j} \left(d_{j,r}^2 + \frac{2r}{L_j} d_{j,r} \right). \tag{S9.35}$$

Since $a_{j+1} = a_j + L_j$ and $a_j \geq u$,

$$3\mathfrak{C} \geq \sum_j \frac{L_j}{a_j} - \frac{3}{2} \sum_j \frac{1}{a_j} \geq \log \frac{t_H}{u} - \frac{3H}{2u}, \tag{S9.36}$$

which is (9.122).

For the second finite inequality, the exact mandatory term gives $\mathcal{Q}_j^\circ \geq (L_j - 1)/4$ for every integer $L_j \geq 1$. Moreover

$$\log \frac{a_{j+1}}{a_j} = \log \left(1 + \frac{1}{a_j} \right) + \log \left(1 + \frac{L_j - 1}{a_j + 1} \right), \tag{S9.37}$$

and the second logarithm is at most $(L_j - 1)/(a_j + 1) \leq 4\mathcal{Q}_j^\circ / a_j$. Since $a_j = u + \tau_j \geq u + j$,

$$\sum_{j<H} \log \left(1 + \frac{1}{a_j} \right) \leq \sum_{j<H} \log \frac{u + j + 1}{u + j} = \log \frac{u + H}{u}. \tag{S9.38}$$

Summing proves (9.123). If $\mathfrak{C} \leq b$ and $H = o(t_H)$, this first yields $t_H/(u + H) \leq e^{4b}$; hence $t_H/u = O_b(1)$ and then $H/u = o(1)$. Equation (9.122) now gives (9.124).

Finally, $\mathcal{X}_j \geq \sum_r d_{j,r}^2 \geq \frac{1}{2} \sum_r d_{j,r}(d_{j,r} + 1)$, and $a_j \leq t_H$, hence

$$\sum_j \frac{\mathcal{X}_j}{a_j} \geq \frac{\mathcal{V}(u, H)}{2t_H}. \tag{S9.39}$$

In the stated linear regime $h = t_H - u \asymp u$. Hence, for any fixed $\iota \in (0, 2/5)$ and all sufficiently large u, $u^{3/5+\iota} \leq h \leq u$. Proposition 9.43, Lemma 9.44, $g_u = O(\sqrt{u})$ and $E''' \geq 0$ therefore give $\mathcal{V}(u, H) \geq h - H - o(h) = h - o(u)$. Combining the last two estimates with the mandatory part of the centered energy proves (9.125). The final assertion follows from the strict monotonicity of g. □

Corollary 9.49 (Mean centered cost forces an early subfamily)**.** *Let $\mathcal{K}(X)$ be a family of first arrivals with $H = o(t_H)$ uniformly along the family. Suppose*

$$\limsup_{X \to \infty} \frac{1}{\#\mathcal{K}(X)} \sum_{\nu \in \mathcal{K}(X)} \mathfrak{C}_\nu = c. \tag{S9.40}$$

If $3c < \log \lambda$ for some fixed $\lambda > 1$, then a positive proportion of the members of $\mathcal{K}(X)$ satisfy $t_H < \lambda u$ for all sufficiently large X.

Proof. Choose b with $c < b < \frac{1}{3} \log \lambda$. Markov's inequality leaves a positive proportion with $\mathfrak{C}_\nu \leq b$. On that subfamily, (9.123) first gives $t_H/u = O_b(1)$; the hypothesis $H = o(t_H)$ then gives $H/u = o(1)$, and (9.122) yields $\log(t_H/u) \leq 3b + o(1) < \log \lambda$. □

Proposition 9.50 (Post-horizon centered surplus cost)**.** *Let $t_H = u + r$ be the first arrival at height H in an oriented excursion, and fix $1 < \lambda < 2$ with $t_H > \lambda u$. Put*

$$r_\lambda := \lfloor \lambda u \rfloor - u, \qquad K_\lambda := S_{r_\lambda}, \qquad R := r - r_\lambda. \tag{S9.41}$$

For $0 \leq j \leq R$ define the deviation from the chord joining the horizon to the first arrival,

$$\widetilde{A}_j := S_{r_\lambda + j} - \left(K_\lambda + \frac{j(H - K_\lambda)}{R} \right), \tag{S9.42}$$

and put

$$\mathcal{V}_\lambda^{\mathrm{pre}}(t_H) := \sum_{j=1}^{R-1} \widetilde{A}_j^2, \qquad \mathcal{W}_\lambda(t_H) := \frac{\mathcal{V}_\lambda^{\mathrm{pre}}(t_H)}{uH^2}. \tag{S9.43}$$

Then

$$\log \frac{v}{u} - \log \lambda \geq \log\left(1 + \frac{\mathcal{W}_\lambda(t_H)}{\lambda} - O\left(\frac{1}{u}\right)\right). \tag{9.126}$$

Moreover, with

$$p_j := \sigma\mu(u + r_\lambda + j) - \frac{H - K_\lambda}{R}, \qquad 1 \leq j \leq R, \tag{S9.44}$$

one has $\sum_{j=1}^{R} p_j = 0$ *and the exact centered correlation identity*

$$\mathcal{W}_\lambda(t_H) = -\frac{1}{uH^2} \sum_{g=1}^{R-1} g \sum_{j=1}^{R-g} p_j p_{j+g}. \tag{9.127}$$

Proof. Because t_H is the first arrival at H, both the path and the chord lie between 0 and H on the horizon–arrival segment. Hence $|\widetilde{A}_j| \leq H$ and $\mathcal{V}_\lambda^{\mathrm{pre}} \leq RH^2$, so $R/u \geq \mathcal{W}_\lambda$. Since $u + r_\lambda = \lfloor \lambda u \rfloor$,

$$\frac{t_H}{\lambda u} \geq 1 + \frac{\mathcal{W}_\lambda}{\lambda} - O(1/u). \tag{S9.45}$$

As $v > t_H$, this proves (9.126). The increments p_j have total sum zero and partial sums $\widetilde{A}_j$, so the bridge case of Lemma 5.1 gives (9.127). □

Remark 9.51 (Intrinsic normalization of the post-horizon cost). The normalization $\mathcal{W}_\lambda$ is positional, not intrinsic. Put

$$D_\lambda := H - K_\lambda \geq 1, \qquad \mathcal{E}_\lambda := \frac{\mathcal{V}_\lambda^{\mathrm{pre}}(t_H)}{RD_\lambda^2}. \tag{S9.46}$$

Then exactly

$$\mathcal{W}_\lambda = \frac{R}{u}\left(\frac{D_\lambda}{H}\right)^2 \mathcal{E}_\lambda. \tag{9.128}$$

Thus $\mathcal{W}_\lambda \to 0$ can occur for three different reasons: $D_\lambda/H \to 0$ (the height is already almost attained at the horizon), $R/u \to 0$ (the remaining rise occurs in a short segment), or $\mathcal{E}_\lambda \to 0$ (the remaining rise is intrinsically close to its chord). Only the third alternative is a genuine affine-rigidity statement.

For example, in a purely geometric unit-step model start the segment at height 1, take n zero increments and then n increments $+1$. Then $D_\lambda = n$, $R = 2n$ and

$$\mathcal{V}_\lambda^{\mathrm{pre}} = \frac{n(2n^2+1)}{12}, \qquad \mathcal{E}_\lambda \longrightarrow \frac{1}{12}, \qquad \mathcal{W}_\lambda \sim \frac{n}{6u} \tag{S9.47}$$

when $H = n+1$. Hence $\mathcal{W}_\lambda \to 0$ if $n = o(u)$ although the intrinsic chord deviation does not vanish. This is a geometric counterexample to the inference "small $\mathcal{W}_\lambda$ implies an almost affine rise"; it is not a claim about the actual Möbius path.

There is also an exact signed-correlation diagnostic for the third alternative. Write, on the horizon–arrival segment,

$$x_j := \sigma\mu(u + r_\lambda + j), \qquad T_r := \sum_{j \leq r} x_j, \qquad B_r := T_r - \frac{rD_\lambda}{R}, \tag{S9.48}$$

and

$$C_g^{\mathrm{raw}} := \sum_{j=1}^{R-g} x_j x_{j+g}. \tag{S9.49}$$

Then Lemma 5.1 in its non-bridge form gives

$$\sum_{g=1}^{R-1} gC_g^{\mathrm{raw}} = \frac{D_\lambda^2(R^2-1)}{6R} + D_\lambda \sum_{r=1}^{R-1} \left(1 - \frac{2r}{R}\right) B_r - \mathcal{V}_\lambda^{\mathrm{pre}}. \tag{9.129}$$

Consequently, if $\mathcal{E}_\lambda = o(1)$ along a sequence with $R \to \infty$, then Cauchy–Schwarz gives

$$\sum_{g<R} gC_g^{\mathrm{raw}} = \left(\frac{1}{6} + o(1)\right) D_\lambda^2 R > 0. \tag{S9.50}$$

Thus an intrinsically almost affine rise forces a large *positive* lag-weighted raw correlation in this convention. Neither (9.128) nor (9.129) supplies the missing arithmetic estimate; they only separate the three possible mechanisms.

Proposition 9.52 (Three-source critical packing)**.** *Fix $1 < \lambda < \lambda_0$ and order a bad witness subfamily as $(u_j, v_j]$, $1 \le j \le N$, with integer $N \ge 2$, $2 \le u_1 < \cdots < u_N \le X$ and $t_j > \lambda u_j$. For $j < N$ put*

$$A_j := \log \frac{t_j}{\lambda u_j}, \qquad R_j := \log \frac{v_j}{t_j}, \qquad G_j := \log \frac{u_{j+1}}{v_j}. \tag{9.130}$$

Then $A_j, R_j, G_j \ge 0$ and

$$(N-1)\log\lambda + \sum_{j=1}^{N-1} (A_j + R_j + G_j) \le \log \frac{X}{2}. \tag{9.131}$$

Writing $d_0 := T_0/\pi = 1/\log\lambda_0$, if for some fixed $\delta > 0$

$$\frac{1}{N-1} \sum_{j<N} (A_j + R_j + G_j) \ge \log \frac{\lambda_0}{\lambda} + \delta - o(1), \tag{9.132}$$

then

$$N \le (d_0 - c_\delta + o(1)) \log X \tag{9.133}$$

for some $c_\delta > 0$. Conversely, if $N = (d_0 - o(1)) \log X$, then

$$\frac{1}{N-1} \sum_{j<N} (A_j + R_j + G_j) \le \log \frac{\lambda_0}{\lambda} + o(1). \tag{9.134}$$

Proof. Disjointness gives $u_{j+1} \ge v_j$, and by definition

$$\log \frac{u_{j+1}}{u_j} = \log\lambda + A_j + R_j + G_j. \tag{9.135}$$

Summing for $j < N$ and using $u_1 \ge 2$, $u_N \le X$ proves (9.131). The first consequence follows because the assumed mean makes every bad member consume asymptotically more than $\log\lambda_0 + \delta$ of the logarithmic budget. The converse follows by rearranging (9.131) and using $(N-1)^{-1}\log(X/2) \le d_0^{-1} + o(1) = \log\lambda_0 + o(1)$. □

Remark 9.53 (Reference scale of the centered costs). This is a model calculation, not a theorem about μ. In a drifted-walk reference model with drift $m = H/h$ and active-step variance $\rho_{\mathrm{sf}} = 6/\pi^2$, a typical record block has duration $\asymp 1/m$ and centered bridge energy of order ρ_{sf}/m^2. Summing over H record levels therefore suggests

$$\mathfrak{C}(u, H) \asymp \frac{\rho_{\mathrm{sf}} h^2}{Hu}. \tag{9.136}$$

For a post-horizon segment of length R, a centered bridge instead has energy of order $\rho_{\mathrm{sf}} R^2/6$, suggesting

$$\mathcal{W}_\lambda \asymp \frac{\rho_{\mathrm{sf}} R^2}{6uH^2}. \tag{9.137}$$

Thus, when $h, R \asymp u$ and $H \asymp u^{\alpha}$ with $\alpha > 1/2$, the first scale grows like $u^{1-\alpha}$ whereas the second decays like $u^{1-2\alpha}$. This model therefore does not predict that either a uniform upper bound on $\mathfrak{C}$ or a uniform positive lower bound on $\mathcal{W}_\lambda$ will close the KP-scale problem. Their role is to identify exact adaptive correlation profiles and critical-packing rigidity.

Remark 9.54 (Scope of the centered route). The centered quantity removes the positive excess E''' from the statement of the target, but it creates a different analytic problem: the record blocks are adaptive and may be much shorter than the interval on which the Kaczorowski–Pintz witness is localized. In the difficult regime $h \asymp u$ and $H \asymp u^{\alpha}$ with $\alpha > 1/2$, their mean length is $h/H \asymp u^{1-\alpha}$, below the current all-interval Fourier range $u^{3/5+\iota}$ used in Theorem 12.13. Thus the centered identity is an internal diagnostic for improving a chosen linear counting threshold, but the present Fourier input does not close it block by block.

9.8. **Wide peaks from cancellation in all short intervals.** The only arithmetic input of the following lemma is the quantitative short-interval theorem of Matomäki–Teräväinen [35, Theorem 1.1]: for every fixed $\theta > 0.55$ and every fixed $\varepsilon > 0$, uniformly for the relevant intervals,

$$\sum_{x<n\le x+H} \mu(n) \ll_{\theta,\varepsilon} \frac{H}{(\log x)^{1/3-\varepsilon}} \qquad (H \ge x^{\theta}). \tag{9.138}$$

Lemma 9.55 (Mesoscopic Lipschitz bound). *Fix $\Lambda > 1$ and $\theta_{\mathrm{w}} \in (0.55, 1)$. For large u put $h_u = \lceil (\Lambda u)^{\theta_{\mathrm{w}}} \rceil$ and*

$$\omega_u := \max\Big\{ h_u^{-1},\ \max_{u\le a\le \Lambda u} \frac{|M(a+h_u) - M(a)|}{h_u} \Big\}. \tag{9.139}$$

Then $\omega_u \to 0$, and for all integers $u \le a < b \le \Lambda u$,

$$|M(b) - M(a)| \le \omega_u (b-a) + h_u. \tag{9.140}$$

Proof. Fix $\theta' \in (0.55, \theta_{\mathrm{w}})$. For all sufficiently large u, $a + h_u \le 2\Lambda u$ uniformly for $u \le a \le \Lambda u$. By [35, Theorem 1.1], for every fixed $\varepsilon > 0$ there is a constant C such that, uniformly for $u \le b \le 2\Lambda u$,

$$\Big| M\Big(b + \lceil b^{\theta'} \rceil\Big) - M(b) \Big| \le C \frac{b^{\theta'}}{(\log u)^{1/3-\varepsilon}} \tag{9.141}$$

for all sufficiently large u. Theorem 1.1 of [35] is uniform in the starting point x for intervals of length $x^{\theta'}$; here $b \asymp_\Lambda u$, so $\log b = \log u + O_\Lambda(1)$, and replacing $x^{\theta'}$ by $\lceil b^{\theta'} \rceil$ changes the sum by $O(1)$, absorbed in C.

For a given a, define recursively

$$b_0 = a, \qquad b_{j+1} = b_j + \lceil b_j^{\theta'} \rceil, \tag{9.142}$$

and stop at the first J for which $b_{J+1} > a + h_u$. Since $b_j \asymp u$ throughout the construction, each full block has length $\asymp u^{\theta'}$, whence

$$J \ll \frac{h_u}{u^{\theta'}} + 1, \qquad \sum_{j<J} b_j^{\theta'} \ll h_u. \tag{9.143}$$

All starting points b_j lie in $[u, 2\Lambda u]$. Summing the full-block estimates and bounding the final incomplete piece trivially gives, uniformly for $u \le a \le \Lambda u$,

$$\frac{|M(a+h_u) - M(a)|}{h_u} \ll_{\varepsilon,\theta',\theta_{\mathrm{w}},\Lambda} (\log u)^{-1/3+\varepsilon} + u^{\theta'-\theta_{\mathrm{w}}}. \tag{9.144}$$

Consequently

$$\omega_u \ll_{\varepsilon,\theta',\theta_{\mathrm{w}},\Lambda} (\log u)^{-1/3+\varepsilon} + u^{\theta'-\theta_{\mathrm{w}}} + h_u^{-1} = o(1), \tag{9.145}$$

and the same estimate is uniform for $u \in [U, 2U]$ as $U \to \infty$.

For (9.140), write $b-a=qh_u+r$ with $0\le r<h_u$. Every complete h_u-block starts at a point in $[u,\Lambda u]$ and contributes at most $\omega_u h_u$, while the terminal remainder contributes at most $r<h_u$. Thus $|M(b)-M(a)|\le \omega_u(b-a)+h_u$. □

Proposition 9.56 (Wide peaks)**.** *With $\Lambda,\theta_{\mathrm{w}},h_u,\omega_u$ as in Lemma 9.55, let $I_\nu=(u,v]$ be an excursion of orientation σ_ν with $v\le \Lambda u$ and u large, let t be an integer with $u<t<v$ and $|M(t)|=H$, and put $G=(H-h_u)_+$ and $m=\lfloor G/\omega_u\rfloor$. Then*

$$t-u\ge \frac{G}{\omega_u},\qquad v-t\ge \frac{G}{\omega_u}, \tag{9.146}$$

$$\mathfrak{b}\ge H+2\sum_{j=1}^{m}(G-\omega_u j),\qquad Q\ge H^2+2\sum_{j=1}^{m}(G-\omega_u j)^2. \tag{9.147}$$

Consequently, uniformly for such excursions with $H/h_u\to\infty$,

$$v-u\ge (2+o(1))\frac{H}{\omega_u},\qquad \mathfrak{b}\ge (1+o(1))\frac{H^2}{\omega_u},\qquad Q\ge \Big(\frac{2}{3}+o(1)\Big)\frac{H^3}{\omega_u}, \tag{9.148}$$

so that $(v-u)/H$, $\mathfrak{b}/H^2$ and Q/H^3 tend to infinity, whereas the unit-step bounds of Lemma 3.3 and Proposition 5.6 give only 2, 1 and $\frac{2}{3}$.

Proof. By (9.140) applied to $(u,t]$ and $(t,v]$, $H\le \omega_u(t-u)+h_u$ and $H\le \omega_u(v-t)+h_u$. Thus, since the two distances are integers,

$$t-u,\ v-t\ge \left\lceil \frac{G}{\omega_u}\right\rceil, \tag{9.149}$$

and therefore $m=\lfloor G/\omega_u\rfloor\le t-u, v-t$. For $1\le j\le m$ the points $t\pm j$ lie in $[u,v]$, and (9.140) gives $\sigma_\nu M(t\pm j)\ge G-\omega_u j\ge 0$. Summing t and the pairs $t\pm j$ gives the bounds for $\mathfrak{b}$ and Q. Finally $2\sum_{j\le m}(G-\omega_u j)=G^2/\omega_u+O(G)$ and $2\sum_{j\le m}(G-\omega_u j)^2=\frac{2}{3}G^3/\omega_u+O(G^2)$. Under $H/h_u\to\infty$ we have $G=H-h_u=(1+o(1))H$ and $G\to\infty$. Equation (9.145) gives $\omega_u=o(1)$, hence eventually $\omega_u\le 1$ and the relative errors satisfy $\omega_u/G\le 1/G=o(1)$. This proves the stated uniform asymptotics. □

Corollary 9.57 (Packing of high excursions)**.** *Let $Y=Y(U)$ with $Y/U^{\theta_{\mathrm{w}}}\to\infty$. The number of excursions with $U\le u\le 2U$, $v\le \Lambda u$ and height at least Y is $o(U/Y)$, whereas the unit-step bound gives $O(U/Y)$.*

Proof. They are disjoint and contained in $[U,2\Lambda U]$. By Proposition 9.56, each has length at least

$$(2+o(1))\frac{Y}{\max_{U\le u\le 2U}\omega_u}. \tag{9.150}$$

If $N(U,Y)$ denotes their number, disjointness therefore gives

$$N(U,Y)(2+o(1))\frac{Y}{\max_{U\le u\le 2U}\omega_u}\le (2\Lambda-1)U. \tag{9.151}$$

By the uniform estimate (9.145),

$$\omega_U^{*}:=\max_{U\le u\le 2U}\omega_u=o(1). \tag{9.152}$$

Therefore

$$N(U,Y)\ll_\Lambda \omega_U^{*}\frac{U}{Y}=o(U/Y), \tag{9.153}$$

which completes the packing argument. □

Remark 9.58 (Scope). Proposition 9.56 is informative only for heights above $u^{0.55+o(1)}$. If $\Theta > 0.55$ and $0.55 < \theta_{\mathrm{w}} < \Theta - \eta$, it applies to the Kaczorowski–Pintz witness excursions of Corollary 9.18 with $v \le \lambda u$: their peaks are wide in the sense above. It does not by itself improve the linear horizon: the resulting lower bound for $\log(v/u)$ is of order $H/(\omega_u u)$, which may still tend to 0, whereas Remark 9.38 requires a fixed surplus. The published quantitative estimate of Matomäki–Teräväinen [35, Theorem 1.1] makes the gain explicit. If $0.55 < \theta' < \theta_{\mathrm{w}}$, then the blocking argument in Lemma 9.55 gives, for every fixed $\varepsilon > 0$,

$$\omega_u \ll_{\varepsilon,\theta',\theta_{\mathrm{w}},\Lambda} (\log u)^{-1/3+\varepsilon} + u^{\theta'-\theta_{\mathrm{w}}}. \tag{9.154}$$

Consequently, in the regime $H/h_u \to \infty$ of Proposition 9.56,

$$Q \gg_{\varepsilon} H^3 (\log u)^{1/3-\varepsilon}. \tag{9.155}$$

10. Arithmetic selection on the Catalan reflection fibre

The fixed-mask Catalan description separates two finite degrees of freedom of a completed excursion of M: its positive support-time shape and its orientation. The Catalan estimates developed above act on the first coordinate. The present section isolates the second coordinate and shows that it is strongly constrained by divisor inversion. The resulting arithmetic selector is nevertheless insufficient, by itself, to observe a fixed-shift Chowla form on the ambient squarefree space. This produces a sharp separation between orientation selection and two-point cancellation.

Throughout this section, let N be an integer with $M(N) = 0$, so that the portion of the path up to N consists only of completed excursions and zero runs. Write

$$I_\nu = (x_{0,\nu}, x_{1,\nu}], \qquad 1 \le \nu \le J_N, \tag{10.1}$$

for the completed excursions up to N, and let

$$E_\nu := \{d \in I_\nu : \mu(d) \neq 0\} \tag{10.2}$$

be the active support of the νth excursion. For a word $w \in \{-1,0,1\}^N$ define

$$M_w(n) := \sum_{d \le n} w(d), \qquad \Pi_N(w) := \big(M_w(n)^2\big)_{1 \le n \le N}. \tag{10.3}$$

We restrict to words having the same zero mask as μ.

Theorem 10.1 (Exact reflection fibre)**.** *Let $w \in \{-1,0,1\}^N$ have the same support as μ. Then*

$$\Pi_N(w) = \Pi_N(\mu) \tag{10.4}$$

if and only if, on each completed excursion I_ν, the word w agrees either with μ or with $-\mu$. Consequently the fibre of the complete quadratic profile has exactly

$$2^{J_N} \tag{10.5}$$

elements.

Proof. Both partial-sum paths start from the same value, $M_w(0) = M(0) = 0$, and $w, \mu \in \{-1,0,1\}^N$ have the same zero mask. Independent reflection of a completed excursion multiplies M by -1 on the interior of that excursion and leaves it zero at both endpoints; hence it preserves $M(n)^2$ at every integer. Conversely, equality of the quadratic profiles gives the same zero set and the same absolute path. On the interior of each completed excursion one has

$$|M_w(n)| = |M(n)| > 0, \tag{10.6}$$

so M_w has no interior zero and therefore has constant sign there. Since M also has constant sign on that excursion, the ratio M_w/M is a constant in $\{\pm 1\}$. Taking first differences successively across that component shows that the same constant relates w and μ on the active support of the excursion. The choices are independent because every completed excursion begins and ends at zero. □

Theorem 10.1 identifies the precise information lost by every statistic determined by the full quadratic profile: after the absolute excursion shapes are fixed, the remaining ambiguity is a Boolean orientation vector.

For $\mathbf{z} = (\mathbf{z}_1, \dots, \mathbf{z}_{J_N}) \in \{0,1\}^{J_N}$, let $w_{\mathbf{z}}$ be obtained from μ by reflecting precisely those excursions for which $\mathbf{z}_\nu = 1$. Fix arbitrary weights $\varrho_n > 0$ and define the divisor-defect functional

$$\mathfrak{D}_N(w) := \varrho_1 |w(1) - 1|^2 + \sum_{n=2}^{N} \varrho_n \left| \sum_{d \mid n} w(d) \right|^2 . \tag{10.7}$$

For every excursion put

$$g_\nu(n) := \sum_{\substack{d \mid n \\ d \in E_\nu}} \mu(d), \qquad 1 \le n \le N, \tag{10.8}$$

and define the $J_N \times J_N$ matrix

$$(G_N)_{\nu\kappa} := \sum_{n=1}^{N} \varrho_n g_\nu(n) g_\kappa(n). \tag{10.9}$$

Theorem 10.2 (Positive Gram selector for excursion orientations)**.** *The matrix G_N is positive semidefinite and*

$$\mathfrak{D}_N(w_{\mathbf{z}}) = 4\, \mathbf{z}^{\mathsf{T}} G_N \mathbf{z} \qquad (\mathbf{z} \in \{0,1\}^{J_N}). \tag{10.10}$$

Moreover,

$$\mathbf{z}^{\mathsf{T}} G_N \mathbf{z} > 0 \qquad \text{for every } \mathbf{z} \neq 0. \tag{10.11}$$

Thus the divisor equations select the arithmetic orientation uniquely inside the complete quadratic-profile fibre.

Proof. For $n \ge 2$, Möbius inversion gives

$$\sum_{d \mid n} \mu(d) = 0. \tag{10.12}$$

Reflecting the excursions indexed by $\mathbf{z}$ therefore gives

$$\sum_{d \mid n} w_{\mathbf{z}}(d) = -2 \sum_{\nu=1}^{J_N} \mathbf{z}_\nu g_\nu(n). \tag{10.13}$$

At $n = 1$ one has explicitly

$$w_{\mathbf{z}}(1) - 1 = -2 \sum_{\nu=1}^{J_N} \mathbf{z}_\nu g_\nu(1), \tag{10.14}$$

with only the excursion containing 1 able to contribute. Substitution into (10.7) yields (10.10), and (10.9) is a Gram representation, so $G_N \succeq 0$.

If a nonzero Boolean vector $\mathbf{z}$ satisfied $\mathbf{z}^{\mathsf{T}} G_N \mathbf{z} = 0$, then $\mathfrak{D}_N(w_{\mathbf{z}}) = 0$. Hence $w_{\mathbf{z}}(1) = 1$ and

$$\sum_{d \mid n} w_{\mathbf{z}}(d) = 0 \qquad (2 \le n \le N). \tag{10.15}$$

Finite Möbius inversion determines the coefficients recursively and forces $w_{\mathbf{z}}(d) = \mu(d)$ for every $d \le N$, contradicting the reflection of a nonempty set of excursions. □

The Gram form is exact but uses all divisor equations. A much smaller family of specially placed divisor equations already compresses the orientation fibre from exponential to polynomial size.

For the νth excursion list its active support as

$$a_{\nu,1} < \cdots < a_{\nu, L_\nu^{\#}}, \qquad L_\nu^{\#} = 2r_\nu, \tag{10.16}$$

and define its support centre

$$m_\nu := a_{\nu,r_\nu}. \tag{10.17}$$

Set

$$\phi_\nu(d) := \sigma_\nu \mu(d) \qquad (d \in E_\nu), \tag{10.18}$$

so that ϕ_ν is the oriented increment sequence of the positive support-time excursion. Define the central divisor matrix

$$\mathscr{C}^{\mathrm{div}}_{\nu\kappa} := \sum_{\substack{d \mid m_\nu \\ d \in E_\kappa}} \phi_\kappa(d). \tag{10.19}$$

Theorem 10.3 (Polynomial compression of the orientation fibre)**.** *The matrix $\mathscr{C}^{\mathrm{div}}$ is lower triangular. Its diagonal entry in row ν is nonzero whenever*

$$x_{0,\nu} \geq \frac{m_\nu}{2}. \tag{10.20}$$

The condition (10.20) *can fail for at most $\lfloor \log_2 N \rfloor$ completed excursions.*

After the affine normalization $w(1) = 1$ is imposed, the solution set of the central divisor equations is an affine subset of the orientation-coordinate space whose linear direction has dimension at most $\lfloor \log_2 N \rfloor + 1$. More concretely, once the exceptional orientation coordinates have been specified, every nonexceptional coordinate not already fixed by the affine normalization is determined successively by a central equation with nonzero diagonal coefficient. Consequently, among the 2^{J_N} Boolean orientations in the quadratic-profile fibre, at most

$$2^{\lfloor \log_2 N \rfloor + 1} \leq 2N \tag{10.21}$$

satisfy the central divisor equations. Furthermore, at most $\lfloor \log_2 N \rfloor + 1$ additional divisor rows suffice to select the Möbius orientation uniquely.

Proof. If $\kappa > \nu$, every $d \in E_\kappa$ exceeds m_ν, so no such d can divide m_ν; hence $\mathscr{C}^{\mathrm{div}}_{\nu\kappa} = 0$. If (10.20) holds, every proper divisor of m_ν is at most $m_\nu/2 \leq x_{0,\nu}$ and therefore lies outside E_ν. Thus only $d = m_\nu$ contributes to the diagonal, giving

$$\mathscr{C}^{\mathrm{div}}_{\nu\nu} = \phi_\nu(m_\nu) \in \{\pm 1\}. \tag{10.22}$$

Since $m_\nu \leq x_{1,\nu}$, failure of (10.20) implies $x_{1,\nu} > 2x_{0,\nu}$. Count the initial excursion separately. If $2 \leq a_1 < \cdots < a_r$ are the positive starts of the remaining failing completed excursions, then disjointness gives

$$N \geq x_{1,r} > 2a_r > 2^2 a_{r-1} > \cdots > 2^r a_1 \geq 2^{r+1}. \tag{10.23}$$

Thus the total number of failures, including the possible initial one, is strictly smaller than $\log_2 N$ when N is not a power of two and at most $\log_2 N - 1$ when it is; in either case it is at most $\lfloor \log_2 N \rfloor$.

Let $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_{J_N})^{\mathsf{T}}$ be the arithmetic orientation vector. For every central row with $m_\nu > 1$,

$$(\mathscr{C}^{\mathrm{div}} \boldsymbol{\sigma})_\nu = \sum_{d \mid m_\nu} \mu(d) = 0. \tag{10.24}$$

The row $n = 1$ is imposed separately as the affine normalization $w(1) = 1$. Let $\mathcal{E}_N$ be the set of indices for which (10.20) fails. Among the central rows with $m_\nu > 1$, every nonexceptional row has a nonzero new diagonal coefficient. Thus their coefficient matrix has rank at least

$$J_N - |\mathcal{E}_N| - 1, \tag{10.25}$$

the possible loss of one row accounting only for the distinguished index $m_\nu = 1$. Hence the associated homogeneous linear system has nullity at most $|\mathcal{E}_N| + 1$. Intersecting this linear solution space with the affine normalization $w(1) = 1$ gives an affine solution set whose linear direction has dimension at most $|\mathcal{E}_N| + 1 \leq \lfloor \log_2 N \rfloor + 1$.

For the Boolean count we use the triangular structure directly rather than a dimension argument. Choose first the coordinates indexed by $\mathcal{E}_N$ and, if it is not already among them, the single coordinate attached to the row $m_\nu = 1$. Then process every remaining nonexceptional row in increasing order. All earlier coordinates are already known, while the coefficient of the new coordinate is $\mathscr{C}^{\mathrm{div}}_{\nu\nu} \neq 0$, so that coordinate is uniquely determined. The affine equation $w(1) = 1$ can only remove choices, not create them. Therefore there are at most $2^{|\mathcal{E}_N|+1} \leq 2^{\lfloor \log_2 N \rfloor + 1} \leq 2N$ Boolean solutions. In particular the Boolean estimate is not inferred from real dimension alone.

For the final assertion, for a generic real orientation vector $\boldsymbol{\sigma}' = (\sigma'_1, \dots, \sigma'_{J_N})$ define $w_{\boldsymbol{\sigma}'}(d) = \sigma'_\nu \phi_\nu(d)$ for $d \in E_\nu$ and $w_{\boldsymbol{\sigma}'}(d) = 0$ off the active support. Factor the complete divisor readout as

$$\boldsymbol{\sigma}' \longmapsto w_{\boldsymbol{\sigma}'} \longmapsto \left(\sum_{d \mid n} w_{\boldsymbol{\sigma}'}(d) \right)_{1 \leq n \leq N}. \tag{10.26}$$

The first map is injective: the supports E_ν are disjoint and every completed excursion contains an active increment, so two distinct orientation vectors give distinct words. The second map is injective on $\mathbb{R}^N$ because its matrix is $(\mathbf{1}_{d|n})_{1 \leq n,d \leq N}$, which is lower unitriangular. Hence the composition has full column rank J_N on the orientation coordinates. The central rows with $m_\nu > 1$ have linear rank at least $J_N - |\mathcal{E}_N| - 1$, independently of the separate affine normalization $w(1) = 1$. Extending a basis of their row space to a basis of the full readout therefore needs at most $|\mathcal{E}_N| + 1 \leq \lfloor \log_2 N \rfloor + 1$ additional divisor rows. The full-rank system has at most one orientation solution, and the arithmetic orientation $\boldsymbol{\sigma}$ is such a solution; it is therefore unique. □

The preceding theorem concerns selection inside the highly structured reflection fibre. It does not imply that a fixed-shift correlation can be recovered from a comparably small family of divisor measurements. In fact a linear obstruction holds on the ambient squarefree space.

Let $\mathbf{D}_N$ be the divisor-incidence matrix

$$(\mathbf{D}_N)_{r,d} := \mathbf{1}_{d|r}, \qquad 1 \leq r, d \leq N, \tag{10.27}$$

and, for $\mathcal{T} \subseteq \{1, \dots, N\}$, let $\mathbf{D}_\mathcal{T}$ denote the restriction to rows in $\mathcal{T}$. Let A_h be the symmetric fixed-shift matrix defined by

$$x^{\mathsf{T}} \mathsf{A}_h x = \sum_{n \leq N-h} x_n x_{n+h}. \tag{10.28}$$

Write

$$\mathfrak{S}_h := \prod_{p^2 \mid h} \left(1 - \frac{1}{p^2}\right) \prod_{p^2 \nmid h} \left(1 - \frac{2}{p^2}\right) > 0, \tag{10.29}$$

so that, by the classical squarefree-pair theorem of Mirsky [37],

$$\#\{n \leq X : n, n+h \text{ squarefree}\} = \mathfrak{S}_h X + o_h(X) \tag{10.30}$$

for each fixed nonzero h.

Theorem 10.4 (Linear obstruction to divisor-row observability)**.** *Fix $h \geq 1$. Suppose that for some $c_{\mathrm{obs}} \in [0, \infty)$, some $\varepsilon \in [0, 1/2)$, and some row set $\mathcal{T} \subseteq \{1, \dots, N\}$ one has*

$$|x^{\mathsf{T}} \mathsf{A}_h x| \leq c_{\mathrm{obs}} \|\mathbf{D}_\mathcal{T} x\|_2^2 + \varepsilon \|x\|_2^2 \tag{10.31}$$

for every real vector x supported on the squarefree integers in $[1, N]$. Then

$$|\mathcal{T}| \geq \frac{\mathfrak{S}_h}{4} N + o_h(N). \tag{10.32}$$

In particular, no $o(N)$ family of divisor rows can uniformly observe the fixed-shift quadratic form with error below $1/2$.

Proof. Let $N/2 < b \le N - h$ be such that both b and $b+h$ are squarefree. Since $b > N/2$, column b of $\mathbf{D}_N$ is nonzero only in row b; the same holds for $b+h$. If neither row belongs to $\mathcal{T}$, then for

$$x = e_b \pm e_{b+h} \tag{10.33}$$

one has $\mathbf{D}_{\mathcal{T}} x = 0$, while

$$|x^{\mathsf{T}} \mathsf{A}_h x| = 1, \qquad \|x\|_2^2 = 2. \tag{10.34}$$

Thus (10.31) would force $1 \le 2\varepsilon$, contrary to $\varepsilon < 1/2$. Hence $\mathcal{T}$ must meet every edge $\{b, b+h\}$ of the graph of squarefree shifted pairs in $(N/2, N]$.

By (10.29), this graph has

$$\frac{\mathfrak{S}_h}{2} N + o_h(N) \tag{10.35}$$

edges. Its maximum degree is at most 2, so every vertex cover has cardinality at least half the number of edges. This gives (10.32). □

Corollary 10.5 (Selection–observability separation)**.** *For cutoffs N with $M(N) = 0$, the complete quadratic profile has an orientation fibre of size 2^{J_N}. The central divisor equations reduce that fibre to at most $2N$ candidates, and $O(\log N)$ further divisor equations select the arithmetic orientation uniquely. In contrast, for every fixed $h \ge 1$, uniform observation of the ambient fixed-shift form $x^{\mathsf{T}} \mathsf{A}_h x$ with error strictly below $1/2$ requires $\Omega_h(N)$ divisor rows.*

Consequently, the unresolved difficulty in fixed-shift Möbius cancellation is not the finite identification of the orientation bits once the absolute excursion geometry is known. It is the collective cancellation of the shift interaction across the arithmetic path. The comparison is deliberately limited: the observability theorem is a worst-case statement on the ambient squarefree-supported vector space, whereas the selection theorem concerns the Möbius orientation fibre. What is proved is the contrast between triangular compression to at most $2N$ candidates inside that fibre and the linear number of divisor rows required for uniform ambient fixed-shift observation; no intrinsic complexity-class separation is asserted.

11. Prefix structure of the localized window

11.1. **The linear window and the one-step recursion.** Once the horizon is below $2x_0$, the square-range bilinear structure of Proposition 11.41 is no longer needed: on $(u, 2u]$ the Möbius function is a *linear* expression in the prefix $\mu \mathbf{1}_{[1,u]}$.

Proposition 11.1 (Linear window, classical)**.** *Let $u \ge 1$. For every integer $u < n \le 2u$,*

$$\mu(n) = -\sum_{\substack{d \mid n \\ d \le u}} \mu(d). \tag{11.1}$$

Consequently, if $M(u) = 0$ and $u < t \le 2u$,

$$M(t) = -\sum_{d \le u} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right). \tag{11.2}$$

Proof. Since $n \ge 2$, $\sum_{d \mid n} \mu(d) = 0$. Every proper divisor of n is at most $n/2 \le u$, and every divisor $d \le u$ is proper because $n > u$. Summing (11.1) over $u < n \le t$, the number of multiples of d in $(u, t]$ is $\lfloor t/d \rfloor - \lfloor u/d \rfloor$. □

Corollary 11.2 (Localized prefix formula)**.** *Let $M(u) = 0$, let $1 < \lambda < 2$, and let $u < t \le \lambda u$. Then*

$$M(t) = -\sum_{d \le \lambda u/2} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right). \tag{11.3}$$

Indeed, if $\lambda u/2 < d \le u$, then $d > u/2$ and the first multiple of d strictly larger than u is $2d > \lambda u \ge t$.

Corollary 11.3 (Prefix-dependency cutoff)**.** *Put*

$$\chi_0 := \frac{1}{2}e^{\pi/T_0}. \tag{11.4}$$

For every fixed χ with $\chi_0 < \chi < 1$, RH is equivalent to Littlewood's bound on every excursion starting at u throughout

$$u < t \le \min\{x_1, 2\chi u\}, \tag{11.5}$$

where, throughout that window, $M(t)$ is given by the sum (11.3) *using only the prefix $\mu(d)$ with $d \le \chi u$. In particular, the choice $\chi = 5/8$ corresponds to $\lambda = 5/4$ and gives the RH-equivalent window*

$$1 \le r = t - u \le \min\{\ell - 1, u/4\} \tag{11.6}$$

using only $d \le 5u/8$. The return point $r = \ell$, when it belongs to the window, is omitted because $M(u + \ell) = 0$.

The number χ_0 is an infimum, not an attained endpoint, because the horizon theorem requires $\lambda > \lambda_0$.

Proof. Take $\lambda = 2\chi > \lambda_0$ in Corollary 9.17(iii), then use Corollary 11.2. □

Corollary 11.4 (Algebraic dependency gap)**.** *Let $\chi_0 < \chi < 1$. If $u \in \mathbb{N}$ satisfies $M(u) = 0$, then every increment $\mu(n)$ with*

$$u < n \le 2\chi u \tag{11.7}$$

is determined by the prefix $\mu(1), \dots, \mu(\lfloor \chi u \rfloor)$. Consequently the intermediate block

$$(\chi u, u] \tag{11.8}$$

does not occur as a direct input in the divisor recursion determining the localized future window. As $\chi \downarrow \chi_0$, the omitted block has relative length tending to $1 - \chi_0$.

Proof. For $u < n \le 2\chi u$, every proper divisor $d \mid n$ satisfies $d \le n/2 \le \chi u$. Hence

$$\mu(n) = -\sum_{\substack{d \mid n \\ d < n}} \mu(d) \tag{11.9}$$

uses only $d \le \chi u$. □

In convolution form, let $\mathbf{1}(n) \equiv 1$ and $\mu_u := \mu \mathbf{1}_{[1,u]}$. Then the coordinatewise statement is

$$((\mu - \mu_u) * \mathbf{1})(n) = \mu(n)\mathbf{1}_{\{u < n \le 2u\}} \qquad (n \le 2u). \tag{11.10}$$

Indeed, for $n \le 2u$ the only divisor of n that can exceed u is n itself. This is the order-one analogue of (11.103), which for $u < n \le u^2$ reads $\mu(n) = -(\mu_u * \mu_u * 1)(n)$; on $(u, 2u]$ both give $\mu(n)$, the bilinear form being the quadratic continuation of the linear one. Equivalently, (11.2) follows from $\sum_{d \le t} \mu(d)\lfloor t/d \rfloor = 1$, since $\lfloor t/d \rfloor = 1$ for $u < d \le t \le 2u$.

Proposition 11.5 (One-step recursion, classical)**.** *For real $x \ge 1$,*

$$\sum_{k \le x} M\left(\frac{x}{k}\right) = 1. \tag{11.11}$$

Hence, if $M(u) = 0$ and $t > u$,

$$M(t) = -\sum_{2 \le k \le t} \left[M\left(\frac{t}{k}\right) - M\left(\frac{u}{k}\right) \right], \tag{11.12}$$

with $M(y) = 0$ for $y < 1$. Every argument on the right is at most $t/2$; hence on a localized window $u < t \le \lambda u$ all arguments are at most $\lambda u/2$. For $\lambda = 5/4$ this is $5u/8$.

Proof. $\sum_{k\le x} M(x/k) = \sum_{k\le x}\sum_{d\le x/k}\mu(d) = \sum_{m\le x}\sum_{d\mid m}\mu(d) = 1$. Here and below $M(x/k) = M(\lfloor x/k\rfloor)$ by the global real-argument convention. Subtract the identity at u, where $M(u) = 0$, and extend the second sum to $k \le t$ using $M(u/k) = 0$ for $k > u$. □

Proposition 11.6 (Scale cascade in the localized excursion window)**.** *Let $(u, v]$ be an excursion, let $\lambda_0 < \lambda < 2$, and let $u < t \le \min\{v, \lambda u\}$. For $k \ge 2$ put*

$$\Delta_k(u,t) := M(t/k) - M(u/k), \tag{11.13}$$

and for an integer $K \ge 2$ put

$$\mathcal{R}_K(u,t) := -\sum_{K<k\le t} \Delta_k(u,t). \tag{11.14}$$

Then, exactly,

$$M(t) = -\sum_{k=2}^{K} \Delta_k(u,t) + \mathcal{R}_K(u,t), \tag{11.15}$$

and $\mathcal{R}_K(u,t)$ uses only values $M(y)$ with

$$y \le \frac{\lambda u}{K+1}. \tag{11.16}$$

Equivalently, all dependence on values above $\lambda u/(K+1)$ is concentrated in the $K-1$ increments $\Delta_2, \dots, \Delta_K$. Moreover

$$\Delta_k(u,t) = \sum_{u/k<n\le t/k} \mu(n), \tag{11.17}$$

so the supporting real interval has length $(t-u)/k$ and contains $\lfloor t/k\rfloor - \lfloor u/k\rfloor$ integers.

Proof. Equation (11.15) is Proposition 11.5 with the terms $2 \le k \le K$ separated. If $k > K$, then $t/k \le \lambda u/(K+1)$ and $u/k \le \lambda u/(K+1)$. The last identity is the definition of the summatory function on the interval $(u/k, t/k]$. □

Corollary 11.7 (First channels)**.** *For $K = 2$,*

$$M(t) = -\big(M(t/2) - M(u/2)\big) + \mathcal{R}_2(u,t), \tag{11.18}$$

where $\mathcal{R}_2$ uses only $y \le \lambda u/3$. With the convenient localized choice $\lambda = 5/4$, the only channel above $5u/12$ is

$$\Delta_2(u, u+r) = \sum_{u/2<n\le u/2+r/2} \mu(n), \qquad 0 < r \le u/4, \tag{11.19}$$

whose supporting interval has length at most $u/8$. In the limiting sense $\lambda \downarrow \lambda_0$, the successive remainder cutoffs are $\lambda_0 u/2, \lambda_0 u/3, \lambda_0 u/4, \dots$; in particular the first cutoff approaches $u/2$ as the verified height increases.

Lemma 11.8 (Channel persistence extracted from a high excursion)**.** *Let $S_r = \sigma_\nu M(u+r) > 0$ on $1 \le r \le h$ with $S_0 = 0$ and $|S_r - S_{r-1}| \le 1$. Fix $K \ge 2$ and write, using Proposition 11.6,*

$$S_r = \sum_{k=2}^{K} X_k(r) + Y_K(r), \quad X_k(r) := -\sigma_\nu \Delta_k(u, u+r), \quad Y_K(r) := \sigma_\nu \mathcal{R}_K(u, u+r). \tag{11.20}$$

If $H = \max_{r\le h} S_r$ and τ is a point with $S_\tau = H$, then among the last $\lfloor H/2\rfloor + 1$ indices

$$\tau - \lfloor H/2\rfloor, \dots, \tau \tag{11.21}$$

there is one fixed component among $X_2, \dots, X_K, Y_K$ which is at least $H/(2K)$ at no fewer than $\lceil (\lfloor H/2\rfloor + 1)/K\rceil$ indices.

Proof. Since $S_0 = 0$ and the path has unit steps, $\tau \geq H$ and $S_{\tau-j} \geq H - j \geq H/2$ for $0 \leq j \leq \lfloor H/2 \rfloor$. At every one of these indices the sum of the K displayed components is at least $H/2$, hence some component is at least $H/(2K)$. Pigeonholing the $\lfloor H/2 \rfloor + 1$ indices among the K components gives at least $\lceil (\lfloor H/2 \rfloor + 1)/K \rceil$ occurrences, proving the claim. □

Remark 11.9 (What the persistence lemma adds). Lemma 11.8 converts a large value attained inside one excursion into a quantified persistence requirement: either one of the finitely many scaled interval sums $\Delta_2, \ldots, \Delta_K$ repeatedly carries a positive fraction of the height, or the responsibility lies repeatedly in the deeper remainder, which is supported entirely below $\lambda u/(K+1)$. This support statement is not a smallness estimate for the remainder. This avoids the iteration defect of Remark 11.22: no new point is claimed to be the start-window point of another excursion.

Corollary 11.10 (Channel descent with explicit loss). *Let $u \in \mathbb{N}$ satisfy $M(u) = 0$, let $t > u$, $2 \leq K \leq u$, and $0 < \vartheta < 1$. Then either*

(i) *there are $2 \leq k \leq K$ and $y \in \{\lfloor u/k \rfloor, \lfloor t/k \rfloor\}$ with $|M(y)| \geq |M(t)|/(2K)$, hence*

$$\frac{|M(y)|}{y^\vartheta} \geq \frac{k^\vartheta}{2K} \cdot \frac{|M(t)|}{t^\vartheta}, \qquad y \leq \left\lfloor \frac{t}{2} \right\rfloor; \tag{11.22}$$

or

(ii) *$|\mathcal{R}_K(u,t)| \geq |M(t)|/K$, where $\mathcal{R}_K$ depends only on $M(y)$ with $y \leq \lfloor t/(K+1) \rfloor$.*

The statement holds at every $t > u$, not only in a localized window.

Proof. The proof of Proposition 11.6 applies verbatim to every $t > u$, with $\mathcal{R}_K$ supported on $y \leq t/(K+1)$. By (11.15), one of the K terms $\Delta_2, \ldots, \Delta_K, \mathcal{R}_K$ has modulus at least $|M(t)|/K$. If it is $\Delta_k = M(\lfloor t/k \rfloor) - M(\lfloor u/k \rfloor)$, one of its two values has modulus at least $|M(t)|/(2K)$. Since $k \leq K \leq u$, both candidate arguments are positive integers; hence $y \geq 1$. Moreover $y \leq \lfloor t/k \rfloor \leq \lfloor t/2 \rfloor$, and therefore

$$\frac{|M(y)|}{y^\vartheta} \geq \frac{|M(t)|}{2K} \left(\frac{t}{k}\right)^{-\vartheta} = \frac{k^\vartheta}{2K} \frac{|M(t)|}{t^\vartheta}. \tag{11.23}$$

□

Remark 11.11 (Channels do not iterate to a bound). Corollary 11.10 has no iteration gap: it applies at the new point y relative to the last zero of M before y. Its defect is the constant loss $k^\vartheta/(2K) < 1$ per step. If alternative (i) held at every step, a chain of length $m \asymp \log t$ would give only $|M(t)| \leq (2K)^m \, O(1) = t^{O(\log K)}$, weaker than the trivial bound. For fixed K and $0 < \vartheta < 1$ the guaranteed factor can never reach 1, since

$$\frac{k^\vartheta}{2K} \leq \frac{K^\vartheta}{2K} = \frac{1}{2K^{1-\vartheta}} < \frac{1}{2}. \tag{11.24}$$

Thus no iteration of the same pigeonhole estimate can remove the loss at an individual step. A closing argument must instead recover the loss collectively from the persistence supplied by Lemma 11.8, or prove cancellation in the deep remainder. Both require information beyond dependency alone.

Proposition 11.12 (Quantified persistence along channel descent). *Fix $K \geq 2$, $0 < \vartheta < 1$ and $\delta > 0$, and put*

$$c_{\vartheta,K} := \log\left(\frac{2K}{2^\vartheta}\right) > 0. \tag{11.25}$$

Suppose $t_0 \to \infty$ through points satisfying

$$Z_\vartheta(t_0) := \frac{|M(t_0)|}{t_0^\vartheta} \geq t_0^\delta. \tag{11.26}$$

Since $|M(t_0)| \leq t_0$, any nonempty asymptotic regime of this form has $\delta \leq 1 - \vartheta$.

At stage j, let u_j be the last zero of M before t_j. Continue the descent only while $u_j \geq K$ and alternative (i) *of Corollary 11.10 occurs; stop if either the deep-remainder alternative occurs or $u_j < K$. Whenever a descent step is available there are $2 \leq k_j \leq K$ and $t_{j+1} \leq \lfloor t_j/2 \rfloor$ such that*

$$Z_\vartheta(t_{j+1}) \geq \frac{k_j^\vartheta}{2K} Z_\vartheta(t_j). \tag{11.27}$$

Consequently, after m successful steps,

$$Z_\vartheta(t_m) \geq Z_\vartheta(t_0) \prod_{j<m} \frac{k_j^\vartheta}{2K} \geq t_0^\delta e^{-m c_{\vartheta,K}}. \tag{11.28}$$

Set

$$m_* := \left\lfloor \frac{\delta \log t_0}{2c_{\vartheta,K}} \right\rfloor, \qquad \eta_{\vartheta,K,\delta} := \frac{\delta \log 2}{2c_{\vartheta,K}}. \tag{11.29}$$

Then, for all sufficiently large t_0, either a deep-remainder alternative occurs before step m_*, or the descent is valid through step m_* and there is a descendant satisfying

$$t_{m_*} \leq t_0^{1-\eta_{\vartheta,K,\delta}+o(1)}, \qquad t_{m_*} \geq t_0^{\delta/(2(1-\vartheta))}, \qquad Z_\vartheta(t_{m_*}) \geq t_0^{\delta/2}. \tag{11.30}$$

The parameter in this conclusion always satisfies $0 < \eta_{\vartheta,K,\delta} < 1/2$; in particular the formerly stated case $\eta_{\vartheta,K,\delta} \geq 1$ is empty.

Proof. Each successful application of Corollary 11.10 gives $t_{j+1} \leq \lfloor t_j/2 \rfloor$ and multiplies the normalized value by at least $2^\vartheta/(2K) = e^{-c_{\vartheta,K}}$, proving (11.28). Hence for every $m \leq m_*$ reached without a deep remainder,

$$Z_\vartheta(t_m) \geq t_0^{\delta/2}, \qquad t_m \leq t_0/2^m. \tag{11.31}$$

It remains to justify that the stopping condition $u_j < K$ cannot occur before m_*. By Subsection 1.5, $M(n) = 0$ for arbitrarily large integers n; choose once and for all a zero $z_K \geq K$. If the last zero before t_j were $< K$, then necessarily $t_j < z_K$; otherwise z_K itself would be a preceding zero. Hence

$$Z_\vartheta(t_j) \leq z_K^{1-\vartheta}, \tag{11.32}$$

a constant depending only on K and ϑ. This contradicts $Z_\vartheta(t_j) \geq t_0^{\delta/2}$ for large t_0. Therefore the only possible premature stop is the deep-remainder alternative. If it does not occur, the iteration reaches m_*. Finally $t_{m_*} \leq t_0 2^{-m_*} = t_0^{1-\eta_{\vartheta,K,\delta}+o(1)}$. The trivial estimate $Z_\vartheta(t) \leq t^{1-\vartheta}$ also gives $t_{m_*} \geq t_0^{\delta/(2(1-\vartheta))}$. Finally, $c_{\vartheta,K} = \log(2K) - \vartheta \log 2 > \log 2$ and $\delta \leq 1 - \vartheta < 1$, whence $0 < \eta_{\vartheta,K,\delta} < \delta/2 < 1/2$. □

Remark 11.13 (What remains after quantified persistence). Proposition 11.12 shows that a polynomial normalized violation cannot disappear immediately under channel descent. The pointwise loss accumulates, and bounding the deep remainder by absolute values loses a factor of the number of channels. A closing argument must therefore exploit the multi-index persistence of Lemma 11.8 collectively, or establish cancellation among the channels entering $\mathcal{R}_K$.

Corollary 11.14 (RH in the localized linear window)**.** *For every excursion starting at u and every $u < t \leq 5u/4$, the signed identities are*

$$M(t) = -\sum_{d \leq 5u/8} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right), \tag{11.33}$$

$$M(t) = -\sum_{2 \leq k \leq t} \left[M\left(\frac{t}{k}\right) - M\left(\frac{u}{k}\right) \right]. \tag{11.34}$$

Consequently RH holds if and only if, for every $\varepsilon > 0$,

$$\left|\sum_{d\le 5u/8}\mu(d)\left(\left\lfloor\frac{t}{d}\right\rfloor-\left\lfloor\frac{u}{d}\right\rfloor\right)\right| = \left|\sum_{2\le k\le t}\left[M\left(\frac{t}{k}\right)-M\left(\frac{u}{k}\right)\right]\right| \ll_\varepsilon t^{1/2+\varepsilon} \tag{11.35}$$

for every excursion starting at u and every

$$u < t \le \min\left\{x_1, \frac{5}{4}u\right\}. \tag{11.36}$$

Equivalently, in intra-excursion coordinates $t = u + r$ it is enough to control

$$1 \le r \le \min\{\ell - 1, u/4\}. \tag{11.37}$$

Proof. Use Corollary 11.3 with $\chi = 5/8$, together with Proposition 11.5. □

Proposition 11.15 (Kernel form of the recursion)**.** *Let $M(u) = 0$ and let t be an integer with $u < t < 2u$. For $1 \le y \le \lfloor t/2\rfloor$ put*

$$c_y(t,u) = \left(\left\lfloor\frac{t}{y}\right\rfloor-\left\lfloor\frac{t}{y+1}\right\rfloor\right)-\left(\left\lfloor\frac{u}{y}\right\rfloor-\left\lfloor\frac{u}{y+1}\right\rfloor\right). \tag{11.38}$$

Then:

(i) $M(t) = -\sum_{1\le y\le\lfloor t/2\rfloor} c_y(t,u)M(y)$;

(ii) $c_y(t,u) = \dfrac{t-u}{y(y+1)} + e_y$ *with $|e_y| < 2$; for $y > \sqrt{t}$ one has $c_y \in \{-1,0,1\}$, and $c_y \ne 0$ only if $y = \lfloor t/k\rfloor$ or $y = \lfloor u/k\rfloor$ for some $2 \le k < \sqrt{t}$ (the value $k = 1$ gives $y \in \{t,u\} > \lfloor t/2\rfloor$);*

(iii) $c_1(t,u) \ge \frac{1}{2}(t-u-1)$;

(iv) $M(t) = -(t-u)\,\Upsilon(\sqrt{t}) + \mathrm{Err}(t,u)$, *where Υ is the coefficient* (11.117) *and*

$$|\mathrm{Err}(t,u)| \le 2\sum_{y\le\sqrt{t}}|M(y)| + \sum_{2\le k<\sqrt{t}}\left(\left|M\left(\frac{t}{k}\right)\right|+\left|M\left(\frac{u}{k}\right)\right|\right). \tag{11.39}$$

Proof. (i) In (11.12) group the indices $k \ge 2$ by the value $y = \lfloor t/k\rfloor$, respectively $y = \lfloor u/k\rfloor$. The number of $k \ge 1$ with $\lfloor t/k\rfloor = y$ is $\lfloor t/y\rfloor - \lfloor t/(y+1)\rfloor$, and for $y \le \lfloor t/2\rfloor$ none of them is $k = 1$. The same holds for u, because $k = 1$ gives $y = u > \lfloor t/2\rfloor$. (ii) For $x \in \{t,u\}$ the bracket $\lfloor x/y\rfloor - \lfloor x/(y+1)\rfloor$ equals $x/(y(y+1))$ minus a difference of two fractional parts, so it differs from $x/(y(y+1))$ by less than 1. For $y > \sqrt{t} \ge \sqrt{u}$ both $t/y - t/(y+1)$ and $u/y - u/(y+1)$ are < 1, so each bracket is 0 or 1, and it is 1 only if $y = \lfloor x/k\rfloor$ with $k \le x/y < \sqrt{t}$. (iii) $c_1 = (t-u) - (\lfloor t/2\rfloor - \lfloor u/2\rfloor)$ and $\lfloor t/2\rfloor - \lfloor u/2\rfloor \le \lceil (t-u)/2\rceil$. (iv) Insert (ii) into (i) for $y \le \sqrt{t}$; the smooth part is $(t-u)\sum_{y\le\sqrt{t}} M(y)/(y(y+1)) = (t-u)\Upsilon(\sqrt{t})$. For $y > \sqrt{t}$ use the last assertion of (ii). □

Corollary 11.16 (Loss of the pointwise induction)**.** *Let $0 < \vartheta < 1$ and suppose $|M(y)| \le Cy^\vartheta$ for all $y \le \lfloor t/2\rfloor$. Then, for $u < t < 2u$ with $M(u) = 0$,*

$$|M(t) + (t-u)\Upsilon(\sqrt{t})| \ll_\vartheta C\,t^{(1+\vartheta)/2}, \tag{11.40}$$

while the full absolute kernel can, in general, have size comparable to the window length $t - u$ and yields at best $|M(t)| \ll_\vartheta C\,t$ from this information alone. For $\vartheta = \frac{1}{2} + \varepsilon$ the pointwise induction therefore loses $t^{1/2-\varepsilon}$, and $t^{1/4-\varepsilon/2}$ after the smooth component $-(t-u)\Upsilon(\sqrt{t})$ is extracted.

Proof. By Proposition 11.15(iv), $|\mathrm{Err}| \le 2C\sum_{y\le\sqrt{t}} y^\vartheta + 2Ct^\vartheta\sum_{2\le k<\sqrt{t}} k^{-\vartheta} \ll_\vartheta Ct^{(1+\vartheta)/2}$. The full kernel satisfies $\sum_y |c_y| y^\vartheta \ge c_1 \ge \frac{1}{2}(t-u-1)$ by (iii), and $\sum_y |c_y| y^\vartheta \ll_\vartheta t$ by (ii). □

The smooth component is not small for termwise reasons: by (11.119),

$$\Upsilon(\sqrt{t}) = -\sum_{n>\sqrt{t}} \frac{M(n)}{n(n+1)}, \tag{11.41}$$

so its size is governed by the global identity $\sum_{n\geq 1} M(n)/(n(n+1)) = 0$, i.e. by the pole of ζ at $s = 1$ (Remark 11.62), and involves values of M beyond $t/2$. Even under RH (Littlewood's bound $M(n) \ll_\varepsilon n^{1/2+\varepsilon}$ inserted in the tail) the tail gives only $(t-u)\Upsilon(\sqrt{t}) \ll t^{3/4+\varepsilon/2}$, the same order as the bound for $\mathrm{Err}(t,u)$; the RH scale for $M(t)$ is a statement about their sum.

Remark 11.17 (Localized intra-excursion system). Let $I = (u, x_1]$ be an excursion of length $\ell = x_1 - u$ and orientation σ_ν, and put

$$S_r := \sigma_\nu M(u+r) > 0 \qquad (1 \leq r < \ell), \qquad S_0 = S_\ell = 0. \tag{11.42}$$

For a fixed $\lambda \in (\lambda_0, 2)$ set

$$h_\lambda(u) := \min\{\ell - 1, (\lambda - 1)u\}. \tag{11.43}$$

Then the entire RH-relevant initial segment satisfies simultaneously

$$S_r = -\sigma_\nu \sum_{y \leq \lambda u/2} c_y(u+r, u) M(y) > 0, \qquad 1 \leq r \leq h_\lambda(u), \tag{11.44}$$

and $|S_r - S_{r-1}| \leq 1$. For $\lambda = 5/4$ this is the segment $r \leq \min\{\ell - 1, u/4\}$ and all dependencies lie below $5u/8$.

Definition 11.18 (Relaxed divisor-sum window class)**.** Fix $u \geq 2$ and $1 \leq h < u$. For $\mathbf{a} = (a_d)_{d\leq u} \in \{-1, 0, 1\}^u$ with $a_1 = 1$ and $\sum_{d\leq u} a_d = 0$, put

$$D_{\mathbf{a}}(n) := \sum_{\substack{d|n \\ d\leq u}} a_d, \qquad u < n \leq u + h. \tag{11.45}$$

We call $\mathbf{a}$ window-admissible with orientation $\sigma_\nu \in \{-1, 1\}$ if $D_{\mathbf{a}}(n) \in \{-1, 0, 1\}$ throughout the window and

$$\sigma_\nu \sum_{u<n\leq u+r} D_{\mathbf{a}}(n) < 0 \qquad (1 \leq r \leq h). \tag{11.46}$$

The induced oriented path is $S_r = -\sigma_\nu \sum_{u<n\leq u+r} D_{\mathbf{a}}(n) > 0$. For the genuine Möbius prefix, these conditions reproduce the future-window increment constraint and the excursion-sign constraint coming from $\mu * 1 = \delta_1$, but they do not impose the full family of internal divisor identities below u.

Proposition 11.19 (Ballistic counterexample to the relaxed window class)**.** *The relaxed class of Definition 11.18 admits ballistic paths. Namely, take*

$$a_1 = 1, \qquad a_u = -1, \qquad a_d = 0 \quad (1 < d < u). \tag{11.47}$$

Then for every $u < n < 2u$ one has $D_{\mathbf{a}}(n) = 1$, and with $\sigma_\nu = -1$,

$$S_r = r \qquad (1 \leq r < u). \tag{11.48}$$

If $m_y := \sum_{d\leq y} a_d$, then $m_y = 1$ for every $y < u$. Hence for any $0 < \vartheta < 1$, any fixed $1 < \lambda < 2$, and $h = \lfloor(\lambda - 1)u\rfloor$,

$$\sup_{y\leq \lambda u/2} \frac{|m_y|}{y^\vartheta} = 1, \qquad \frac{S_h}{(u+h)^\vartheta} \asymp_\lambda u^{1-\vartheta} \longrightarrow \infty. \tag{11.49}$$

Thus a maximum principle comparing the normalized localized-window height to the normalized history is false on this relaxed class.

Proof. For $u < n < 2u$, the integer u does not divide n. Hence among the two nonzero coordinates of $\mathbf{a}$ only $d = 1$ contributes to $D_{\mathbf{a}}(n)$, so $D_{\mathbf{a}}(n) = 1$. The remaining assertions are immediate. □

Remark 11.20 (What the relaxation loses). The counterexample violates the internal divisor identities of the Möbius prefix. Already for every prime $p < u$ the genuine identity requires $a_1 + a_p = 0$, whereas the ballistic vector above has $a_p = 0$. Consequently any successful localized argument of this type must use some of the relations

$$\sum_{d|n} a_d = 0 \qquad (2 \le n \le u), \tag{11.50}$$

not merely the future-window constraints.

Remark 11.21 (The prefix correlation identity). The completed-excursion bridge identity must not be applied to a truncated localized window. For $1 \le h < \ell$, put $x_j = \sigma_\nu \mu(u+j)$, $S_r = \sum_{j \le r} x_j$, and $C_g^{\langle h \rangle} = \sum_{j=1}^{h-g} x_j x_{j+g}$. Lemma 5.1 applied to the prefix of length h gives exactly

$$\sum_{r=1}^{h-1} S_r^2 + \sum_{g=1}^{h-1} g C_g^{\langle h \rangle} = S_h \sum_{r=1}^{h-1} S_r. \tag{11.51}$$

The right-hand side vanishes only for a completed bridge. Thus $Q_\nu = -\sum_g g C_g^{(\nu)}$ cannot be used as an identity internal to a truncated horizon unless the excursion actually returns there. In particular, passing from a large localized value to the energy of the whole excursion mixes the retained window with the part deliberately discarded by the horizon theorem.

Remark 11.22 (The one-step iteration gap). Even a one-step implication sending a large value in the localized window to a large value at some $y \le \lambda u/2$ would not automatically iterate. The point y need not lie in the localized window of the excursion that contains y; it may lie beyond $\lambda u'$ relative to that excursion's start u'. Sparsity of the family of λ-long excursions does not repair a single descent chain. Iteration therefore requires either a principle valid for all t, or a mechanism that places the new point inside an admissible window at every step.

Proposition 11.23 (Termwise absolute-value contraction gives no nontrivial gain). *Let $(u, x_1]$ be an excursion, $1 \le h < \min\{x_1 - u, u\}$, and let $w_1, \dots, w_h \ge 0$, not all zero. Put $K_y = \sum_{r \le h} w_r c_y(u+r, u)$, so that $\sum_r w_r M(u+r) = -\sum_y K_y M(y)$. If, for some $0 < \vartheta < 1$, $\alpha \ge 0$ and $C_\alpha > 0$,*

$$\sum_y |K_y| y^\vartheta \le C_\alpha \sum_{r \le h} w_r (u+r)^\alpha, \tag{11.52}$$

then

$$\sum_{r \le h} w_r r \le (1 + 2C_\alpha) \sum_{r \le h} w_r (u+r)^\alpha, \qquad \sum_{r \le h} w_r |M(u+r)| \le (1 + 2C_\alpha) \sum_{r \le h} w_r (u+r)^\alpha. \tag{11.53}$$

Thus the contraction hypothesis can hold only on windows where the trivial bound $|M(u+r)| \le r$ already gives the conclusion.

Proof. Proposition 11.15(iii) gives $c_1(u+r, u) \ge (r-1)/2 \ge 0$, hence $\frac{1}{2}\sum_r w_r (r-1) \le K_1 \le C_\alpha \sum_r w_r (u+r)^\alpha$. Since $M(u) = 0$ and M has unit steps, $|M(u+r)| \le r$; since $(u+r)^\alpha \ge 1$, $\sum w_r \le \sum w_r (u+r)^\alpha$, and both inequalities follow. □

Remark 11.24 (No strict coercivity from the first coefficient). Writing $t = u + r$ gives the exact formula

$$c_1(u+r, u) = \begin{cases} \lceil r/2 \rceil, & u \text{ even}, \\ \lfloor r/2 \rfloor, & u \text{ odd}. \end{cases} \tag{11.54}$$

In particular $c_1(u+1, u) = 0$ when u is odd. The preceding proposition therefore rules out only this particular termwise absolute-value mechanism; it does not rule out a global signed contraction or cancellation among the coefficients K_y.

Definition 11.25 (Multiplicative layers of the prefix)**.** Let $\Omega(n)$ denote the number of prime factors of n, counted with multiplicity. For $P \geq 2$ and $k \geq 1$, a vector $(a_d)_{d \leq P}$ satisfies the first k Möbius layers if $a_1 = 1$ and

$$\sum_{d|n} a_d = 0 \qquad (2 \leq n \leq P,\ \Omega(n) \leq k). \tag{11.55}$$

Thus layer 1 imposes the identities at primes, layer 2 adds prime squares and products of two primes, and so on.

Proposition 11.26 (One-layer propagation into the localized future)**.** *Suppose $(a_d)_{d \leq P}$ satisfies the first k Möbius layers. Then*

$$a_d = \mu(d) \qquad (d \leq P,\ \Omega(d) \leq k). \tag{11.56}$$

Moreover, for every integer n with

$$P < n \leq 2P, \qquad \Omega(n) \leq k+1, \tag{11.57}$$

one has

$$-\sum_{\substack{d|n \\ d \leq P}} a_d = \mu(n). \tag{11.58}$$

Thus the first k multiplicative layers of the prefix determine the future increments at every integer n in the stated window $P < n \leq 2P$ with $\Omega(n) \leq k+1$.

Proof. If $d \mid n$ and $d < n$, then $n/d \geq 2$, hence $\Omega(d) \leq \Omega(n) - 1$. We induct on $\Omega(n)$. For $\Omega(n) = 0$, $n = 1$ and $a_1 = 1 = \mu(1)$. If $2 \leq n \leq P$, $\Omega(n) \leq k$, and $a_d = \mu(d)$ for every $d \leq P$ with $\Omega(d) < \Omega(n)$, then every proper divisor d of n is such a d, and the layer equation gives $a_n = -\sum_{d|n,\, d<n} a_d = -\sum_{d|n,\, d<n} \mu(d) = \mu(n)$. For the second, every proper divisor of $n \leq 2P$ is at most $n/2 \leq P$ and has $\Omega(d) \leq k$. Hence the displayed sum equals $-\sum_{d|n,\ d<n} \mu(d) = \mu(n)$. □

Remark 11.27 (Multiplicative-layer depth as a quantitative invariant). Fix $\chi \in (\chi_0, 1)$ and the localized window $u < n \leq 2\chi u$. Let $k = k(u)$ be a depth up to which the internal Möbius divisor identities are imposed below $P = \chi u$. What growth of $k(u)$, if any, suffices to combine those identities with $M(u) = 0$ and the unit-step and excursion-sign constraints so as to force $\max_{u<t\leq 2\chi u} |M(t)| \ll_\varepsilon u^{1/2+\varepsilon}$? A bounded depth would give a compressed arithmetic mechanism; imposing all layers reconstructs the Möbius prefix. Proposition 11.26 makes the depth a quantifiable intermediate invariant; the results below treat only fixed k.

Proposition 11.28 (Layer sums in the localized window)**.** *Fix $1 < \lambda < 2$ and an integer $j \geq 1$ before letting $u \to \infty$. Then*

$$\sum_{\substack{u<n\leq \lambda u \\ \Omega(n)=j}} \mu(n) = (-1)^j\,(1+o(1))\,\frac{(\lambda-1)\,u\,(\log\log u)^{j-1}}{(j-1)!\,\log u}. \tag{11.59}$$

Proof. If $\Omega(n) = j$ and $\mu(n) \neq 0$, then n is squarefree with exactly j prime factors and $\mu(n) = (-1)^j$. By Landau's theorem for squarefree integers with exactly j distinct prime factors, the number $\pi_j^\flat(x)$ of such $n \leq x$ satisfies $\pi_j^\flat(x) \sim f_j(x) := x(\log\log x)^{j-1}/((j-1)!\log x)$ [28]. This is the squarefree count $\omega(n) = \Omega(n) = j$, not the count with repeated prime factors. For fixed j and fixed $\lambda > 1$,

$$\frac{f_j(\lambda u)}{f_j(u)} = \lambda\,\frac{(\log\log(\lambda u))^{j-1}}{(\log\log u)^{j-1}}\,\frac{\log u}{\log(\lambda u)} \longrightarrow \lambda. \tag{11.60}$$

Hence $f_j(\lambda u) - f_j(u) = (\lambda - 1 + o(1))f_j(u)$, as claimed. □

Corollary 11.29 (Bounded layer depth: fixed k)**.** *Fix an integer $k \geq 1$ before letting $u \to \infty$, $\chi \in (\chi_0, 1)$, $P = \chi u$, and the window $u < n \leq 2\chi u$.*

(i) *The increments fixed by the first k layers (Proposition 11.26), namely those with $\Omega(n) \le k+1$, form a set of $o(u)$ integers of the window.*

(ii) *Their summed contribution over the window is*

$$\sum_{\substack{u<n\le 2\chi u\\ \Omega(n)\le k+1}} \mu(n) = (-1)^{k+1}(1+o(1))\frac{(2\chi-1)\,u\,(\log\log u)^k}{k!\,\log u}, \tag{11.61}$$

so, for every fixed $\varepsilon>0$ and all sufficiently large u, it exceeds $u^{1-\varepsilon}$ in modulus and has the fixed sign $(-1)^{k+1}$.

(iii) *Let $a_d=\mu(d)$ for $\Omega(d)\le k$ and $a_d=0$ otherwise. For squarefree n in the window with $\omega(n)=w\ge k+2$,*

$$D_{\mathbf{a}}(n)=\sum_{\substack{d\mid n\\ d\le P}} a_d=\sum_{j=0}^{k}(-1)^j\binom{w}{j}=(-1)^k\binom{w-1}{k}, \tag{11.62}$$

which has modulus at least $k+1\ge 2$. Thus the zero extension of a k-layer prefix violates the unit-step constraint at every such n, i.e. at all but $o(u)$ squarefree integers of the window.

Proof. (ii) follows from Proposition 11.28 summed over $j\le k+1$, the term $j=k+1$ dominating. For (i), Landau's asymptotic $\#\{n\le x:\Omega(n)=j\}\sim x(\log\log x)^{j-1}/((j-1)!\log x)$ for fixed j (all integers, not only squarefree ones) [28] shows that the window contains $o(u)$ integers with $\Omega(n)\le k+1$. For (iii), all divisors of n other than n are $\le n/2\le P$, and n itself has $\Omega(n)=w>k$, so $D_{\mathbf{a}}(n)$ is the truncated alternating binomial sum, evaluated by induction on k via $\binom{w-1}{k}=\binom{w}{k}-\binom{w-1}{k-1}$. Finally, by Landau's theorem for squarefree integers with exactly j distinct prime factors [28], $\#\{n\le x:\mu(n)^2=1,\ \omega(n)\le k+1\}=o(x)$, whereas $\#\{n\le x:\mu(n)^2=1\}\sim 6x/\pi^2$; hence all but $o(u)$ squarefree n in the window have $\omega(n)\ge k+2$. □

Remark 11.30 (What fixed layer depth proves). For every fixed k, Corollary 11.29 shows that the first k layers determine only a density-zero subset of the localized window, while the signed sum of the determined increments has modulus $(1+o(1))(2\chi-1)u(\log\log u)^k/(k!\log u)$, larger than $u^{1-\varepsilon}$ for every fixed $\varepsilon>0$, once u is sufficiently large. Hence, if the full sum on the window is $O(u^{1/2+\varepsilon})$, substantial cancellation must occur between the increments determined by any fixed set of initial layers and the remaining ones. The results proved here do not determine how the required depth grows with u; in particular they do not show that $\asymp\log\log u$ layers (the typical size of Ω) are necessary or sufficient, which would require estimates uniform in $k=k(u)$. Part (iii) shows that the unit-step constraint is far from vacuous on a k-layer relaxation: it forbids the naive completion at almost every integer of the window, so the free coordinates a_d with $\Omega(d)>k$ are strongly constrained.

11.2. **Complete excursions, past forcing, and channel transforms.** The localized formulas above use only an initial part of an excursion. A completed excursion supplies more information: its path has one fixed sign until the first return to zero. The next identities keep that no-crossing condition instead of compressing the excursion to a single energy.

Proposition 11.31 (Past-forcing identity)**.** *Let $(u,v]$ be a completed excursion with $u\ge 1$, and let $u<t\le v$ be an integer. Define the past-forcing function*

$$P_u(t):=-\sum_{d\le u}\mu(d)\left(\left\lfloor\frac{t}{d}\right\rfloor-\left\lfloor\frac{u}{d}\right\rfloor\right) \tag{11.63}$$

and the local channel cutoff

$$q_u(t):=\left\lfloor\frac{t}{u+1}\right\rfloor. \tag{11.64}$$

Then

$$M(t) = P_u(t) - \sum_{u<d\le t/2} \mu(d)\left(\left\lfloor \frac{t}{d}\right\rfloor - 1\right), \tag{11.65}$$

and equivalently

$$P_u(t) = \sum_{k=1}^{q_u(t)} M(t/k). \tag{11.66}$$

In particular, the second sum in (11.65) *is empty for* $t \le 2u+1$*, and hence*

$$P_u(t) = M(t) \qquad (u < t \le \min\{v, 2u+1\}). \tag{11.67}$$

Proof. For $u < n \le t$, the divisor identity gives

$$\mu(n) = -\sum_{\substack{d\mid n\\ d<n}} \mu(d). \tag{11.68}$$

After summing over n and interchanging the two finite sums, a divisor $d \le u$ contributes once for every multiple in $(u,t]$, giving the first term of (11.65). For $d > u$, the multiple $n = d$ is excluded, so the number of remaining multiples is $\lfloor t/d\rfloor - 1$, which is nonzero only for $d \le t/2$.

For the second form,

$$\sum_{u<d\le t/2} \mu(d)\left(\left\lfloor \frac{t}{d}\right\rfloor - 1\right) = \sum_{k=2}^{q_u(t)} \sum_{u<d\le t/k} \mu(d) \tag{11.69}$$

$$= \sum_{k=2}^{q_u(t)} M(t/k), \tag{11.70}$$

where $M(u) = 0$ and real arguments follow the global floor convention. Moving this sum to the other side gives (11.66). Finally, if $t \le 2u+1$, no integer d satisfies $u < d \le t/2$. □

Corollary 11.32 (Sign-coherent channel representation)**.** *Let* σ *be the orientation of the completed excursion* $(u,v]$*. For every integer* $u < t < v$*,*

$$\sigma P_u(t) = \sum_{k=1}^{q_u(t)} |M(t/k)| \ge q_u(t), \tag{11.71}$$

and at the return point

$$\sigma P_u(v) = \sum_{k=2}^{q_u(v)} |M(v/k)| \ge q_u(v) - 1. \tag{11.72}$$

Consequently

$$|M(t)| \le |P_u(t)| \qquad (u < t < v). \tag{11.73}$$

Proof. If $1 \le k \le q_u(t)$, then $u < \lfloor t/k\rfloor \le t < v$. Every term in (11.66) therefore lies in the same no-crossing excursion and has sign σ. This gives (11.71). At $t = v$ the $k = 1$ term is $M(v) = 0$, whereas every term with $k \ge 2$ remains strictly inside the excursion, giving (11.72). The $k = 1$ summand in the interior is $M(t)$ itself, proving (11.73). □

The same forcing function can be written entirely in terms of the past summatory path. Proposition 11.5 and (11.66) give

$$P_u(t) = 1 - \sum_{m=1}^{u} w_t(m) M(m), \qquad w_t(m) := \left\lfloor \frac{t}{m}\right\rfloor - \left\lfloor \frac{t}{m+1}\right\rfloor \ge 0. \tag{11.74}$$

Thus, when u is an excursion boundary, the new excursion is forced by a nonnegative weighted superposition of the completed excursions and zero runs already present in $[1,u]$.

Proposition 11.33 (Past-forcing criterion)**.** *RH is equivalent to the following statement: for every $\varepsilon > 0$ there is C_ε such that, on every sufficiently late completed excursion $(u, v]$,*

$$|P_u(t)| \le C_\varepsilon t^{1/2+\varepsilon} \qquad (u < t < v). \tag{11.75}$$

Proof. If (11.75) holds, then (11.73) gives Littlewood's bound at every nonzero point of every sufficiently late excursion; zero runs are automatic, and finitely many early values are absorbed into the constant. Hence RH follows.

Conversely assume RH and fix $\varepsilon > 0$. Choose $0 < \varepsilon_1 < \min\{\varepsilon, 1/2\}$ and then $\delta > 0$ so small that

$$\frac{1}{2} + \varepsilon_1 + \delta\left(\frac{1}{2} - \varepsilon_1\right) < \frac{1}{2} + \varepsilon. \tag{11.76}$$

By return localization, every sufficiently late excursion satisfies $v \le u^{1+\delta}$. Littlewood's bound under RH and (11.66) give, with $\alpha = 1/2 + \varepsilon_1$,

$$|P_u(t)| \le C_{\varepsilon_1} t^\alpha \sum_{k \le q_u(t)} k^{-\alpha} \tag{11.77}$$

$$\ll_{\varepsilon_1} t^\alpha q_u(t)^{1-\alpha} \le t^{\alpha+\delta(1-\alpha)} \ll_\varepsilon t^{1/2+\varepsilon}, \tag{11.78}$$

using $q_u(t) \le u^\delta \le t^\delta$. This proves the converse. □

Remark 11.34 (Complete-excursion sign cone). The preceding identities turn the no-crossing condition into a linear feasibility problem on the past. To make this explicit, order the divisor matrix $\mathbf{D}_v$ of Section 10 by the split $[1, u] \cup (u, v]$:

$$\mathbf{D}_v = \begin{pmatrix} \mathbf{D}_u & 0 \\ \mathbf{B}_{u,v} & \mathbf{C}_{u,v} \end{pmatrix}. \tag{11.79}$$

For a relaxed past vector $\mathbf{a} \in \mathbb{R}^u$ satisfying $\mathbf{1}^\mathsf{T}\mathbf{a} = 0$ and a future vector $\mathbf{y} \in \mathbb{R}^{v-u}$, the divisor equations on the future rows are

$$\mathbf{B}_{u,v}\mathbf{a} + \mathbf{C}_{u,v}\mathbf{y} = 0. \tag{11.80}$$

The matrix $\mathbf{C}_{u,v}$ is lower triangular with diagonal 1, hence

$$\mathbf{y} = -\mathbf{C}_{u,v}^{-1}\mathbf{B}_{u,v}\mathbf{a}. \tag{11.81}$$

After applying the lower-triangular cumulative-sum matrix, every future value $M(u+r)$ is therefore a linear functional of the past vector. A completed excursion of orientation σ imposes the half-space constraints $\sigma M(u + r) \ge 1$ for $1 \le r < v - u$ together with the closing hyperplane $M(v) = 0$. For $v \le 2u + 1$ no future integer can be a proper divisor of another future integer, so $\mathbf{C}_{u,v} = I$ and this linear map reduces exactly to (11.63). The genuine Möbius past is a discrete arithmetic point inside this sign cone; relaxing some of its divisor rows produces the polyhedral feasibility problem suggested by the completed excursion method.

Proposition 11.35 (Dirichlet transform of one excursion's channel vector)**.** *Let $M(u) = 0$, let $t > u$ be an integer, extend the channel notation by $\Delta_1(u, t) := M(t)$, and let $s = \varsigma + i\tau \in \mathbb{C}$. Then the finite identity*

$$\sum_{k \ge 1} k^s \Delta_k(u, t) = \sum_{u < n \le t} \Phi_s(n) \tag{11.82}$$

holds, where

$$\Phi_s(n) := \sum_{k \mid n} k^s \mu(n/k) = n^s \prod_{p \mid n} (1 - p^{-s}). \tag{11.83}$$

The sum over k is finite because $\Delta_k(u, t) = 0$ for $k > t$.

Proof. Expand each channel and group by $n = km$:

$$\sum_{k\geq 1} k^s \Delta_k(u,t) = \sum_{k\geq 1} k^s \sum_{u/k<m\leq t/k} \mu(m) \tag{11.84}$$

$$= \sum_{u<n\leq t} \sum_{k|n} k^s \mu(n/k). \tag{11.85}$$

The product formula in (11.83) follows by multiplicativity; it is the generalized Jordan totient $\mathrm{id}_s * \mu$. □

At $s = 0$, (11.82) is exactly the one-step channel identity because $\Phi_0(n) = 0$ for $n > 1$. At $s = 1$, $\Phi_1 = \varphi$ is Euler's totient. For a local damping parameter $\alpha_{\mathrm{D}} > 0$,

$$M(t) = \sum_{u<n\leq t} \Phi_{-\alpha_{\mathrm{D}}}(n) - \sum_{k\geq 2} k^{-\alpha_{\mathrm{D}}} \Delta_k(u,t), \tag{11.86}$$

with

$$\Phi_{-\alpha_{\mathrm{D}}}(n) = (-1)^{\omega(n)} \left(\frac{\operatorname{rad} n}{n}\right)^{\alpha_{\mathrm{D}}} \prod_{p|n} (1 - p^{-\alpha_{\mathrm{D}}}). \tag{11.87}$$

For squarefree n this is $\mu(n)\prod_{p|n}(1-p^{-\alpha_{\mathrm{D}}})$. As $\alpha_{\mathrm{D}} \to \infty$, the channels $k \geq 2$ vanish and $\Phi_{-\alpha_{\mathrm{D}}}(n) \to \mu(n)$, so the identity degenerates to the direct window sum. The family therefore redistributes, but does not create, information.

Remark 11.36 (No information gain from the transform). For any fixed real ς, the function of τ

$$F_{u,t}(\varsigma + i\tau) := \sum_{k=1}^{t} k^{\varsigma + i\tau} \Delta_k(u,t) \tag{11.88}$$

determines the complete finite channel vector. Indeed, the distinct frequencies $\log k$ give the coefficient formula

$$k^{\varsigma} \Delta_k(u,t) = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} F_{u,t}(\varsigma + i\tau) e^{-i\tau \log k}\, d\tau. \tag{11.89}$$

Thus (11.82) is a change of coordinates on a single excursion's channel vector. A gain can only come from choosing a parameter for which either the smoothed window side or the weighted channel side admits an estimate compatible with the no-crossing constraints.

Remark 11.37 (Global logarithmic phases: two centers and their limits). A formal global zero mode x^{ρ}, $\rho = \beta + i\gamma$, enters the kth scale channel with factor

$$k^{-\rho} = k^{-\beta} e^{-i\gamma \log k}. \tag{S11.1}$$

This is useful for interpreting channel geometry but is not an additional identity of a completed excursion.

For

$$\mathscr{S}_K(s) := \sum_{k=1}^{K} k^{-s}, \qquad s = \varsigma + i\tau, \tag{S11.2}$$

Euler–Maclaurin gives, for a fixed complex parameter $s \neq 1$ with $\varsigma > -1$ (no uniformity in $\tau = \Im s$ is asserted),

$$\mathscr{S}_K(s) = \frac{K^{1-s}}{1-s} + \zeta(s) + \frac{1}{2} K^{-s} + O_s(K^{-\varsigma-1}). \tag{11.90}$$

If ρ is a zero, this yields

$$\sum_{k=2}^{K} k^{-\rho} = -1 + \frac{K^{1-\rho}}{1-\rho} + \frac{1}{2} K^{-\rho} + O_{\rho}(K^{-\beta-1}). \tag{11.91}$$

The ordinary infinite series $\sum k^{-\rho}$ does *not* converge for $0 < \beta < 1$; the equation $\zeta(\rho) = 0$ is an analytic-continuation statement, or equivalently the vanishing of the renormalized constant term in (11.90). The comparison $K^{1-\beta} \asymp |\gamma|$ is therefore diagnostic: below that scale the boundary term is small compared with the $k = 1$ mode, while above it the truncated tail cannot be ignored.

There are two different "centers" in the elementary exponential sum, and they should not be confused. With the positive-phase convention

$$S_K^+(\tau) := \sum_{k=1}^{K} e^{i\tau \log k} = \sum_{k=1}^{K} k^{i\tau}, \tag{S11.3}$$

the *centroid of the points* is

$$\frac{S_K^+(\tau)}{K} = \frac{e^{i\tau \log K}}{1 + i\tau} + O_\tau(K^{-1}), \qquad \tau \neq 0. \tag{11.92}$$

It does not converge in the plane: it rotates asymptotically on the circle of radius $(1 + \tau^2)^{-1/2}$. This behavior holds for every nonzero frequency and contains no special information about zeta zeros.

By contrast, the *center of the Euler–Maclaurin spiral* is the constant term. Equation (11.90) with $s = -i\tau$ gives

$$S_K^+(\tau) = \zeta(-i\tau) + \frac{K^{1+i\tau}}{1 + i\tau} + \frac{1}{2}K^{i\tau} + O_\tau(K^{-1}). \tag{11.93}$$

For $\tau \neq 0$, the functional equation has a nonzero prefactor at $s = -i\tau$, and therefore

$$\zeta(-i\tau) \neq 0 \quad \Longleftrightarrow \quad \zeta(1 + i\tau) \neq 0. \tag{S11.4}$$

Thus nonvanishing of this spiral center for every $\tau \neq 0$ is the classical zero-free line $\Re s = 1$, equivalent to the prime number theorem; it is not an RH criterion.

The RH-sensitive version uses a real weight. For $0 \leq \varsigma < 1$,

$$\sum_{k=1}^{K} k^{-\varsigma} e^{-i\tau \log k} = \zeta(\varsigma + i\tau) + \frac{K^{1-\varsigma-i\tau}}{1 - \varsigma - i\tau} + \frac{1}{2}K^{-\varsigma-i\tau} + O_{\varsigma,\tau}(K^{-\varsigma-1}). \tag{11.94}$$

Hence the renormalized spiral center is $\zeta(\varsigma + i\tau)$. The statement that this center never vanishes for *all* $1/2 < \varsigma < 1$ and all real τ is exactly RH; on the critical line $\varsigma = 1/2$, it vanishes precisely at ordinates of critical-line zeros. For one fixed vertical line $1/2 < \varsigma < 1$, nonvanishing is only the corresponding zero-free-line statement, not RH by itself.

The associated weighted point centroid is

$$\frac{\sum_{k \leq K} k^{-\varsigma} e^{-i\tau \log k}}{\sum_{k \leq K} k^{-\varsigma}} = \frac{1 - \varsigma}{1 - \varsigma - i\tau} e^{-i\tau \log K} + o(1), \tag{11.95}$$

so it also rotates rather than converging. After derotation it converges, while logarithmic averaging gives

$$\frac{1}{\log K} \sum_{k \leq K} \frac{e^{-i\tau \log k}}{k} \longrightarrow 0 \qquad (\tau \neq 0). \tag{11.96}$$

Formally, a spectral mode x^ρ inserted into the scale-channel identity carries the factor $k^{-\rho}$, so the vanishing renormalized center at a zero explains the familiar mode-by-mode cancellation picture. This is *global zero analysis*: it does not turn $\sum k^{-\rho}$ into a convergent series and does not create a new constraint on one excursion. The exact object internal to the method is instead the finite transform of Proposition 11.35.

Remark 11.38 (Excursion-weighted logarithmic channel polynomial). Let A_k be local real or complex coefficients produced by a specified functional of one excursion; the letter A_k has no connection

with the global power-horizon exponent A of Section 9. The corresponding logarithmic channel polynomial is

$$\mathfrak{P}_K(\varsigma,\tau;A) := \sum_{k=2}^{K} A_k k^{-\varsigma} e^{-i\tau\log k}. \tag{11.97}$$

A normalized angular centroid, when the denominator is nonzero, is

$$\frac{\mathfrak{P}_K(\varsigma,\tau;A)}{\sum_{k=2}^{K}|A_k|k^{-\varsigma}}. \tag{11.98}$$

Three logically different questions are then available: angular cancellation of this centroid, convergence after the explicit derotation $e^{i\tau\log K}$, and a square-function estimate. A bound of the schematic form

$$|\mathfrak{P}_K(\varsigma,\tau;A)|^2 \ll \sum_{k=2}^{K}|A_k|^2 k^{-2\varsigma} \tag{11.99}$$

is not a general property of Dirichlet polynomials and is false for arbitrary coefficients and frequencies. Even $A_k \equiv 1$ is obstructed by the boundary term in (11.90). Writing $H_K(2\varsigma) := \sum_{k\le K} k^{-2\varsigma}$, the pure-coefficient model shows that a square-root scale could only begin, at the level of this boundary term, in a regime roughly satisfying

$$|\tau|\, H_K(2\varsigma)^{1/2} \gtrsim K^{1-\varsigma} \tag{11.100}$$

(up to the harmless $1-\varsigma$ part of $|1-\varsigma-i\tau|$). Accordingly, (11.99) is recorded only as a possible arithmetic target for the highly structured coefficients generated by a completed excursion, with an explicit frequency range to be supplied. Establishing such a range would provide genuine signed cancellation among channels; merely knowing the rotating centroid or its derotated limit would not.

Proposition 11.39 (Exact correlations between centered scale channels)**.** *Let $a \ge 1$, $L \ge 1$, and for $1 \le r \le L$ define, for every $k \ge 1$,*

$$b_{k,r} := M\Big(\frac{a+r}{k}\Big) - M\Big(\frac{a}{k}\Big) - \frac{r}{L}\left[M\Big(\frac{a+L}{k}\Big) - M\Big(\frac{a}{k}\Big)\right]. \tag{S11.5}$$

All sums over k are finite. Then

$$b_{1,r} = -\sum_{k\ge 2} b_{k,r} \qquad (1 \le r \le L), \tag{11.101}$$

and hence

$$\sum_{r=1}^{L-1} b_{1,r}^2 = \sum_{k,\ell\ge 2}\sum_{r=1}^{L-1} b_{k,r}b_{\ell,r}. \tag{11.102}$$

Proof. The divisor identity $\sum_{k\ge1} M(x/k) = 1$ holds for every $x \ge 1$. Subtracting it at a from the identities at $a+r$ and $a+L$, and then taking the displayed affine combination, gives $\sum_{k\ge1} b_{k,r} = 0$. This is (11.101); squaring and summing over r gives (11.102). □

Remark 11.40 (What the channel energy still needs). Proposition 11.39 is exact and stays inside one finite block; it introduces no zeros of ζ. Its unresolved content is entirely in the cross terms

$$\sum_{k\ne\ell}\sum_{r} b_{k,r}b_{\ell,r}. \tag{S11.6}$$

The identity alone merely rewrites the same centered energy. A useful estimate would have to exploit the no-crossing and complete-excursion constraints to control these signed cross-channel correlations. This is also why the phase behavior of a Dirichlet polynomial with arbitrary coefficients A_k cannot substitute for an arithmetic estimate on the actual excursion-generated coefficients.

11.3. **The square range as an exact bilinear form of the prefix.**

Proposition 11.41 (Bilinear square-range truncation identity)**.** *Let $u \geq 1$ and $1 \leq t \leq u^2$. Then*

$$M(t) = 2M(\min(t,u)) - \sum_{d,j \leq u} \mu(d)\mu(j) \left\lfloor \frac{t}{dj} \right\rfloor. \tag{11.103}$$

In particular, if $M(u) = 0$ and $u < t \leq u^2$,

$$M(t) = -\mathcal{B}_u(t), \qquad \mathcal{B}_u(t) := \sum_{d,j \leq u} \mu(d)\mu(j) \left\lfloor \frac{t}{dj} \right\rfloor, \qquad \mathcal{B}_u(u) = 0. \tag{11.104}$$

Proof. Let $\mu_u = \mu \mathbf{1}_{[1,u]}$. Each factor of $(\mu - \mu_u) * (\mu - \mu_u)$ is supported on integers $\geq u+1$, so the convolution, and its convolution with 1, vanish on $n < (u+1)^2$, in particular on $n \leq u^2$. Expanding and using $\mu * 1 = \delta$ gives $\mu(n) = 2\mu_u(n) - (\mu_u * \mu_u * 1)(n)$ for $n \leq u^2$; sum over $n \leq t$. □

Remark 11.42 (Higher-order determinacy). For every $k \geq 1$ and $n \leq u^k$,

$$\mu(n) = \sum_{j=1}^{k} (-1)^{j-1} \binom{k}{j} (\mu_u^{*j} * 1^{*(j-1)})(n). \tag{11.105}$$

Indeed, $(\mu - \mu_u)^{*k}$ is supported on integers at least $(u+1)^k$, so after convolution with $1^{*(k-1)}$ it still vanishes for $n < (u+1)^k$, in particular for $n \leq u^k$. Expanding gives

$$0 = \sum_{j=0}^{k} (-1)^j \binom{k}{j} \mu^{*(k-j)} * \mu_u^{*j} * 1^{*(k-1)}. \tag{11.106}$$

For $j = 0$, $\mu^{*k} * 1^{*(k-1)} = \mu$. For $j \geq 1$, pair $k-j$ factors of μ with $k-j$ of the factors 1 to obtain $\mu^{*(k-j)} * 1^{*(k-j)} = \delta$; the remaining $j-1$ factors of 1 produce the term in (11.105). Rearranging proves the formula. For $k = 1$ this is the trivial truncation on $n \leq u$; for $k = 2$ it recovers the bilinear square-range truncation of Proposition 11.41; and the support argument above extends to the intended range $n \leq u^k$. Thus determination by the prefix is not special to u^2. What is special to $t \leq u^2$ is the order-two form (11.104), with rank-one coefficients $\mu(d)\mu(j)$ and elementary kernel $\lfloor t/(dj) \rfloor$. This is a truncation identity of Heath-Brown/Vaughan type [17]; summed forms of this kind are used in the fast computation of M [18].

Remark 11.43 (Square-excursion bridges). Let $u = x_{0,\nu}$ and $x_{1,\nu} \leq u^2$. Then $\mathcal{B}_u(u) = \mathcal{B}_u(x_{1,\nu}) = 0$ and $-\sigma_\nu \mathcal{B}_u(t) > 0$ for $u < t < x_{1,\nu}$. Can the rank-one coefficients $\mu(d)\mu(j)$, the kernel $\lfloor t/(dj) \rfloor$ and these bridge and sign constraints be combined to prove the bound

$$\max_{u < t \leq x_{1,\nu}} |\mathcal{B}_u(t)| \ll_\varepsilon x_{1,\nu}^{1/2+\varepsilon} \tag{11.107}$$

uniformly over square-ordinary excursions? By Corollary 9.16 this uniform bound is equivalent to RH. By Corollary 11.14 it suffices to consider $t \leq \min\{x_{1,\nu}, \frac{5}{4}u\}$, where $\mathcal{B}_u(t)$ reduces to the localized linear form and uses only $\mu(d)$ with $d \leq 5u/8$.

11.4. **Trial-division horizon of an excursion.**

Lemma 11.44 (Square horizon, classical)**.** *Let $u \geq 2$. Trial division using the primes $p \leq u$ computes $\mu(n)$ for every $n \leq u^2$.*

Proof. Remove the prime factors $p \leq u$ of n; if one occurs twice, $\mu(n) = 0$. The cofactor $q \leq u^2$ has all prime factors $> u$, so it is 1 or a single prime. □

Remark 11.45 (Determinacy is not information). Primes $> u$ are not logically new information, and determination by a finite prefix is not special to u^2: (11.105) propagates a prefix to every power range, and (11.11) determines M from $M(1)$. The ranges $(u, 2u]$ and $(u, u^2]$ are special only in the

concrete sense that the exact summatory formula is of order one ((11.2)) or two ((11.104)) in the prefix, and that one-stage trial division by primes $\le u$ suffices on $(u, u^2]$.

Definition 11.46 (Smooth and rough parts)**.** For $y > 1$ write $n = s_y(n)\, r_y(n)$, where every prime factor of $s_y(n)$ is $\le y$ and every prime factor of $r_y(n)$ is $> y$. Let $P^+(n)$ be the largest prime factor of n, with $P^+(1) = 1$.

Lemma 11.47 (Smooth and rough parts, classical)**.** $\mu(n) = \mu(s_y(n))\,\mu(r_y(n))$. *If $n < y^k$ then $\Omega(r_y(n)) \le k-1$, and if $n \le y^2$ then $r_y(n)$ is 1 or a prime.*

Proof. The two parts are coprime. If $\Omega(r) \ge 1$ then $y^{\Omega(r)} < r \le n$. □

Remark 11.48 (The horizon u^2 is not the multiplicative threshold of an excursion)*.* For a square-ordinary excursion $\sqrt{v_\nu} \le u_\nu$. Every $n \le v_\nu$ has at most one prime factor $> \sqrt{v_\nu}$, so the threshold that governs the multiplicative structure inside I_ν is $\sqrt{v_\nu}$, below u_ν. The regime $t > u_\nu^2$ of Section 11.5 lies beyond v_ν for a square-ordinary excursion; that section concerns $M(t)$ globally, and Section 11.6 gives the local version inside an excursion.

11.5. **Large-prime decomposition at level U^2.** The identities of this section are decompositions by large prime factors of Buchstab/Legendre type. The asymptotic statements use only the prime number theorem with classical error term and the Vinogradov–Korobov bound [39, 60].

Definition 11.49 (Smooth baseline and large-prime part)**.** For $U \ge 2$ and $t \ge 1$ put

$$M_U^{\mathrm{sm}}(t) := \sum_{\substack{n \le t \\ P^+(n) \le U^2}} \mu(n), \qquad M_U^{\mathrm{lp}}(t) := \sum_{\substack{n \le t \\ P^+(n) > U^2}} \mu(n), \qquad M(t) = M_U^{\mathrm{sm}}(t) + M_U^{\mathrm{lp}}(t). \tag{11.108}$$

Proposition 11.50 (Single-prime copy, classical)**.** *For a prime p and $t < p^2$,*

$$\mathcal{D}_p(t) := \sum_{a \le t/p} \mu(ap) = -M(t/p). \tag{11.109}$$

Proof. $a \le t/p < p$, so $p \nmid a$ and $\mu(ap) = -\mu(a)$. □

Proposition 11.51 (Sign content of one prime cone)**.** *Let $\mathcal{S}(y) = \sum_{a \le y} \mu(a)^2$. Among the numbers ap with $a \le y < p$, the numbers of values $+1, -1, 0$ of μ are*

$$N_+ = \frac{\mathcal{S}(y) - M(y)}{2}, \qquad N_- = \frac{\mathcal{S}(y) + M(y)}{2}, \qquad N_0 = \lfloor y \rfloor - \mathcal{S}(y), \tag{11.110}$$

with limiting proportions $3/\pi^2,\ 3/\pi^2,\ 1 - 6/\pi^2$.

Proof. $\mu(ap) = -\mu(a)$, and $\#\{a \le y : \mu(a) = \mp 1\} = (\mathcal{S}(y) \mp M(y))/2$. □

Proposition 11.52 (First-order window, classical)**.** *If $U^2 < t < U^4$, no $n \le t$ has two prime factors $> U^2$ (with multiplicity), and*

$$M_U^{\mathrm{lp}}(t) = -\sum_{U^2 < p \le t} M(t/p). \tag{11.111}$$

Proof. Two such factors would give $n > U^4 > t$. Hence $M_U^{\mathrm{lp}}(t) = \sum_{U^2 < p \le t} \mathcal{D}_p(t)$, and $t < U^4 < p^2$; apply (11.109). □

Proposition 11.53 (Prime-count form)**.** *For $U^2 < t < U^4$ and $W := t/U^2$,*

$$M_U^{\mathrm{lp}}(t) = -\sum_{a \le W} \mu(a)\big(\pi(t/a) - \pi(U^2)\big), \qquad M(t) = M_U^{\mathrm{sm}}(t) - \sum_{a \le W} \mu(a)\pi(t/a) + M(W)\pi(U^2). \tag{11.112}$$

Proof. Expand (11.111) as $-\sum_{U^2<p\le t}\sum_{a\le t/p}\mu(a)$. This is a finite sum, so reversing the order is purely combinatorial. For fixed a the prime runs over $U^2 < p \le t/a$, which is non-empty only if $a < W$. The displayed sum may harmlessly be written with $a \le W$: if W is an integer, the endpoint $a = W$ contributes $\pi(U^2) - \pi(U^2) = 0$. □

Thus the only input from primes $> U^2$ is the value of π at the points t/a, $a \le W$, weighted by $\mu(a)$ at the small scale $W < U^2$.

Definition 11.54 (Copy multiplicities)**.** For $U^2 < t < U^4$, $W = t/U^2$ and $1 \le y \le \lfloor W\rfloor$ put

$$N_y(t,U) = \#\Big\{p:\ \max(U^2, \tfrac{t}{y+1}) < p \le \tfrac{t}{y}\Big\}, \qquad \text{so that} \qquad M_U^{\mathrm{lp}}(t) = -\sum_{1\le y\le \lfloor W\rfloor} N_y(t,U)M(y). \tag{11.113}$$

Remark 11.55 (Asymptotics of the multiplicities)*.* For fixed $W > 1$ and $U \to \infty$,

$$N_y \sim \frac{WU^2}{2\log U}\cdot\frac{1}{y(y+1)} \quad (1 \le y \le \lfloor W\rfloor - 1), \qquad N_{\lfloor W\rfloor} \sim \frac{U^2}{2\log U}\Big(\frac{W}{\lfloor W\rfloor} - 1\Big) \quad (W \notin \mathbb{Z}), \tag{11.114}$$

and $N_W = 0$ if $W \in \mathbb{Z}$. The top multiplicity is truncated because $t/(\lfloor W\rfloor + 1) < U^2$.

Definition 11.56 (Coherent coefficient)**.**

$$\mathcal{C}(W) := \int_1^W \frac{M(w)}{w^2}\,dw = \sum_{a\le W}\mu(a)\Big(\frac{1}{a} - \frac{1}{W}\Big) = \sum_{a\le W}\frac{\mu(a)}{a} - \frac{M(W)}{W}. \tag{11.115}$$

Proposition 11.57 (Fixed W)**.** *For fixed $W > 1$, as $U \to \infty$,*

$$M_U^{\mathrm{lp}}(WU^2) = -\frac{WU^2}{2\log U}(\mathcal{C}(W) + o(1)). \tag{11.116}$$

Proof. In (11.112) the sum is finite; for $a < W$ the prime number theorem gives $\pi(WU^2/a) - \pi(U^2) \sim (W/a - 1)U^2/(2\log U)$, and $\sum_{a\le W}\mu(a)(W/a - 1) = W\,\mathcal{C}(W)$. □

Lemma 11.58 (The multiplicity-weighted coefficient)**.** *The coefficient*

$$\Upsilon(W) := \sum_{1\le y\le W}\frac{M(y)}{y(y+1)} = \sum_{n\le W}\frac{\mu(n)}{n} - \frac{M(W)}{\lfloor W\rfloor + 1} \tag{11.117}$$

satisfies

$$\Upsilon(W) = \mathcal{C}(\lfloor W\rfloor + 1), \qquad \mathcal{C}(W) = \Upsilon(W) - M(W)\Big(\frac{1}{W} - \frac{1}{\lfloor W\rfloor + 1}\Big), \tag{11.118}$$

and

$$\mathcal{C}(W) = -\int_W^\infty \frac{M(w)}{w^2}\,dw, \qquad \Upsilon(W) = -\sum_{n>W}\frac{M(n)}{n(n+1)}. \tag{11.119}$$

Proof. M is constant on $[y, y+1)$ and $\int_y^{y+1} w^{-2}dw = 1/(y(y+1))$, which gives (11.118). For $s > 1$, (8.1) gives $\int_1^\infty M(w)w^{-s-1}\,dw = 1/(s\zeta(s)) \to 0$ as $s \downarrow 1$. For $s \in [1, 1+\delta]$ the bound $M(w) \ll we^{-c\sqrt{\log w}}$ dominates the integrand by an integrable multiple of $e^{-c\sqrt{\log w}}/w$, so dominated convergence gives $\int_1^\infty M(w)w^{-2}dw = 0$. □

Remark 11.59 ($\mathcal{C}$ versus Υ)*.* Summing (11.114) against $M(y)$ suggests the coefficient $\Upsilon(W)$, but the truncated top multiplicity replaces it by $\mathcal{C}(W)$; by (11.118) the two agree exactly when $M(\lfloor W\rfloor) = 0$. For $1 < W < 2$ one has $M_U^{\mathrm{lp}} = -(\pi(t) - \pi(U^2)) \sim -(W-1)U^2/(2\log U)$, i.e. $\mathcal{C}(W) = 1 - 1/W$, whereas $\Upsilon(W) = 1/2$.

Corollary 11.60 (Decay of the coefficients)**.** *By* (11.119) *and* (7.3)*, there is* $c_1 > 0$ *such that, for* $W > e$*,*

$$\mathcal{C}(W),\ \Upsilon(W) \ll \exp(-c_1(\log W)^{3/5}(\log\log W)^{-1/5}). \tag{11.120}$$

Proposition 11.61 (Asymptotic size of the smooth and large-prime components)**.** (i) *Suppose* $W > 1$ *is fixed and* $\mathcal{C}(W) \neq 0$*. Then*

$$M_U^{\mathrm{lp}}(WU^2) \sim -\frac{WU^2}{2\log U}\mathcal{C}(W), \tag{11.121}$$

$$M_U^{\mathrm{sm}}(WU^2) \sim +\frac{WU^2}{2\log U}\mathcal{C}(W). \tag{11.122}$$

For $1 < W < 2$ *one has* $\mathcal{C}(W) = 1 - 1/W > 0$*, so the hypothesis holds.*

(ii) *There is* $C > 0$ *such that, as* $t \to \infty$*, uniformly for* $U^2 < t < U^4$ *with* $W = t/U^2 \geq \exp(C(\log\log t)^2)$*,*

$$M_U^{\mathrm{lp}}(t) = (1+o(1))\frac{t}{(\log t)^2}, \qquad M_U^{\mathrm{sm}}(t) = -(1+o(1))\frac{t}{(\log t)^2}. \tag{11.123}$$

In particular, in both regimes $|M_U^{\mathrm{lp}}(t)|$ *and* $|M_U^{\mathrm{sm}}(t)|$ *exceed* $t^{1/2+\varepsilon}$ *for every fixed* $\varepsilon < 1/2$ *and all large* t*, and only their sum* M *can be compared with the scale* $t^{1/2+\varepsilon}$*.*

Proof. We use the classical form [39]: for some $c > 0$,

$$\pi(y) = \mathrm{li}(y) + O(ye^{-c\sqrt{\log y}}), \quad M(y) \ll ye^{-c\sqrt{\log y}}, \quad \widetilde{m}(y) := \sum_{a\leq y}\frac{\mu(a)}{a} \ll e^{-c\sqrt{\log y}}, \tag{11.124}$$

and the series $\sum_{n\geq 1}\mu(n)\log n/n = -1$ converges conditionally. This follows, for example, from the displayed bound on $\widetilde{m}$ and partial summation (and agrees with the limit at $s = 1$ of $-\frac{d}{ds}\zeta(s)^{-1}$). The integral below converges absolutely by the bound on $\widetilde{m}$, and

$$\int_1^\infty \widetilde{m}(w)\frac{dw}{w} = \lim_{Y\to\infty}\Big(\widetilde{m}(Y)\log Y - \sum_{a\leq Y}\frac{\mu(a)\log a}{a}\Big) = 1. \tag{11.125}$$

(i) Proposition 11.57, and $M(t) = o(t/\log t)$. For $1 < W < 2$, $M = 1$ on $[1, 2)$.

(ii) Since $t < U^4$, $W < \sqrt{t}$, so $\log(t/a) \geq \frac{1}{2}\log t$ for $a \leq W$. By (11.112),

$$M_U^{\mathrm{lp}}(t) = -\sum_{a\leq W}\mu(a)\,\mathrm{li}(t/a) - \sum_{a\leq W}\mu(a)(\pi(t/a) - \mathrm{li}(t/a)) + M(W)\pi(U^2). \tag{11.126}$$

The second term is $\ll t(1+\log W)e^{-c\sqrt{\log t/2}} = o(t/\log^2 t)$. For the third, $\frac{1}{4}\log t < \log U < \frac{1}{2}\log t$, so

$$|M(W)|\pi(U^2) \ll We^{-c\sqrt{\log W}}\frac{U^2}{\log U} \ll e^{-c\sqrt{\log W}}\frac{t}{\log t} = o(t/\log^2 t) \tag{11.127}$$

once $c\sqrt{C} > 1$, where c is the constant in (11.124); we fix such a C. The same condition makes the boundary term of the partial summation below $o(t/\log^2 t)$, and the second term of (11.126) is $o(t/\log^2 t)$ for any C. For the first, with $F(w) = w\,\mathrm{li}(t/w)$,

$$\sum_{a\leq W}\mu(a)\,\mathrm{li}(t/a) = \widetilde{m}(W)F(W) - \int_1^W \widetilde{m}(w)F'(w)\,dw, \tag{11.128}$$

$$F'(w) = \mathrm{li}(t/w) - \frac{t/w}{\log(t/w)} = \frac{t/w}{\log^2(t/w)}\Big(1 + O\Big(\frac{1}{\log t}\Big)\Big). \tag{11.129}$$

The boundary term is $\ll e^{-c\sqrt{\log W}}t/\log t = o(t/\log^2 t)$. The integral equals

$$\frac{t}{\log^2 t}\left(\int_1^W \widetilde{m}(w)g_t(w)\frac{dw}{w} + O(1/\log t)\right), \qquad g_t(w) = \frac{\log^2 t}{\log^2(t/w)} \in [1,4]. \tag{11.130}$$

The required uniformity in (U,W) follows by an explicit ε–W argument. Given $\varepsilon_1 > 0$, choose W_0 so that

$$\int_{W_0}^{\infty} \frac{|\widetilde{m}(w)|}{w}\,dw < \varepsilon_1. \tag{11.131}$$

For all sufficiently large t, the lower bound on W gives $W > W_0$. On $[1, W_0]$,

$$\sup_{1\le w\le W_0} |g_t(w) - 1| \ll \frac{\log W_0}{\log t} \to 0, \tag{11.132}$$

while on $[W_0, W]$ one has $1 \le g_t(w) \le 4$. The omitted interval (W,∞) is also contained in the chosen tail. Therefore

$$\int_1^W \widetilde{m}(w)g_t(w)\frac{dw}{w} = \int_1^{\infty} \widetilde{m}(w)\frac{dw}{w} + o(1) = 1 + o(1), \tag{11.133}$$

uniformly over all admissible pairs (U,W). Hence $M_U^{\mathrm{lp}}(t) = (1+o(1))t/\log^2 t$, and $M_U^{\mathrm{sm}} = M - M_U^{\mathrm{lp}}$ with $M(t) = o(t/\log^2 t)$. □

Remark 11.62 (Why $t/\log^2 t$). $\mathcal{C}(W)$ captures only the $1/\log t$ term, whose coefficient tends to zero by Corollary 11.60. The $1/\log^2 t$ term comes from $\zeta'(s)/\zeta(s)^2 \to -1$, that is, from the pole of ζ at $s = 1$, and does not decay in W.

Remark 11.63 (First copy transitions). For $1 < W < 2$ every copy has length 1 and $M_U^{\mathrm{lp}}(t) = -(\pi(t) - \pi(U^2))$. For $2 < W < 3$ copies of length 2 appear but $M(2) = 0$. For $3 < W < 4$ copies of length 3 enter with $M(3) = -1$, of opposite net sign.

11.6. **Local large-prime decomposition of an excursion.**

Proposition 11.64 (Local decomposition)**.** *Let $I_\nu = (u,v]$ be a completed excursion and $y = \sqrt{v}$. For $u < t \le v$,*

$$M(t) = \sum_{\substack{u<n\le t\\ P^+(n)\le y}} \mu(n) - \sum_{a<y} \mu(a)\,\Pi_a(t), \qquad \Pi_a(t) := \#\Big\{p:\ \max\big(y, \tfrac{u}{a}\big) < p \le \tfrac{t}{a}\Big\}, \tag{11.134}$$

and the bridge condition $M(v) = 0$ reads

$$\sum_{\substack{u<n\le v\\ P^+(n)\le \sqrt{v}}} \mu(n) = \sum_{a<\sqrt{v}} \mu(a)\,\Pi_a(v). \tag{11.135}$$

Proof. $M(u) = 0$. An $n \le v = y^2$ has at most one prime factor $p > y$; then $n = ap$ with $a < v/y = y < p$, so $\mu(n) = -\mu(a)$, and $(a,p) \mapsto ap$ is a bijection onto these n. □

Remark 11.65 (Primes in short intervals). The decomposition holds for every excursion, with threshold $\sqrt{v_\nu}$. The large-prime input consists of prime counts in the intervals $(u/a, t/a]$, of length at most ℓ_ν/a at height about v_ν/a. Huxley proved the prime number theorem uniformly in intervals of length $h \ge x^{7/12+\varepsilon}$ [21]; the zero-density estimates of Guth–Maynard give the corresponding result for every fixed $\theta > 17/30$ and intervals of length x^θ [12]. For Möbius sums, Matomäki–Teräväinen prove $\sum_{x<n\le x+x^\theta} \mu(n) = o(x^\theta)$ for every $\theta > 0.55$ [35]. The cited results do not provide, below these ranges, an asymptotic for Π_a uniform in a.

Lemma 11.66 (Inverse logarithms)**.** *Let C be as in Proposition 11.61. Uniformly for*

$$\exp(C(\log\log x)^2) \le z \le \sqrt{x}, \tag{11.136}$$

$$\sum_{a\le z} \frac{\mu(a)}{a\log(x/a)} = -\frac{1+o(1)}{(\log x)^2}. \tag{11.137}$$

Proof. Partial summation gives

$$\frac{\widetilde{m}(z)}{\log(x/z)} - \int_1^z \frac{\widetilde{m}(w)}{w\log^2(x/w)}\,dw. \tag{11.138}$$

Since $z \le \sqrt{x}$, the boundary term is uniformly $O(e^{-c\sqrt{\log z}}/\log x)$, hence $o(1/\log^2 x)$. Write $\log^{-2}(x/w) = g_x(w)\log^{-2} x$, where $1 \le g_x(w) \le 4$ for $1 \le w \le z$. To justify the moving cut-off uniformly, fix $\varepsilon > 0$ and choose W_0 so large that

$$\int_{W_0}^{\infty} \frac{|\widetilde{m}(w)|}{w}\,dw < \varepsilon \tag{11.139}$$

by (11.125). On $[1, W_0]$ one has $g_x(w) \to 1$ uniformly as $x \to \infty$, while on $[W_0, z]$ the bound $|g_x(w)| \le 4$ makes the contribution at most 4ε. Hence

$$\int_1^z \frac{\widetilde{m}(w)}{w}\,g_x(w)\,dw = \int_1^{\infty} \frac{\widetilde{m}(w)}{w}\,dw + o(1) = 1 + o(1), \tag{11.140}$$

uniformly in the stated range of z, which proves (11.137). □

Heuristic 11.67 (Opposite drifts)**.** Let $u \ge \sqrt{v}\exp(C(\log\log v)^2)$. Near $t \in I_\nu$ the counts Π_a grow for $a < t/\sqrt{v}$, at rate $1/(a\log(t/a))$ if primes near t/a have their expected local density uniformly in a. Over $(t, t+h]$ the large-prime part then changes by about $-h\sum_{a<t/\sqrt{v}} \mu(a)/(a\log(t/a)) = (1+o(1))h/\log^2 t$ by Lemma 11.66. Since $M(v) - M(u) = 0$, the $\sqrt{v}$-smooth part drifts at rate about $-1/\log^2 v$, independently of σ_ν. As

$$\frac{\ell_\nu}{\log^2 v_\nu} \gg \sqrt{\ell_\nu} \iff \ell_\nu \gg \log^4 v_\nu, \tag{11.141}$$

for $\ell_\nu \gg (\log v_\nu)^4$ both drift contributions exceed $\sqrt{\ell_\nu}$, and (11.135) is a balance between two opposite drifts: the local analogue of Proposition 11.61.

12. Two-point correlations and the energy of excursions

For integers $0 \le u < X$ and $1 \le h < X - u$ put

$$C_h^{(u,X]} := \sum_{u<n\le X-h} \mu(n)\mu(n+h), \qquad C_h^{[X]} := C_h^{(0,X]}. \tag{12.1}$$

For an excursion $I_\nu = (x_{0,\nu}, x_{1,\nu}]$ one has $C_h^{(\nu)} = C_h^{(x_{0,\nu},x_{1,\nu}]}$, since $\sigma_\nu^2 = 1$. Chowla's conjecture asserts $C_h^{[X]} = o(X)$ for each fixed $h \ge 1$. Matomäki, Radziwiłł and Tao [33] prove, for $K = K(X) \to \infty$, $K \le X$,

$$\sum_{h\le K} \left| \sum_{n\le X} \mu(n)\mu(n+h) \right| = o(KX). \tag{12.2}$$

Since replacing the upper limit X by $X - h$ changes the h-th sum by at most h, we shall use the triangular consequence

$$\sum_{h\le K} |C_h^{[X]}| = o(KX) \qquad (K \to \infty,\ K = o(X)). \tag{12.3}$$

Their stronger averaged Elliott form permits translated shifts of size $O(X)$ and covers μ.

12.1. **Signed lag averages and the weighted RH criterion.**

Remark 12.1 (Unweighted signed averages are degenerate). By Lemma 5.3, every excursion satisfies $\sum_h C_h^{(\nu)} = -q_\nu/2$, with $q_\nu \le \ell_\nu$. Hence $\ell_\nu^{-2}\sum_h C_h^{(\nu)} = O(\ell_\nu^{-1})$ for purely algebraic reasons: a signed average over lags of the correlations inside a bridge is small without any pseudorandomness of μ. Statements of Chowla type inside an excursion must therefore carry absolute values inside the lag average, or impose a genuinely weighted condition.

The weighted sum behaves differently. By (5.8), $Q_\nu = -\sum_h hC_h^{(\nu)}$: the weight h records at which lags the correlation responsible for the excursion energy sits. The same identity holds globally at every zero of M, and the cross terms between different excursions cancel exactly.

Proposition 12.2 (Weighted correlations at zeros of M)**.** *Let $v \ge 2$ be a zero of M. Then*

$$\sum_{n<v} M(n)^2 = -\sum_{h<v} h\, C_h^{[v]}, \qquad \sum_{h<v} h\, C_h^{[v]} = \sum_{\nu:\, x_{1,\nu}\le v} \sum_h h\, C_h^{(\nu)}. \tag{12.4}$$

Thus, in the h-weighted sum, the correlations between increments lying in different completed excursions cancel exactly; internal zero-run increments contribute nothing because $\mu(n) = 0$ there.

Proof. Apply Lemma 5.1 to the block $x_n = \mu(n)$, $1 \le n \le v$: then $S_t = M(t)$ and $S_v = M(v) = 0$, which gives the first identity. Because $M(v) = 0$, $\sum_{n\le v} M(n)^2 = \sum_{n<v} M(n)^2$. By (5.12), this equals $\sum_\nu Q_\nu$ over the completed excursions ending at or before v, and $Q_\nu = -\sum_h hC_h^{(\nu)}$ by (5.8). □

Theorem 12.3 (Weighted-correlation criterion)**.** *Using the unconditional return localization (RL) of Corollary 9.11, RH holds if and only if, for every $\varepsilon > 0$,*

$$-\sum_{h<v} h\, C_h^{[v]} \ll_\varepsilon v^{2+\varepsilon} \qquad \text{for every integer } v \text{ with } M(v) = 0. \tag{12.5}$$

Proof. If RH holds, Theorem 8.6 and (12.4) give the bound. Conversely, fix a target $\varepsilon_0 > 0$ and choose $\varepsilon > 0$ so small that $(1+\varepsilon)(2+\varepsilon) \le 2+\varepsilon_0$. Let x be large and put $N = \lfloor x \rfloor$. If $M(N) = 0$, then $\sum_{n\le x} M(n)^2 = B_2(N)$ is already evaluated at a zero. Otherwise, let $u = \max\{n \le N : M(n) = 0\}$ and let $v > N$ be the next zero. Then $(u, v]$ is the completed excursion whose open prefix contains N, and Corollary 9.11 gives $v \le u^{1+\varepsilon} \le x^{1+\varepsilon}$ for all sufficiently large x. In either case, with $v = N$ in the zero case,

$$\sum_{n\le x} M(n)^2 \le \sum_{n<v} M(n)^2 \ll_\varepsilon v^{2+\varepsilon} \le x^{2+\varepsilon_0}. \tag{12.6}$$

Theorem 8.6 now applies. □

Proposition 12.4 (Square-root lag cutoff at zeros)**.** *Let $v \to \infty$ through integers satisfying $M(v) = 0$, and let $K = K(v) \to \infty$ with $K \le v^{1/2}$. Then*

$$\sum_{h\le K} h\, C_h^{[v]} = o(K^2 v). \tag{12.7}$$

In particular,

$$\sum_{n<v} M(n)^2 = -\sum_{\sqrt{v}<h<v} h\, C_h^{[v]} + o(v^2), \tag{12.8}$$

where the lower limit may be replaced by $\lfloor\sqrt{v}\rfloor$ without changing the statement.

Proof. By (12.3), $\sum_{h\le K} |C_h^{[v]}| = o(Kv)$. Therefore

$$\left|\sum_{h\le K} hC_h^{[v]}\right| \le K \sum_{h\le K} |C_h^{[v]}| = o(K^2 v). \tag{12.9}$$

Taking $K = \lfloor \sqrt{v} \rfloor$ and using Proposition 12.2 gives (12.8). □

Corollary 12.5 (RH after deleting all small lags)**.** *RH holds if and only if, for every $\varepsilon > 0$,*

$$-\sum_{\sqrt{v}<h<v} h\, C_h^{[v]} \ll_\varepsilon v^{2+\varepsilon} \qquad \text{for every sufficiently large zero } v \text{ of } M. \tag{12.10}$$

Thus the lags $h \le \sqrt{v}$ are unconditionally negligible at the critical v^2 scale of the weighted-correlation criterion.

Theorem 12.6 (Zero-edge lag localization)**.** *Let $v \to \infty$ through integers satisfying $M(v) = 0$. Then*

$$-\sum_{h<v} hC_h^{[v]} = \sum_{n<v} M(n)^2 = v^{1+2\Theta+o(1)}. \tag{12.11}$$

Moreover, for every fixed $0 \le \alpha < \Theta$,

$$\sum_{h\le v^\alpha} hC_h^{[v]} = o\left(\sum_{n<v} M(n)^2\right), \tag{12.12}$$

and hence

$$-\sum_{v^\alpha<h<v} hC_h^{[v]} = (1+o(1))\sum_{n<v} M(n)^2. \tag{12.13}$$

Thus the signed weighted correlation total is forced beyond every power v^α with $\alpha < \Theta$.

Proof. The first equality is Proposition 12.2. Since v is an integer zero of M, the final summand vanishes and $B_2(v) = \sum_{n\le v} M(n)^2 = \sum_{n<v} M(n)^2$; hence the asymptotic exponent follows by applying Corollary 8.11 along the subsequence of such zeros $v \to \infty$. For $K = \lfloor v^\alpha \rfloor$ with $\alpha > 0$, (12.3) gives

$$\left|\sum_{h\le K} hC_h^{[v]}\right| \le K \sum_{h\le K} |C_h^{[v]}| = o(K^2 v) = o(v^{1+2\alpha}). \tag{12.14}$$

Choose a fixed δ_0 with $0 < \delta_0 < 2(\Theta - \alpha)$. Along zeros $v \to \infty$, Corollary 8.11 gives $B_2(v) \ge v^{1+2\Theta-\delta_0}$ for all sufficiently large v, while the small-lag contribution is $o(v^{1+2\alpha})$. Their ratio is therefore $o(v^{-2(\Theta-\alpha)+\delta_0}) = o(1)$. The case $\alpha = 0$ follows trivially from the fixed finite number of lags and $|C_h^{[v]}| \le v$. □

Remark 12.7 (Endpoint versus the universal square-root cutoff)*.* Proposition 12.4 is uniform without knowing Θ and reaches the endpoint $K = \sqrt{v}$ at the absolute v^2 scale. Theorem 12.6 is stronger in relative terms but requires a strict power gap $\alpha < \Theta$; it does not settle the endpoint $\alpha = \Theta$. Under RH it says that every range $h \le v^{1/2-\delta}$ carries $o(1)$ of the total signed weighted correlation, while if RH fails the negligible range extends correspondingly beyond the square-root scale.

Remark 12.8 (Where the cancellation must occur)*.* The trivial bound is $\sum_h h|C_h^{[v]}| \le \sum_h h(v-h) \le v^3/6$. RH requires the signed weighted sum to be $v^{2+o(1)}$. Absolute values cannot give this at the natural scale: even square-root cancellation $|C_h^{[v]}| \ll v^{1/2+o(1)}$ in every single lag would give only $\sum_h h|C_h^{[v]}| \ll v^{5/2+o(1)}$. Corollary 12.5 sharpens the location of the unresolved cancellation: at the v^2 scale, only lags $h > \sqrt{v}$ remain. By Proposition 12.2 the signed weighted identity itself decomposes excursion by excursion.

12.2. **Lag localization of the energy of an excursion.**

Proposition 12.9 (Energy beyond a lag cutoff)**.** *For every excursion I_ν of length $\ell = \ell_\nu$ and every $1 \le h_0 < \ell$,*

$$\Big|Q_\nu + \sum_{h_0<h<\ell} h\, C_h^{(\nu)}\Big| \le h_0 \sum_{h\le h_0} |C_h^{(\nu)}|. \tag{12.15}$$

Consequently, if $\sum_{h\le h_0} |C_h^{(\nu)}| \le \delta h_0 \ell$, then

$$\sum_{h_0<h<\ell} h\, C_h^{(\nu)} = -Q_\nu + O(\delta h_0^2 \ell). \tag{12.16}$$

Proof. Split $Q_\nu = -\sum_{h<\ell} h C_h^{(\nu)}$ at h_0. □

Corollary 12.10 (Averaged Chowla on linear-scale intervals)**.** *Let $(u_j, u_j + L_j]$ be intervals with $L_j \to \infty$ and $u_j \le BL_j$ for a fixed B, and let $h_j \to \infty$ with $h_j = o(L_j)$. Then*

$$\sum_{h\le h_j} |C_h^{(u_j, u_j+L_j]}| = o(h_j L_j). \tag{12.17}$$

Proof. Put $X_j = L_j - h_j$. Then $X_j \sim L_j$ and, for all large j, $u_j \le 2BX_j$. Apply the shifted form of [33, Theorem 1.6], whose hypotheses explicitly allow translation parameters $b_r \le AX_j$, to the two-point Möbius correlation. The same paper states immediately after Theorem 1.6 that the argument proving its averaged Chowla theorem also applies with λ replaced by μ. Taking one relative shift fixed and averaging the other over $1 \le h \le h_j$, with the common translation $u_j = O(X_j)$ absorbed into the allowed b_r, gives

$$\sum_{h\le h_j} \left| \sum_{1\le n\le X_j} \mu(u_j + n)\mu(u_j + n + h) \right| = o(h_j X_j) = o(h_j L_j). \tag{12.18}$$

For each $h \le h_j$, extending the inner sum from $X_j = L_j - h_j$ to $L_j - h$ adds at most $h_j - h$ terms. Hence the total triangular boundary cost is

$$\sum_{h\le h_j} (h_j - h) = O(h_j^2) = o(h_j L_j), \tag{12.19}$$

which proves the claim. □

Corollary 12.11 (Signed large-lag concentration in linear-scale excursions)**.** *Let $B > 0$ be fixed and let I_j be excursions with $\ell_j \to \infty$, $x_{0,j} \le B\ell_j$ and heights H_j with $H_j^3/\ell_j \to \infty$. For every $h_j \to \infty$ with $h_j^2 \ell_j = o(H_j^3)$,*

$$\sum_{h_j<h<\ell_j} h\, C_h^{(I_j)} = -(1 + o(1))Q_{I_j}. \tag{12.20}$$

If in addition $\Theta > 1/2$, $0 < \eta < \Theta - \frac{1}{2}$ is fixed, there is a fixed $\lambda' > 1$ with $x_{1,j} \le \lambda' x_{0,j}$, and these excursions carry Kaczorowski–Pintz witnesses $H_j \ge \frac{1}{2} x_{0,j}^{\Theta-\eta}$, then for every fixed

$$0 < \theta_{\mathrm{lag}} < \frac{3(\Theta - \eta) - 1}{2} \tag{12.21}$$

one may take $h_j = x_{0,j}^{\theta_{\mathrm{lag}}}$, and hence

$$\sum_{x_{0,j}^{\theta_{\mathrm{lag}}}<h<\ell_j} h\, C_h^{(I_j)} = -(1 + o(1))Q_{I_j}. \tag{12.22}$$

Proof. By Corollary 12.10, $\sum_{h\le h_j}|C_h^{(I_j)}| = \delta_j h_j \ell_j$ with $\delta_j \to 0$. Proposition 12.9 gives an error at most $\delta_j h_j^2 \ell_j = o(H_j^3) = o(Q_{I_j})$. In the witness case the hypotheses $x_{0,j} \le B\ell_j$ and $x_{1,j} \le \lambda' x_{0,j}$ give $\ell_j \asymp x_{0,j}$. Hence $H_j^3/\ell_j \gg x_{0,j}^{3\Theta-3\eta-1}$, so the displayed condition on θ_{lag} gives $h_j^2\ell_j = o(H_j^3)$. □

Remark 12.12 (Direction of the lag statement). Corollary 12.11 says that the signed weighted correlation total representing the energy lies beyond every cutoff h_j satisfying $h_j = o(H_j^{3/2}\ell_j^{-1/2})$; it does *not* say that the energy is concentrated only at lags larger than the endpoint scale $H_j^{3/2}\ell_j^{-1/2}$ itself.

12.3. **Uniform two-point cancellation in sublinear intervals.** The linear-scale restriction can be removed for mesoscopic lags by a Fourier-uniform estimate valid in every interval in the range displayed below. Theorem 1.5 of Matomäki–Teräväinen [35] gives, for every fixed $\iota > 0$ and every fixed $0 < A < 1/3$, a bound uniform in the additive frequency and in every interval with $L \ge x^{3/5+\iota}$. We use exactly that published range; no stronger logarithmic exponent is assumed here. Thus, for every fixed $0 < A < 1/3$ and $\iota > 0$,

$$\sup_{\xi\in\mathbb{R}}\Big|\sum_{x<n\le x+L}\mu(n)e(\xi n)\Big| \ll_{A,\iota} \frac{L}{(\log x)^A} \qquad (x^{3/5+\iota} \le L \le x). \tag{12.23}$$

Theorem 12.13 (Uniform two-point Cesàro cancellation). *Fix $0 < A < 1/3$ and $\iota > 0$. Uniformly for all integers u, L, R with $u^{3/5+\iota} \le L \le u$ and $1 \le R \le L/2$,*

$$\frac{1}{R}\sum_{h=1}^{R}\frac{|C_h^{(u,u+L]}|}{L-h} \ll_{A,\iota} \Big(\frac{L}{R}\Big)^{1/2}(\log u)^{-A}. \tag{12.24}$$

In particular, if $R \ge L(\log u)^{-2A+2\kappa}$ with $0 < \kappa < A$, the left side is $O((\log u)^{-\kappa})$.

Proof. Put $a_i = \mu(u+i)$ for $1 \le i \le L$ and $A_I(\xi) = \sum_i a_i e(i\xi)$. For $h > 0$ define $C_{-h}^{(u,u+L]} := \sum_{i=1}^{L-h} a_{i+h}a_i = C_h^{(u,u+L]}$ and put $C_0 = q = \sum_i a_i^2$. These are the Fourier coefficients of $|A_I(\xi)|^2$. Parseval gives the precise identity

$$\int_0^1 |A_I(\xi)|^4\,d\xi = q^2 + 2\sum_{h=1}^{L-1}|C_h^{(u,u+L]}|^2 \le \|A_I\|_\infty^2 q \le L\|A_I\|_\infty^2. \tag{12.25}$$

Hence Cauchy–Schwarz gives $\sum_{h\le R}|C_h| \le \sqrt{RL}\,\|A_I\|_\infty$, and since $L - h \ge L/2$ the left side of (12.24) is at most $2\|A_I\|_\infty/\sqrt{RL}$. Apply (12.23). □

Corollary 12.14 (Published mesoscopic logarithmic range). *Fix $\iota > 0$ and $0 < \delta < 2/3$. Uniformly for*

$$u^{3/5+\iota} \le L \le u, \qquad \frac{L}{(\log u)^{2/3-\delta}} \le R \le \frac{L}{2}, \tag{12.26}$$

one has

$$\frac{1}{R}\sum_{h\le R}\frac{|C_h^{(u,u+L]}|}{L-h} \ll_{\iota,\delta} (\log u)^{-\delta/4}. \tag{12.27}$$

Proof. In Theorem 12.13 take $A = 1/3 - \delta/4$ and $\kappa = \delta/4$. □

Remark 12.15 (Scope of the sublinear estimate). Theorem 12.13 is a short Parseval–Cauchy–Schwarz corollary of the published Fourier estimate (12.23); it does not use the excursion condition. Its role here is to provide an arithmetic input on every adaptively selected interval, something that "almost all intervals" results cannot supply. The factor $(L/R)^{1/2}$ is inherent in this direct L^2-to-L^1 Cauchy–Schwarz step using only the displayed second-moment control of the C_h; additional information on the distribution of the correlations could in principle improve it.

Corollary 12.16 (Sublinear large-lag estimate)**.** *Fix $\iota > 0$ and $0 < \delta < 2/3$, and let $I = (u, u+L]$ be a completed excursion with $u^{3/5+\iota} \le L \le u$. For all sufficiently large u, put*

$$R = \left\lceil \frac{L}{(\log u)^{2/3-\delta}} \right\rceil. \tag{12.28}$$

For sufficiently large u, $(\log u)^{2/3-\delta} \ge 3$, and hence $1 \le R \le L/2$; therefore the preceding two-point estimate applies. Then

$$\sum_{R<h<L} hC_h^{(I)} = -Q_I + O_{\iota,\delta}\left(\frac{L^3}{(\log u)^{4/3-7\delta/4}} \right). \tag{12.29}$$

Consequently, if $Q_I L^{-3} (\log u)^{4/3-7\delta/4} \to \infty$, then the left side is $-(1+o(1))Q_I$.

Proof. Corollary 12.14 gives $\sum_{h\le R} |C_h^{(I)}| \ll RL(\log u)^{-\delta/4}$. Hence

$$\left| \sum_{h\le R} hC_h^{(I)} \right| \le R \sum_{h\le R} |C_h^{(I)}| \ll R^2 L (\log u)^{-\delta/4}, \tag{12.30}$$

which is the error in (12.29) after inserting the chosen value of R into $Q_I = -\sum_{h<L} hC_h^{(I)}$. □

Remark 12.17 (Connection with wide peaks). The quantitative form of the short-interval estimate in [35, Theorem 1.1] gives, after the blocking argument of Lemma 9.55, $\omega_u \ll_\varepsilon (\log u)^{-1/3+\varepsilon} + u^{-c}$ for some $c > 0$ (depending on the two fixed short-interval exponents used there). Thus, in the regime $H/h_u \to \infty$ of Proposition 9.56,

$$Q_I \gg_\varepsilon H^3 (\log u)^{1/3-\varepsilon}. \tag{12.31}$$

Combining this with Corollary 12.16, a sufficient condition for signed large-lag concentration is

$$\left(\frac{H}{L} \right)^3 (\log u)^{5/3-7\delta/4-\varepsilon} \longrightarrow \infty. \tag{12.32}$$

For linearly localized Kaczorowski–Pintz excursions satisfying $v \le \lambda u$ with a fixed $\lambda < 2$, this machinery applies when $\Theta > 3/5$ (after choosing the loss parameters so that $\Theta - \eta > 3/5 + \iota$); the remaining obstruction is that the excursion may be much longer than its height.

12.4. **Fejér reduction and the weaker local-variance target.**

Remark 12.18 (Excursion geometry alone does not give cancellation). For the tent excursion $a_i = +1$ $(i \le k)$, $a_i = -1$ $(k < i \le 2k)$, which satisfies the bridge and one-sign constraints of a primitive excursion, $C_h = 2k - 3h$ for $1 \le h < k$. Hence for $R \to \infty$, $R = o(k)$,

$$\frac{1}{R} \sum_{h\le R} \frac{|C_h|}{2k-h} \longrightarrow 1, \qquad \sum_{h<R} (R-h) C_h \asymp kR^2, \tag{12.33}$$

whereas $RL \asymp kR$. Arithmetic input is indispensable.

Lemma 12.19 (Finite Fejér identity, classical)**.** *For a real sequence $(a_i)_{1\le i\le L}$, extended by 0 outside $[1, L]$, put*

$$q = \sum_{i=1}^{L} a_i^2, \qquad C_h = \sum_{i=1}^{L-h} a_i a_{i+h}, \qquad F_R := \sum_{h=1}^{R-1} (R-h) C_h. \tag{12.34}$$

Then for $1 \le R \le L$,

$$Rq + 2F_R = \sum_{m\in\mathbb{Z}} \left(\sum_{j=1}^{R} a_{m+j} \right)^2. \tag{12.35}$$

Proof. Expand the square. Each diagonal term occurs R times, and a pair at lag h, $1 \le h < R$, occurs $2(R-h)$ times. □

Proposition 12.20 (Bridge–Fejér identity and interior variance)**.** *Let $I = (u, u+L]$ be a completed excursion, put $a_i = \mu(u+i)$ and $S_t = \sum_{i=1}^{t} a_i = M(u+t)$ for $0 \le t \le L$. Then $S_0 = S_L = 0$, and for every $1 \le R < L$,*

$$Rq + 2F_R = 2Q_I - 2\sum_{t=1}^{L-1-R} S_t S_{t+R}. \tag{12.36}$$

Moreover

$$Rq + 2F_R = E_{\mathrm{bd}}(R) + \sum_{m=0}^{L-R} \big(M(u+m+R) - M(u+m)\big)^2, \tag{12.37}$$

where

$$E_{\mathrm{bd}}(R) = \sum_{t=1}^{R-1} S_t^2 + \sum_{t=L-R+1}^{L-1} S_t^2 = O(R^3). \tag{12.38}$$

Consequently, if $R = o(\sqrt{L})$, then

$$F_R = o(RL) \quad \Longleftrightarrow \quad \sum_{m=0}^{L-R} \big(M(u+m+R) - M(u+m)\big)^2 = Rq + o(RL). \tag{12.39}$$

If in addition $L \ge u^{1/2+\varepsilon}$ and $L \le u$, then

$$q = \left(\frac{6}{\pi^2} + o(1)\right) L, \tag{12.40}$$

and the right side of (12.39) *is equivalently*

$$\sum_{m=0}^{L-R} \big(M(u+m+R) - M(u+m)\big)^2 = \left(\frac{6}{\pi^2} + o(1)\right) RL. \tag{12.41}$$

Proof. Extend S_t by 0 outside $0 \le t \le L$. Since $\sum_{j=1}^{R} a_{m+j} = S_{m+R} - S_m$, (12.35) gives

$$Rq + 2F_R = \sum_{m} (S_{m+R} - S_m)^2 = 2\sum_{m} S_m^2 - 2\sum_{m} S_m S_{m+R}, \tag{12.42}$$

The natural range of the last sum is $0 \le m \le L-R$, but the terms $m = 0$ and $m = L-R$ vanish because $S_0 = S_L = 0$, leaving $1 \le t \le L-1-R$; this is (12.36). Splitting the same sum into $m < 0$, $0 \le m \le L-R$, and $m > L-R$ gives (12.37): for $-R < m < 0$ the contribution is S_{m+R}^2, for $L-R < m < L$ it is S_m^2, and the endpoint terms vanish because $S_0 = S_L = 0$. Since $|a_i| \le 1$ and the path returns to zero, $|S_t| \le t$ and $|S_{L-t}| \le t$, whence $E_{\mathrm{bd}}(R) \le 2\sum_{t<R} t^2 = O(R^3)$. Thus $E_{\mathrm{bd}}(R) = o(RL)$ when $R = o(\sqrt{L})$, proving the equivalence. Finally the classical estimate $\sum_{n \le x} \mu(n)^2 = (6/\pi^2)x + O(\sqrt{x})$ gives $q = (6/\pi^2)L + O(\sqrt{u+L})$. Under $u^{1/2+\varepsilon} \le L \le u$,

$$\frac{\sqrt{u+L}}{L} \le \frac{\sqrt{2u}}{u^{1/2+\varepsilon}} \ll u^{-\varepsilon}, \tag{12.43}$$

so the error is uniform and (12.40) follows. □

Remark 12.21 (Fejér density at frequency zero). With $A_I(\xi) = \sum_{i=1}^{L} a_i e(i\xi)$ and

$$K_R(\xi) = \frac{1}{R}\left|\sum_{j=1}^{R} e(j\xi)\right|^2, \tag{12.44}$$

one has

$$q + \frac{2F_R}{R} = \int_0^1 |A_I(\xi)|^2 K_R(\xi)\, d\xi. \tag{12.45}$$

Thus $F_R = o(RL)$ says that the Fejér average of the local spectral density at frequency 0, at resolution $1/R$, equals $q + o(L)$. It is not a statement that $|A_I(\xi)|^2$ is spectrally flat on the whole circle.

Remark 12.22 (Three distinct targets). The following conditions must not be conflated.

First, absolute averaged Chowla, $\sum_{h<R}|C_h| = o(RL)$, implies only $|F_R| \le R\sum_{h<R}|C_h| = o(R^2L)$, not the sharper $F_R = o(RL)$ when $R \to \infty$. Conversely, $F_R = o(RL)$ allows cancellation between the C_h and does not control their absolute values.

Second, F_R does not by itself control the small-lag contribution to the excursion energy. If

$$V_R := \sum_{h<R} hC_h, \tag{12.46}$$

then exactly

$$F_R + V_R = R\sum_{h<R} C_h. \tag{12.47}$$

Thus a Fejér estimate alone does not imply $V_R = o(Q_I)$.

Third, Proposition 12.20 requires both endpoints of the block to be zeros of M. For a block beginning at a zero but ending before the return, additional boundary terms remain.

Remark 12.23 (Why the variance target may go below 3/5). For the Fejér problem one does not need the stronger absolute estimate $\sum|C_h|$. By Proposition 12.20, on a completed excursion and for $R = o(\sqrt{L})$ it is enough to prove the diagonal local-variance law

$$\sum_{m=0}^{L-R}\big(M(u+m+R) - M(u+m)\big)^2 = Rq + o(RL). \tag{12.48}$$

This is a different signed target from absolute averaged Chowla: neither implication is supplied by the estimates above. It is a plausible target below the Fourier-uniform threshold $L \ge u^{3/5+\iota}$. The all-interval U^2/Fourier estimates currently used above do not, by themselves, supply this asymptotic for every adaptively selected block in the range $L \ge u^{1/2+\varepsilon}$. Results holding for almost all starting points cannot be substituted, because excursion endpoints are selected by M itself.

No fixed-prefix formula for the increments in (12.41) is asserted here. In particular, the identity $\sum_{d\le x}\mu(d)\lfloor x/d\rfloor = 1$ introduces lower-scale terms such as $M(x/2)$ when one truncates at $x/2$; omitting those terms would give an invalid reduction.

12.5. **Dyadic multiscale Fejér forcing on completed excursions.** The preceding Fejér identity treats one scale at a time. A completed one-sign excursion permits a second operation: close the zero-padded block into a cycle. The wrap-around terms then have a favorable sign, and the whole dyadic tower can be combined without paying an unsigned boundary error.

Let $I = (u, u+L]$ be a completed excursion with orientation σ. Put

$$a_i = \sigma\mu(u+i) \quad (1 \le i \le L), \qquad a_i = 0 \quad (L < i \le \widehat{L}), \tag{12.49}$$

where $\widehat{L} = 2^{m_{\mathrm{dy}}}$ is the least power of two not smaller than L. Write

$$S_r = \sum_{i=1}^{r} a_i \quad (0 \le r \le \widehat{L}), \qquad q = \sum_{i=1}^{\widehat{L}} a_i^2. \tag{12.50}$$

Then $S_r \ge 0$, $S_L = 0$, and in fact $S_r = 0$ for $L \le r \le \widehat{L}$. For $1 \le h < \widehat{L}$ let

$$C_h = \sum_{i=1}^{\widehat{L}-h} a_i a_{i+h}, \qquad F_R = \sum_{h=1}^{R-1}(R-h)C_h. \tag{12.51}$$

All variables introduced here are local except the already defined F_R and the multiscale sum $\mathfrak{K}_I$ below.

Lemma 12.24 (Cyclic Fejér boundary sign)**.** *Extend (a_i) periodically modulo $\widehat{L}$ and define*

$$C_h^{\mathrm{cyc}} = \sum_{i=1}^{\widehat{L}} a_i a_{i+h}, \qquad F_R^{\mathrm{cyc}} = \sum_{h=1}^{R-1} (R-h) C_h^{\mathrm{cyc}}, \tag{12.52}$$

where the index $i+h$ is read modulo $\widehat{L}$. For every $1 \le R \le \widehat{L}/2$,

$$\boxed{F_R^{\mathrm{cyc}} = F_R - \sum_{r=1}^{R-1} S_r S_{\widehat{L}-R+r} \le F_R.} \tag{12.53}$$

If

$$W_R^{\mathrm{cyc}} := \sum_{s=0}^{\widehat{L}-1} \left(\sum_{j=1}^{R} a_{s+j} \right)^2, \tag{12.54}$$

with cyclic indices, then

$$\boxed{W_R^{\mathrm{cyc}} = Rq + 2F_R^{\mathrm{cyc}}.} \tag{12.55}$$

Proof. For $1 \le h < R$ the only terms added by cyclic closure are the pairs that cross the cut between $\widehat{L}$ and 1, hence

$$C_h^{\mathrm{cyc}} - C_h = \sum_{i=\widehat{L}-h+1}^{\widehat{L}} a_i a_{i+h-\widehat{L}}. \tag{12.56}$$

Therefore

$$F_R^{\mathrm{cyc}} - F_R = \sum_{h=1}^{R-1} (R-h) \sum_{i=\widehat{L}-h+1}^{\widehat{L}} a_i a_{i+h-\widehat{L}} \tag{12.57}$$

$$= \sum_{j=1}^{R-1} \sum_{i=\widehat{L}-R+j+1}^{\widehat{L}} (R - (\widehat{L} - i + j)) a_i a_j, \tag{12.58}$$

where in the second line $j = i + h - \widehat{L}$ and the inequalities $1 \le h < R$ have been rewritten in (i, j) coordinates. On the other hand, $S_{\widehat{L}} = 0$ gives

$$S_{\widehat{L}-R+r} = - \sum_{i=\widehat{L}-R+r+1}^{\widehat{L}} a_i, \tag{S12.1}$$

and hence

$$- \sum_{r=1}^{R-1} S_r S_{\widehat{L}-R+r} = \sum_{r=1}^{R-1} \sum_{j=1}^{r} \sum_{i=\widehat{L}-R+r+1}^{\widehat{L}} a_j a_i \tag{12.59}$$

$$= \sum_{j=1}^{R-1} \sum_{i=\widehat{L}-R+j+1}^{\widehat{L}} (R - (\widehat{L} - i + j)) a_i a_j. \tag{12.60}$$

The two displayed double sums are identical, proving (12.53). Because the zero-padded oriented excursion has $S_r \ge 0$ and also $S_{\widehat{L}-R+r} \ge 0$, the boundary term is nonnegative and $F_R^{\mathrm{cyc}} \le F_R$.

Finally expand W_R^{cyc}. Each a_i^2 occurs in exactly R cyclic windows. For a cyclic pair at lag $h < R$, the product $a_i a_{i+h}$ occurs in exactly $R-h$ windows in each orientation, hence with total coefficient $2(R-h)$. This gives $W_R^{\mathrm{cyc}} = Rq + 2\sum_{h<R}(R-h)C_h^{\mathrm{cyc}}$ and proves (12.55). □

Define the dyadic scale set

$$\mathcal{R}_I := \{1, 2, 4, \dots, \widehat{L}/2\} \tag{12.61}$$

and the signed multiscale Fejér sum

$$\mathfrak{K}_I := \sum_{R \in \mathcal{R}_I} \frac{F_R}{R}. \tag{12.62}$$

There are exactly m_{dy} scales in $\mathcal{R}_I$.

Proposition 12.25 (Dyadic multiscale Fejér inequality)**.** *With the preceding notation and* $H = \max_{0 \le r \le L} S_r$*,*

$$\boxed{H^2 \le m_{\mathrm{dy}} \sum_{R \in \mathcal{R}_I} \frac{W_R^{\mathrm{cyc}}}{R} \le m_{\mathrm{dy}}(m_{\mathrm{dy}} q + 2\mathfrak{K}_I).} \tag{12.63}$$

Consequently,

$$\boxed{\mathfrak{K}_I \ge \frac{H^2}{2m_{\mathrm{dy}}} - \frac{m_{\mathrm{dy}} q}{2}.} \tag{12.64}$$

Proof. View S periodically on $\mathbb{Z}/\widehat{L}\mathbb{Z}$ and put

$$\mathcal{A}_R S(s) := \frac{1}{R} \sum_{t=0}^{R-1} S_{s+t} \qquad (R \in \mathcal{R}_I). \tag{12.65}$$

For $2R \le \widehat{L}$,

$$\mathcal{A}_R S(s) - \mathcal{A}_{2R} S(s) = -\frac{1}{2R} \sum_{t=0}^{R-1} (S_{s+t+R} - S_{s+t}). \tag{12.66}$$

By Cauchy–Schwarz and the definition of W_R^{cyc},

$$|\mathcal{A}_R S(s) - \mathcal{A}_{2R} S(s)|^2 \le \frac{W_R^{\mathrm{cyc}}}{4R}. \tag{12.67}$$

Telescoping from $R = 1$ to $R = \widehat{L}$ and writing $\overline{S} = \widehat{L}^{-1} \sum_{s=0}^{\widehat{L}-1} S_s$ gives

$$|S_s - \overline{S}| \le \frac{1}{2} \sum_{R \in \mathcal{R}_I} \left(\frac{W_R^{\mathrm{cyc}}}{R} \right)^{1/2} \le \frac{1}{2} \left(m_{\mathrm{dy}} \sum_{R \in \mathcal{R}_I} \frac{W_R^{\mathrm{cyc}}}{R} \right)^{1/2}. \tag{12.68}$$

Apply this once at $s = 0$, where $S_0 = 0$, and once at an apex s_* with $S_{s_*} = H$. The triangle inequality then yields

$$H \le |H - \overline{S}| + |\overline{S}| \le \left(m_{\mathrm{dy}} \sum_{R \in \mathcal{R}_I} \frac{W_R^{\mathrm{cyc}}}{R} \right)^{1/2}, \tag{12.69}$$

which is the first inequality in (12.63). By Lemma 12.24,

$$\sum_{R \in \mathcal{R}_I} \frac{W_R^{\mathrm{cyc}}}{R} = m_{\mathrm{dy}} q + 2 \sum_{R \in \mathcal{R}_I} \frac{F_R^{\mathrm{cyc}}}{R} \le m_{\mathrm{dy}} q + 2\mathfrak{K}_I. \tag{12.70}$$

This proves the second inequality and, after rearrangement, (12.64). □

Corollary 12.26 (RH violation forces multiscale correlation excess)**.** *Assume* $\Theta > 1/2$*. Fix*

$$\frac{1}{2} < \alpha < \Theta \qquad \text{and} \qquad \lambda > \lambda_0. \tag{12.71}$$

Then there is an unbounded family of completed excursions $I = (u, v]$ with $v \le \lambda u$ for which

$$\boxed{\mathfrak{K}_I \gg_{\alpha,\lambda} \frac{u^{2\alpha}}{\log u}.} \tag{12.72}$$

In particular, failure of RH forces a positive correlation excess across the dyadic Fejér scales; the diagonal term $m_{\mathrm{dy}} q = O_\lambda(u \log u)$ cannot sustain the required heights.

Proof. Choose $\eta > 0$ so that $\alpha < \Theta - \eta$. By Corollary 9.10, the number of large-witness excursions starting below X is at least $(T_0/\pi - o(1)) \log X$. By Lemma 9.4, the number among them with $v > \lambda u$ is at most $(1/\log\lambda + o(1)) \log X$. Since $\lambda > e^{\pi/T_0}$, the difference of these two coefficients is positive; hence an unbounded family remains with $v \le \lambda u$. On each such excursion a witness $t \in (u, v]$ satisfies $|M(t)| \ge \frac{1}{2} t^{\Theta-\eta}$, and therefore, for all sufficiently large u, its height satisfies $H \ge u^\alpha$. Also $L = v - u \le (\lambda - 1)u$, so $q \le L = O_\lambda(u)$ and, because $L \le \widehat{L} < 2L$, one has $m_{\mathrm{dy}} = O_\lambda(\log u)$. Insert these bounds into (12.64); since $2\alpha > 1$, the $u^{2\alpha}/\log u$ term eventually dominates $u \log u$, proving (12.72). □

Remark 12.27 (Open multiscale Fejér target). The preceding corollary is a reduction, not a proof of RH. A concrete arithmetic estimate that would close this route is the following: for some fixed $\lambda > \lambda_0$ and some finite A, prove uniformly for every sufficiently late completed excursion $I = (u, v]$ with $v \le \lambda u$ that

$$\boxed{\mathfrak{K}_I \ll_{\lambda,A} u(\log u)^A.} \tag{12.73}$$

Together with Corollary 12.26, this would rule out every $\Theta > 1/2$. The Fejér weights themselves are classical, but (12.73) is not a theorem used in this paper and is not claimed here as a known or bibliographically new result. It is the open signed multiscale estimate isolated by the completed-excursion reduction. Existing small-lag or one-scale controls do not automatically imply it.

13. Extreme values in stochastic models of excursions

The counts of Section 9.1 bound the number of excursions in a family, not their heights. They acquire a second function when combined with a tail hypothesis on a normalized excursion statistic: they fix the scale of the family maximum, with no independence assumption.

Let $(\mathfrak{X}, \mathcal{G}, \Pr)$ be a probability space carrying a random sequence of disjoint integer intervals $I_k = (x_{0,k}, x_{1,k}]$ with $x_{1,k} \le x_{0,k+1}$, lengths at least 2, starts in $\{0, 2, 3, \dots\}$, and real random variables Z_k (for instance a model of the Möbius walk with $Z_k = H_k/\sqrt{\ell_k}$). A family $\mathcal{F}$ is given by a rule selecting indices from $(x_{0,k}, x_{1,k})_k$, such as $\mathcal{F}_\mathfrak{h}$; let $k_\mathcal{F}(1) < k_\mathcal{F}(2) < \cdots$ enumerate its members, and put

$$Z_{\max,\mathcal{F}}(X) := \max\{Z_k : k \in \mathcal{F},\ x_{0,k} \le X\} \qquad (\max \emptyset = -\infty). \tag{13.1}$$

The lemmas of Section 9.1 hold for every realization.

Hypothesis 13.1 (SG_ϖ)**.** There are $C, \kappa_0, \varpi > 0$ such that for every $j \ge 1$ and $z \ge 0$

$$\Pr(k_\mathcal{F}(j) \text{ exists and } Z_{k_\mathcal{F}(j)} > z) \le C e^{-\kappa_0 z^\varpi}. \tag{13.2}$$

For $\varpi = 2$ this is a uniform sub-Gaussian tail, denoted (SG).

Theorem 13.2 (Extreme scale from a count)**.** *Assume* (SG_ϖ) *for $\mathcal{F}$ and suppose that $E_\mathcal{F}(X) \le \mathcal{N}(X)$ for every realization, with $\mathcal{N}(X) \ge 1$ deterministic. Then*

$$\Pr(Z_{\max,\mathcal{F}}(X) > z) \le \mathcal{N}(X)\, C e^{-\kappa_0 z^\varpi}. \tag{13.3}$$

Consequently, for every $\delta > 0$,

$$\Pr\Big(Z_{\max,\mathcal{F}}(X) > \Big(\frac{1+\delta}{\kappa_0} \log \mathcal{N}(X)\Big)^{1/\varpi}\Big) \le C\, \mathcal{N}(X)^{-\delta}, \tag{13.4}$$

and $\max\{0, Z_{\max,\mathcal{F}}(X)\} = O_{\Pr}\big((\log \mathcal{N}(X))^{1/\varpi}\big)$ *as* $\mathcal{N}(X) \to \infty$.

Proof. The members of $\mathcal{F}$ starting at most X are the first $E_{\mathcal{F}}(X) \le \mathcal{N}(X)$ members of the enumeration, so $\{Z_{\max,\mathcal{F}}(X) > z\} \subset \bigcup_{1\le j\le \lfloor \mathcal{N}(X)\rfloor}\{k_{\mathcal{F}}(j) \text{ exists},\ Z_{k_{\mathcal{F}}(j)} > z\}$, and (13.3) is the union bound. Substituting $\kappa_0 z^{\varpi} = (1+\delta)\log \mathcal{N}(X)$ gives (13.4). □

Corollary 13.3 (Triple-logarithmic scale on the square family)**.** *Assume (SG) for the square-exceptional family* $\mathcal{F}_{x^2}$ *of the model, and write* $Z_{\max,\square} := Z_{\max,\mathcal{F}_{x^2}}$. *Then, for* $X \ge e^e$ *and every* $\delta > 0$,

$$\Pr\big(Z_{\max,\square}(X) > z\big) \le 4C(\log\log X)e^{-\kappa_0 z^2}. \tag{13.5}$$

For all sufficiently large X,

$$\Pr\Big(Z_{\max,\square}(X) > \sqrt{\tfrac{1+\delta}{\kappa_0}\log\log\log X}\Big) \le 4C(\log\log X)^{-\delta}. \tag{13.6}$$

Thus, for all sufficiently large X, $\max\{0, Z_{\max,\square}(X)\} = O_{\Pr}(\sqrt{\log\log\log X})$. *The same holds for the family* $\mathcal{F}_{cx^2}$ *of Lemma 9.2 with a constant depending on* c.

Proof. Use (9.4) in (13.3) and note that $e^{-\kappa_0 z^2} = (\log\log X)^{-(1+\delta)}$ at the stated z. □

The union bound gives convergence in probability; a Borel–Cantelli argument along a sequence adapted to $\mathcal{N}$ gives an almost sure bound with the same constant.

Theorem 13.4 (Almost sure extreme scale)**.** *Assume* (SG_{ϖ}) *for* $\mathcal{F}$ *and* $E_{\mathcal{F}}(X) \le \mathcal{N}(X)$ *for every realization, where* $\mathcal{N}$ *is continuous, nondecreasing and unbounded. Then almost surely*

$$\limsup_{X\to\infty} \frac{Z_{\max,\mathcal{F}}(X)}{(\log \mathcal{N}(X))^{1/\varpi}} \le \kappa_0^{-1/\varpi}. \tag{13.7}$$

Proof. Fix $\delta > 0$. For $j \ge j_0$ let $X_j := \inf\{X : \mathcal{N}(X) \ge 2^j\}$. By continuity, monotonicity and unboundedness, after increasing j_0 if necessary one has $\mathcal{N}(X_j) = 2^j$. Put $z_j = \big((1+\delta)(j+1)\log 2/\kappa_0\big)^{1/\varpi}$ and $A_j = \{Z_{\max,\mathcal{F}}(X_{j+1}) > z_j\}$. By (13.3), $\Pr(A_j) \le 2^{j+1}Ce^{-(1+\delta)(j+1)\log 2} = C2^{-(j+1)\delta}$, which is summable, so almost surely A_j occurs only finitely often. For $X_j \le X \le X_{j+1}$, monotonicity of $Z_{\max,\mathcal{F}}$ in X gives $Z_{\max,\mathcal{F}}(X) \le z_j$ outside A_j, while $\log \mathcal{N}(X) \ge j\log 2$. Hence the lim sup is at most $\lim_j\big((1+\delta)(j+1)/(\kappa_0 j)\big)^{1/\varpi} = ((1+\delta)/\kappa_0)^{1/\varpi}$. Let $\delta \downarrow 0$ along a countable sequence. □

With $\mathcal{N}_{\square}(X) = 2 + \log_2(\log X/\log 2)$, a continuous increasing majorant of the integer counting function, Theorem 13.4 gives under (SG)

$$\limsup_{X\to\infty} \frac{Z_{\max,\square}(X)}{\sqrt{\log\log\log X}} \le \kappa_0^{-1/2} \quad \text{almost surely.} \tag{13.8}$$

Applied to the three counts available for any sequence of disjoint intervals (all excursions: at most $X+1$; Lemma 9.4; Lemma 9.1), Theorems 13.2 and 13.4 give three levels:

family	members starting $\le X$	extreme scale under (SG)
all excursions	$\le X+1$	$\sqrt{\log X}$
$x_1 > \lambda x_0$	$O(\log X)$	$\sqrt{\log\log X}$
$x_1 > cx_0^2$ (or x_0^A)	$O(\log\log X)$	$\sqrt{\log\log\log X}$

(13.9)

Each level of lacunarity removes one logarithm from the extreme scale. Under (SG_{ϖ}) the last column reads $(\log X)^{1/\varpi}$, $(\log\log X)^{1/\varpi}$, $(\log\log\log X)^{1/\varpi}$.

Remark 13.5 (Scope). For M itself the square-exceptional family is finite by Corollary 9.11, so Theorems 13.2–13.4 concern stochastic models of the Möbius walk, in which the pathwise counts of Section 9.1 persist while no analogue of the Kaczorowski–Pintz localization is available. Hypothesis (SG) is an assumption on the model; the theorems state exactly what the counts contribute to its extremes.

14. A one-frequency model for the net area curve

Heuristic 14.1 (Scale and period)**.** Motivated by (6.1), model $M(x) \approx \sqrt{x}\,F(\log x)$ with F almost periodic, and retain only the lowest zero γ_1 (RH does not imply that its coefficient dominates). Within an excursion of length $\asymp x$ this gives $H_\nu \asymp \sqrt{x}$ and $\sqrt{H_\nu}\ell_\nu \asymp x^{5/4}$, so $\mathcal{A}$ oscillates at scale

$$|\mathcal{A}(x)| \asymp x^{5/4}, \tag{14.1}$$

the scale of Corollary 7.4, with period $2\pi/\gamma_1$ in $\log x$, i.e. a factor $e^{2\pi/\gamma_1} \approx 1.5597$ in x. In this model consecutive zeros of M are separated by the factor $e^{\pi/\gamma_1} = \lambda_*$. This is the uniform Corollary 4 benchmark of the counting comparison, not the current verified-height horizon λ_0; in the one-frequency model a model excursion has endpoint ratio exactly λ_*, at the boundary of the strict λ_*-long condition.

15. Column, row, and diagonal phase sums indexed by zeta zeros

This section is diagnostic and is logically separate from the excursion identities above. Write $\rho_j = \beta_j + i\gamma_j$ for the nontrivial zeros with $\gamma_j > 0$, ordered by ordinate and counted with multiplicity, and put

$$\mathscr{P}_{m,j} := m^{i\gamma_j}, \qquad m, j \geq 1. \tag{S15.1}$$

A column fixes a zero and varies the integer; a row fixes an integer and sums over zeros; the diagonal pairs the integer k with the kth zero. These are three mathematically different operations.

For a fixed zero on the critical line, $\beta_j = 1/2$, Euler–Maclaurin gives (with γ_j fixed; this is not a uniform estimate as $\gamma_j \to \infty$) (the same expansion used in (11.90))

$$\sum_{m \leq K} m^{-1/2}\mathscr{P}_{m,j} - \frac{K^{1/2+i\gamma_j}}{1/2 + i\gamma_j} = \zeta\Big(\tfrac{1}{2} - i\gamma_j\Big) + O_{\gamma_j}(K^{-1/2}) \longrightarrow 0. \tag{15.1}$$

Thus the zero is detected by the renormalized constant term, not by the unsubtracted partial sum, whose boundary term grows like $K^{1/2}$. For an off-line zero the corresponding zeta weight is $m^{-\beta_j}$, not the fixed weight $m^{-1/2}$.

For a fixed integer $m \geq 2$, summing a row over zeros is governed by Landau's formula. In the fixed-m form recorded by Kaczorowski–Languasco–Perelli [24],

$$\sum_{0 < \gamma_j \leq T} m^{\rho_j} = -\frac{T}{2\pi}\Lambda(m) + O_m(\log T) \qquad (T \to \infty). \tag{15.2}$$

Here $\Lambda(m)$ is nonzero precisely for prime powers. Under RH, $m^{\rho_j} = m^{1/2}\mathscr{P}_{m,j}$, and hence

$$\sum_{0 < \gamma_j \leq T} \mathscr{P}_{m,j} = -\frac{T}{2\pi}\frac{\Lambda(m)}{\sqrt{m}} + O_m(\log T). \tag{15.3}$$

The pure-phase form (15.3) therefore carries an RH hypothesis; the unconditional row formula is (15.2).

15.1. **The diagonal is ordinal, not arbitrary.** Neither the column identity nor Landau's row formula evaluates the diagonal, but the pairing $k \leftrightarrow \gamma_k$ is not arbitrary: k is the counting index of the zero. If $N_\zeta(T)$ denotes the positive-ordinate zero-counting function, then $N_\zeta(\gamma_k^-) < k \leq N_\zeta(\gamma_k)$ for the right-continuous count. Equality $k = N_\zeta(\gamma_k)$ holds for the last occurrence at that ordinate; the difference for other occurrences is bounded by the total number of zero occurrences at that height. The Riemann–von Mangoldt formula [58] gives

$$N_\zeta(T) = \mathcal{N}_{\mathrm{RVM,sm}}(T) + O(\log T), \qquad \mathcal{N}_{\mathrm{RVM,sm}}(T) := \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}. \tag{15.4}$$

Consequently $\gamma_k \sim 2\pi k/\log k$. The inverse of the smooth main term is explicit through the principal Lambert W-function (see DLMF §4.13): for real $x > 7/8$ set

$$\bar{\gamma}^{\mathrm{RVM}}(x) := \frac{2\pi(x-7/8)}{W_0((x-7/8)/e)}, \qquad \bar{\gamma}_k^{\mathrm{RVM}} := \bar{\gamma}^{\mathrm{RVM}}(k), \tag{15.5}$$

so that $\mathcal{N}_{\mathrm{RVM,sm}}(\bar{\gamma}_k^{\mathrm{RVM}}) = k$. Thus the diagonal mixes a strong deterministic ordinal component with the fluctuations of the actual zero ordinates around this smooth clock.

With a weight exponent $\alpha_{\mathrm{ph}} \geq 0$, chosen independently of the real parts β_k, define

$$\mathscr{D}_{\alpha_{\mathrm{ph}}}(N) := \sum_{k=1}^{N} k^{-\alpha_{\mathrm{ph}}} \mathscr{P}_{k,k} = \sum_{k=1}^{N} k^{-\alpha_{\mathrm{ph}}} e^{i\gamma_k \log k}. \tag{15.6}$$

No analogue of the column or row explicit formula is asserted here for (15.6). In particular, $\alpha_{\mathrm{ph}} = 1/2$ is an experimental normalization; it is not, by itself, a test of RH or of $\Theta = 1/2$.

The smooth clock permits an exact separation of the two ingredients. Define, locally in this section,

$$\chi_k^{\mathrm{RVM,ph}} := e^{i(\gamma_k - \bar{\gamma}_k^{\mathrm{RVM}})\log k}. \tag{15.7}$$

Then

$$e^{i\gamma_k \log k} = e^{i\bar{\gamma}_k^{\mathrm{RVM}} \log k} \chi_k^{\mathrm{RVM,ph}}. \tag{15.8}$$

The factor $e^{i\bar{\gamma}_k^{\mathrm{RVM}} \log k}$ is forced by the smooth inverse zero-counting law; $\chi_k^{\mathrm{RVM,ph}}$ carries the displacement of the actual ordinate from that clock.

Two different averages of the diagonal path must also be distinguished:

$$C^{\mathrm{pts}}_{\alpha_{\mathrm{ph}}}(N) := \frac{\mathscr{D}_{\alpha_{\mathrm{ph}}}(N)}{N}, \qquad C^{\mathrm{path}}_{\alpha_{\mathrm{ph}}}(N) := \frac{1}{N}\sum_{n=1}^{N} \mathscr{D}_{\alpha_{\mathrm{ph}}}(n) = \sum_{k=1}^{N} \frac{N-k+1}{N} k^{-\alpha_{\mathrm{ph}}} e^{i\gamma_k \log k}. \tag{15.9}$$

The first averages the points (equivalently the terminal sum divided by N); the second averages all partial sums of the path. At $\alpha_{\mathrm{ph}} = 1/2$, the supplied computation at $N = 2\,001\,052$ gives

$$C^{\mathrm{pts}}_{1/2}(N) \approx -8.626 \cdot 10^{-7} + 1.291 \cdot 10^{-6} i, \qquad C^{\mathrm{path}}_{1/2}(N) \approx -2.461 + 2.493 i. \tag{S15.2}$$

The small point average is forced by normalization:

$$|C^{\mathrm{pts}}_{1/2}(N)| \leq \frac{1}{N}\sum_{k \leq N} k^{-1/2} = O(N^{-1/2}), \tag{S15.3}$$

independently of the zeta ordinates. The displayed value of $C^{\mathrm{path}}_{1/2}(N)$ is a finite-N observation, not a proved limiting value; its regeneration file and checksum are recorded in the v52 computational audit package.

15.2. **The smooth clock and the obstruction for the actual ordinates.** The smooth clock itself has enough phase drift to make the weighted diagonal series converge. This fact is useful mainly because it isolates what remains to be controlled for the actual ordinates.

Lemma 15.1 (Derivative of the smooth diagonal phase)**.** *Let $\bar{\gamma}^{\mathrm{RVM}}(x)$ be the inverse of the smooth Riemann–von Mangoldt main term and put*

$$\mathscr{H}_{\mathrm{RVM,ph}}(x) := \frac{\bar{\gamma}^{\mathrm{RVM}}(x)\log x}{2\pi}. \tag{S15.4}$$

Then, as $x \to \infty$,

$$\mathscr{H}'_{\mathrm{RVM,ph}}(x) = 1 + \frac{\log\log x + 1}{\log x} + O\left(\frac{(\log\log x)^2}{(\log x)^2}\right), \tag{15.10}$$

and

$$\frac{d}{dx}\big(\mathscr{H}'_{\mathrm{RVM,ph}}(x)-1\big)=-\frac{\log\log x+O(1)}{x(\log x)^2}<0 \tag{15.11}$$

for all sufficiently large x.

Proof. Writing $y=(x-7/8)/e$ and using the principal Lambert function in the exact inverse formula for $\bar{\gamma}^{\mathrm{RVM}}(x)$, the standard expansion $W_0(y)=\log y-\log\log y+O(\log\log y/\log y)$ gives

$$\bar{\gamma}^{\mathrm{RVM}}(x)=\frac{2\pi x}{\log x}\left(1+\frac{\log\log x+1+O(1/\log\log x)}{\log x}\right). \tag{S15.5}$$

To justify differentiation, use the exact identities, with $y=x-7/8$ and $w=W_0(y/e)$,

$$w'=\frac{w}{y(1+w)},\qquad \mathscr{H}'_{\mathrm{RVM,ph}}(x)=\frac{\log x}{1+w}+\frac{y}{xw}. \tag{S15.6}$$

Differentiating this exact formula and then expanding w gives both displays, with derivative error $O((\log\log x)^2/(x(\log x)^3))$. No differentiation of a bare O-term is being assumed. In particular $\mathscr{H}'_{\mathrm{RVM,ph}}(x)>1$ and decreases to 1 from above eventually. □

Proposition 15.2 (Convergence for the smooth Riemann–von Mangoldt clock)**.** *After an arbitrary choice of finitely many initial terms, the series*

$$\mathscr{D}_{\mathrm{RVM,sm}}:=\sum_{k\geq 1}k^{-1/2}e^{i\bar{\gamma}_k^{\mathrm{RVM}}\log k} \tag{15.12}$$

converges. Consequently its path averages

$$C^{\mathrm{path}}_{\mathrm{RVM,sm}}(N):=\sum_{k=1}^{N}\frac{N-k+1}{N}k^{-1/2}e^{i\bar{\gamma}_k^{\mathrm{RVM}}\log k} \tag{15.13}$$

converge to the same limit $\mathscr{D}_{\mathrm{RVM,sm}}$.

Proof. Put

$$\mathscr{H}_{\mathrm{RVM,ph}}(x):=\frac{\bar{\gamma}^{\mathrm{RVM}}(x)\log x}{2\pi},\qquad \Psi_{\mathrm{RVM,ph}}(x):=2\pi\big(\mathscr{H}_{\mathrm{RVM,ph}}(x)-x\big). \tag{S15.7}$$

Since $e^{2\pi ik}=1$, one has $e^{i\bar{\gamma}_k^{\mathrm{RVM}}\log k}=e^{i\Psi_{\mathrm{RVM,ph}}(k)}$. By Lemma 15.1,

$$\mathscr{H}'_{\mathrm{RVM,ph}}(x)-1=\frac{\log\log x+1}{\log x}+O\left(\frac{(\log\log x)^2}{(\log x)^2}\right)>0 \tag{15.14}$$

for large x, and this difference is eventually decreasing. Hence on every dyadic block $X\leq k\leq 2X$ the increments $\Psi_{\mathrm{RVM,ph}}(k+1)-\Psi_{\mathrm{RVM,ph}}(k)$ are eventually positive, monotone, and comparable to $\log\log X/\log X$. The elementary first-difference estimate for exponential sums therefore gives, uniformly for $X\leq Y\leq 2X$,

$$\left|\sum_{X<k\leq Y}e^{i\Psi_{\mathrm{RVM,ph}}(k)}\right|\ll\frac{\log X}{\log\log X}. \tag{S15.8}$$

For completeness, this estimate follows by writing $e^{i\Psi_{\mathrm{RVM,ph}}(k)}=(e^{i\Delta_k}-1)^{-1}(e^{i\Psi_{\mathrm{RVM,ph}}(k+1)}-e^{i\Psi_{\mathrm{RVM,ph}}(k)})$, where $\Delta_k=\Psi_{\mathrm{RVM,ph}}(k+1)-\Psi_{\mathrm{RVM,ph}}(k)$, and applying summation by parts; monotonicity of Δ_k makes the total variation $O(1/\min|\Delta_k|)$. Partial summation now yields

$$\sum_{X<k\leq 2X}k^{-1/2}e^{i\Psi_{\mathrm{RVM,ph}}(k)}\ll\frac{\log X}{\sqrt{X}\,\log\log X}. \tag{S15.9}$$

The right side is summable over dyadic X, proving convergence of (15.12). The assertion for (15.13) is then the Cesàro convergence of the partial sums of a convergent series. □

The next finite construction shows why neither the zero-counting law nor pointwise proximity to its smooth inverse can be substituted for control of the phase displacement.

Proposition 15.3 (A near-clock sequence with coherent diagonal phases)**.** *For sufficiently large integers k, put*

$$m_k^{\mathrm{RVM,rnd}} := \mathrm{round}\big(\mathscr{H}_{\mathrm{RVM,ph}}(k)\big), \qquad \widetilde{\gamma}_k^{\mathrm{RVM,rnd}} := \frac{2\pi m_k^{\mathrm{RVM,rnd}}}{\log k}. \tag{15.15}$$

After altering finitely many initial terms if necessary, the sequence $(\widetilde{\gamma}_k^{\mathrm{RVM,rnd}})$ is strictly increasing and satisfies

$$|\widetilde{\gamma}_k^{\mathrm{RVM,rnd}} - \bar{\gamma}_k^{\mathrm{RVM}}| \le \frac{\pi}{\log k}, \qquad N_{\mathrm{RVM,rnd}}(T) - \mathcal{N}_{\mathrm{RVM,sm}}(T) = O(1), \tag{15.16}$$

where $N_{\mathrm{RVM,rnd}}$ is its counting function. Nevertheless

$$e^{i\widetilde{\gamma}_k^{\mathrm{RVM,rnd}}\log k} = 1, \tag{15.17}$$

and therefore

$$\sum_{k\le N} k^{-1/2} e^{i\widetilde{\gamma}_k^{\mathrm{RVM,rnd}}\log k} \sim 2\sqrt{N}, \qquad \sum_{k\le N} \frac{N-k+1}{N} k^{-1/2} e^{i\widetilde{\gamma}_k^{\mathrm{RVM,rnd}}\log k} \sim \frac{4}{3}\sqrt{N}. \tag{15.18}$$

Proof. The first inequality in (15.16) is immediate from $|m_k^{\mathrm{RVM,rnd}} - \mathscr{H}_{\mathrm{RVM,ph}}(k)| \le 1/2$. By (15.14), $\mathscr{H}_{\mathrm{RVM,ph}}(k+1) - \mathscr{H}_{\mathrm{RVM,ph}}(k) > 1$ eventually, so $m_{k+1}^{\mathrm{rnd}} \ge m_k^{\mathrm{RVM,rnd}} + 1$. Since $m_k^{\mathrm{RVM,rnd}} \asymp k$, one has $m_k^{\mathrm{RVM,rnd}} \log(1+1/k) = O(1) < \log k$ eventually. Together with $m_{k+1}^{\mathrm{rnd}} - m_k^{\mathrm{RVM,rnd}} \ge 1$, this proves $m_{k+1}^{\mathrm{rnd}}/\log(k+1) > m_k^{\mathrm{RVM,rnd}}/\log k$ for all sufficiently large k, hence $\widetilde{\gamma}_{k+1} > \widetilde{\gamma}_k^{\mathrm{RVM,rnd}}$. Furthermore

$$\bar{\gamma}_{k+1}^{\mathrm{RVM}} - \bar{\gamma}_k^{\mathrm{RVM}} = \frac{2\pi}{\log k}(1 + o(1)). \tag{S15.10}$$

Hence, for all sufficiently large k, the first inequality in (15.16) implies $\bar{\gamma}_{k-1}^{\mathrm{RVM}} < \widetilde{\gamma}_k^{\mathrm{RVM,rnd}} < \bar{\gamma}_{k+1}^{\mathrm{RVM}}$. Thus the counting function of the rounded sequence differs by at most a bounded number of indices from the discrete smooth-clock count; since $\mathcal{N}_{\mathrm{RVM,sm}}(\bar{\gamma}_k^{\mathrm{RVM}}) = k$ and $\mathcal{N}_{\mathrm{RVM,sm}}$ is increasing, this gives $N_{\mathrm{RVM,rnd}}(T) - \mathcal{N}_{\mathrm{RVM,sm}}(T) = O(1)$. Equation (15.17) follows directly from the definition, and (15.18) follows from $\sum_{k\le N} k^{-1/2} \sim 2\sqrt{N}$ and $N^{-1}\sum_{k\le N}(N-k+1)k^{-1/2} \sim \frac{4}{3}\sqrt{N}$. □

Thus even an increasing artificial sequence with the same smooth counting law up to $O(1)$ and with individual displacement only $O(1/\log k)$ can make the diagonal completely coherent. This is a deterministic obstruction, not a statement about the zeros of ζ.

For the actual ordinates, define the exact remainder

$$\mathscr{R}_{\mathrm{RVM,ph}}(N) := \sum_{k=1}^{N} \frac{N-k+1}{N} k^{-1/2} e^{i\bar{\gamma}_k^{\mathrm{RVM}}\log k}(\chi_k^{\mathrm{RVM,ph}} - 1). \tag{15.19}$$

Then, identically,

$$C_{1/2}^{\mathrm{path}}(N) = C_{\mathrm{RVM,sm}}^{\mathrm{path}}(N) + \mathscr{R}_{\mathrm{RVM,ph}}(N). \tag{15.20}$$

Since Proposition 15.2 proves convergence of the smooth term, the path average for the genuine ordinates converges if and only if $\mathscr{R}_{\mathrm{RVM,ph}}(N)$ has a finite limit. No such estimate is proved here. In particular, there is no basis here for asserting that the limit of (15.19) is zero, or that the observed finite value $-2.461 + 2.493i$ of $C_{1/2}^{\mathrm{path}}$ is its limiting value. The unresolved problem is therefore the collective phase control of $\chi_k^{\mathrm{RVM,ph}}$, not the smooth zero-counting law by itself.

15.3. **A scale benchmark and what it does not test.** For scale comparison only, replace the phases for $k \geq 2$ by independent uniform unit-circle variables, retaining the deterministic first term. The exact second moments are

$$\mathbb{E}|\mathscr{D}_{1/2}(N)|^2 = \sum_{k\leq N} \frac{1}{k}, \qquad \mathbb{E}|C_{1/2}^{\mathrm{path}}(N)|^2 = \sum_{k\leq N} \frac{(N-k+1)^2}{N^2 k}. \tag{15.21}$$

At the stated N, the corresponding root-mean-square scales are approximately 3.884 and 3.686. The observed $|\mathscr{D}_{1/2}(N)| \approx 3.11$ and $|C_{1/2}^{\mathrm{path}}(N)| \approx 3.50$ are therefore on this scale.

This independent-phase model is only a scale benchmark. It deliberately destroys the intrinsic ordinal relation between k and the zero-counting rank at γ_k and therefore cannot be used to conclude that the actual diagonal contains no arithmetic structure. Likewise, unrestricted permutations of the actual ordinates preserve their multiset but destroy the same ordinal relation; they answer a more limited question about the importance of that pairing. The smooth-clock decomposition (15.8) is the more intrinsic comparison: it preserves the zero-counting law and isolates the fluctuations that remain to be controlled.

The distinction can be summarized as follows. A fixed column is a Dirichlet partial sum whose renormalized constant term detects a zeta zero. A fixed row is a Landau zero-sum that detects prime powers (and whose pure-phase form uses RH). The diagonal is not arbitrary, because its two coordinates are joined by the zero-counting index, but no analogous arithmetic evaluation is proved here. The smooth diagonal converges; an artificial sequence within $O(1/\log k)$ of that clock can nevertheless be made fully coherent; and the convergence of the path average for the genuine ordinates is reduced exactly to the remainder (15.19).

16. Numerical data

The following census covers all $n \leq N$, where $N = 455\,052\,511$; it contains 87 385 excursions and records $(u_\nu, \tau_\nu, v_\nu, H_\nu, \sigma_\nu)$ and the curves $\mathcal{A}_+, \mathcal{A}_-, \mathcal{A}$:

$$\mathcal{A}_+(N) \approx 6.3449 \times 10^9, \quad \mathcal{A}_-(N) \approx -7.0877 \times 10^9, \quad \mathcal{A}(N) \approx -7.4279 \times 10^8, \tag{16.1}$$

$$\max_\nu \frac{H_\nu}{\sqrt{\ell_\nu}} \approx 2.0743, \qquad \max_\nu \frac{\mathfrak{b}_\nu}{\ell_\nu^{3/2}} \approx 1.0214. \tag{16.2}$$

Since $|M(x)| \leq 0.571\sqrt{x}$ for $33 \leq x \leq 10^{16}$ [20], and $|M(n)| \leq \sqrt{n}$ for $n < 33$ with equality only at $n = 1$, the constant $C_0 = 1$ is admissible in Lemma 7.1, which gives $\mathcal{A}_+(N) + |\mathcal{A}_-(N)| \leq \frac{4}{5}N^{5/4} \approx 5.317 \times 10^{10}$ against the observed $\approx 1.3433 \times 10^{10}$, a ratio of approximately 3.96. Heuristic 8.5 with $L = \sqrt{N}$ predicts $\#\{\ell_\nu > \sqrt{N}\} \asymp N^{1/4} \approx 146$. At this range $N^{0.05} \approx 2.7$, so factors x^ε are not separable from constants. These numerical values are computational observations and are not used as premises of any theorem. The v52 audit package contains `COMPUTATIONAL_PROVENANCE_v52.txt` and `MANIFEST_SHA256.txt`, which identify the available regeneration scripts and figures by SHA-256 checksum. Reproducing the full census also requires the original Möbius data file; where a raw input file is not present in the audit package, the corresponding number remains explicitly non-reproducible from the manuscript alone. The conventions are those of Proposition 4.2 and Definition 3.6.

17. The RVM arithmetic system and its relation to the classical one

This section collects, in one namespace, the deterministic arithmetic system obtained from the midpoint Riemann–von Mangoldt clock and distinguishes it from the genuine zeta/Mertens system. There are two RVM Mertens objects and they must not be conflated. The *critical RVM baseline* $M_{\mathrm{RVM}}^{\mathrm{crit}}$ is built only from the synthetic ordinates γ_k^{RVM} and has its nontrivial spectral divisor on $\Re s = 1/2$ by construction. The *ordinate-transported object* $M_{\mathrm{RVM}}^{\mathrm{ord}}$ keeps the genuine real parts

β_k and replaces only the genuine ordinates γ_k by γ_k^{RVM}. Its pole abscissa is therefore the genuine quantity Θ.

The same separation is used for the associated canonical products: $P_{\mathrm{RVM}}^{\mathrm{crit}}$ is the purely synthetic critical product, whereas $P_{\mathrm{RVM}}^{\mathrm{ord}}$ is the true-β rigid-ordinate product. The first system has its own ζ_{RVM}, prime-power counting function J_{RVM}, prime counting function π_{RVM} and integer RVM-prime realizations. The second system is an analytic transport of the true zeta divisor. The section first develops the pure RVM system, then states the exact defects relative to M, ζ, the ordinary primes and π, and only afterwards returns to the true-β ordinate transport already used in the previous version.

17.1. **The midpoint Riemann–von Mangoldt clock.** Let the nontrivial zeros in the upper half-plane, counted with multiplicity, be

$$\rho_k = \beta_k + i\gamma_k, \qquad 0 < \beta_k < 1, \qquad 0 < \gamma_1 \le \gamma_2 \le \cdots . \tag{17.1}$$

$\beta_k \in (0,1) \subset \mathbb{R}$, $\gamma_k \in (0,\infty)$ and $k \in \mathbb{N}$. No critical-line assumption is made. When two zeros belong to the same functional pair at a common positive ordinate, we fix the otherwise immaterial tie by placing $1/2 + d$ immediately before $1/2 - d$, with $d > 0$. Occurrences on the critical line retain their multiplicities. This convention does not change P_ζ; it only fixes which consecutive rigid ordinates receive the two members of an off-line pair.

Put

$$\mathcal{N}_{\mathrm{RVM}}(T) := \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}, \qquad \mathcal{N}_{\mathrm{RVM}}(\gamma_k^{\mathrm{RVM}}) = k - \frac{1}{2} \quad (k \in \mathbb{N},\ \gamma_k^{\mathrm{RVM}} > 2\pi). \tag{17.2}$$

Solving with the principal branch of the Lambert function gives

$$\boxed{\gamma_k^{\mathrm{RVM}} = \frac{2\pi(k - 11/8)}{W_0((k-11/8)/e)}.} \tag{17.3}$$

The quotient in (17.3) is positive also for $k = 1$. This is the midpoint convention used by the file `zLamb.bin`. It differs from the clock $\bar\gamma_k^{\mathrm{RVM}}$ of Section 15, where $\mathcal{N}_{\mathrm{RVM}}(\bar\gamma_k^{\mathrm{RVM}}) = k$.

Lemma 17.1 (Bounded vertical displacement)**.** *One has*

$$\gamma_k - \gamma_k^{\mathrm{RVM}} = O(1). \tag{17.4}$$

The assertion is unconditional and includes multiplicities.

Proof. The Riemann–von Mangoldt formula gives

$$N_\zeta(T) = \mathcal{N}_{\mathrm{RVM}}(T) + O(\log(2+T)). \tag{17.5}$$

At an ordinate γ_k, either endpoint convention for N_ζ gives $N_\zeta(\gamma_k) = k + O(m_k)$, where m_k is the multiplicity of the zero at that ordinate. The same counting formula gives $m_k = O(\log(2+\gamma_k))$. Consequently

$$\mathcal{N}_{\mathrm{RVM}}(\gamma_k) = k + O(\log(2+\gamma_k)). \tag{17.6}$$

On the other hand,

$$\mathcal{N}'_{\mathrm{RVM}}(T) = \frac{1}{2\pi}\log\frac{T}{2\pi} \asymp \log T \tag{17.7}$$

for large T. The mean-value theorem applied between γ_k and γ_k^{RVM} therefore proves (17.4); finitely many initial ordinates are absorbed into the constant. □

17.2. **The critical RVM zeta function.** The midpoint clock itself defines a complete synthetic zeta system without using the genuine real parts β_k. Put

$$\rho_k^{\mathrm{RVM}} := \frac{1}{2} + i\gamma_k^{\mathrm{RVM}}, \qquad \overline{\rho_k^{\mathrm{RVM}}} = \frac{1}{2} - i\gamma_k^{\mathrm{RVM}}. \tag{17.8}$$

The real part $1/2$ in (17.8) is an input of the synthetic model; it is not a conclusion about the genuine zeta zeros. Define

$$P_{\mathrm{RVM}}^{\mathrm{crit}}(s) := \prod_{k\geq 1}\left(1 + \frac{s(s-1)}{1/4 + (\gamma_k^{\mathrm{RVM}})^2}\right), \tag{17.9}$$

and

$$\xi_{\mathrm{RVM}}(s) := \frac{1}{2}P_{\mathrm{RVM}}^{\mathrm{crit}}(s), \qquad \zeta_{\mathrm{RVM}}(s) := \frac{P_{\mathrm{RVM}}^{\mathrm{crit}}(s)}{G(s)}, \qquad G(s) := s(s-1)\Gamma(s/2)\pi^{-s/2}. \tag{17.10}$$

Theorem 17.2 (Canonical characterization of ζ_{RVM})**.** *The product* (17.9) *converges absolutely and locally uniformly. The function $P_{\mathrm{RVM}}^{\mathrm{crit}}$ is entire,*

$$P_{\mathrm{RVM}}^{\mathrm{crit}}(s) = P_{\mathrm{RVM}}^{\mathrm{crit}}(1-s), \tag{17.11}$$

and its zeros, with multiplicity, are exactly ρ_k^{RVM} and $\overline{\rho_k^{\mathrm{RVM}}}$. Consequently ζ_{RVM} has a simple pole at $s = 1$ of residue 1*, the usual trivial zeros at the negative even integers, and nontrivial zeros exactly at*

$$s = \frac{1}{2} \pm i\gamma_k^{\mathrm{RVM}}. \tag{17.12}$$

Moreover $\xi_{\mathrm{RVM}}(s) = \xi_{\mathrm{RVM}}(1-s)$.

Proof. The clock law gives $\gamma_k^{\mathrm{RVM}} \asymp k/\log k$, hence $\sum_k (\gamma_k^{\mathrm{RVM}})^{-2} < \infty$. Each paired factor in (17.9) differs from 1 by $O_K((\gamma_k^{\mathrm{RVM}})^{-2})$ on a compact set K, proving normal convergence. The factor depends only on $s(s-1)$, which proves (17.11). The zero statement follows factor by factor. Finally $P_{\mathrm{RVM}}^{\mathrm{crit}}(1) = 1$ and $G(s) = (s-1)(1+O(s-1))$ near $s = 1$, while the poles of $\Gamma(s/2)$ produce the trivial zeros after division by G. □

Remark 17.3 (Synthetic criticality is not classical RH)*.* The zero line in (17.12) is fixed by (17.8). Thus any square-root exponent proved for the pure RVM system is a theorem about this synthetic system; it cannot be used as evidence that the genuine real parts β_k equal $1/2$.

17.3. **The RVM prime-power count and prime count.** For $x > 1$ define the symmetric RVM explicit-formula value

$$J_{\mathrm{RVM}}(x) := \mathrm{li}(x) - 2\Re\sum_{k\geq 1}\mathrm{Ei}\Big(\rho_k^{\mathrm{RVM}}\log x\Big) - \log 2 + \int_x^\infty \frac{dt}{t(t^2-1)\log t}, \tag{17.13}$$

where the zero sum is taken in the symmetric limiting sense supplied by the clock ordering. Put

$$J_{\mathrm{RVM}}^*(x) := \begin{cases} J_{\mathrm{RVM}}(x), & x \geq 2, \\ 0, & 0 < x < 2. \end{cases} \tag{17.14}$$

For comparisons that use only integer samples, also define the distinct right-continuous step extension

$$J_{\mathrm{RVM}}^{*,\mathbb{Z}}(x) := J_{\mathrm{RVM}}^*(\lfloor x\rfloor), \qquad x \in [1,\infty). \tag{S16.1}$$

The symbols J_{RVM}^* and $J_{\mathrm{RVM}}^{*,\mathbb{Z}}$ are not interchangeable: the first retains the continuous variation inside a unit cell, whereas the second moves that entire variation to the next integer. Set $\pi_{\mathrm{RVM}}(x) = 0$ for $0 < x < 2$. For $x \geq 2$, the RVM prime count is the exact finite Möbius inversion

$$\boxed{\pi_{\mathrm{RVM}}(x) := \sum_{1\leq m\leq \log_2 x}\frac{\mu(m)}{m}J_{\mathrm{RVM}}^*(x^{1/m}) \qquad (x \geq 2).} \tag{17.15}$$

Proposition 17.4 (Exact RVM prime-power inversion)**.** *For every $x \geq 2$,*

$$\boxed{J^*_{\mathrm{RVM}}(x) = \sum_{1 \leq m \leq \log_2 x} \frac{1}{m} \pi_{\mathrm{RVM}}(x^{1/m}).} \tag{17.16}$$

Thus J_{RVM} and π_{RVM} determine one another exactly once the support convention (17.14) *is fixed.*

Proof. This is ordinary Möbius inversion in the exponent variable. Both sums are finite because $x^{1/m} < 2$ for $m > \log_2 x$. □

Hypothesis 17.5 (Lambert-clock square-root estimate)**.** The compact-uniform existence of the symmetric Ei sum is proved in Theorem 17.27. The additional analytic hypothesis here is only the quantitative statement: the Lambert-clock exponential sums satisfy the uniform blockwise van der Corput bounds, endpoint estimates and tail bounds needed to give

$$J_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O\left(\frac{\sqrt{x}(\log\log x)^2}{\log x} \right). \tag{17.17}$$

Under this hypothesis, finite Möbius inversion gives

$$\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O\left(\frac{\sqrt{x}(\log\log x)^2}{\log x} \right). \tag{17.18}$$

Remark 17.6 (Analytic proof obligation). The stationary-phase calculation identifies the expected square-root scale, but it does not by itself prove Hypothesis 17.5. A complete proof must control the symmetric meaning and branches of Ei, uniform tails, endpoint terms under integration by parts, and the van der Corput constants uniformly over the Lambert-clock blocks. No later unconditional theorem uses (17.17)–(17.18).

Remark 17.7 (What the exponent means). The factor $\sqrt{x}$ in Hypothesis 17.5 comes from the input $\Re\rho_k^{\mathrm{RVM}} = 1/2$. The hypothesis asks that the deterministic clock should not worsen the inserted critical exponent. It is not a deduction of classical RH.

17.4. **Integer RVM primes: weighted and** $0/1$ **realizations.** Define the integer RVM-prime weight by

$$c_n^{\mathrm{RVM}} := \pi_{\mathrm{RVM}}(n) - \pi_{\mathrm{RVM}}(n-1), \qquad n \geq 2, \tag{17.19}$$

and the corresponding integer-supported Euler object, initially for $\Re s > 1$, by

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) := \prod_{n \geq 2} (1 - n^{-s})^{-c_n^{\mathrm{RVM}}}. \tag{17.20}$$

The associated prime-power count is

$$J_{\mathbb{Z},\mathrm{RVM}}(x) := \sum_{1 \leq m \leq \log_2 x} \frac{1}{m} \pi_{\mathrm{RVM}}(\lfloor x^{1/m} \rfloor). \tag{17.21}$$

For real $x \geq 1$ write

$$\Delta^{\mathrm{cont}}_{\mathrm{RVM}}(x) := J_{\mathbb{Z},\mathrm{RVM}}(x) - J^*_{\mathrm{RVM}}(x), \qquad \Delta^{\mathrm{step}}_{\mathrm{RVM}}(x) := J_{\mathbb{Z},\mathrm{RVM}}(x) - J^{*,\mathbb{Z}}_{\mathrm{RVM}}(x). \tag{S16.2}$$

Both defects are real-valued. A Stieltjes variation hypothesis must name one of them; specifying values only on $\mathbb{N}$ does not by itself define the measure $d\Delta$ on $[1, \infty)$.

Hypothesis 17.8 (Uniform unit-cell control)**.** For x sufficiently large and $0 < h \leq 1$,

$$|J_{\mathrm{RVM}}(x+h) - J_{\mathrm{RVM}}(x)| \ll h + \frac{\log x}{\sqrt{x}}, \tag{17.22}$$

uniformly in the unit interval. Consequently the finite inversion would give

$$\pi_{\mathrm{RVM}}(y) - \pi_{\mathrm{RVM}}(\lfloor y \rfloor) = O(1), \qquad c_n^{\mathrm{RVM}} = O(1). \tag{17.23}$$

Remark 17.9 (Why the unit-cell estimate is separate). The global square-root estimate does not imply (17.22). Proving this stronger statement requires a uniform estimate for increments of the truncated Ei sum, with the high-clock tail and all endpoint terms controlled uniformly for $0 < h \le 1$. It is therefore kept as an explicit hypothesis rather than being inferred from Hypothesis 17.5.

Corollary 17.10 (Conditional integer discretization of the zero divisor)**.** *Assume Hypothesis 17.8 and the Mellin–Stieltjes identification used in the proof below. One has*

$$J_{\mathbb{Z},\mathrm{RVM}}(x) - J^{*}_{\mathrm{RVM}}(x) = O(\log\log x). \tag{17.24}$$

Consequently

$$\boxed{\zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \zeta_{\mathrm{RVM}}(s)e^{h_{\mathrm{RVM}}(s)},} \tag{17.25}$$

where

$$h_{\mathrm{RVM}}(s) := s\int_1^{\infty} (J_{\mathbb{Z},\mathrm{RVM}}(x) - J^{*}_{\mathrm{RVM}}(x))x^{-s-1}\,dx \tag{17.26}$$

is holomorphic for $\Re s > 0$. Hence the weighted integer system and ζ_{RVM} have the same zero divisor in $\Re s > 0$.

Proof. Under the unit-cell hypothesis, each summand in (17.21) changes by $O(1/m)$ under the unit-cell replacement, and $\sum_{m\le\log_2 x} m^{-1} = O(\log\log x)$. The Mellin–Stieltjes logarithm of an Euler product converts the difference of prime-power counts into (17.26); the logarithmic growth in (17.24) makes the integral locally normally convergent for $\Re s > 0$. Exponentiation introduces no zeros. □

The factorization above uses the continuous defect $\Delta^{\mathrm{cont}}_{\mathrm{RVM}}$. If one assumes weighted total variation for $\Delta^{\mathrm{step}}_{\mathrm{RVM}}$ instead, comparison with ζ_{RVM} additionally requires control of the grid correction

$$H^{\mathrm{grid}}_{\mathrm{RVM}}(s) := s\int_1^{\infty} (J^{*,\mathbb{Z}}_{\mathrm{RVM}}(x) - J^{*}_{\mathrm{RVM}}(x))x^{-s-1}\,dx \tag{S16.3}$$

on the stated half-plane, or an equivalent direct Mellin–Stieltjes identification. This is part of hypothesis 3b in the integer-step reading; it is not supplied merely by 3a.

Remark 17.11 (Positivity is a separate theorem target). Equation (17.20) is an analytic weighted Euler object. It is a positive Beurling-type generalized-prime system only when the weights c_n^{RVM} are nonnegative. Boundedness in (17.23) does not imply positivity. This is distinct from existence constructions of Beurling systems satisfying RH-type prime-counting bounds; see, for example, [62].

For the Poisson expansion of the Lambert clock, let $j \ge 1$ denote the Poisson harmonic. The stationary point, phase and leading amplitude are

$$T_j = 2\pi x^{1/j}, \tag{17.27}$$

$$\Phi_j(T_j) = 2\pi j x^{1/j} - \frac{11\pi j}{4}, \tag{17.28}$$

$$\mathcal{A}_j(x) = \frac{\log x}{j^{3/2}} x^{1/(2j)}. \tag{17.29}$$

For $j = 1$ the induced derivative term is exactly

$$2\cos(2\pi x), \tag{17.30}$$

so that on every integer cell

$$\int_{n-1}^{n} 2\cos(2\pi x)\,dx = 0. \tag{17.31}$$

The next stationary term has scale $x^{-1/4}$.

Hypothesis 17.12 (Uniform RVM cell remainder)**.** The Poisson/stationary-phase remainder after removing the first harmonic satisfies

$$c_n^{\mathrm{RVM}} - \int_{n-1}^{n} \left(\frac{1}{\log x} + 2\cos(2\pi x) \right) dx = o\Big(\frac{1}{\log n} \Big) \qquad (n \to \infty). \tag{17.32}$$

This increment formulation includes any jump at an integer endpoint; it does not presuppose differentiability or absolute continuity of π_{RVM}.

Proposition 17.13 (Consequences of the cell remainder)**.** *Hypothesis 17.12 implies that, for all sufficiently large n,*

$$0 < c_n^{\mathrm{RVM}} < 1. \tag{17.33}$$

Proof. Integrating the main density over $[n-1, n]$ gives $1/\log n + O(1/(n \log^2 n))$. The first Poisson harmonic vanishes exactly by (17.31), and the assumed remainder is $o(1/\log n)$. Thus $c_n^{\mathrm{RVM}} \sim 1/\log n$, which proves both inequalities eventually. □

Assuming (17.33), define

$$R_{\mathrm{RVM}}(n) := \left\lfloor \pi_{\mathrm{RVM}}(n) + \frac{1}{2} \right\rfloor, \qquad d_n^{\mathrm{RVM}} := R_{\mathrm{RVM}}(n) - R_{\mathrm{RVM}}(n-1). \tag{17.34}$$

After changing finitely many initial values one obtains the deterministic integer set

$$\mathcal{P}_{\mathrm{RVM}} := \{ n \geq 2 : d_n^{\mathrm{RVM}} = 1 \}. \tag{17.35}$$

Corollary 17.14 (Conditional 0/1 RVM prime realization)**.** *Under Hypotheses 17.12 and 17.8, and the Mellin–Stieltjes identification of Corollary 17.10, the set $\mathcal{P}_{\mathrm{RVM}}$ can be chosen so that*

$$\pi_{\mathcal{P}_{\mathrm{RVM}}}(n) = \pi_{\mathrm{RVM}}(n) + O(1). \tag{17.36}$$

Its Euler product differs from ζ_{RVM} by an exponential of a holomorphic function on $\Re s > 0$ and by finitely many elementary factors. Consequently its nontrivial zero divisor in $\Re s > 0$ is exactly (17.12)*.*

17.5. **The critical RVM Mertens object.** Define the normalized Mellin transform of the critical RVM baseline by

$$\widehat{M}_{\mathrm{RVM}}^{\mathrm{crit}}(s) := \frac{1}{\zeta_{\mathrm{RVM}}(s)}. \tag{17.37}$$

Whenever ordinary inversion is justified put

$$M_{\mathrm{RVM}}^{\mathrm{crit}}(x) := \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \widehat{M}_{\mathrm{RVM}}^{\mathrm{crit}}(s) \frac{x^s}{s}\, ds, \qquad c > 1, \tag{17.38}$$

and otherwise keep it as a formal inverse-Mellin symbol until an admissible function or distribution space and an inversion theorem are specified.

Proposition 17.15 (Spectral characterization of $M_{\mathrm{RVM}}^{\mathrm{crit}}$)**.** *The nontrivial poles of $\widehat{M}_{\mathrm{RVM}}^{\mathrm{crit}}$ are exactly $1/2 \pm i\gamma_k^{\mathrm{RVM}}$. Hence its nontrivial pole abscissa is exactly $1/2$ by construction.*

Proof. This is immediate from Theorem 17.2 and (17.37). □

Proposition 17.16 (Gaussian square-root bound for the pure RVM baseline, conditional on vertical growth)**.** *Fix $c > 1$ and $0 < \varepsilon < c - 1/2$. Assume that, uniformly for $1/2 + \varepsilon \leq \sigma \leq c$,*

$$\frac{1}{\zeta_{\mathrm{RVM}}(\sigma + it)} = \exp\big(o(t^2)\big) \qquad (|t| \to \infty), \tag{17.39}$$

with the same bound on the horizontal segments used in the contour rectangle. For

$$M_{\mathrm{RVM,G}}^{\mathrm{crit}}(x) := \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{s^2} \widehat{M}_{\mathrm{RVM}}^{\mathrm{crit}}(s) \frac{x^s}{s}\, ds, \tag{17.40}$$

one has

$$M^{\mathrm{crit}}_{\mathrm{RVM,G}}(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{17.41}$$

Proof. On $s = \sigma + it$ one has $|e^{s^2}| = e^{\sigma^2 - t^2}$. Hence (17.39) makes both vertical integrals absolutely convergent and forces the two horizontal sides of a height-T rectangle to tend to zero as $T \to \infty$. The strip $1/2 + \varepsilon \le \Re s \le c$ contains no nontrivial pole of $1/\zeta_{\mathrm{RVM}}$; the point $s = 1$ is a zero, not a pole, of that reciprocal, and no trivial pole is crossed in this strip. Cauchy's theorem therefore moves the contour to $\Re s = 1/2 + \varepsilon$. On the shifted line $x^{\Re s} = x^{1/2+\varepsilon}$, while the remaining integral is finite by the same Gaussian domination, proving the stated bound. □

Remark 17.17 (Unsmoothed pointwise control is separate). Proposition 17.16 is a regularized statement. A pointwise square-root bound for the unsmoothed inverse (17.38) requires vertical-growth and Mellin multiplier control; it is not asserted merely from the location of the synthetic poles.

17.6. **Exact symmetric Hadamard normalization of the genuine completed factor.** Put

$$P_\zeta(s) := 2\xi(s) = G(s)\zeta(s). \tag{17.42}$$

For the positive ordinates, counted with multiplicity, define the paired product

$$P^{\mathrm{pair}}_\zeta(s) := \prod_{k\ge 1} \left(1 - \frac{s}{\rho_k}\right)\left(1 - \frac{s}{\overline{\rho}_k}\right). \tag{17.43}$$

The pairing in (17.43) is by conjugation, while the full upper-half-plane multiset also contains the functional mirror $1 - \overline{\rho}_k$ of every ρ_k.

Theorem 17.18 (The paired Hadamard product has no residual exponential gauge)**.** *The product* (17.43) *converges absolutely and locally uniformly and*

$$\boxed{P_\zeta(s) = P^{\mathrm{pair}}_\zeta(s).} \tag{17.44}$$

Thus the possible factor e^{a+bs} in an abstract order-one Hadamard factorization is forced to be identically 1 *by the functional symmetry and the normalization at $s = 0$.*

Proof. For $\rho = \beta + i\gamma$ with $0 < \beta < 1$,

$$\left(1 - \frac{s}{\rho}\right)\left(1 - \frac{s}{\bar\rho}\right) = 1 - \frac{2\beta s}{\beta^2 + \gamma^2} + \frac{s^2}{\beta^2 + \gamma^2} = 1 + O_K(\gamma^{-2}) \tag{S17.1}$$

uniformly on a fixed compact set K. Since $N_\zeta(T) = O(T \log T)$, one has $\sum_{\gamma_k > 0} \gamma_k^{-2} < \infty$, so the product converges absolutely and locally uniformly.

Write

$$f_{\beta,\gamma}(s) := \frac{(s-\beta)^2 + \gamma^2}{\beta^2 + \gamma^2}. \tag{S17.2}$$

If $\beta \ne 1/2$, functional symmetry supplies at the same positive ordinate the mirror $1 - \beta + i\gamma$. A direct calculation gives

$$f_{\beta,\gamma}(1-s) f_{1-\beta,\gamma}(1-s) = f_{\beta,\gamma}(s) f_{1-\beta,\gamma}(s), \tag{S17.3}$$

and for $\beta = 1/2$ the single factor is already invariant. Absolute convergence permits grouping the factors in these functional pairs, hence

$$P^{\mathrm{pair}}_\zeta(1-s) = P^{\mathrm{pair}}_\zeta(s). \tag{S17.4}$$

For $|s| \le R$, the factor bound above gives $\log \max_{|s|\le R} |P^{\mathrm{pair}}_\zeta(s)| \le \sum_k \log(1 + (R^2 + 2R)/\gamma_k^2) = O(R \log R)$, by partial summation using $N_\zeta(T) = O(T \log T)$. Thus P^{pair}_ζ has order at most one. Both P_ζ and P^{pair}_ζ are entire of order at most one and have exactly the same zeros with multiplicity. Hadamard factorization therefore gives $P_\zeta = e^{a+bs} P^{\mathrm{pair}}_\zeta$ for constants a, b. At $s = 0$ both functions

equal 1, so $e^a = 1$. Applying $s \mapsto 1-s$ and using the two functional symmetries gives $e^{bs} = e^{b(1-s)}$ for every s, hence $b = 0$. Therefore the ratio is the constant 1. □

Remark 17.19 (Why this matters for the transport identities). The exponential ambiguity is not discarded by convention: it is eliminated by Theorem 17.18. Consequently every product formula below for $\mathscr{Q}_{\mathrm{RVM}}$ and $\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}$ is literally exact; no hidden factor e^{bs} remains in a Mellin multiplier or logarithmic derivative.

17.7. **Exact defects relative to M, ζ, the ordinary primes and π.** Let

$$\boxed{\mathscr{Q}_{\mathrm{RVM}}(s) := \frac{P_\zeta(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)} = \frac{\zeta(s)}{\zeta_{\mathrm{RVM}}(s)}.} \tag{17.45}$$

By Theorem 17.18 it has the exact convergent paired product representation

$$\mathscr{Q}_{\mathrm{RVM}}(s) = \prod_{k\ge 1} \frac{(1-s/\rho_k)(1-s/\overline{\rho}_k)}{1+s(s-1)/(1/4+(\gamma_k^{\mathrm{RVM}})^2)}. \tag{17.46}$$

Proposition 17.20 (Exact transform relation between the true and critical RVM Mertens objects)**.** *For $\Re s > 1$ and by meromorphic continuation wherever both sides are defined,*

$$\boxed{\widehat{M}^{\mathrm{crit}}_{\mathrm{RVM}}(s) = \mathscr{Q}_{\mathrm{RVM}}(s)\widehat{M}(s), \qquad \widehat{M}(s) = \frac{1}{\zeta(s)}.} \tag{17.47}$$

If the inverse-Mellin distributions of $\mathscr{Q}_{\mathrm{RVM}}$ and of $\widehat{M}^{\mathrm{crit}}_{\mathrm{RVM}}/s$ exist in a common space, and convolution with M is defined there, write the multiplier kernel as $\kappa^{\mathrm{crit}}_{\mathrm{RVM}}$. Then

$$M^{\mathrm{crit}}_{\mathrm{RVM}} = \kappa^{\mathrm{crit}}_{\mathrm{RVM}} *_{\mathrm{Mel}} M \tag{17.48}$$

as Mellin distributions. Moreover

$$\widehat{(M - M^{\mathrm{crit}}_{\mathrm{RVM}})}(s) = \frac{1-\mathscr{Q}_{\mathrm{RVM}}(s)}{\zeta(s)}. \tag{17.49}$$

Thus a genuine zero not cancelled by a coincidence with the RVM divisor survives as a pole of the difference transform; ordinate closeness alone does not imply a smaller pointwise exponent for $M - M^{\mathrm{crit}}_{\mathrm{RVM}}$.

Define the true-prime/RVM-prime defect at the integers by

$$\varepsilon_n^{\mathrm{RVM}} := \mathbf{1}_{\{n \text{ prime}\}} - c_n^{\mathrm{RVM}}, \quad (n \ge 2), \qquad E_{\mathrm{RVM}}(N) := \sum_{2\le n\le N} \varepsilon_n^{\mathrm{RVM}}. \tag{17.50}$$

Then exactly

$$\boxed{E_{\mathrm{RVM}}(N) = \pi(N) - \pi_{\mathrm{RVM}}(N).} \tag{17.51}$$

For $\Re s > 1$,

$$\frac{\zeta(s)}{\zeta_{\mathbb{Z},\mathrm{RVM}}(s)} = \prod_{n\ge 2} (1-n^{-s})^{-\varepsilon_n^{\mathrm{RVM}}}, \tag{17.52}$$

and with (17.25),

$$\boxed{\frac{\zeta(s)}{\zeta_{\mathrm{RVM}}(s)} = e^{h_{\mathrm{RVM}}(s)} \prod_{n\ge 2} (1-n^{-s})^{-\varepsilon_n^{\mathrm{RVM}}}} \qquad (\Re s > 1). \tag{17.53}$$

Thus the same discrepancy is visible either in the zero divisor through $\mathscr{Q}_{\mathrm{RVM}}$ or on the integer-prime side through $\varepsilon_n^{\mathrm{RVM}}$.

Let

$$N_{\mathrm{RVM}}(T) := \#\{k : \gamma_k^{\mathrm{RVM}} \le T\} = \left\lfloor \mathcal{N}_{\mathrm{RVM}}(T) + \frac{1}{2} \right\rfloor \quad (T \ge 2\pi), \tag{17.54}$$

Writing the classical Riemann–von Mangoldt formula in the form $N_\zeta(T) = \mathcal{N}_{\mathrm{RVM}}(T) + S(T) + O(T^{-1})$ away from ordinates and putting

$$\Psi_{\mathrm{RVM}}(T) := \mathcal{N}_{\mathrm{RVM}}(T) - \left\lfloor \mathcal{N}_{\mathrm{RVM}}(T) + \frac{1}{2} \right\rfloor, \tag{17.55}$$

one obtains

$$\boxed{N_\zeta(T) - N_{\mathrm{RVM}}(T) = S(T) + \Psi_{\mathrm{RVM}}(T) + O(T^{-1}).} \tag{17.56}$$

The bounded sawtooth Ψ_{RVM} is the clock quantization error; the arithmetical zero-count fluctuation is carried by $S(T)$.

Remark 17.21 (First-order ordinate displacement). Linearizing the two midpoint equations gives, at first order,

$$\gamma_k - \gamma_k^{\mathrm{RVM}} \approx -\frac{2\pi S(\gamma_k)}{\log(\gamma_k^{\mathrm{RVM}}/2\pi)}. \tag{17.57}$$

This formula is diagnostic, not an exact replacement for the zero-counting identity (17.56).

Remark 17.22 (The classical boundary). Because ζ_{RVM} has no zeros in $\Re s > 1/2$, asking that the true quotient $\zeta/\zeta_{\mathrm{RVM}}$ be zero-free there is equivalent to asking that ζ itself be zero-free there. With the usual vertical-growth hypotheses this is the classical von Koch/RH boundary. The RVM construction does not make that defect estimate easier by itself; its value is that the synthetic baseline and the arithmetic defect are separated exactly.

17.8. **A Hamburger endpoint criterion.** Let the positive genuine ordinates γ_k be counted with multiplicity and define the critical-line projection

$$P_{\mathrm{line}}(s) := \prod_{k\geq 1} \left(1 + \frac{s(s-1)}{1/4 + \gamma_k^2}\right), \qquad \zeta_{\mathrm{line}}(s) := \frac{P_{\mathrm{line}}(s)}{G(s)}. \tag{17.58}$$

The product converges locally normally, satisfies $P_{\mathrm{line}}(s) = P_{\mathrm{line}}(1-s)$, is normalized by $P_{\mathrm{line}}(0) = P_{\mathrm{line}}(1) = 1$, and has all nontrivial zeros on $\Re s = 1/2$ with exactly the genuine ordinates. Hence ζ_{line} has the same gamma factor, functional equation and residue 1 at $s = 1$ as ζ.

Hamburger's converse theorem [2] says, in the form used here, that if a Dirichlet series $F(s) = \sum a_n n^{-s}$ is absolutely convergent for $\Re s > 1$, $(s-1)^m F(s)$ is entire of finite order for some m, and F satisfies the zeta functional equation, then $F = c\zeta$; the residue normalization fixes $c = 1$. Therefore the single endpoint condition

$$\boxed{\zeta_{\mathrm{line}}(s) = \sum_{n\geq 1} a_n n^{-s} \quad \text{absolutely for } \Re s > 1} \tag{D}$$

is equivalent to RH. This is a Category 3 reformulation, not a proof: the genuine ordinates have already been inserted into ζ_{line}.

Under the Hamburger hypotheses, condition (D) forces the logarithmic prime-power measure of the endpoint to be the classical atomic measure $\sum_{p,m} m^{-1}\delta_{p^m}$. In contrast, the formal RVM explicit-formula picture starts from a continuous clock density. The unresolved transition is therefore continuous-to-atomic. The arithmetic information enters through the genuine zero-count fluctuation $S(T)$. Selberg's prime approximation to $S(T)$ is a mean statement, not pointwise control at every γ_k; proving (D) by this route would require enough pointwise control at those ordinates to produce only prime-power frequencies and to exclude cross-frequencies such as $\log p_1 \pm \log p_2$. No such estimate is proved here.

17.9. **A critical-line ordinate homotopy from the RVM clock to the genuine ordinates.** This subsection expands the critical-line path that was only summarized in the preceding version. It moves the ordinates and *never* the real part of a zero. To remain consistent with the notation of this paper, the τ_k of the separate working note are written here as γ_k^{RVM}. Put

$$\delta_k^{\mathrm{RVM}\to\mathrm{line}} := \gamma_k - \gamma_k^{\mathrm{RVM}}, \qquad \gamma_{k,\kappa}^{\mathrm{RVM,path}} := \gamma_k^{\mathrm{RVM}} + \kappa\delta_k^{\mathrm{RVM}\to\mathrm{line}}, \qquad 0 \le \kappa \le 1. \tag{17.59}$$

Lemma 17.1 gives a constant D with $|\delta_k^{\mathrm{RVM}\to\mathrm{line}}| \le D$ for every k after enlarging D to absorb finitely many initial ordinates.

Theorem 17.23 (Critical-line path and transported analytic class)**.** *For $0 \le \kappa \le 1$ define*

$$P_{\mathrm{RVM},\kappa}^{\mathrm{path}}(s) := \prod_{k\ge1}\left(1 + \frac{s(s-1)}{1/4 + (\gamma_{k,\kappa}^{\mathrm{RVM,path}})^2}\right), \qquad \zeta_{\mathrm{RVM},\kappa}^{\mathrm{path}}(s) := \frac{P_{\mathrm{RVM},\kappa}^{\mathrm{path}}(s)}{G(s)}. \tag{17.60}$$

Then the product converges absolutely and locally uniformly jointly in (s,κ). For every κ, $P_{\mathrm{RVM},\kappa}^{\mathrm{path}}$ is entire of order one, $P_{\mathrm{RVM},\kappa}^{\mathrm{path}}(s) = P_{\mathrm{RVM},\kappa}^{\mathrm{path}}(1-s)$, and $\zeta_{\mathrm{RVM},\kappa}^{\mathrm{path}}$ has residue 1 at $s=1$, the usual trivial zeros, and nontrivial zeros exactly

$$\frac{1}{2} \pm i\gamma_{k,\kappa}^{\mathrm{RVM,path}}. \tag{17.61}$$

For $0 \le \kappa < 1$ these nontrivial zeros are simple; at $\kappa = 1$ their multiplicities are exactly the multiplicities of the genuine ordinates. Moreover

$$\boxed{\zeta_{\mathrm{RVM},0}^{\mathrm{path}} = \zeta_{\mathrm{RVM}}, \qquad \zeta_{\mathrm{RVM},1}^{\mathrm{path}} = \zeta_{\mathrm{line}}.} \tag{17.62}$$

Proof. Because $\gamma_{k,\kappa}^{\mathrm{RVM,path}}$ lies between the two positive ordinates γ_k^{RVM} and γ_k, the bounded-displacement lemma and $\gamma_k^{\mathrm{RVM}} \asymp k/\log k$ imply $\gamma_{k,\kappa}^{\mathrm{RVM,path}} \asymp k/\log k$ uniformly in κ for large k. Hence $\sum_k (\gamma_{k,\kappa}^{\mathrm{RVM,path}})^{-2}$ converges uniformly in κ. On a compact set $K \subset \mathbb{C}$ the kth factor differs from 1 by $O_K(t_k^{-2})$, so the Weierstrass criterion gives joint local normal convergence. The bound $\sum_k \log(1 + (R^2+R)/(1/4 + (\gamma_{k,\kappa}^{\mathrm{RVM,path}})^2)) = O(R\log R)$ gives order at most one, while the zero count $\#\{k : \gamma_{k,\kappa}^{\mathrm{RVM,path}} \le T\} \asymp T\log T$ gives order at least one. Hence the order is exactly one. Dependence only on $s(s-1)$ gives the functional symmetry. The assertions at $s=1$, at the negative even integers and at the nontrivial zeros follow exactly as in Theorem 17.2.

For $\kappa < 1$,

$$\gamma_{k+1,\kappa}^{\mathrm{RVM,path}} - \gamma_{k,\kappa}^{\mathrm{RVM,path}} = (1-\kappa)(\gamma_{k+1}^{\mathrm{RVM}} - \gamma_k^{\mathrm{RVM}}) + \kappa(\gamma_{k+1} - \gamma_k) > 0, \tag{S17.5}$$

because the RVM clock is strictly increasing and the genuine ordinate list is nondecreasing. At $\kappa = 1$ repeated genuine ordinates give precisely their multiplicities. The endpoint identities are the definitions of $P_{\mathrm{RVM}}^{\mathrm{crit}}$ and P_{line}. □

Theorem 17.24 (Endpoint identification)**.** *With the path above,*

$$\boxed{\mathbf{P}:\ \zeta_{\mathrm{RVM},1}^{\mathrm{path}} = \zeta \quad\Longleftrightarrow\quad \mathrm{RH}.} \tag{17.63}$$

The symbol $\mathbf{P}$ is used below only when the endpoint is identified with the classical zeta function; the construction of the path itself does not assume it.

Proof. If $\zeta_{\mathrm{RVM},1}^{\mathrm{path}} = \zeta$, the nontrivial zeros of ζ are those of $\zeta_{\mathrm{RVM},1}^{\mathrm{path}} = \zeta_{\mathrm{line}}$ and therefore lie on $\Re s = 1/2$. Conversely, assume RH. Then the genuine paired product $P_\zeta = P_\zeta^{\mathrm{pair}}$ from Theorem 17.18 has factors $1 + s(s-1)/(1/4+\gamma_k^2)$, hence $P_\zeta = P_{\mathrm{line}}$. Division by the common factor G gives $\zeta = \zeta_{\mathrm{line}} = \zeta_{\mathrm{RVM},1}^{\mathrm{path}}$. □

Critical-line input versus endpoint identification. Putting the synthetic zeros on $\Re s = 1/2$ is a definition of the RVM path, not an assumption that the genuine zeros lie there. If RH is false, the endpoint product ζ_{line} still exists, but its zero divisor differs from that of ζ, so the path does not arrive at the classical zeta function. The mathematical target is therefore an independent proof of the endpoint equality in **P**. Using that equality as a premise gives a conditional consequence equivalent to RH; merely constructing the critical-line path does not assume RH.

The next results move to the explicit-formula side. Write $u = \log x > 0$ and put

$$\mathcal{T}(t,u) := \Re\,\mathrm{Ei}((1/2+it)u), \qquad C_0(x) := -\log 2 + \int_x^\infty \frac{dv}{v(v^2-1)\log v}. \tag{17.64}$$

Initially let $\varphi \in C_c^1((0,\infty);\mathbb{R})$; extend all paired identities complex-linearly to complex test functions. Set

$$g_\varphi(u) := e^{u/2}\varphi(u), \qquad Q_\varphi(t) := \int_0^\infty \mathcal{T}(t,u)\varphi(u)\,du. \tag{17.65}$$

Lemma 17.25 (Smoothing converts the ordinate derivative into a summable kernel)**.** *One has*

$$\partial_t \mathcal{T}(t,u) = \Re\frac{ie^{(1/2+it)u}}{1/2+it} = \sqrt{x}\,\frac{t\cos(tu) - \frac{1}{2}\sin(tu)}{1/4+t^2}. \tag{17.66}$$

Moreover

$$Q_\varphi'(t) = \Re\frac{i\widehat{g}_\varphi(-t)}{1/2+it}, \qquad |Q_\varphi'(t)| \le \frac{\|g_\varphi'\|_1}{t^2}, \tag{17.67}$$

and $Q_\varphi(t) = O_\varphi(t^{-2})$ as $t \to \infty$.

Proof. Since $\mathrm{Ei}'(z) = e^z/z$, differentiation of $z = (1/2+it)u$ gives the first identity. Integrating against φ gives the Fourier transform in (17.67). Integration by parts yields $|\widehat{g}_\varphi(-t)| \le \|g_\varphi'\|_1/t$, which proves the t^{-2} bound. Finally, for the real part of the principal branch, uniformly on the compact u-support of φ,

$$\mathcal{T}(t,u) = \frac{e^{u/2}}{u}\frac{\sin(tu)}{t} + O_\varphi(t^{-2}). \tag{S17.6}$$

The possible imaginary constant in the asymptotic of Ei does not contribute to $\mathcal{T}$. Integration by parts in the leading term, with $e^{u/2}\varphi(u)/u \in C_c^1((0,\infty))$, supplies one more factor t^{-1}. This proves $Q_\varphi(t) = O_\varphi(t^{-2})$. In particular $\sum_k |Q_\varphi(\gamma_{k,\kappa}^{\mathrm{RVM,path}})|$ converges uniformly in κ; the same estimates are uniform on bounded sets of test functions with a common compact support. □

Theorem 17.26 (Distributional transport on the prime-counting side)**.** *Define $J_{\mathrm{RVM},\kappa}^{\mathrm{path}}$ as a distribution in $u = \log x$ by*

$$J_{\mathrm{RVM},\kappa}^{\mathrm{path}}(x) := \mathrm{li}(x) - 2\sum_{k\ge1} \mathcal{T}(\gamma_{k,\kappa}^{\mathrm{RVM,path}}, \log x) + C_0(x), \tag{17.68}$$

where the zero sum is interpreted after testing against $C_c^1((0,\infty))$. Then $\kappa \mapsto J_{\mathrm{RVM},\kappa}^{\mathrm{path}}$ is C^1 with values in $\mathcal{D}'((0,\infty))$, and for every test function

$$\sum_{k\ge1} |\delta_k^{\mathrm{RVM}\to\mathrm{line}}| \int_0^1 \left|\left\langle \partial_t \mathcal{T}(\gamma_{k,\kappa}^{\mathrm{RVM,path}}, \cdot), \varphi \right\rangle\right| d\kappa < \infty. \tag{17.69}$$

Consequently

$$\partial_\kappa J_{\mathrm{RVM},\kappa}^{\mathrm{path}}(x) = -2\sqrt{x}\sum_{k\ge1} \delta_k^{\mathrm{RVM}\to\mathrm{line}} \frac{\gamma_{k,\kappa}^{\mathrm{RVM,path}}\cos(\gamma_{k,\kappa}^{\mathrm{RVM,path}}u) - \frac{1}{2}\sin(\gamma_{k,\kappa}^{\mathrm{RVM,path}}u)}{1/4 + (\gamma_{k,\kappa}^{\mathrm{RVM,path}})^2}, \tag{17.70}$$

$$J_{\mathrm{RVM},1}^{\mathrm{path}} - J_{\mathrm{RVM},0}^{\mathrm{path}} = -2\sum_{k\ge1}(\mathcal{T}(\gamma_k,\cdot) - \mathcal{T}(\gamma_k^{\mathrm{RVM}},\cdot)) = \int_0^1 \partial_\kappa J_{\mathrm{RVM},\kappa}^{\mathrm{path}}\,d\kappa \tag{17.71}$$

in $\mathcal{D}'$.

Proof. The last estimate in Lemma 17.25 first justifies the defining tested zero sum itself, absolutely and uniformly in κ. For every k, the fundamental theorem of calculus in κ gives the endpoint difference as the integral of $\delta_k^{\mathrm{RVM}\to\mathrm{line}}\partial_t\mathcal{T}$. By Lemma 17.25, bounded displacement and $\gamma_{k,\kappa}^{\mathrm{RVM,path}} \asymp \gamma_k^{\mathrm{RVM}}$,

$$|\delta_k^{\mathrm{RVM}\to\mathrm{line}}| \int_0^1 \left|\langle \partial_t\mathcal{T}(\gamma_{k,\kappa}^{\mathrm{RVM,path}},\cdot),\varphi\rangle\right| d\kappa \ll_\varphi (\gamma_k^{\mathrm{RVM}})^{-2}. \tag{S17.7}$$

The right-hand series converges. Tonelli–Fubini therefore justifies the interchange, and the same majorant gives continuous dependence on κ and differentiation term by term in the distribution topology. □

The distributional theorem is deliberately stronger in rigor but weaker in pointwise content than a formal interchange of the two infinite operations. No pointwise convergence at an intermediate $0<\kappa<1$ is needed below.

Quadratic transport and the corrected logarithmic interpolation. For fixed $b>0$, the quadratic Taylor majorant converges at every fixed height; bounded displacement alone gives an $O_b(\log(t+2))$ bound, not a divergent series. The proposed affine-geodesic argument has a valid transport core, but its normalization term cannot be dropped. Write $\tau_k^{\mathrm{RVM}} := \gamma_k^{\mathrm{RVM}}$, $\delta_k^{\mathrm{RVM}\to\mathrm{line}} := \gamma_k - \tau_k^{\mathrm{RVM}}$, and $\tau_{k,\kappa}^{\mathrm{RVM}} := \tau_k^{\mathrm{RVM}} + \kappa\delta_k^{\mathrm{RVM}\to\mathrm{line}}$. The first sequence is strictly increasing; the genuine ordinates are nondecreasing and retain multiplicity. All these quantities are real, $k\in\mathbb{N}$, and $D_{\mathrm{RVM}} := \sup_k|\delta_k^{\mathrm{RVM}\to\mathrm{line}}| < \infty$. Counts are right-continuous. A time at an endpoint is included on the left and excluded on the right of a transport interval.

Audit proposition A3 (Monotone transport and the quadratic displacement). Let $N_{\mathrm{RVM}}(t) = \#\{k : \tau_k^{\mathrm{RVM}} \le t\}$ and $E_{\mathrm{RVM}}^{\mathrm{tr}}(t) := N_{\mathrm{RVM}}(t) - N_\zeta(t)$. Let $\mathcal{B}_{\mathrm{RVM}}(t)$ contain the indices whose two endpoints straddle t under the preceding half-open convention. At every t all these intervals have the same orientation, and their indices form exactly the consecutive block between the two counts. Moreover, for every finite $K\in\mathbb{N}$,

$$\frac{1}{2}\sum_{k\le K}(\delta_k^{\mathrm{RVM}\to\mathrm{line}})^2 = \int_0^\infty \sum_{k\in\mathcal{B}_{\mathrm{RVM}}(t),\ k\le K} |t-\tau_k^{\mathrm{RVM}}|\,dt. \tag{QT.1}$$

For the midpoint RVM clock, unconditionally,

$$A_{\mathrm{RVM}}^{\mathrm{tr}}(T) := \sum_{\tau_k^{\mathrm{RVM}}\le T}(\delta_k^{\mathrm{RVM}\to\mathrm{line}})^2 \sim \frac{T\log\log T}{\pi\log T} \qquad (T\to\infty). \tag{QT.2}$$

In particular the stated upper bound follows without sampling S at zeros.

Proof. If $\tau_k^{\mathrm{RVM}} \le t < \gamma_k$ and $\gamma_j \le t < \tau_j^{\mathrm{RVM}}$, monotonicity forces both $j<k$ and $k<j$, a contradiction. The block is therefore $\{N_\zeta(t)+1,\dots,N_{\mathrm{RVM}}(t)\}$ for positive discrepancy and the reversed-count block for negative discrepancy. This includes repeated genuine ordinates. Integrating the distance over a single transport interval gives $\frac{1}{2}(\delta_k^{\mathrm{RVM}\to\mathrm{line}})^2$; finite additivity, or Tonelli for the infinite identity in $[0,\infty]$, proves the formula.

Put $m = |E_{\mathrm{RVM}}^{\mathrm{tr}}(t)|$ and $h_{\mathrm{RVM}}^{\mathrm{gap}}(t) = 2\pi/\log(t/2\pi)$ for large t. Every clock point in the block lies within D_{RVM} of t. The adjacent clock point is at distance at most one clock gap. The inverse clock derivative gives, uniformly for gaps in this fixed-width neighborhood,

$$\tau_{j+1}^{\mathrm{RVM}} - \tau_j^{\mathrm{RVM}} = h_{\mathrm{RVM}}^{\mathrm{gap}}(t)\{1 + O_{D_{\mathrm{RVM}}}(1/(t\log t))\}. \tag{QT.3}$$

Summing the arithmetic progression of distances, with an arbitrary fractional first gap, yields

$$H_{\mathrm{RVM}}^{\mathrm{tr}}(t) := \sum_{k\in\mathcal{B}_{\mathrm{RVM}}(t)}|t-\tau_k^{\mathrm{RVM}}| = \frac{\pi m^2}{\log(t/2\pi)} + O_{D_{\mathrm{RVM}}}\left(\frac{m}{\log t} + \frac{m^2}{t(\log t)^2}\right). \tag{QT.4}$$

Here $m = O(\log t)$ by bounded displacement and the clock count. The Riemann–von Mangoldt formula gives $E^{\mathrm{tr}}_{\mathrm{RVM}}(t) = -S(t) + O(1)$ almost everywhere; the sign is immaterial for the square. Selberg's unconditional second-moment input is

$$\int_0^T S(t)^2\,dt \sim \frac{T\log\log T}{2\pi^2} \quad [58,\ \S9.25,\ (9.25.1)]. \tag{QT.5}$$

This uses the unconditional asymptotic, not an unverified $O(T)$ remainder. Cauchy–Schwarz shows that replacing S^2 by $(E^{\mathrm{tr}}_{\mathrm{RVM}})^2$ changes the integral by $O(T\sqrt{\log\log T} + T)$. Partial summation therefore gives

$$\int_{t_0}^T \frac{(E^{\mathrm{tr}}_{\mathrm{RVM}}(t))^2}{\log(t/2\pi)}\,dt \sim \frac{T\log\log T}{2\pi^2\log T}, \tag{QT.6}$$

for any fixed sufficiently large t_0. The integrated $m/\log t$ error is $O(T\sqrt{\log\log T}/\log T)$ by Cauchy–Schwarz on dyadic blocks; the other error is $O(\log T)$ using $m = O(\log t)$. Finally $\int_0^T H^{\mathrm{tr}}_{\mathrm{RVM}}$ differs from $A^{\mathrm{tr}}_{\mathrm{RVM}}(T)/2$ only through intervals crossing T. There are $O_{D_{\mathrm{RVM}}}(\log T)$ such clock indices, each contributing $O(D^2_{\mathrm{RVM}})$; the initial bounded range contributes $O(1)$. These boundary errors are negligible. The asymptotic follows. □

Audit proposition A4 (Logarithmic interpolation with its normalization defect). Fix a real $\sigma \in (1/2, 1)$, put $s = \sigma + it$, $b = \sigma - 1/2 > 0$, and take $t > 2D_{\mathrm{RVM}} + 1$. Choose logarithms coherently from the paired products on the zero-free quadrant $1/2 < \Re s < 1$, $\Im s > 0$; write $L_{\mathrm{RVM},\kappa}(s) = \log\zeta^{\mathrm{path}}_{\mathrm{RVM},\kappa}(s)$. Set $q_{\mathrm{RVM}}(u) = \log(u^2 + 1/4)$ for real $u > 0$, and define

$$C_{\mathrm{RVM}}(\kappa) := \sum_{k\ge1}\left\{(1-\kappa)q_{\mathrm{RVM}}(\tau^{\mathrm{RVM}}_k) + \kappa q_{\mathrm{RVM}}(\gamma_k) - q_{\mathrm{RVM}}(\tau^{\mathrm{RVM}}_{k,\kappa})\right\}. \tag{QT.7}$$

This real series and its first two κ derivatives converge uniformly. It vanishes at the endpoints and is strictly negative for $0 < \kappa < 1$ if any displacement is nonzero. There is a measurable set $\mathcal{Z}_{\mathrm{RVM},\sigma} \subset [1,\infty)$ with

$$\lim_{T\to\infty}\frac{\mathrm{meas}(\mathcal{Z}_{\mathrm{RVM},\sigma}\cap[1,T])}{T} = 0. \tag{QT.8}$$

Outside this one set of times, uniformly for $\kappa \in [0,1]$,

$$L_{\mathrm{RVM},\kappa}(\sigma+it) = (1-\kappa)L_{\mathrm{RVM},0}(\sigma+it) + \kappa L_{\mathrm{RVM},1}(\sigma+it) + C_{\mathrm{RVM}}(\kappa) + o(1), \tag{QT.9}$$

$$\partial_\kappa L_{\mathrm{RVM},\kappa}(\sigma+it) = L_{\mathrm{RVM},1}(\sigma+it) - L_{\mathrm{RVM},0}(\sigma+it) + C'_{\mathrm{RVM}}(\kappa) + o(1). \tag{QT.10}$$

Thus the renormalized logarithm $L_{\mathrm{RVM},\kappa} - C_{\mathrm{RVM}}(\kappa)$ is asymptotically affine in density one. The original normalized path is not an exact logarithmic geodesic.

Proof. Put $z = s - 1/2$ and $g_{\mathrm{RVM,s}}(u) = \log(z^2 + u^2) - q_{\mathrm{RVM}}(u)$, with the numerator logarithm chosen continuously (its imaginary part is positive here). Normal convergence of the paired product gives the exact master identity

$$L_{\mathrm{RVM},\kappa}(s) - L_{\mathrm{RVM},0}(s) = \sum_{k\ge1}\{g_{\mathrm{RVM,s}}(\tau^{\mathrm{RVM}}_{k,\kappa}) - g_{\mathrm{RVM,s}}(\tau^{\mathrm{RVM}}_k)\}, \qquad g'_{\mathrm{RVM,s}}(u) = \frac{2u(1/4 - z^2)}{(z^2+u^2)(1/4+u^2)}. \tag{QT.11}$$

In particular $|z^2 + u^2| = \sqrt{(u-t)^2+b^2}\sqrt{(u+t)^2+b^2}$, a product rather than a single square root. For $p_s(u) = \log(z^2 + u^2)$, factoring the numerator gives

$$|p''_s(u)| \le \frac{1}{(u-t)^2+b^2} + \frac{1}{(u+t)^2+b^2}. \tag{QT.12}$$

Use the integral Taylor remainder, which is valid for complex-valued functions; a real Lagrange point is not asserted. The affine-interpolation error for the numerator, and the error for its κ

derivative, are bounded uniformly by a constant times

$$K_{\mathrm{RVM,b}}(t) := \sum_{k\ge 1} (\delta_k^{\mathrm{RVM}\to\mathrm{line}})^2 \left\{ \frac{1}{((|\tau_k^{\mathrm{RVM}} - t| - D_{\mathrm{RVM}})_+)^2 + b^2} + \frac{1}{(\tau_k^{\mathrm{RVM}} + t - D_{\mathrm{RVM}})^2 + b^2} \right\}. \tag{QT.13}$$

The denominator contribution is exactly $C_{\mathrm{RVM}}(\kappa)$. Since $q''_{\mathrm{RVM}}(u) = 2(1/4-u^2)/(u^2+1/4)^2 < 0$ for $u > 1/2$, and all endpoints exceed $1/2$, concavity gives the stated strict sign. Uniform convergence and differentiation follow from the summable majorant $O((\delta_k^{\mathrm{RVM}\to\mathrm{line}})^2/(\tau_k^{\mathrm{RVM}})^2)$.

For $\tau_k^{\mathrm{RVM}} \in [T/2, 3T]$, the integral of the first kernel over $[T, 2T]$ is at most $2D_{\mathrm{RVM}}/b^2+\pi/b$. Audit proposition A3 bounds the total squared displacement there by $O(T \log\log T/\log T)$. Outside this range, bounded displacement and the clock density bound the averaged contribution by $O(\log T/T)$; the second kernel has the same bound, by summing against $N_{\mathrm{RVM}}(u) = O(u\log(u+2))$. Tonelli applies to these nonnegative majorants. Hence

$$\frac{1}{T}\int_T^{2T} K_{\mathrm{RVM,b}}(t)\,dt \ll_{b, D_{\mathrm{RVM}}} \frac{\log\log T}{\log T} + \frac{\log T}{T} = o(1). \tag{QT.14}$$

On dyadic blocks, Markov with a threshold equal to the square root of the right-hand bound constructs a single exceptional set of density zero; outside it $K_{\mathrm{RVM,b}}(t) \to 0$. This works for every κ simultaneously. The claimed expansions follow. The proposed error with an additional $\sum_k (\delta_k^{\mathrm{RVM}\to\mathrm{line}})^2/(\tau_k^{\mathrm{RVM}})^2$ cannot tend to zero: that sum is a fixed positive constant unless all displacements vanish. It is the normalization defect, not a disappearing tail. □

Audit proposition A5 (Conditional loss of vertical rigidity). Assume the endpoint premise **P** (hence $\zeta^{\mathrm{path}}_{\mathrm{RVM},1} = \zeta$ and RH), and additionally a proved vertical limit $\log|\zeta_{\mathrm{RVM}}(\sigma+it)| \to c_{\mathrm{RVM}}$ for the chosen $\sigma \in (1/2, 1)$. Then for every fixed $\kappa \in (0, 1]$, $\log|\zeta^{\mathrm{path}}_{\mathrm{RVM},\kappa}(\sigma+it)|$ is unbounded above. More precisely, in density one,

$$\log|\zeta^{\mathrm{path}}_{\mathrm{RVM},\kappa}(\sigma+it)| = \kappa\log|\zeta(\sigma+it)| + (1-\kappa)c_{\mathrm{RVM}} + C_{\mathrm{RVM}}(\kappa) + o(1). \tag{QT.15}$$

Proof. Take real parts in the preceding proposition. Classical value-distribution results, or the positive-lower-density form of Voronin universality, imply that for each real V the set of times with $\log|\zeta(\sigma+it)| > V$ has positive lower density [58, Ch. XI]. A density-zero exception cannot remove that set. Since $\kappa > 0$, arbitrary V prove unboundedness. The fluctuation scale is multiplied by κ, with the deterministic normalization shift explicitly retained. □

Consequently, *if the vertical-limit premise is proved*, either RH is false or this loss occurs at every positive path parameter. Without that premise the disjunction is not an unconditional consequence of this audit. The conclusion obstructs propagation of a uniform vertical bound along the path under these premises; it does not exclude every conceivable RH proof. Route independence for the smoothed J endpoint difference remains the separate exact statement already proved above. It does not turn the normalized logarithmic path into an exact segment.

Theorem 17.27 (Uniform convergence at the RVM endpoint)**.** *The series*

$$\sum_{k\ge 1} \mathcal{T}(\gamma_k^{\mathrm{RVM}}, u) \tag{17.72}$$

converges uniformly for u in every compact interval $[a, b] \subset (0, \infty)$. In particular the symmetric explicit-formula object $J^{\mathrm{path}}_{\mathrm{RVM},0}$ is an ordinary continuous function and equals the J_{RVM} of (17.13)*.*

Proof. Put $r_k^{\mathrm{RVM}} := \gamma_k^{\mathrm{RVM}}$ and $\ell_k^{\mathrm{RVM}} := \log(r_k^{\mathrm{RVM}}/2\pi)$. The inverse-clock relation gives

$$r_k^{\mathrm{RVM}}\ell_k^{\mathrm{RVM}} \ge 2\pi(k - 11/8), \qquad \frac{2\pi}{\ell_{k+1}^{\mathrm{RVM}}} \le r_{k+1}^{\mathrm{RVM}} - r_k^{\mathrm{RVM}} \le \frac{2\pi}{\ell_k^{\mathrm{RVM}}}, \tag{S17.8}$$

and, by differentiating the inverse clock once more,

$$|r_{k+2}^{\mathrm{RVM}} - 2r_{k+1}^{\mathrm{RVM}} + r_k^{\mathrm{RVM}}| \le \frac{4\pi^2}{r_k^{\mathrm{RVM}}(\ell_k^{\mathrm{RVM}})^3}. \tag{S17.9}$$

For $E_k^{\mathrm{RVM}} = e^{iur_k^{\mathrm{RVM}}}$ and $A_K^{\mathrm{RVM}}(u) = \sum_{k\le K} E_k^{\mathrm{RVM}}$, choose $k_0 = k_0(b)$ so that $u(r_{k+1}^{\mathrm{RVM}} - r_k^{\mathrm{RVM}}) \le \pi$ for $u \in [a,b]$ and $k \ge k_0$. Writing $\Delta_k^{\mathrm{RVM,path}} = u(r_{k+1}^{\mathrm{RVM}} - r_k^{\mathrm{RVM}})$ and

$$E_k^{\mathrm{RVM}} = \frac{E_{k+1}^{\mathrm{RVM}} - E_k^{\mathrm{RVM}}}{e^{i\Delta_k^{\mathrm{RVM,path}}} - 1} \tag{S17.10}$$

and summing by parts gives, uniformly on $[a,b]$,

$$|A_K^{\mathrm{RVM}}(u)| \ll_{a,b} 1 + \log K. \tag{17.73}$$

Indeed the endpoint coefficient is $O_{a,b}(\log K)$ and the successive coefficient differences are $O_{a,b}(1/k)$ by the displayed second-difference bound. A second Abel summation then yields uniform convergence of $\sum e^{iur_k^{\mathrm{RVM}}}/r_k^{\mathrm{RVM}}$, because $r_k^{\mathrm{RVM}} \asymp k/\log k$ and the resulting majorant is $\sum \log^2 k/k^2$.

Finally, uniformly for $u \in [a,b]$, the right-half-plane asymptotic of Ei gives

$$\mathcal{T}(r_k^{\mathrm{RVM}}, u) = \frac{e^{u/2}}{u}\frac{\sin(r_k^{\mathrm{RVM}}u)}{r_k^{\mathrm{RVM}}} + O_{a,b}((r_k^{\mathrm{RVM}})^{-2}). \tag{S17.11}$$

The first series converges uniformly by the preceding paragraph and the error series absolutely. Hence the Ei series defining $J_{\mathrm{RVM},0}^{\mathrm{path}}$ converges uniformly on compact u-intervals. □

Accordingly, Hypothesis 17.5 below concerns the *quantitative square-root estimate*; existence and compact-uniform convergence of the RVM zero sum are already theorems.

For the endpoint $\kappa = 1$, define

$$B(T;u) := \sum_{0<\gamma\le T} e^{i\gamma u}, \tag{17.74}$$

where the genuine upper-half-plane zeros are counted with multiplicity. Also define

$$D(T;u) := \sum_{0<\gamma\le T} e^{i\gamma u}\left(\cosh((\beta-1/2)u) - 1\right). \tag{17.75}$$

Lemma 17.28 (Exact off-line defect)**.** *For every fixed $u > 0$,*

$$\boxed{B(T;u) = \sum_{0<\gamma\le T} x^{\rho-1/2} - D(T;u), \qquad x = e^u.} \tag{17.76}$$

Moreover the standard Selberg zero-density estimate implies

$$\sum_{0<\gamma\le T} (\beta-1/2)^2 \ll \frac{T}{\log T}, \qquad |D(T;u)| \ll_u \frac{T}{\log T}. \tag{17.77}$$

Proof. The upper-half-plane zero multiset is invariant under $\beta + i\gamma \mapsto 1 - \beta + i\gamma$. Averaging the two equal sums $\sum e^{i\gamma u}e^{\pm(\beta-1/2)u}$ gives the hyperbolic cosine and hence (17.76). For the second statement, the standard Selberg estimate $N(1/2+a,T) \ll T^{1-a/4}\log T$ gives, by layer-cake integration,

$$\sum_{\beta>1/2} (\beta-1/2)^2 = \int_0^{1/2} 2a\, N(1/2+a,T)\, da \ll \frac{T}{\log T}; \tag{S17.12}$$

the left half is identical by symmetry. Finally $\cosh y - 1 \le \frac{1}{2}y^2\cosh|y|$ proves the bound for D. □

The Landau–Gonek explicit formula [1] gives, uniformly when $x = e^u$ ranges over a compact subset of $(1, \infty)$ at positive distance from every prime power,

$$\sum_{0<\gamma\le T} x^\rho = O_x(\log T). \tag{17.78}$$

At a prime power it has the familiar linear main term $-(T/2\pi)\Lambda(x)$.

Theorem 17.29 (Pointwise endpoint criterion and the role of **P**)**.** *Let $x = e^u > 1$ be not a prime power.*

(a) *Without* **P**, *the series $\sum_k e^{i\gamma_k u}/\gamma_k$ converges if and only if the finite limit*

$$\lim_{T\to\infty} \int_{14}^{T} \frac{D(t;u)}{t^2}\, dt \tag{17.79}$$

exists.

(b) *Under* **P** *(equivalently RH), $D \equiv 0$ and both $\sum e^{i\gamma_k u}/\gamma_k$ and $\sum \mathcal{T}(\gamma_k, u)$ converge uniformly when x ranges over a compact set at positive distance from the prime powers.*

Proof. Partial summation gives

$$\sum_{\gamma_k\le T} \frac{e^{i\gamma_k u}}{\gamma_k} = \frac{B(T;u)}{T} + \int_{14}^{T} \frac{B(t;u)}{t^2}\, dt. \tag{S17.13}$$

Lemma 17.28 implies $B(T;u)/T \to 0$. For non-prime-power x, (17.78) makes the integral of the $x^{\rho-1/2}$ part absolutely convergent at infinity; by (17.76), the only remaining issue is exactly (17.79). This proves (a).

Under **P**, Theorem 17.24 gives $\beta = 1/2$ for every zero, so $D = 0$. The local Landau–Gonek bound is uniform on compact sets separated from the prime powers, and partial summation gives a uniformly Cauchy series. The asymptotic

$$\mathcal{T}(\gamma, u) = \frac{e^{u/2}}{u}\frac{\sin(\gamma u)}{\gamma} + O_K(\gamma^{-2}) \tag{S17.14}$$

then gives the same conclusion for the Ei series. □

At a prime power, under **P**, the real Landau–Gonek main term makes $\sum e^{i\gamma_k u}/\gamma_k$ logarithmically divergent through its cosine part. This does not imply divergence of $\sum \mathcal{T}(\gamma_k, u)$: its leading term involves the sine part. The classical symmetric explicit formula assigns the midpoint value at a prime-power jump. The preceding uniform-convergence theorem is stated away from prime powers; it makes no assertion of uniform convergence across a jump. Distributional statements remain available there.

Theorem 17.30 (Identity between the two endpoints)**.** *Let $K \subset (1, \infty)$ be compact and separated from the prime powers. Under* **P**,

$$\boxed{J - J_{\mathrm{RVM}} = -2\sum_{k\ge 1}\big(\mathcal{T}(\gamma_k, \log x) - \mathcal{T}(\gamma_k^{\mathrm{RVM}}, \log x)\big)} \tag{17.80}$$

pointwise and uniformly on K, and the same identity holds in $\mathcal{D}'(U)$ on every open $U \subset (1, \infty)$ containing no prime power. No pointwise statement at an intermediate $0 < \kappa < 1$ is needed.

Proof. The RVM endpoint converges uniformly by Theorem 17.27, and the genuine-ordinate endpoint converges uniformly under **P** by Theorem 17.29. Their difference is therefore the pointwise series in (17.80). Under **P**, the classical explicit formula groups the genuine zeros as $1/2 \pm i\gamma_k$, so its zero term is exactly $2\sum_k \mathcal{T}(\gamma_k, \log x)$ and hence $J^{\mathrm{path}}_{\mathrm{RVM},1} = J$. The distributional identity follows independently from Theorem 17.26 and uniqueness of the distributional limit. □

The next identity does not use $\mathbf{P}$; it compares the two ordinate sets as distributions. Put

$$N_{\mathrm{RVM}}(t) := \#\{k : \gamma_k^{\mathrm{RVM}} \le t\} = \left\lfloor \mathcal{N}_{\mathrm{RVM}}(t) + \frac{1}{2} \right\rfloor \qquad (t \ge 2\pi), \tag{17.81}$$

let $N_\zeta(t)$ be the genuine zero count, define

$$\mathcal{H}_\zeta(t) := 1 + \frac{\theta(t)}{\pi}, \qquad S(t) := N_\zeta(t) - \mathcal{H}_\zeta(t), \qquad \mathfrak{s}(y) := \{y\} - \frac{1}{2}, \tag{17.82}$$

and recall from Stirling's expansion that

$$\mathcal{H}_\zeta(t) - \mathcal{N}_{\mathrm{RVM}}(t) = \frac{1}{48\pi t} + O(t^{-3}). \tag{17.83}$$

Proposition 17.31 (Exact route-independent endpoint formula)**.** *For every $\varphi \in C_c^1((0,\infty))$,*

$$\boxed{\langle J^{\mathrm{path}}_{\mathrm{RVM},1} - J^{\mathrm{path}}_{\mathrm{RVM},0}, \varphi\rangle = 2\int_{10}^{\infty} Q'_\varphi(t)\left[S(t) + \mathfrak{s}\Big(\mathcal{N}_{\mathrm{RVM}}(t) + \frac{1}{2}\Big) + \mathcal{H}_\zeta(t) - \mathcal{N}_{\mathrm{RVM}}(t)\right] dt.} \tag{17.84}$$

The integral is absolutely convergent and contains no κ. Hence every admissible path with the same two endpoint ordinate sets gives the same smoothed endpoint difference.

Proof. For $T > 10$, Stieltjes integration gives

$$\sum_{\gamma_k \le T} Q_\varphi(\gamma_k) - \sum_{\gamma_k^{\mathrm{RVM}} \le T} Q_\varphi(\gamma_k^{\mathrm{RVM}}) = Q_\varphi(T)(N_\zeta - N_{\mathrm{RVM}})(T) - \int_{10}^{T} Q'_\varphi(t)(N_\zeta - N_{\mathrm{RVM}})(t)\, dt. \tag{S17.15}$$

Since

$$N_{\mathrm{RVM}}(t) = \mathcal{N}_{\mathrm{RVM}}(t) - \mathfrak{s}(\mathcal{N}_{\mathrm{RVM}}(t) + 1/2), \tag{S17.16}$$

one has

$$N_\zeta - N_{\mathrm{RVM}} = S + (\mathcal{H}_\zeta - \mathcal{N}_{\mathrm{RVM}}) + \mathfrak{s}(\mathcal{N}_{\mathrm{RVM}} + 1/2) = O(\log t). \tag{S17.17}$$

Lemma 17.25 gives $Q_\varphi(T) = O(T^{-1})$ and $Q'_\varphi(t) = O(t^{-2})$, so the boundary term tends to zero and the integral is absolutely convergent. Matching the finite-height sums with the index-paired difference costs only $O(\log T)$ terms of size $O(T^{-1})$, hence tends to zero. Finally Theorem 17.26 supplies the factor -2 in the endpoint series, which reverses the sign from the Stieltjes integration and gives (17.84). □

Corollary 17.32 (Crude localized bound for the smoothed endpoint difference)**.** *Let $\varphi_\eta(u) = \eta^{-1}\rho((u-u_0)/\eta)$ with $\rho \in C_c^1([-1,1])$, $\int \rho = 1$, $x = e^{u_0}$ and $0 < \eta \le 1$ with $\eta < u_0$, so that the test-function support lies in $(0,\infty)$. Then*

$$|\langle J^{\mathrm{path}}_{\mathrm{RVM},1} - J^{\mathrm{path}}_{\mathrm{RVM},0}, \varphi_\eta\rangle| \ll_\rho \sqrt{x}\left(1 + \log^2\frac{1}{\eta}\right). \tag{17.85}$$

In particular, as $x \to \infty$, $\eta \asymp (\log x)/x$ gives $O(\sqrt{x}\log^2 x)$. Under $\mathbf{P}$ the same estimate applies to the smoothed difference $J - J_{\mathrm{RVM}}$.

Proof. The Fourier integration by parts in Lemma 17.25 gives

$$|Q'_{\varphi_\eta}(t)| \ll_\rho \sqrt{x}\,\frac{\min(1, (\eta t)^{-1})}{t}. \tag{S17.18}$$

Combine this with $|N_\zeta(t) - N_{\mathrm{RVM}}(t)| \ll \log t$ in the proof of Proposition 17.31. Splitting the integral at $t = \max(10, 1/\eta)$ gives $O(1 + \log^2(1/\eta))$ for the first part and $O(1 + \log(1/\eta))$ for the tail. The constant term keeps the bound valid throughout $0 < \eta \le 1$. □

Proposition 17.33 (Formal stationary-phase expansion of the sawtooth term)**.** *This proposition is a* formal asymptotic calculation*, not a proved uniform expansion. If the Fourier series of the sawtooth in* (17.84) *is truncated and stationary phase is applied term by term away from a fixed lower cutoff* $T_{\mathrm{cut}} \geq 10$*, the resonant contribution is*

$$Z_{\mathrm{res}}(x) \sim \frac{1}{\pi} \sum_{\substack{m\geq 1 \\ 2\pi x^{1/m} > T_{\mathrm{cut}}}} (-1)^m m^{-3/2} x^{(m-1)/(2m)} \sin\!\left(2\pi m x^{1/m} + \frac{\pi(m-1)}{4}\right). \tag{17.86}$$

The stationary point of the m*th harmonic is* $t_m = 2\pi x^{1/m}$*. The first term is* $-\pi^{-1}\sin(2\pi x)$*. Consequently its derivative has the same frequency and magnitude* $2\cos(2\pi x)$ *as the first RVM Poisson harmonic, with the opposite sign because* (17.86) *is written for* $J^{\mathrm{path}}_{\mathrm{RVM},1} - J^{\mathrm{path}}_{\mathrm{RVM},0}$ *rather than* $J^{\mathrm{path}}_{\mathrm{RVM},0} - J^{\mathrm{path}}_{\mathrm{RVM},1}$*. The derivative of the* $m = 2$ *term has size* $2^{-1/2}x^{-1/4}$*, again with the corresponding endpoint-difference sign.*

Proof. Only the phase calculation is asserted. Use

$$\mathfrak{s}(y) = -\sum_{m\geq 1} \frac{\sin(2\pi m y)}{\pi m} \tag{S17.19}$$

with a finite harmonic cutoff. Since $2\pi\mathcal{N}'_{\mathrm{RVM}}(t) = \log(t/2\pi)$, the two exponential phases are $tu + 2\pi m\mathcal{N}_{\mathrm{RVM}}(t)$ and $2\pi m\mathcal{N}_{\mathrm{RVM}}(t) - tu$. The first has no stationary point; the second has

$$\Phi'_m(t) = m\log(t/2\pi) - u, \qquad t_m = 2\pi x^{1/m}, \qquad \Phi''_m(t_m) = m/t_m. \tag{S17.20}$$

At the stationary point,

$$\Phi_m(t_m) = -2\pi m x^{1/m} + \frac{7\pi m}{4}. \tag{S17.21}$$

The amplitude in Q'_φ contributes $\frac{1}{2}\sqrt{2\pi/(mt_m)}$, and multiplication by the sawtooth coefficient $(-1)^{m+1}/(\pi m)$ and the outer factor $2\sqrt{x}$ gives exactly the coefficient and phase in (17.86). A complete theorem would still have to bound the nonstationary pieces, the stationary-phase remainders uniformly in both m and x, the harmonic tail, and the lower cutoff terms. Those estimates are not supplied here. □

Remark 17.34 (Status of the ordinate-homotopy chain)*.* The status is deliberately separated.

Status	Content
Unconditional theorem	Theorems 17.23, 17.26, 17.27, Theorem 17.29(a), Proposition 17.31 and Corollary 17.32.
Theorem under **P**	Theorem 17.29(b) and Theorem 17.30. By Theorem 17.24, **P** is exactly RH, so these are not independent proofs of RH.
Formal calculation	Proposition 17.33; the uniform remainder and lower-cutoff analysis are open.
Open targets	Pointwise convergence for $0 < \kappa < 1$ (not needed for the endpoint identity), positivity of the sampled RVM weights, the Mellin identification of the weighted Euler object, and the continuous-to-atomic endpoint problem (D).

The next analytic target specific to the formal resonance calculation is a uniform stationary-phase remainder in m and x, followed by the same analysis after the finite Möbius inversion defining π_{RVM}.

Remark 17.35 (Other deformations are logically separate)*.* Integer sampling of the RVM continuous count produces a weighted Euler object but requires the unit-cell and Mellin-identification hypotheses. Fractional interpolation between an RVM Euler measure and the true prime measure

can introduce branch monodromy at a genuine zero. Neither issue is needed for the critical-line ordinate path above, whose endpoint difference is already determined by Proposition 17.31.

17.10. **The true-β RVM ordinate transport.** The preceding critical system must be distinguished from the following transported system. Here the genuine real parts β_k are retained and only the ordinates are replaced by the deterministic RVM clock. Thus this object keeps the unknown horizontal zero geometry and is the one whose pole abscissa remains Θ.

Recall from Theorem 17.18 that

$$P_\zeta(s) = 2\xi(s) = G(s)\zeta(s) = P_\zeta^{\mathrm{pair}}(s). \tag{17.87}$$

Define the true-β rigid-ordinate product

$$P_{\mathrm{RVM}}^{\mathrm{ord}}(s) := \prod_{k\geq 1}\left(1 - \frac{s}{\beta_k + i\gamma_k^{\mathrm{RVM}}}\right)\left(1 - \frac{s}{\beta_k - i\gamma_k^{\mathrm{RVM}}}\right). \tag{17.88}$$

$P_{\mathrm{RVM}}^{\mathrm{ord}}(0) = P_\zeta(0) = 1$; the genuine paired product is exactly P_ζ by Theorem 17.18.

Lemma 17.36 (Absolute convergence of the paired rigid product)**.** *The product* (17.88) *converges absolutely and locally uniformly in s. It defines an entire function whose zeros, with multiplicity, are exactly $\beta_k \pm i\gamma_k^{\mathrm{RVM}}$.*

Proof. For $\rho = \beta + iy$ with $0 < \beta < 1$,

$$\left(1 - \frac{s}{\rho}\right)\left(1 - \frac{s}{\bar\rho}\right) = 1 - \frac{2\beta s}{\beta^2 + y^2} + \frac{s^2}{\beta^2 + y^2} = 1 + O_K(y^{-2}) \tag{17.89}$$

uniformly for s in a compact set K. The Riemann–von Mangoldt law for the deterministic clock gives $\sum_k (\gamma_k^{\mathrm{RVM}})^{-2} < \infty$. The standard criterion for infinite products proves absolute local uniform convergence. The same estimate shows that no additional zeros are introduced. □

17.11. **A multiplicative true-β transport with no derivative of zeta.** For s away from the displayed poles define

$$\begin{aligned}\mathscr{L}_{\mathrm{RVM}}(s) := \sum_{k\geq 1}\Bigg(&\frac{1}{s - \beta_k - i\gamma_k} + \frac{1}{s - \beta_k + i\gamma_k} \\ &- \frac{1}{s - \beta_k - i\gamma_k^{\mathrm{RVM}}} - \frac{1}{s - \beta_k + i\gamma_k^{\mathrm{RVM}}}\Bigg).\end{aligned} \tag{17.90}$$

The zeros in (17.1) occur with their multiplicities, so no separate multiplicity factor is needed.

Lemma 17.37 (Normal convergence of the logarithmic transport)**.** *The series* (17.90) *converges normally on every compact set avoiding its poles. It therefore defines a meromorphic function.*

Proof. For $0 < \beta < 1$ put

$$A(s;\beta,y) := \frac{1}{s - \beta - iy} + \frac{1}{s - \beta + iy} = \frac{2(s-\beta)}{(s-\beta)^2 + y^2}. \tag{17.91}$$

Uniformly for s in a fixed compact set and $0 < \beta < 1$,

$$\frac{\partial}{\partial y} A(s;\beta,y) = O(y^{-3}) \qquad (y \to \infty). \tag{17.92}$$

Lemma 17.1 and the mean-value theorem make the kth paired difference in (17.90) $O(\gamma_k^{-3})$. Finally the Riemann–von Mangoldt estimate implies $\sum_k \gamma_k^{-3} < \infty$. The Weierstrass test proves normal convergence away from the finitely many poles meeting a given compact set. □

The residues of $\mathscr{L}_{\mathrm{RVM}}$ are integers: $+1$ at each genuine zero occurrence and -1 at its transported occurrence. Hence the exponential of a primitive is single-valued.

Definition 17.38 (Transport multiplier and transported Mertens transform)**.** The Riemann–von Mangoldt transport multiplier is

$$\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}(s) := \frac{P_\zeta(s)}{P^{\mathrm{ord}}_{\mathrm{RVM}}(s)}. \tag{17.93}$$

Its logarithmic derivative is exactly $\mathscr{L}_{\mathrm{RVM}}(s)$. Equivalently, it is the exponential of a primitive of $\mathscr{L}_{\mathrm{RVM}}$, normalized by $\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}(0) = 1$; integer residues make that exponential single-valued. Define

$$\boxed{\widehat{M}^{\mathrm{ord}}_{\mathrm{RVM}}(s) := \frac{G(s)}{P^{\mathrm{ord}}_{\mathrm{RVM}}(s)} = \frac{\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}(s)}{\zeta(s)}.} \tag{17.94}$$

Here the hat denotes the normalized Mellin–Perron transform, not a second summatory function.

Equivalently, the multiplier is the normally convergent product

$$\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}(s) = \prod_{k\ge1} \frac{((s-\beta_k)^2+\gamma_k^2)/(\beta_k^2+\gamma_k^2)}{((s-\beta_k)^2+(\gamma_k^{\mathrm{RVM}})^2)/(\beta_k^2+(\gamma_k^{\mathrm{RVM}})^2)}. \tag{17.95}$$

Indeed, logarithmic differentiation gives $\mathscr{L}_{\mathrm{RVM}}$; normal convergence follows by differentiating each normalized logarithmic factor with respect to its ordinate and using the proof of Lemma 17.37.

Proposition 17.39 (Exact divisor transport)**.** *As divisors, with multiplicities and including possible coincidences,*

$$\mathrm{div}(\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}) = \sum_{k\ge1}[\beta_k \pm i\gamma_k] - \sum_{k\ge1}[\beta_k \pm i\gamma_k^{\mathrm{RVM}}]. \tag{17.96}$$

Consequently all genuine nontrivial poles of $1/\zeta$ are cancelled in $\widehat{M}^{\mathrm{ord}}_{\mathrm{RVM}}$, and its nontrivial pole divisor is exactly

$$\boxed{\sum_{k\ge1}[\beta_k \pm i\gamma_k^{\mathrm{RVM}}].} \tag{17.97}$$

In particular,

$$\sup\{\Re s : s \text{ is a nontrivial pole of } \widehat{M}^{\mathrm{ord}}_{\mathrm{RVM}}\} = \sup_k \beta_k = \Theta. \tag{17.98}$$

No simplicity hypothesis and no value of $\zeta'(\rho_k)$ is used.

Proof. At a pole of $\mathscr{L}_{\mathrm{RVM}}$ with residue $m \in \mathbb{Z}$, a local primitive is $m\log(s-s_0)$ plus a holomorphic function. Thus $\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}$ has divisor order m at s_0, proving (17.96). The nontrivial pole divisor of $1/\zeta$ is the negative of the genuine-zero divisor. Divisors add under multiplication, so (17.94) leaves precisely the negative transported divisor, which is (17.97). Vertical transport does not change any β_k, and (17.98) follows. □

Proposition 17.40 (Functional symmetry of the rigid product)**.** *The product $P^{\mathrm{ord}}_{\mathrm{RVM}}$ depends on the genuine zeros only through the ordered list of real parts (β_k) and their multiplicities; the true ordinates are used only to assign that order. Moreover,*

$$\boxed{P^{\mathrm{ord}}_{\mathrm{RVM}}(s) = P^{\mathrm{ord}}_{\mathrm{RVM}}(1-s) \ \text{ for all } s \quad\Longleftrightarrow\quad \mathrm{RH}.} \tag{17.99}$$

Proof. If RH holds, every $\beta_k = 1/2$, and each paired factor in (17.88) equals

$$\frac{(s-1/2)^2+(\gamma_k^{\mathrm{RVM}})^2}{1/4+(\gamma_k^{\mathrm{RVM}})^2}, \tag{17.100}$$

which is invariant under $s \mapsto 1-s$.

Conversely, suppose $P^{\mathrm{ord}}_{\mathrm{RVM}}(s) = P^{\mathrm{ord}}_{\mathrm{RVM}}(1-s)$. Its zero divisor is then invariant under $s \mapsto 1-s$ and, because $P^{\mathrm{ord}}_{\mathrm{RVM}}$ is real on the real axis, under conjugation. Hence a zero $\beta_k + i\gamma_k^{\mathrm{RVM}}$ forces a zero $1-\beta_k+i\gamma_k^{\mathrm{RVM}}$. The sequence γ_k^{RVM} is strictly increasing, so the only upper-half-plane zero of

$P^{\mathrm{ord}}_{\mathrm{RVM}}$ at that ordinate is the occurrence indexed by k. Therefore $1-\beta_k=\beta_k$ for every k, proving RH. □

This symmetry is an exact reformulation, not an independent proof of RH. Its possible value is that $P^{\mathrm{ord}}_{\mathrm{RVM}}$ has a rigid and explicitly known ordinate set; one would still need an argument, not already equivalent to RH, forcing the symmetry or the required growth bound.

17.12. **An exact reflection defect.** The tie convention fixed above gives a second exact way to read horizontal displacement from the rigid product. Let $\mathcal{J}_{\mathrm{off}}$ index the functional pairs away from the critical line. For $j\in\mathcal{J}_{\mathrm{off}}$ write their real parts as $1/2+d_j$ and $1/2-d_j$, where $0<d_j<1/2$. By convention the right member is assigned the lower of the two consecutive rigid ordinates. Denote those ordinates by

$$g^{\mathrm{RVM}}_{j,0}:=\gamma^{\mathrm{RVM}}_{k_j},\qquad g^{\mathrm{RVM}}_{j,1}:=\gamma^{\mathrm{RVM}}_{k_j+1},\qquad g^{\mathrm{RVM}}_{j,0}<g^{\mathrm{RVM}}_{j,1}. \tag{17.101}$$

Proposition 17.41 (Positive reflection defect)**.** *For $x>0$ put*

$$\Delta^{\mathrm{ord}}_{\mathrm{RVM}}(1/2+x):=\log\frac{P^{\mathrm{ord}}_{\mathrm{RVM}}(1/2-x)}{P^{\mathrm{ord}}_{\mathrm{RVM}}(1/2+x)}. \tag{17.102}$$

Then

$$\boxed{\Delta^{\mathrm{ord}}_{\mathrm{RVM}}(1/2+x)=8x\sum_{j\in\mathcal{J}_{\mathrm{off}}}d_j\int_{g^{\mathrm{RVM}}_{j,0}}^{g^{\mathrm{RVM}}_{j,1}}\frac{t\,dt}{(t^2+(x-d_j)^2)(t^2+(x+d_j)^2)}\geq 0.} \tag{17.103}$$

Equality for one, and hence for every, $x>0$ holds if and only if RH. In particular,

$$\boxed{P^{\mathrm{ord}}_{\mathrm{RVM}}(1)\leq 1,\qquad P^{\mathrm{ord}}_{\mathrm{RVM}}(1)=1\iff\text{RH},} \tag{17.104}$$

and

$$-\frac{(P^{\mathrm{ord}}_{\mathrm{RVM}})'(1/2)}{P^{\mathrm{ord}}_{\mathrm{RVM}}(1/2)}=2\sum_{j\in\mathcal{J}_{\mathrm{off}}}d_j\left(\frac{1}{d_j^2+(g^{\mathrm{RVM}}_{j,0})^2}-\frac{1}{d_j^2+(g^{\mathrm{RVM}}_{j,1})^2}\right)\geq 0, \tag{17.105}$$

with equality if and only if RH.

Proof. For real r, the paired conjugate factor belonging to a zero with real part b and positive ordinate g is

$$\left(1-\frac{r}{b+ig}\right)\left(1-\frac{r}{b-ig}\right)=\frac{(r-b)^2+g^2}{b^2+g^2}>0. \tag{17.106}$$

Set

$$F_{x,d}(g):=\log\frac{g^2+(x+d)^2}{g^2+(x-d)^2}. \tag{17.107}$$

The pair indexed by j contributes $F_{x,d_j}(g^{\mathrm{RVM}}_{j,0})-F_{x,d_j}(g^{\mathrm{RVM}}_{j,1})$ to (17.102). Direct differentiation gives

$$-\frac{d}{dg}F_{x,d}(g)=\frac{8x\,dg}{(g^2+(x-d)^2)(g^2+(x+d)^2)}. \tag{17.108}$$

Integration between $g^{\mathrm{RVM}}_{j,0}$ and $g^{\mathrm{RVM}}_{j,1}$ proves (17.103). Every summand is strictly positive when $d_j>0$, while critical-line occurrences contribute zero. Taking $x=1/2$ and using $P^{\mathrm{ord}}_{\mathrm{RVM}}(0)=1$ gives (17.104). Dividing (17.103) by $2x$ and letting $x\downarrow 0$ gives (17.105); normal convergence justifies the passage to the limit. □

The sign in Proposition 17.41 uses the fixed within-pair ordering. Reversing that ordering reverses the sign but does not change the equality criterion. Thus the vanishing is intrinsic to the rigid divisor, whereas positivity records the chosen orientation of each split functional pair.

17.13. **Horizontal zero energy encoded by the rigid product.** Put

$$\mathfrak{d}_k^{\mathrm{RVM,ord}} := \beta_k - \frac{1}{2}, \qquad I_{\mathrm{RVM}}^{\mathrm{ord}}(T) := \{k : T < \gamma_k^{\mathrm{RVM}} \le T+1\}. \tag{17.109}$$

Define the horizontal energy and its level counts by

$$E_{\mathrm{RVM}}^{\mathrm{ord}}(T) := \sum_{k \in I_{\mathrm{RVM}}^{\mathrm{ord}}(T)} (\mathfrak{d}_k^{\mathrm{RVM,ord}})^2, \tag{17.110}$$

and, for $0 < a < 1/2$, by

$$Q_{\mathrm{RVM,a}}^{\mathrm{ord}}(T) := \#\{k \in I_{\mathrm{RVM}}^{\mathrm{ord}}(T) : |\mathfrak{d}_k^{\mathrm{RVM,ord}}| > a\}. \tag{17.111}$$

These quantities are not extra data attached to $P_{\mathrm{RVM}}^{\mathrm{ord}}$: they can be recovered exactly from the logarithmic potential of the product.

Proposition 17.42 (Riesz representation of the horizontal energy)**.** *In the sense of distributions on the $s = \varsigma + it$ plane,*

$$\boxed{\Delta_s \log |P_{\mathrm{RVM}}^{\mathrm{ord}}(\varsigma + it)| = 2\pi \sum_{k \ge 1} \left(\delta_{(\beta_k, \gamma_k^{\mathrm{RVM}})} + \delta_{(\beta_k, -\gamma_k^{\mathrm{RVM}})} \right),} \qquad \Delta_s := \partial_\varsigma^2 + \partial_t^2. \tag{17.112}$$

Let $\psi \in C_c^\infty((0,\infty))$ and let $\chi \in C_c^\infty(\mathbb{R})$ equal 1 on $[0,1]$. Then

$$\begin{aligned} \mathscr{E}_{\mathrm{RVM}}^{\mathrm{ord}}[\psi] &:= \sum_{k \ge 1} (\mathfrak{d}_k^{\mathrm{RVM,ord}})^2 \psi(\gamma_k^{\mathrm{RVM}}) \\ &= \frac{1}{2\pi} \iint_{\mathbb{R}^2} \chi(\varsigma) \left(\varsigma - \frac{1}{2}\right)^2 \psi(t) \Delta_s \log |P_{\mathrm{RVM}}^{\mathrm{ord}}(\varsigma + it)| \, d\varsigma \, dt \\ &= \frac{1}{2\pi} \iint_{\mathbb{R}^2} \log |P_{\mathrm{RVM}}^{\mathrm{ord}}(\varsigma + it)| \Delta_s \left[\chi(\varsigma) \left(\varsigma - \frac{1}{2}\right)^2 \psi(t) \right] d\varsigma \, dt. \end{aligned} \tag{17.113}$$

In particular $\mathscr{E}_{\mathrm{RVM}}^{\mathrm{ord}}[\psi] \ge 0$ when $\psi \ge 0$.

Proof. For an entire function F, the distributional identity $\Delta \log |F| = 2\pi \sum_z m_z \delta_z$ counts its zeros with multiplicity. Lemma 17.36 identifies the zero divisor of $P_{\mathrm{RVM}}^{\mathrm{ord}}$, proving (17.112). Testing that identity against $\chi(\varsigma)(\varsigma - 1/2)^2 \psi(t)$ gives the second line of (17.113). Distributional integration by parts gives the third line. All test functions have compact support, so there are no boundary terms. □

Thus the two-dimensional surface $\log |P_{\mathrm{RVM}}^{\mathrm{ord}}|$ determines the horizontal energy without locating the zeros one at a time. Equivalently, it is the second moment, about the critical line, of the Riesz mass of that surface.

Theorem 17.43 (Horizontal-energy criteria for RH and Lindelöf)**.** *One has*

$$\boxed{\mathrm{RH} \iff E_{\mathrm{RVM}}^{\mathrm{ord}}(T) = 0 \text{ for every } T > 0.} \tag{17.114}$$

Moreover, the Lindelöf hypothesis is equivalent to

$$\boxed{E_{\mathrm{RVM}}^{\mathrm{ord}}(T) = o(\log T) \qquad (T \to \infty).} \tag{17.115}$$

Proof. The first assertion follows because every summand in (17.110) is nonnegative and the unit windows cover all positive rigid ordinates.

For the second assertion, put $n_{\mathrm{RVM}}^{\mathrm{ord}}(T) := \#I_{\mathrm{RVM}}^{\mathrm{ord}}(T)$. The defining relation $\mathcal{N}_{\mathrm{RVM}}(\gamma_k^{\mathrm{RVM}}) = k - 1/2$ gives

$$n_{\mathrm{RVM}}^{\mathrm{ord}}(T) = \mathcal{N}_{\mathrm{RVM}}(T+1) - \mathcal{N}_{\mathrm{RVM}}(T) + O(1) = \frac{1}{2\pi} \log \frac{T}{2\pi} + O(1). \tag{17.116}$$

For every $0 < a < 1/2$,

$$\boxed{a^2 Q^{\mathrm{ord}}_{\mathrm{RVM,a}}(T) \le E^{\mathrm{ord}}_{\mathrm{RVM}}(T) \le a^2 n^{\mathrm{ord}}_{\mathrm{RVM}}(T) + \frac{1}{4} Q^{\mathrm{ord}}_{\mathrm{RVM,a}}(T).} \tag{17.117}$$

The lower bound uses the terms with $|\mathfrak{d}_k^{\mathrm{RVM,ord}}| > a$. For the upper bound, the remaining terms contribute at most a^2 each and the exceptional terms at most $1/4$ each. Consequently

$$E^{\mathrm{ord}}_{\mathrm{RVM}}(T) = o(\log T) \quad \Longleftrightarrow \quad Q^{\mathrm{ord}}_{\mathrm{RVM,a}}(T) = o(\log T) \quad \text{for every fixed } a > 0. \tag{17.118}$$

Indeed the forward implication follows from the lower bound. Conversely, first choose a so small that the first term in the upper bound is an arbitrarily small multiple of $\log T$, and then use the assumed estimate for $Q^{\mathrm{ord}}_{\mathrm{RVM,a}}(T)$.

Backlund's local zero-density criterion states that the Lindelöf hypothesis is equivalent, for every fixed $a > 0$, to

$$N(1/2 + a, T + 1) - N(1/2 + a, T) = o(\log T), \tag{17.119}$$

where zeros are counted with multiplicity [58, Theorem 13.5]. Functional symmetry supplies the matching count to the left of $1/2 - a$. Finally, Lemma 17.1 says that true and rigid ordinates differ by at most a fixed constant. A unit window for either sequence is therefore contained in a fixed-length window for the other sequence; that window is covered by finitely many unit windows. Hence the local $o(\log T)$ property is unchanged by the RVM transport. Combining this observation with (17.118) proves (17.115). □

There is also the exact layer-cake identity

$$\boxed{E^{\mathrm{ord}}_{\mathrm{RVM}}(T) = 2 \int_0^{1/2} a\, Q^{\mathrm{ord}}_{\mathrm{RVM,a}}(T)\, da.} \tag{17.120}$$

Indeed $d^2 = 2 \int_0^{1/2} a \mathbf{1}_{\{|d|>a\}}\, da$ for $|d| < 1/2$. The count $Q^{\mathrm{ord}}_{\mathrm{RVM,a}}(T)$ itself is an argument-principle count of the zeros of $P^{\mathrm{ord}}_{\mathrm{RVM}}$ in the two rectangles outside $|\Re s - 1/2| \le a$. Thus (17.120) is another relation involving only the rigid product.

The distinction between the two criteria is essential. Exact compatibility with RH requires zero horizontal energy in every window. The weaker bound $E^{\mathrm{ord}}_{\mathrm{RVM}}(T) = o(\log T)$ is exactly Lindelöf and still permits infinitely many off-line zeros. Even $E^{\mathrm{ord}}_{\mathrm{RVM}}(T) = O(1)$ would not force RH, since it could allow a bounded number of horizontally displaced zeros in every window.

Define the global critical-line rigid reference by

$$P^{1/2}_{\mathrm{RVM}}(s) := \prod_{k \ge 1} \frac{(s - 1/2)^2 + (\gamma_k^{\mathrm{RVM}})^2}{1/4 + (\gamma_k^{\mathrm{RVM}})^2}, \qquad \widehat{M}^{1/2}_{\mathrm{RVM}}(s) := \frac{G(s)}{P^{1/2}_{\mathrm{RVM}}(s)}. \tag{17.121}$$

17.14. **The vertical resonant accumulation.** The horizontal energy also identifies precisely why a real-axis comparison with the critical rigid reference does not automatically extend uniformly up vertical lines. For a fixed $a > 0$ define the Poisson-smoothed resonant energy

$$\mathcal{R}^{\mathrm{ord}}_{\mathrm{RVM,a}}(T) := \sum_{k \ge 1} \frac{(\mathfrak{d}_k^{\mathrm{RVM,ord}})^2}{a^2 + (T - \gamma_k^{\mathrm{RVM}})^2}. \tag{17.122}$$

The conjugate ordinates give an additional nonresonant term with $T + \gamma_k^{\mathrm{RVM}}$ in place of $T - \gamma_k^{\mathrm{RVM}}$; it is absorbed in the bounded remainders below.

Lemma 17.44 (Local and Poisson-smoothed horizontal energy)**.** *For every fixed $a > 0$,*

$$\boxed{\mathcal{R}^{\mathrm{ord}}_{\mathrm{RVM,a}}(T) = o(\log T) \quad \Longleftrightarrow \quad E^{\mathrm{ord}}_{\mathrm{RVM}}(T) = o(\log T).} \tag{17.123}$$

Unconditionally,

$$\mathcal{R}^{\mathrm{ord}}_{\mathrm{RVM,a}}(T) = O_a(\log(2+T)). \tag{17.124}$$

Proof. For $\gamma_k^{\mathrm{RVM}} \in (T, T+1]$, the kernel in (17.122) is at least $(a^2+1)^{-1}$. Hence

$$E^{\mathrm{ord}}_{\mathrm{RVM}}(T) \le (a^2+1)\mathcal{R}^{\mathrm{ord}}_{\mathrm{RVM,a}}(T), \tag{17.125}$$

which proves one implication.

For the converse, partition the positive rigid ordinates into unit windows. The contribution of the window at vertical distance between m and $m+1$ from T is at most a constant multiple of

$$\frac{E^{\mathrm{ord}}_{\mathrm{RVM}}(T+m) + E^{\mathrm{ord}}_{\mathrm{RVM}}(T-m-1)}{a^2+m^2}. \tag{17.126}$$

If $E^{\mathrm{ord}}_{\mathrm{RVM}}(U) = o(\log U)$, the windows with $|m| \le T/2$ contribute $o(\log T)$ because $\sum_{m\ge 0}(a^2+m^2)^{-1} < \infty$. The remaining windows contribute $o(\log T)$ by the trivial estimate $E^{\mathrm{ord}}_{\mathrm{RVM}}(U) \le n^{\mathrm{ord}}_{\mathrm{RVM}}(U)/4 = O(\log(2+U))$ and the square-summable kernel. The same partition with the trivial estimate in every window proves (17.124). □

Lemma 17.45 (Vertical comparison with the critical rigid reference)**.** *For $s = \varsigma + iT$ with real $T \ge 0$, $\varsigma \ge \Theta + a$ and fixed real $a > 0$,*

$$\boxed{\left|\log\left|\frac{\widehat{M}^{\mathrm{ord}}_{\mathrm{RVM}}(\varsigma+iT)}{\widehat{M}^{1/2}_{\mathrm{RVM}}(\varsigma+iT)}\right|\right| \ll_a 1 + \mathcal{R}^{\mathrm{ord}}_{\mathrm{RVM,a}}(T).} \tag{17.127}$$

Proof. For a paired conjugate factor put

$$h_b(s,g) := \log\left|\frac{(s-b)^2+g^2}{b^2+g^2}\right|. \tag{17.128}$$

The contribution of an off-line functional pair to $\log|P^{\mathrm{ord}}_{\mathrm{RVM}}/P^{1/2}_{\mathrm{RVM}}|$ is

$$h_{1/2+d_j}(s, g^{\mathrm{RVM}}_{j,0}) - h_{1/2}(s, g^{\mathrm{RVM}}_{j,0}) + h_{1/2-d_j}(s, g^{\mathrm{RVM}}_{j,1}) - h_{1/2}(s, g^{\mathrm{RVM}}_{j,1}). \tag{17.129}$$

Taylor's formula with integral remainder separates this expression into

$$d_j\left(\partial_b h_b(s, g^{\mathrm{RVM}}_{j,0}) - \partial_b h_b(s, g^{\mathrm{RVM}}_{j,1})\right)_{b=1/2} \tag{17.130}$$

and a remainder bounded by d_j^2 times the supremum of $|\partial_b^2 h_b|$ along $|b-1/2| \le d_j$. Because the intervals $[g^{\mathrm{RVM}}_{j,0}, g^{\mathrm{RVM}}_{j,1}]$ are disjoint, the total first-order contribution is at most

$$\frac{1}{2}\int_0^\infty \left|\partial_g\partial_b h_b(s,g)\big|_{b=1/2}\right| \, dg \ll_a 1. \tag{17.131}$$

Here the last estimate follows by factoring $(s-b)^2+g^2 = (s-b-ig)(s-b+ig)$: each resulting Cauchy kernel has horizontal distance at least a from its singularity.

The same factorization gives, uniformly on the Taylor segment,

$$|\partial_b^2 h_b(s,g)| \ll_a \frac{1}{a^2+(T-g)^2} + \frac{1}{a^2+(T+g)^2} + \frac{1}{1+g^2}. \tag{17.132}$$

After multiplication by $(\mathfrak{d}^{\mathrm{RVM,ord}}_k)^2$ and summation, the first term is $\mathcal{R}^{\mathrm{ord}}_{\mathrm{RVM,a}}(T)$, the second is bounded because it is nonresonant, and the third is bounded because $\sum_k(\gamma_k^{\mathrm{RVM}})^{-2} < \infty$. Since $\widehat{M}^{\mathrm{ord}}_{\mathrm{RVM}}/\widehat{M}^{1/2}_{\mathrm{RVM}} = P^{1/2}_{\mathrm{RVM}}/P^{\mathrm{ord}}_{\mathrm{RVM}}$, the same estimate holds with the sign reversed and proves (17.127). □

The unconditional consequence of (17.127) is only polynomial comparison with the critical reference. To obtain a subpolynomial comparison from this argument one needs $\mathcal{R}^{\mathrm{ord}}_{\mathrm{RVM,a}}(T) = o(\log T)$. By Lemma 17.44 and Theorem 17.43, that requirement is equivalent to Lindelöf. Thus the moving resonant layer contains exactly the obstruction missed by the bounded real-axis comparison; it cannot be discarded merely because all rigid ordinates are explicit.

17.15. **Poisson geometry and the interpretation of rigid-product plots.** The geometry seen in numerical plots of $P^{\mathrm{ord}}_{\mathrm{RVM}}$ is explained by a local Poisson calculation. Put

$$\varrho_{\mathrm{RVM}}(T) := \mathcal{N}'_{\mathrm{RVM}}(T) = \frac{1}{2\pi}\log\frac{T}{2\pi}. \tag{17.133}$$

Freeze this density near a large height T and replace the local clock by an exact lattice of spacing $1/\varrho_{\mathrm{RVM}}(T)$. If the real parts in that local lattice all equal a fixed value b, the sine-product identity gives, after absorbing the nonresonant conjugate factors and normalization into a smooth term $B_{\mathrm{RVM};\varsigma,\mathrm{b}}(t)$,

$$\begin{aligned}\log|P^{\mathrm{loc}}_{\mathrm{RVM,b}}(\varsigma+it)| &\simeq B_{\mathrm{RVM};\varsigma,\mathrm{b}}(t) + \frac{1}{2}\log\bigl(2\cosh(2\pi\varrho_{\mathrm{RVM}}(T)|\varsigma-b|) - 2\cos\vartheta(t)\bigr)\\ &= B^{\mathrm{sm}}_{\mathrm{RVM};\varsigma,\mathrm{b}}(t) - \sum_{m\ge1}\frac{e^{-2\pi m\varrho_{\mathrm{RVM}}(T)|\varsigma-b|}}{m}\cos\bigl(m\vartheta(t)\bigr),\end{aligned} \tag{17.134}$$

where $\vartheta(t)$ is the unfolded clock phase. This is a frozen-density local model, not an exact global identity for the nonlinear RVM clock. It shows that the first harmonic has envelope

$$\boxed{A_{\mathrm{RVM},1}(T;\varsigma,b) \asymp e^{-2\pi\varrho_{\mathrm{RVM}}(T)|\varsigma-b|} = \left(\frac{T}{2\pi}\right)^{-|\varsigma-b|}.} \tag{17.135}$$

For the global critical-line reference (17.121), the Poisson calculation, combined formally with the Stirling factor in G, predicts for every fixed $\varsigma > 1/2$ the two-sided reference bound

$$|\widehat{M}^{1/2}_{\mathrm{RVM}}(\varsigma+iT)| \asymp_\varsigma 1. \tag{17.136}$$

This is recorded as the rigid-reference prediction. A global proof with uniform upper and lower constants is not supplied here, and none of the proved criteria in this section uses (17.136) as an input.

This formula accounts for the two numerical surfaces associated with the rigid product. When $d = 0$, every displayed zero has real part $1/2$, $\Theta = 1/2$, and the normalized surface is a regular corrugation in the unfolded vertical coordinate whose amplitude decreases with the distance from the critical line. All base points lie on that line and $P^{\mathrm{ord}}_{\mathrm{RVM}}(s) = P^{\mathrm{ord}}_{\mathrm{RVM}}(1-s)$.

When five functional pairs are displaced by $d = 0.45$, their real parts are 0.95 and 0.05, ten displayed zeros lie off the critical line, and $\Theta = 0.95$. The finite displaced block changes the distance from a fixed cut to the nearby zeros. The previously regular crests acquire different local heights, deep cusps appear near the displaced zeros, and the modulation is confined to the affected portion of the clock. Because the two members of each functional pair occupy distinct consecutive RVM ordinates, the transported divisor is no longer invariant under $s \mapsto 1-s$, exactly as Proposition 17.40 predicts.

Two limitations prevent this picture from becoming an automatic proof of RH. First, a single modulus cut at $\varsigma = \varsigma_0$ sees to leading order only $|\varsigma_0 - \beta_k|$. Reflection $\beta_k \mapsto 2\varsigma_0 - \beta_k$ preserves that distance, so regularity of one cut cannot determine on which side the zero lies. Second, the alternating configuration

$$\beta_{2j-1} = \frac{1}{2} + d, \qquad \beta_{2j} = \frac{1}{2} - d \tag{17.137}$$

has the same rigid clock and $\Theta = 1/2 + d$, but is periodic with a doubled cell in the unfolded coordinate. It can therefore be regular without all real parts being equal. The finite perturbation in the displayed plot is a localized defect, not a slowly varying version of (17.137).

After removing the smooth carrier, the correctly normalized first-harmonic diagnostic is

$$\mathcal{D}_{\mathrm{RVM};\varsigma_0}(T) := -\frac{\log |A_{\mathrm{RVM},1}(T;\varsigma_0)|}{2\pi \varrho_{\mathrm{RVM}}(T)} \simeq |\varsigma_0 - \beta(T)|. \tag{17.138}$$

One cut recovers only a distance. Two distinct exact distances $\ell_j^{\mathrm{cut}} := |\varsigma_j - \beta|$, $\varsigma_1 \neq \varsigma_2$, give the algebraic identity

$$\boxed{\beta = \frac{\varsigma_2^2 - \varsigma_1^2 - (\ell_2^{\mathrm{cut}})^2 + (\ell_1^{\mathrm{cut}})^2}{2(\varsigma_2 - \varsigma_1)}.} \tag{17.139}$$

If the measured diagnostics satisfy $\mathcal{D}_j = \ell_j^{\mathrm{cut}} + \eta_j$, then the reconstructed value $\widetilde{\beta}$ obtained by replacing ℓ_j^{cut} with $\mathcal{D}_j$ obeys

$$|\widetilde{\beta} - \beta| \leq \frac{2\ell_1^{\mathrm{cut}}|\eta_1| + 2\ell_2^{\mathrm{cut}}|\eta_2| + \eta_1^2 + \eta_2^2}{2|\varsigma_2 - \varsigma_1|}. \tag{17.140}$$

Thus the two-cut diagnostic is stable only to the extent that the individual cut errors η_j are controlled; the symbol $\simeq$ is not used as an exact reconstruction formula. The full two-dimensional surface is stronger still: by (17.112), its logarithmic singularities are located exactly at $(\beta_k, \gamma_k^{\mathrm{RVM}})$. Consequently the plots read the horizontal displacements already encoded in a given $P_{\mathrm{RVM}}^{\mathrm{ord}}$; by themselves they do not supply an independent bound on those displacements.

17.16. **The exact umbilical identity with the Mertens function.** The Mertens function enters before any explicit formula over zeros. For $\Re s > 1$, partial summation gives

$$\widehat{M}(s) := s\int_1^\infty M(x)x^{-s-1}\,dx = \sum_{n\geq 1}\frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}. \tag{17.141}$$

Combining this identity with Definition 17.38 gives the central relation

$$\boxed{\widehat{M}_{\mathrm{RVM}}^{\mathrm{ord}}(s) = \mathscr{T}_{\mathrm{RVM}}^{\mathrm{ord}}(s)\widehat{M}(s) = \mathscr{T}_{\mathrm{RVM}}^{\mathrm{ord}}(s)\, s\int_1^\infty M(x)x^{-s-1}\,dx.} \tag{17.142}$$

Thus M is not inferred from a similarity between graphs: it occurs exactly as one factor of the transformed construction.

Definition 17.46 (Unsmoothed transported object)**.** Whenever the following inverse exists as an ordinary function, define

$$M_{\mathrm{RVM}}^{\mathrm{ord}}(x) := \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\widehat{M}_{\mathrm{RVM}}^{\mathrm{ord}}(s)\frac{x^s}{s}\,ds, \qquad c > 1. \tag{17.143}$$

Without an inversion theorem in a specified function or distribution space, (17.143) denotes only a formal inverse-Mellin symbol. This convention prevents a formal residue expansion from being mistaken for a pointwise-defined function.

Assume that the inverse-Mellin distributions in question exist in a common space and that their convolution is defined there (for example, via a weighted Fourier multiplier realization after $x = e^y$). Let $\kappa_{\mathrm{RVM}}^{\mathrm{ord}}$ be the inverse multiplier kernel of $\mathscr{T}_{\mathrm{RVM}}^{\mathrm{ord}}$. In the multiplicative convolution notation

$$(f *_{\mathrm{Mel}} g)(x) := \int_0^\infty f(x/u)g(u)\,\frac{du}{u}, \tag{17.144}$$

the Mellin convolution theorem turns (17.142) into the distributional identity

$$\boxed{M_{\mathrm{RVM}}^{\mathrm{ord}} = \kappa_{\mathrm{RVM}}^{\mathrm{ord}} *_{\mathrm{Mel}} M.} \tag{17.145}$$

If the inverse multiplier is admissible on the same space, then also

$$\boxed{M = (\kappa_{\mathrm{RVM}}^{\mathrm{ord}})^{[-1]} *_{\mathrm{Mel}} M_{\mathrm{RVM}}^{\mathrm{ord}},} \qquad \mathcal{M}((\kappa_{\mathrm{RVM}}^{\mathrm{ord}})^{[-1]})(s) = \mathscr{T}_{\mathrm{RVM}}^{\mathrm{ord}}(s)^{-1}. \tag{17.146}$$

Equations (17.142)– (17.146) are the precise relation between M and $M_{\mathrm{RVM}}^{\mathrm{ord}}$. The forward identity is exact in every common Mellin-distribution realization satisfying the stated existence and convolution premises. The reverse pointwise identity requires a multiplier theorem and is not silently assumed.

The exact transform of their difference is equally simple:

$$\widehat{(M - M_{\mathrm{RVM}}^{\mathrm{ord}})}(s) = \frac{1 - \mathscr{T}_{\mathrm{RVM}}^{\mathrm{ord}}(s)}{\zeta(s)}. \tag{17.147}$$

At a genuine zero which is not a transported location, $\mathscr{T}_{\mathrm{RVM}}^{\mathrm{ord}}$ vanishes and hence $1 - \mathscr{T}_{\mathrm{RVM}}^{\mathrm{ord}} = 1$. Therefore the corresponding pole of $1/\zeta$ survives in (17.147). This proves that vertical displacements need not make $M - M_{\mathrm{RVM}}^{\mathrm{ord}}$ smaller in exponent. More precisely, if for every $\eta > 0$ there is a genuine zero with $\beta_k > \Theta - \eta$ which is not a transported location, then the pole abscissa of (17.147) is also Θ.

17.17. **What the poles prove, and what they do not.** For an ordinary function F with normalized Mellin transform, put

$$\vartheta(F) := \inf\{a : \ F(x) = O_\varepsilon(x^{a+\varepsilon}) \text{ for every } \varepsilon > 0\}. \tag{17.148}$$

Proposition 17.47 (The unconditional pole obstruction)**.** *Assume that the ordinary function* $M_{\mathrm{RVM}}^{\mathrm{ord}}$ *exists, that* (17.94) *is its normalized Mellin transform in some right half-plane, and that the usual Mellin continuation implication is valid there. Then*

$$\boxed{\vartheta(M_{\mathrm{RVM}}^{\mathrm{ord}}) \geq \Theta.} \tag{17.149}$$

Under the noncoincidence condition stated after (17.147)*, the same argument gives*

$$\vartheta(M - M_{\mathrm{RVM}}^{\mathrm{ord}}) \geq \Theta. \tag{17.150}$$

Proof. If $M_{\mathrm{RVM}}^{\mathrm{ord}}(x) = O_\varepsilon(x^{a+\varepsilon})$, its normalized Mellin transform is holomorphic in $\Re s > a$. Proposition 17.39 exhibits poles with real parts arbitrarily close to Θ, forcing $a \geq \Theta$. The difference follows from (17.147) in the same way. □

The reverse inequality $\vartheta(M_{\mathrm{RVM}}^{\mathrm{ord}}) \leq \Theta$ does not follow from the locations of the poles. It is a growth and inversion statement. In particular, the identity

$$\vartheta(M_{\mathrm{RVM}}^{\mathrm{ord}}) = \Theta \tag{17.151}$$

must not be inserted between two equivalences until that upper bound has been proved. This distinction is exactly analogous to the distinction between locating singularities of a Mellin transform and controlling its inverse on vertical lines.

17.18. **A Gaussian-regularized criterion with explicit inversion.** The Gaussian regularization separates divisor location from unsmoothed pointwise inversion, but the transform convention must be kept explicit. The normalized unilateral transform used for Mertens-type functions is

$$\widehat{F}(s) := s \int_1^\infty F(x) x^{-s-1}\, dx, \tag{17.152}$$

whereas the bilateral Mellin transform used below is

$$\mathcal{M}_{\mathrm{bil}} f(s) := \int_0^\infty f(x) x^{-s-1}\, dx. \tag{17.153}$$

They are not interchangeable without the factor s and without controlling the part $0 < x < 1$.

The zero count $N_\zeta(T) = O(T\log T)$ and the bounded-displacement lemma give the following standard finite-order estimate on any fixed vertical line at positive distance from the zero and pole divisors: for every $0 < \delta < 1$,

$$\log^+ |\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}(\sigma+it)| + \log^+ |\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}(\sigma+it)^{-1}| = O_{\sigma,\delta}((2+|t|)^{1+\delta}). \tag{17.154}$$

The same type of bound is assumed/used for $\widehat{M}^{\mathrm{ord}}_{\mathrm{RVM}}$ on the closed zero-free strips through which a contour is moved. The restriction $\delta < 1$ is what makes the factor e^{-t^2} dominate the vertical growth.

Definition 17.48 (Gaussian-regularized transported baseline)**.** For $x > 0$ and $c > 1$, define

$$M^{\mathrm{ord}}_{\mathrm{RVM,G}}(x) := \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} e^{s^2}\,\widehat{M}^{\mathrm{ord}}_{\mathrm{RVM}}(s)\frac{x^s}{s}\,ds. \tag{17.155}$$

The factor $1/s$ is part of the inversion of the normalized unilateral transform. On lines $c > 1$ the integral is absolutely convergent.

Lemma 17.49 (Gaussian bilateral inversion)**.** *Let $H(s) := e^{s^2}\widehat{M}^{\mathrm{ord}}_{\mathrm{RVM}}(s)/s$. Suppose that in a vertical strip $0 < a < b$ the function H is holomorphic and satisfies the pre-Gaussian finite-order bound $|e^{-s^2}H(s)| \le \exp(C(1+|\Im s|)^{1+\delta})$, with $0 < \delta < 1$, uniformly on every closed substrip. Then the inverse integral on any line $c \in (a,b)$ defines the same function $M^{\mathrm{ord}}_{\mathrm{RVM,G}}(x)$, the horizontal sides of the displacement rectangles vanish, and whenever*

$$M^{\mathrm{ord}}_{\mathrm{RVM,G}}(x) = O(x^{a+\eta}) \quad (x\to\infty), \qquad M^{\mathrm{ord}}_{\mathrm{RVM,G}}(x) = O(x^{b-\eta}) \quad (x\downarrow 0) \tag{S17.22}$$

for some $\eta > 0$, one has

$$\mathcal{M}_{\mathrm{bil}} M^{\mathrm{ord}}_{\mathrm{RVM,G}}(s) = H(s) \tag{17.156}$$

throughout $a+\eta < \Re s < b-\eta$, when this strip is nonempty.

Proof. On a vertical line, $|e^{(\sigma+it)^2}| = e^{\sigma^2-t^2}$, which dominates $\exp(O(|t|^{1+\delta}))$ because $0 < \delta < 1$. The same domination makes the horizontal sides of a rectangular contour tend to zero and proves line independence whenever no pole is crossed. The two endpoint bounds make the bilateral Mellin integral absolutely convergent in the stated strip. On a common vertical line, put $x = e^y$. The inverse is the Fourier inverse of the integrable Gaussian-decaying function $H(c+it)$, and the endpoint bounds make $e^{-cy}M^{\mathrm{ord}}_{\mathrm{RVM,G}}(e^y)$ integrable. Fourier inversion and uniqueness therefore identify its Fourier transform with $H(c+it)$; holomorphy extends the equality through the common strip. This avoids an unjustified absolute interchange over the entire multiplicative group. □

Theorem 17.50 (Gaussian transported-divisor criterion)**.** *Assume the vertical finite-order bounds required in Lemma 17.49. Then the following are equivalent:*

(i) *RH;*
(ii) *for every $\varepsilon > 0$,*

$$M^{\mathrm{ord}}_{\mathrm{RVM,G}}(x) = O_\varepsilon(x^{1/2+\varepsilon}) \qquad (x\to\infty). \tag{17.157}$$

This criterion uses the genuine multiplicities and the transported pairs $(\beta_k, \gamma_k^{\mathrm{RVM}})$ and is therefore a Category 3 divisor criterion, not an independent construction of the real parts β_k.

Proof. Assume RH and fix $\varepsilon > 0$. All nontrivial poles of $\widehat{M}^{\mathrm{ord}}_{\mathrm{RVM}}$ lie on $\Re s = 1/2$. Shift (17.155) from $c > 1$ to $\Re s = 1/2+\varepsilon$. No nontrivial pole is crossed. The point $s = 1$ is a zero, not a pole, of $1/\zeta$ in the factorization $\widehat{M}^{\mathrm{ord}}_{\mathrm{RVM}} = \mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}/\zeta$; the trivial poles coming from negative even zeta zeros and the point $s = 0$ lie to the left of the shifted line. Lemma 17.49 therefore justifies the displacement and gives $O_\varepsilon(x^{1/2+\varepsilon})$.

Conversely assume the displayed bound for every $\varepsilon > 0$. Moving the original line to any fixed $A > 1$ inside the pole-free half-plane $\Re s > 1$ gives, by the same Gaussian estimate, $M^{\mathrm{ord}}_{\mathrm{RVM,G}}(x) = O_A(x^A)$ as $x \downarrow 0$. Hence for each $\varepsilon > 0$ and $A > 1$ the bilateral transform

$$\int_0^\infty M^{\mathrm{ord}}_{\mathrm{RVM,G}}(x) x^{-s-1}\, dx \tag{S17.23}$$

is holomorphic in $1/2+\varepsilon < \Re s < A$. By Lemma 17.49 it equals $e^{s^2}\widehat{M}^{\mathrm{ord}}_{\mathrm{RVM}}(s)/s$ first in the common strip $1 < \Re s < A$ and hence throughout by analytic continuation. Because e^{s^2}/s is nonzero in $\Re s > 1/2$, the transported transform has no nontrivial pole there. Proposition 17.39 then gives $\beta_k \le 1/2$ for every genuine zero. Functional symmetry sends β_k to $1-\beta_k$, forcing $\beta_k = 1/2$ and RH. □

The theorem is deliberately regularized. Removing the Gaussian requires unsmoothed vertical control and a pointwise Mellin multiplier theorem; neither follows from the location of the transported poles.

17.19. **The unsmoothed transfer theorem that would complete the route.** The relation with M becomes pointwise if the transport and its inverse act boundedly on a suitable weighted Mellin space. The following statement records a sufficient hypothesis rather than concealing it in the notation.

Hypothesis 17.51 (Bidirectional unsmoothed Mellin transfer)**.** For every $\varepsilon > 0$, assume that the inverse multiplier kernels exist on a specified common Mellin space. The distributions with Mellin transforms $\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}$ and $(\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}})^{-1}$ act boundedly by multiplicative convolution on the weighted supremum space

$$\mathcal{B}_\varepsilon := \{f : (0,\infty) \to \mathbb{C} :\ f(x) = 0\ (0 < x < 1), \sup_{x \ge 1} x^{-1/2-\varepsilon}|f(x)| < \infty\}, \tag{17.158}$$

and the distributional identities (17.145) and (17.146) hold pointwise there.

A stronger, concrete sufficient condition is that the two inverse-Mellin kernels are signed measures whose weighted total variations are finite on the multiplicative group. Under Hypothesis 17.51,

$$\boxed{M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \quad \Longleftrightarrow \quad M^{\mathrm{ord}}_{\mathrm{RVM}}(x) = O_\varepsilon(x^{1/2+\varepsilon}).} \tag{17.159}$$

Together with Littlewood's criterion, this would give

$$\boxed{\mathrm{RH} \quad \Longleftrightarrow \quad \Theta = \frac{1}{2} \quad \Longleftrightarrow \quad M^{\mathrm{ord}}_{\mathrm{RVM}}(x) = O_\varepsilon(x^{1/2+\varepsilon}) \text{ for every } \varepsilon > 0.} \tag{17.160}$$

The proof of (17.159) is immediate from the two bounded convolution operators. The substantive open step is to prove the hypothesis for the explicit product (17.95). This hypothesis already has nearly the full analytic strength of the desired conclusion and must not be advertised as an easier consequence of the bounded displacement. The earlier shortcut of testing the operator on $f \equiv 1$ is not valid. Although $f \equiv 1$ lies in $\mathcal{B}_\varepsilon$ and its *normalized unilateral* transform is 1, the convolution hypothesis is formulated on the multiplicative group and the norm of $\mathcal{B}_\varepsilon$ only controls $x \ge 1$, while the zero extension is an explicit support convention. Boundedness of that operator on this supremum space does not, by itself, identify a holomorphic Mellin multiplier in the half-plane. To deduce holomorphy of $\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}$ and its inverse from operator boundedness one would need an additional Mellin spectral-multiplier theorem, including compatibility with the imposed $0 < x < 1$ support convention. Such a theorem is not assumed silently here.

Thus (17.159) is a precise sufficient transfer statement, but proving its hypothesis is itself an RH-strength problem about the explicit multiplier. In particular it is not implied by the known

estimate $\gamma_k - \gamma_k^{\mathrm{RVM}} = O(1)$, because a bounded ordinate shift produces the phase factor

$$x^{\beta_k+i\gamma_k} - x^{\beta_k+i\gamma_k^{\mathrm{RVM}}} = x^{\beta_k+i\gamma_k^{\mathrm{RVM}}}\left(e^{i(\gamma_k-\gamma_k^{\mathrm{RVM}})\log x} - 1\right), \tag{17.161}$$

whose parenthesis can have modulus 2.

There is also a rigorous Hilbert-space version. Suppose ordinary functions M and $M^{\mathrm{ord}}_{\mathrm{RVM}}$ vanish for $0 < x < 1$, belong to the displayed weighted L^2 space, and realize the transform identity in the Mellin–Plancherel sense. Suppose further that, on a line $\Re s = \sigma$, one has

$$0 < c_\sigma \le |\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}(\sigma + it)| \le C_\sigma < \infty \qquad (t \in \mathbb{R}). \tag{17.162}$$

For the normalized transform one has

$$\int_1^\infty |M(x)|^2 x^{-2\sigma-1}\,dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{|\widehat{M}(\sigma+it)|^2}{|\sigma+it|^2}\,dt, \tag{17.163}$$

and the identical formula for $M^{\mathrm{ord}}_{\mathrm{RVM}}$. Thus the factor $1/|s|^2$ required by normalized Mellin–Plancherel is explicit. Using (17.142), it cancels on the two sides of the multiplier comparison and gives

$$c_\sigma^2 \int_1^\infty |M(x)|^2 x^{-2\sigma-1}\,dx \le \int_1^\infty |M^{\mathrm{ord}}_{\mathrm{RVM}}(x)|^2 x^{-2\sigma-1}\,dx \le C_\sigma^2 \int_1^\infty |M(x)|^2 x^{-2\sigma-1}\,dx. \tag{17.164}$$

This is Plancherel applied to $e^{-\sigma y}M(e^y)$ and $e^{-\sigma y}M^{\mathrm{ord}}_{\mathrm{RVM}}(e^y)$ on $y \ge 0$. Estimate (17.162) is again a hypothesis, not a consequence of the pole locations.

17.20. **The residue-preserving model and the scope of the excursion theory.** For comparison only, if all zeros are simple and a residue series is given a specified summation procedure, one may form

$$M^{\mathrm{ord,res}}_{\mathrm{RVM}}(x) = Q^{\mathrm{ord,res}}_{\mathrm{RVM}}(x) + 2\Re \sum_{k\ge1} \frac{x^{\beta_k+i\gamma_k^{\mathrm{RVM}}}}{(\beta_k + i\gamma_k^{\mathrm{RVM}})\zeta'(\rho_k)}. \tag{17.165}$$

It has the same transported pole locations but generally different residues from $M^{\mathrm{ord}}_{\mathrm{RVM}}$. Thus both models have the same pole obstruction, while only $M^{\mathrm{ord}}_{\mathrm{RVM}}$ has the exact multiplier identity (17.142) and avoids ζ'.

Finally, the theorems of this paper split into two classes. Mellin continuation, pole obstructions, weighted L^2 arguments, and abstract energy criteria can be formulated for $M^{\mathrm{ord}}_{\mathrm{RVM}}$ once their analytic hypotheses are verified. In contrast, the excursion identities using jumps in $\{-1, 0, 1\}$, zero runs, the divisor recursion, the Catalan selector, and the arithmetic orientation fibre are properties of the genuine Mertens path. They do not transfer merely from (17.142). The transported baseline is therefore an analytic control and a possible multiplier route to RH, not a replacement for the integer excursion path.

18. Conclusions

18.1. **Status ledger.** The conclusions use the same four-way classification as the introduction.

1. Exact algebraic identities. The excursion decomposition, bridge and Fejér identities, prefix/channel formulae, reflection-fibre selection, finite RVM inversion, exact symmetric Hadamard normalization, the route-independent distributional endpoint formula, the true-β transport identity and Riesz/energy identities are exact statements about the objects defined in the paper. This class also includes the consecutive-block transport identity in Audit proposition A3 and the exact normalization defect $C_{\mathrm{RVM}}(\kappa)$ in Audit proposition A4.

2. Theorems proved using published results. The horizon localization and return bounds use Kaczorowski–Pintz after the transform verification in Lemma 9.5; the exact logarithmic exponents use Pintz [45]; and the short-interval and squarefree-pair consequences use the stated Matomäki–Teräväinen and Mirsky theorems with their strict exponent ranges. The quadratic-displacement

asymptotic $A^{\mathrm{tr}}_{\mathrm{RVM}}(T) \sim T \log\log T/(\pi \log T)$ uses Selberg's unconditional second moment of $S(t)$; the density-one logarithmic interpolation then follows from that asymptotic and the explicit Taylor majorant.

3. Equivalent criteria encoding genuine zeros. The endpoint identity $\zeta_1 = \zeta$ of the critical-line ordinate homotopy, the true-β RVM energy criteria, the Gaussian transported-divisor criterion, and the Hamburger endpoint condition (D) are exact equivalences only after the genuine zero divisor or its ordinates have been inserted. They reorganize the RH obstruction but do not constitute an independent proof of RH. The fixed real part $1/2$ in the synthetic RVM construction is not itself an RH assumption: if RH fails, the critical-line endpoint is $\zeta_{\mathrm{line}} \neq \zeta$. The equivalence becomes an RH premise only if the endpoint equality is invoked without an independent proof.

4. Open hypotheses and route-completion targets. These include the one-excursion sublinear first-passage estimate, the multiscale Fejér upper bound, the quantitative Lambert-clock square-root and unit-cell estimates, the uniform stationary-phase remainder for the new ordinate-homotopy expansion, the continuous-to-atomic endpoint problem for ζ_{line}, positivity of the sampled RVM weights, the weighted-variation estimate for the integer-step defect together with its Mellin–Stieltjes grid correction, the asserted vertical limit T_{RVM}, and the unsmoothed bidirectional Mellin multiplier theorem. Numerical data remain evidence about finite objects only.

Accordingly, the manuscript does not claim a new unconditional proof of RH, nor a new unconditional RH equivalence independent of the genuine zero divisor. Its rigorous contribution is the collection of exact excursion identities and reductions, together with the consequences obtained from published oscillation and short-interval theorems; the RVM section separates those results from its still-open analytic continuation and inversion tasks.

18.2. **What the paper does.** The paper replaces the single function M by the sequence of its excursions, and asks which statements about M can be read, and which can be localized, excursion by excursion. Three facts make this possible. First, an excursion is a bridge: it starts and ends at 0, so its quadratic mass is a lag-weighted sum of Möbius correlations internal to it (Lemma 5.1), and its height and length bound its linear and quadratic masses sharply (Proposition 5.6). Second, excursions are disjoint intervals, and disjoint intervals that grow by a factor λ are few: at most $\log X/\log\lambda + O(1)$ of them start below X (Lemma 9.4). Third, the oscillation theorems of Kaczorowski–Pintz and Pintz give, respectively, the number and the exponent of the large values of M. Every main result is obtained by combining two of these three facts.

18.3. **Horizon localization, area and arithmetic selection.** The horizon-localization theorem (Theorem 9.15) uses the extremal-frequency form of Kaczorowski–Pintz. If RH is false, the relevant frequency parameter is the first ordinate γ_Θ on the extremal boundary (or ∞ if the boundary is not attained). Since all zeros up to $T_0 = 3\cdot 10^{12}$ are on the critical line, the contradiction argument has at least $(T_0/\pi - o(1))\log X$ large-witness excursions. Consequently RH is already decided on every fixed linear horizon λx_0 with

$$\lambda > \lambda_0 := e^{\pi/T_0}. \tag{S18.1}$$

The older $\lambda_* = e^{\pi/\gamma_1}$ is retained as the uniform benchmark coming from Corollary 4 of [25]; it is not the best explicit linear factor once the extremal theorem and the verified zero height are used. The corresponding prefix dependency cutoff is $\chi_0 = \lambda_0/2$.

The surplus and first-arrival calculations remain structural. They identify how the disjoint-interval budget, record retrogression and centered adaptive correlations encode the position of a witness, but they are no longer needed merely to pass below the uniform benchmark λ_*. Proposition 9.48 sharpens the centered record calculation in the genuinely linear regime by combining its mandatory logarithmic contribution with half of the exact record-return cost. Proposition 9.50 turns post-horizon chord deviation into logarithmic surplus, while Proposition 9.52 shows that witness overshoot, return tail and inter-bad gaps all draw from the same critical logarithmic budget.

The drifted-walk reference calculation indicates that neither centered normalization is expected to close the KP-scale problem merely by its typical size.

The exact area exponent (Corollary 8.13) identifies the growth of the skeleton area as $x^{1+\Theta/2+o(1)}$. The upper bound is a concavity argument; the lower bound needs the inequality $B_1 \le 2(m^*)^{1/2}\mathcal{A}_{\mathrm{abs}}(v^*)$, the exponent of B_1 from Pintz's theorem and the return localization of Corollary 9.11. Together these give that the area bound $x^{5/4+\varepsilon}$ is equivalent to RH.

The analysis of the localized window (Section 11) shows that the RH-equivalent part of an excursion is an explicit linear function of μ on the prefix up to χu, $\chi > \chi_0 = \frac{1}{2}e^{\pi/T_0}$; it decomposes this function into scale channels, proves a channel-descent inequality with an explicit geometric loss, and splits M into a smooth part and a large-prime part which, in the range of the split considered, have size $(1+o(1))t/\log^2 t$ and opposite signs (Proposition 11.61).

The weighted correlation identity at zeros, combined with averaged Chowla and with Pintz's exponent (Theorem 12.6), locates the signed weighted correlation total representing $\sum_{n<v} M(n)^2$ at lags larger than v^α for every $\alpha < \Theta$. On the reflection fibre (Section 10), divisor equations reduce the 2^J orientations compatible with the profile M^2 to at most $2N$, while Theorem 10.4 shows that a fixed-shift correlation cannot be controlled uniformly by $o(N)$ divisor rows.

18.4. **Structural identities.** The skeleton and its area, energy and inflection formulas (Sections 3–4), the bridge identity $Q_\nu = -\sum_h hC_h^{(\nu)}$ with its sharp bounds, the four counting lemmas for disjoint intervals (Lemmas 9.1–9.4), the dipole identity (Proposition 6.1) and the positivity of the excursion blocks $D_I(\varsigma)$ each follow in a few lines from the definitions. Their role is structural: they turn statements about M into statements about a sequence of disjoint bridges, to which counting and the oscillation theorems can be applied.

18.5. **Equivalent criteria and transported estimates.** The L^1, L^2, energy and correlation criteria (Proposition 8.1, Theorems 8.6 and 8.9) are equivalent to Littlewood's criterion; here they are derived from the excursion decomposition, and the energy criterion is squeezed between the L^1 mass and the maximum. The exponents of $\sum|M|$ and $\sum M^2$ are immediate consequences of Pintz's theorem. The linear window identity follows from $\mu * 1 = \delta$, and the bilinear identity is structurally analogous to the $k = 2$ Heath–Brown/Vaughan truncation pattern. The results on correlations in excursion blocks transport published short-interval and averaged-Chowla theorems to intervals selected by M; the additional ingredient specific to the present formulation is that the selected intervals are completed excursions.

18.6. **The pure critical RVM arithmetic system and the critical-line path.** Section 17 separates a genuinely synthetic arithmetic system from both the critical-line endpoint path and the true-β transport. The midpoint Lambert clock defines $P_{\mathrm{RVM}}^{\mathrm{crit}}$ and ζ_{RVM} with nontrivial zeros exactly $1/2 \pm i\gamma_k^{\mathrm{RVM}}$; the critical real part is an input, not a conclusion about ζ. The explicit formula defines J_{RVM}, and Theorem 17.27 proves that its zero sum converges uniformly on compact x-ranges. Finite Möbius inversion defines π_{RVM}, while Hypothesis 17.5 now records only the stronger quantitative square-root target. Under the stated unit-cell and Mellin–Stieltjes identification hypotheses, sampling π_{RVM} on the integers gives the weights c_n^{RVM} and an integer-supported Euler object with the same zero divisor in $\Re s > 0$ up to a nonvanishing holomorphic factor. The remaining step needed for a positive 0/1 integer-prime realization is the cell-remainder Hypothesis 17.12; it is not silently assumed. On the Mertens side, $\widehat{M}_{\mathrm{RVM}}^{\mathrm{crit}} = 1/\zeta_{\mathrm{RVM}}$ has pole abscissa $1/2$ by construction and is related exactly to M by the quotient $\mathscr{Q}_{\mathrm{RVM}} = \zeta/\zeta_{\mathrm{RVM}}$. The integer prime defect $E_{\mathrm{RVM}}(N) = \pi(N) - \pi_{\mathrm{RVM}}(N)$ gives the corresponding exact prime-side comparison.

The critical-line homotopy of Subsection 17.9 moves γ_k^{RVM} continuously to the genuine ordinates γ_k while keeping every nontrivial zero on $\Re s = 1/2$. The family preserves the functional equation, pole and trivial-zero structure, and its explicit-formula side is C^1 in $\mathcal{D}'$. The RVM endpoint is an ordinary continuous function; the genuine-ordinate endpoint has a precise convergence obstruction

$D(T;u)$, and under the endpoint premise **P** it agrees with the classical prime-power count away from its jumps. Proposition 17.31 shows that the smoothed endpoint difference depends only on $N_\zeta - N_{\mathrm{RVM}}$ and not on the chosen path. Theorem 17.24 states explicitly that **P** is equivalent to RH, so this chain is a Category 3 reformulation rather than a hidden proof. The stationary-phase resonance formula is kept formal until a remainder uniform in the harmonic index and in x is supplied.

The monotone matching of the two ordinate lists has an additional unconditional consequence. Audit proposition A3 proves, with multiplicities and endpoint errors included,

$$\sum_{\gamma_k^{\mathrm{RVM}} \le T} (\gamma_k - \gamma_k^{\mathrm{RVM}})^2 \sim \frac{T \log\log T}{\pi \log T}. \tag{S18.RVM}$$

Audit proposition A4 then shows that the path logarithm is asymptotically affine outside a set of natural density zero only after the deterministic correction $C_{\mathrm{RVM}}(\kappa)$ is subtracted. The uncorrected normalized product is not an exact logarithmic geodesic. Under **P** and the separately assumed vertical limit for ζ_{RVM}, Audit proposition A5 gives loss of vertical boundedness for every fixed $\kappa > 0$. This last statement is conditional on both premises and is not an unconditional RH argument.

18.7. **The rigid-ordinate transported baseline.** Section 17 introduces a second, analytic use of the Riemann–von Mangoldt clock. Its canonical product $P^{\mathrm{ord}}_{\mathrm{RVM}}$ retains the ordered real parts β_k and multiplicities of the genuine zeros and replaces only γ_k by γ_k^{RVM}. The identity

$$\widehat{M}^{\mathrm{ord}}_{\mathrm{RVM}} = \frac{G}{P^{\mathrm{ord}}_{\mathrm{RVM}}} = \mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}} \widehat{M} \tag{S18.2}$$

is exact, so the relation to the genuine Mertens function is not inferred from numerical similarity. The nontrivial pole abscissa remains Θ, and $P^{\mathrm{ord}}_{\mathrm{RVM}}(s) = P^{\mathrm{ord}}_{\mathrm{RVM}}(1-s)$ holds exactly under RH. With an oriented assignment inside every split functional pair, the real-axis reflection defect is nonnegative and vanishes exactly under RH. Independently of that orientation, the Riesz measure of $\log|P^{\mathrm{ord}}_{\mathrm{RVM}}|$ recovers the nonnegative local energy $E^{\mathrm{ord}}_{\mathrm{RVM}}(T) = \sum(\beta_k - 1/2)^2$ exactly. Its vanishing in every window is equivalent to RH, whereas $E^{\mathrm{ord}}_{\mathrm{RVM}}(T) = o(\log T)$ is equivalent to Lindelöf through Backlund's local zero-density criterion. The Poisson-smoothed resonant energy $\mathcal{R}^{\mathrm{ord}}_{\mathrm{RVM,a}}(T)$ is sublogarithmic exactly in the same regime. It controls the vertical comparison with the critical rigid reference and shows why a bounded real-axis comparison does not extend uniformly past a moving layer containing $\asymp \log T$ rigid ordinates per unit height.

The local Poisson model explains why the critical-line rigid product has a regular decaying corrugation and why finitely many displaced pairs create cusps and amplitude modulation. A single fixed modulus cut measures only the distance $|\varsigma - \beta_k|$ and cannot recover its side; an alternating pair $1/2+d, 1/2-d$ also gives a regular doubled cell with $\Theta = 1/2+d$. Two cuts remove that ambiguity, while the complete surface recovers the horizontal divisor through its logarithmic singularities. These observations diagnose a given $P^{\mathrm{ord}}_{\mathrm{RVM}}$ but do not independently bound its real parts.

Subject to the explicit finite-order vertical-growth hypotheses in Theorem 17.50, a fixed Gaussian regularization gives an RH-equivalent divisor criterion. The corresponding unsmoothed equivalence is reduced to a bidirectional Mellin-multiplier theorem for $\mathscr{T}^{\mathrm{ord}}_{\mathrm{RVM}}$ and its inverse. That hypothesis is explicitly recognized as an RH-strength analytic assertion: operator boundedness alone does not permit the earlier shortcut based on the constant function; a separate Mellin spectral-multiplier theorem would be required to infer half-plane holomorphy. The section distinguishes this unproved upper-transfer estimate from the unconditional lower obstruction supplied by the poles.

18.8. **Limits of the method and open directions.** Several negative results delimit the excursion mechanisms: pointwise channel descent loses a fixed factor per step, termwise contraction of the exact kernel gives nothing beyond the unit-step cone, the relaxed divisor class admits ballistic

paths, and geometric inputs alone do not force positional surplus. The verified-height refinement also shows that the fixed linear constant is no longer the principal bottleneck: under failure of RH the extremal Kaczorowski–Pintz density and the verified height T_0 reduce the linear horizon to every fixed $\lambda > \lambda_0 = e^{\pi/T_0}$.

The main forward direction is therefore the one-excursion problem. A completed excursion contains more information than an open prefix because it has one sign until its first return to zero. Proposition 11.31 and Corollary 11.32 turn this into the exact past-only quantity

$$P_u(t) = \sum_{k \le q_u(t)} M(t/k), \qquad \sigma P_u(t) = \sum_{k \le q_u(t)} |M(t/k)| \quad (u < t < v). \tag{S18.3}$$

Thus the future excursion is constrained by the past before u together with the no-crossing inequalities and the closing equation $M(v) = 0$. Remark 11.34 expresses the same fact as a linear sign cone for the divisor recursion. This is the preferred route to a sublinear first-passage theorem: first characterize the feasible geometry of one complete excursion, then extend it to all of M by the exact segmentation into completed excursions and zero runs.

A concrete target is to fix $\theta_{\mathrm{sub}} < 1$ and a Littlewood-scale level $u^{1/2+\varepsilon}$, and to show that the relaxed sign cone of a completed excursion cannot keep

$$1 \le \sigma M(u+r) < u^{1/2+\varepsilon} \qquad (1 \le r \le u^{\theta_{\mathrm{sub}}}) \tag{S18.4}$$

while also satisfying the later no-crossing constraints and the return $M(v) = 0$, unless some explicitly identified arithmetic constraint on the past fails. Because all path inequalities are linear in the relaxed past, Farkas duality or related convex certificates provide a natural finite framework: either the linear relaxation already forces an early high level, or a dual feasible obstruction identifies exactly which arithmetic rows, zero-mask constraints or scale relations are missing. This route retains the complete excursion rather than compressing it to a single energy.

The corresponding first-passage target is now explicit. For some $\varepsilon_{\mathrm{L}} > 0$ and $1/2 + \varepsilon_{\mathrm{L}} < \theta_{\mathrm{sub}} < 1$, one would need an unbounded sequence of excursion starts satisfying

$$\sum_{r \le u^{\theta_{\mathrm{sub}}}} M(u+r)^2 > u^{\theta_{\mathrm{sub}}+1+2\varepsilon_{\mathrm{L}}}. \tag{S18.5}$$

Lemma 9.20 then forces a value of size $u^{1/2+\varepsilon_{\mathrm{L}}}$ inside the sublinear prefix. What is not proved is the production of this *prefix* energy from a later KP witness; local persistence around a late peak does not transport itself back to the start of the excursion. Equation (9.43) identifies the exact signed correlation-plus-boundary quantity that would have to be controlled, and the boundary vanishes only when the complete excursion is used.

The centered record functionals remain useful diagnostics of witness position. In particular,

$$\mathfrak{C}(u,H) = -\sum_{j<H} \frac{1}{a_j} \sum_{g<L_j} g C^{\circ}_{j,g} \tag{S18.6}$$

satisfies both the finite bootstrap

$$\log \frac{t_H}{u+H} \le 4\mathfrak{C}(u,H) \tag{S18.7}$$

and, in the genuinely linear first-arrival regime,

$$\mathfrak{C}(u,H) \ge \frac{1}{3}\log\frac{t_H}{u} + \frac{t_H - u}{2t_H} - o(1). \tag{S18.8}$$

For a sublinear Littlewood-level first passage the required scale is much smaller:

$$\mathfrak{C}(u,H_{\mathrm{L}}) \ll u^{\theta_{\mathrm{sub}}-1}. \tag{S18.9}$$

The post-horizon quantity $\mathcal{W}_\lambda$ must also be interpreted through its intrinsic factorization

$$\mathcal{W}_\lambda = \frac{R}{u}\left(\frac{D_\lambda}{H}\right)^2 \mathcal{E}_\lambda; \tag{S18.10}$$

small $\mathcal{W}_\lambda$ may mean a short remaining segment rather than an almost affine rise. The three-source logarithmic budget remains a useful packing identity, but none of these scalar observables supplies the missing sublinear exponent by itself.

A second route retains all dyadic Fejér scales of a completed excursion simultaneously. Proposition 12.25 uses the one-sign bridge condition to close the zero-padded path into a cycle with a favorable boundary sign and proves

$$H^2 \le m_{\mathrm{dy}}(m_{\mathrm{dy}} q + 2\mathfrak{K}_I). \tag{S18.11}$$

Thus, under a violation of RH, the linearly short KP witness excursions selected by the counting argument must satisfy the superlinear lower bound (12.72). This identifies a single signed collective quantity spanning all dyadic scales. The missing arithmetic input is exactly the quasi-linear upper target (12.73); it is not supplied by the classical Fejér identity or by the present short-interval estimates.

The logarithmic phase picture has a more limited role. A formal global zero mode contributes $k^{-\rho} = k^{-\beta}e^{-i\gamma\log k}$ to the kth channel, and Euler–Maclaurin makes the rotation $e^{-i\gamma\log K}$ explicit. This explains the transition $K^{1-\beta} \asymp |\gamma|$ in a truncated zero-mode channel sum, but it is global zero analysis, not an additional constraint on a completed excursion. In particular, $\sum k^{-\rho} = \zeta(\rho) = 0$ is not an ordinary convergent series in the critical strip; it is an analytic-continuation statement. The useful version inside the method is instead Proposition 11.35, which is a finite transform of the actual channel vector of one excursion. The weighted polynomial

$$\mathfrak{P}_K(\varsigma,\tau;A) = \sum_{k=2}^{K} A_k k^{-\varsigma} e^{-i\tau\log k} \tag{S18.12}$$

can test signed channel cancellation for coefficients A_k generated by the excursion. A square-function estimate for these structured coefficients would be genuinely new, but Remark 11.38 records why no such estimate is available for arbitrary coefficients or frequencies.

The global explicit-formula route is therefore retained only as a diagnostic. At the smoothing orders used by Kaczorowski–Pintz the spectral denominator suppresses nonleading singularities; returning to bounded smoothing order exposes the full residue structure (for simple zeros, coefficients of the shape $1/(\rho\zeta'(\rho))$). The present excursion programme deliberately does not assume control of those coefficients. Its next proof target is instead local and deterministic: characterize the complete no-crossing excursion from its past and its own early part strongly enough to force a sublinear first passage, and only afterwards use the global excursion segmentation to recover a statement for all M.

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